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Collective Dynamics of Colliding Inertial Particles in Prescribed Carrier Flows

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Abstract

Particle-laden flows are inherently multiscale: particles obey local mechanical laws, yet their collective dynamics can generate strongly organised spatial structures. This thesis examines how particle inertia and finite-size interparticle collisions interact with carrier-imposed organisation, progressing from numerical hard-particle modelling and controlled carrier flows to a reduced near-wall turbulent configuration.

First, a timestep-free, event-scheduled hard-disk solver is developed as a force-free reference under its stated assumptions. A forced kick–drift solver then combines continuous Stokes relaxation with chronologically resolved hard-particle contacts. Analytical verification against single-particle reference solutions and timestep-refinement studies support the numerical reliability of the implementations within the tested settings and identify the resolution range used in the subsequent physical analyses.

Next, an alternating sinusoidal shear flow is used to isolate regime-dependent collision effects. At high inertia, exactly elastic collisions suppress the strong spatial inhomogeneity of the collisionless population, whereas at lower inertia the same collision law generates substantial banded inhomogeneity from an approximately homogeneous collisionless state. This reversal of the collision response therefore does not require collisional energy dissipation. A strongly inelastic case instead develops pronounced late-time aggregation without a detected statistically stationary state, whereas a regime in which carrier-induced separation acts faster than successive collisions remains approximately homogeneous despite collision impacts. The population-level effect of collisions therefore depends on the dynamical regime rather than on the collision law alone.

The **principal investigation** concerns a reduced wall-parallel particle model driven by a time-resolved carrier field extracted from a three-dimensional turbulent channel-flow simulation. Matched collision-on and collision-off calculations show that the carrier field establishes the dominant streamwise particle organisation and the anisotropic near-contact population from which collisions are selected. Collisions do not create this preferred direction; instead, they redistribute particle organisation across scales, weakening short-range pair-axis alignment, reducing local particle-number fluctuations, and increasing streamwise density coherence. At the event scale, contact produces an immediate radial rebound. Despite this locally dispersive response, same-pair recurrence is common and predominantly local in resolved separation. The prevalence of local recurrence increases with pair multiplicity, consistent with recurrent sampling of carrier-organised configurations that favour slow separation or renewed approach and thereby provide repeated opportunities for contact. This persistence is pair local rather than indicative of a stable many-particle cage: the wider particle neighbourhood continues to turn over.

Overall, the results support a common physical picture: *carrier-imposed organisation followed by collision-induced, regime-dependent redistribution*. The carrier flow establishes the organised particle population on which collisions act, while finite-size collisions modify the strength, spatial scale, and subsequent evolution of that organisation.

Chapter 1

Introduction

1.1 Motivation and Physical Context

Particle-laden flows arise in atmospheric clouds, environmental transport, sprays, combustion, and industrial processing [1–3]. Their collective behaviour cannot be inferred from the trajectory of a representative particle alone. Spatial concentration, encounter frequency, collision occurrence, and repeated contact depend on how the particle population organises within the carrier flow.

Finite particle response time due to inertia provides one source of this organisation [2, 4, 5]. Inertial particles cannot adjust instantaneously to changes in the velocity of the local carrier flow, so their trajectories depart from fluid pathlines. Consequently, non-uniform particle distributions can develop even in an incompressible carrier. The carrier flow also influences where the particles accumulate, which pairs become available at small separation, and which relative motions bring them towards contact.

Finite particle size introduces a second mechanism. When two particles collide, the resulting impulse changes their relative velocity over a time scale much shorter than that associated with transport by the surrounding carrier flow. Although the interaction is local and nearly instantaneous, its consequences need not remain local: a collision modifies the subsequent trajectories of the participating particles, changes the configurations available for later encounters, and can cumulatively redistribute the particle population [6, 7].

Carrier-induced organisation and collision dynamics must therefore be considered together [8, 9]. Collisions are selected from a particle population already organised by inertial transport and therefore potentially non-uniform and, in anisotropic carrier environments such as near-wall turbulence, directionally structured. Conversely, collision impulses alter the trajectories from which subsequent approach, recurrence, and collective organisation emerge. The central physical problem of this thesis is thus to determine how local hard-particle interactions modify collective organisation across larger spatial and temporal scales.

1.2 Broader Research Programme and Thesis Aim

The broader research programme concerns the prediction and control of collective behaviour in finite-size particle-laden flows. Its long-term objectives are:

1. to construct a microscopic model for inertial particles subject to continuous forcing and instantaneous hard-particle collisions;
2. to determine how microscopic particle dynamics are expressed at the population level through concentration, collision, and transport statistics in controlled carrier-flow regimes;

3. to extend the particle-level description to turbulent environments and identify the physical ingredients required for future reduced-order or stochastic representations;
4. to develop machine-learning strategies for controlling particle concentration, clustering, and collective transport; and
5. to use Bayesian optimisation to investigate how particle shape and flexibility can be selected to influence preferential sampling and transport, first in idealised environments and subsequently in turbulent flows [10, 11].

The complete programme extends beyond the scope of one Master's thesis. The present work therefore concentrates on the first three objectives. It develops the microscopic hard-particle framework, examines collision-modified collective behaviour in a controlled alternating sinusoidal shear flow, and applies the same particle model to a reconstructed wall-parallel carrier field from a turbulent channel-flow DNS. Machine-learning assisted control, and optimisation of non-spherical particle shapes remain prospective directions.

Aim of the present thesis. The immediate aim of this thesis is to determine how finite-size interparticle collisions modify collective organisation generated by inertial transport, and how local contact dynamics contribute to population-level redistribution. This aim is pursued through a connected progression: the hard-particle numerical framework establishes the mechanical basis; the alternating sinusoidal shear isolates regime-dependent collision effects; and the near-wall turbulent configuration provides the principal investigation, where carrier organisation, collision selection, local response, same-pair recurrence, and cumulative redistribution are examined together.

The central thesis question is therefore:

> *How do finite-size interparticle collisions modify the collective organisation generated by inertial transport, from controlled carrier-flow regimes to near-wall turbulence?*

Specific research questions addressed within this thesis. The central question is addressed through three linked thesis questions (TQ1–TQ3), progressing from numerical representation, through controlled collective behaviour, to the principal near-wall turbulent problem.

TQ1 — Numerical representation. *How can finite-size inertial particles subject to continuous forcing and instantaneous hard-particle collisions be simulated while preserving the chronological resolution of first-contact events?*

Chapter 2 addresses TQ1 by developing a common hard-particle formulation, a timestep-free event-driven solver for the force-free hard-disk problem, and a forced kick–drift solver with chronological collision resolution within each ballistic drift.

TQ2 — Controlled-flow collision regimes. *How do finite-size collisions modify collective particle organisation across contrasting regimes of particle inertia, restitution, and carrier-induced separation in an alternating sinusoidal shear flow?*

Chapter 3 addresses TQ2 through particle-level collision and collisionless simulations, full box-occupancy distributions, supporting scalar diagnostics, and qualitative comparison with published regimes. Particular attention is given to the reversed ordering in which the same elastic collision law enhances spatial inhomogeneity at low inertia but suppresses it at high inertia.

TQ3 — Near-wall turbulent collision dynamics. *In near-wall turbulence, how does carrier-imposed organisation determine pair availability and collision selection, and how do local impacts contribute to same-pair recurrence and population-level redistribution?*

Chapter 4 addresses TQ3 through conditioned analyses of carrier organisation, environmental sampling, pair availability, inward-motion supply, collision conversion, event-centred response, same-pair recurrence, neighbourhood turnover, and cumulative redistribution. TQ3 constitutes the principal scientific question of the thesis.

1.3 Principal Contributions

The principal contributions follow the progressive structure of the thesis.

1. **A numerically assessed foundation for forced hard-particle dynamics.**

   In Chapter 2, I developed and implemented a common computational framework for finite-size particles subject to continuous forcing and instantaneous hard-particle collisions. The framework comprises two complementary numerical methods: a timestep-free, event-scheduled solver for the force-free hard-particle problem, used as a reference under its stated assumptions, and a forced kick–drift solver with chronological localisation of contacts within each ballistic drift. I assessed these methods through conservation diagnostics, comparisons with analytical single-particle solutions, and regime-specific timestep refinement. These tests establish the numerical scope within which collision-associated differences are interpreted in the subsequent physical analyses.

2. **A controlled-flow demonstration of regime-dependent collision effects.** In Chapter 3, I designed and analysed collision and collisionless simulations in an alternating sinusoidal shear flow. I varied particle response, restitution, and carrier parameters, thereby examining regimes with different collision frequencies and carrier-induced separation times. I quantified the resulting population organisation using particle configurations, full box-occupancy distributions, the variance-to-mean ratio (VMR), and stationarity diagnostics. I compared the resulting regimes qualitatively with those reported by López and Puglisi [8, 9], while accounting for the different numerical treatment of collisions. The simulations demonstrate that the same exactly elastic collision law can enhance spatial inhomogeneity at low inertia while suppressing it at high inertia.

3. **A conditioned multiscale analysis of finite-size collision dynamics in near-wall turbulence.**

   This constitutes the principal original contribution of the thesis. In Chapter 4, I reconstructed a time-resolved wall-parallel carrier field from turbulent-channel simulations and performed matched collision and collisionless particle calculations from the same labelled initial state. I have developed diagnostics for carrier sampling, particle-density organisation, pair anisotropy, collision-angle selection, inward-motion exposure, environment-conditioned collision conversion, event-centred pair response, same-pair recurrence and neighbourhood turnover.

   These diagnostics distinguish the carrier-organised encounter population from the configurations selected for collision and from the subsequent response to contact. The analysis shows that the carrier establishes the dominant streamwise particle architecture, while collisions redistribute particle organisation across spatial scales within this pre-existing anisotropic structure.

Taken together, these contributions establish a consistent framework linking local hard-particle contact to population-level organisation. The central outcome is that collisions do not act independently of the state organised by the carrier flow: they modify a pre-existing particle organisation, and their cumulative effect depends on particle response, encounter dynamics, and spatial scale. This provides the unifying physical interpretation of the thesis.

Chapter 2

Numerical Methods for Collisional Particle Dynamics

2.1 Introduction

The particle systems considered in this thesis have a *hybrid* dynamical structure. Between contacts, particle positions and velocities evolve continuously, whereas at discrete contact times the velocities undergo instantaneous jumps through impulsive momentum exchange. In the force-free limit, the inter-collision motion reduces to ballistic free flight. Under fluid drag or other continuous forcing, however, particle velocities evolve between impacts and the trajectories are no longer ballistic. The numerical task is therefore to combine continuous forced transport with discrete hard-particle impacts while preserving the first-contact geometry and impulsive collision law.

Two standard numerical paradigms address complementary aspects of this problem. A *time-driven* method advances the complete particle system through prescribed time increments and therefore accommodates continuous forcing naturally [12]. For hard particles, however, contacts occur asynchronously and over a negligible physical duration, so their localisation and resolution within each timestep require particular care. Instead, an *event-driven* method exploits the analytically known inter-collision trajectories between events: candidate contact times are predicted directly and the simulation clock advances to the earliest admissible event. For force-free hard disks, the relative motion is linear and the first-contact condition reduces to a quadratic equation, making the event-driven formulation particularly well-suited for the contact mechanics.

The difficulty addressed in this thesis is that the physical systems of interest are not generally force free. Coupling particles to a spatially and temporally varying carrier flow removes the global ballistic structure required for direct event scheduling. So a fully event-driven formulation would require first-contact prediction along forced trajectories, for which a closed algebraic solution is generally unavailable. Conversely, a conventional time-driven update without explicit contact localisation may represent first contact only through an end-of-step overlap. The adopted kick–drift strategy separates these difficulties through two stages: by advancing continuous forcing during *kick* stages and resolving transport and hard-particle contacts during finite force-free *drift* intervals. Motion within each drift is piecewise ballistic, allowing first-contact times to be determined algebraically and processed chronologically. Therefore, the method is time-driven with respect to the forcing but event-resolved within each drift; the timestep then controls the coupling between forcing and transport rather than prescribing the resolved collision times.

The remainder of the chapter first establishes the mechanical operations shared by both solvers—first-contact prediction, ballistic transport, and impulsive collision response—before presenting their distinct time-advancement strategies. The event-scheduled implementation (Section 2.4) adapted from Lubachevsky [13], provides a timestep-free treatment of the force-free hard-disk problem within its stated

scope. The kick–drift formulation (section 2.5) then incorporates continuous forcing while retaining the same contact geometry and collision law within each ballistic drift. These common mechanical and diagnostic conventions provide a consistent numerical foundation for the controlled-flow and near-wall turbulent analyses developed in Chapters 3 and 4.

2.2 Two-Dimensional Hard-Particle Model

The two numerical engines introduced in Section 2.1 use different time-advancement strategies but they represent the same underlying mechanical system. This section defines the particle geometry, continuous dynamics between collisions, instantaneous binary collision law with different elasticities, and boundary conditions independently of either numerical method. These physical definitions form the basis of the microscopic model used throughout the thesis. Problem-specific forcing, carrier fields, and parameter choices are introduced in the corresponding physical chapters.

2.2.1 Particle State and Admissible Configurations

Consider N rigid disks in the two-dimensional rectangular domain

$$\Omega = [0, L) \times [0, W). \tag{2.2.1}$$

where L and W denote the domain length and width, respectively. For a square domain, $W = L$, so that $\Omega = [0, L)^2$. Each particle i is characterised by the position of its centre $\mathbf{x}_i(t) \in \Omega$, translational velocity $\mathbf{v}_i(t) \in \mathbb{R}^2$, radius $a_i > 0$, and mass $m_i > 0$. When Stokes forcing is present, particle i is additionally assigned a response time $\tau_{p,i} > 0$.

The particles are rigid, smooth, non-deformable, and non-rotating. Finite size is represented through excluded-volume geometry, and contacts transmit only normal impulses along the line joining the particle centres. For the unordered pair $i < j$, the centre-to-centre contact distance is given by:

$$\sigma_{ij} = a_i + a_j. \tag{2.2.2}$$

For monodisperse particles of common radius a,

$$a_i = a, \qquad \sigma_{ij} = \sigma = 2a. \tag{2.2.3}$$

Let $\mathbf{r}_{ij}$ denote the displacement from particle j to particle i, evaluated according to the applicable boundary convention (see 2.2.4), and define the pair separation as:

$$r_{ij} = \left\|\mathbf{r}_{ij}\right\|. \tag{2.2.4}$$

For periodic domains, $\mathbf{r}_{ij}$ is the minimum-image displacement defined in Section 2.2.4. The hard-particle constraint is $r_{ij} \geq \sigma_{ij}$, $i \neq j$. Equivalently, the surface-to-surface gap

$$h_{ij} = r_{ij} - \sigma_{ij} \tag{2.2.5}$$

must satisfy $h_{ij} \geq 0$. Thus $h_{ij} > 0$ denotes a separated pair, $h_{ij} = 0$ denotes geometric contact, and $h_{ij} < 0$ corresponds to a pair overlapping each other. The unit pair normal is defined throughout the thesis as

$$\mathbf{n}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}}, \qquad \mathbf{n}_{ji} = -\mathbf{n}_{ij}, \tag{2.2.6}$$

so that $\mathbf{n}_{ij}$ points from particle j towards particle i. At contact, this orientation defines the normal used in the impulsive collision law.

2.2.2 Continuous Dynamics Between Collisions

Between collision events, the position and velocity of particle i evolve according to

$$\frac{\mathrm{d}\mathbf{x}_i}{\mathrm{d}t} = \mathbf{v}_i, \qquad \frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = \mathbf{f}_i^{\mathrm{cont}}\big(\mathbf{x}_i, \mathbf{v}_i, t\big), \qquad i = 1, \ldots, N, \tag{2.2.7}$$

where $\mathbf{f}_i^{\mathrm{cont}}$ denotes the finite-duration, non-impulsive acceleration acting between contacts. Collision impulses are excluded from Eq. (2.2.7) and are applied through the instantaneous velocity map of Section 2.2.3. The acceleration may be smooth or piecewise smooth; the term *continuous dynamics* distinguishes this finite-duration evolution from the instantaneous contact response.

Force-free specialisation. When $\mathbf{f}_i^{\mathrm{cont}} = \mathbf{0}$, the velocities remain constant between interactions and the motion is ballistic:

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \Delta t\, \mathbf{v}_i(t), \qquad \mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t). \tag{2.2.8}$$

This free-flight map is used between successive events in the event-scheduled force-free solver and within each force-free drift of the forced kick–drift solver.

Forced specialisation. When $\mathbf{f}_i^{\mathrm{cont}} \neq \mathbf{0}$, the particle velocity evolves between collisions and the trajectory is no longer ballistic in general. The continuous forcing and its numerical subflows are specified in Section 2.5; the carrier fields and problem-specific assumptions are introduced in the corresponding physical chapters.

2.2.3 Instantaneous Frictionless Binary Collisions

When an approaching pair reaches contact, the continuous between-collision evolution is replaced instantaneously by the hard-particle collision map. The pair separation r_{ij}, contact distance σ_{ij}, and unit normal $\mathbf{n}_{ij}$ are those defined in Section 2.2.1. Geometric contact occurs when $r_{ij} = \sigma_{ij}$.

Let superscripts $-$ and $+$ denote the particle states immediately before and after impact, respectively. The pre-impact relative velocity is

$$\mathbf{g}_{ij}^{-} = \mathbf{v}_i^{-} - \mathbf{v}_j^{-}, \tag{2.2.9}$$

with normal component

$$g_{n,ij}^{-} = \mathbf{g}_{ij}^{-} \cdot \mathbf{n}_{ij}. \tag{2.2.10}$$

Because $g_{n,ij}^{-} = \dot{r}_{ij}^{-}$, a contacting pair is approaching when

$$g_{n,ij}^{-} < 0. \tag{2.2.11}$$

Only an approaching contact receives a collision impulse. Within the smooth, frictionless contact model adopted here, a separating contact, $g_{n,ij}^{-} > 0$, and an exactly grazing contact, $g_{n,ij}^{-} = 0$, receive no impulsive update. Contact models incorporating surface roughness, tangential friction, or particle rotation lie outside the present scope.

Since the particles are smooth and frictionless, the collision impulse acts only along the contact normal. Introducing the reduced mass

$$\mu_{ij} = \frac{m_i m_j}{m_i + m_j}, \tag{2.2.12}$$

the impulse applied to particle i is

$$\mathbf{J}_{ij} = -(1 + e)\, \mu_{ij}\, g_{n,ij}^{-}\, \mathbf{n}_{ij}, \qquad 0 \le e \le 1, \tag{2.2.13}$$

where e is the normal coefficient of restitution. Particle j receives the equal and opposite impulse.

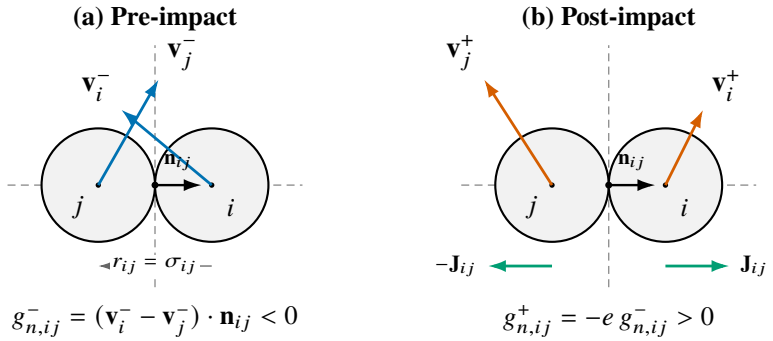

Figure 2.1: **Geometry and sign convention for an isolated frictionless binary collision.** The unit normal $\mathbf{n}_{ij}$ points from particle j towards particle i, and the geometric contact occurs at $r_{ij} = \sigma_{ij}$. An approaching pair satisfies $g_{n,ij}^- < 0$. The equal and opposite normal impulses $\pm\mathbf{J}_{ij}$ conserve the total momentum of the pair, leave the tangential relative velocity unchanged, and impose $g_{n,ij}^+ = -e\, g_{n,ij}^-$. The arrows illustrate an equal-mass collision with $e = 0.7$; the dashed vertical line denotes the tangent at contact.

Particle positions remain continuous through the impact,

$$\mathbf{x}_i^+ = \mathbf{x}_i^-, \qquad \mathbf{x}_j^+ = \mathbf{x}_j^-, \tag{2.2.14}$$

while the velocities undergo the instantaneous update

$$\begin{aligned} \mathbf{v}_i^+ &= \mathbf{v}_i^- + \frac{\mathbf{J}_{ij}}{m_i}, \\ \mathbf{v}_j^+ &= \mathbf{v}_j^- - \frac{\mathbf{J}_{ij}}{m_j}. \end{aligned} \tag{2.2.15}$$

Substituting Eq. (2.2.13) into Eq. (2.2.15) gives

$$\begin{aligned} \mathbf{v}_i^+ &= \mathbf{v}_i^- - (1+e)\frac{m_j}{m_i + m_j} g_{n,ij}^- \mathbf{n}_{ij}, \\ \mathbf{v}_j^+ &= \mathbf{v}_j^- + (1+e)\frac{m_i}{m_i + m_j} g_{n,ij}^- \mathbf{n}_{ij}. \end{aligned} \tag{2.2.16}$$

The normal relative velocity after impact therefore satisfies

$$g_{n,ij}^+ = -e\, g_{n,ij}^-. \tag{2.2.17}$$

Because no tangential impulse is applied, the tangential relative velocity

$$\mathbf{g}_{t,ij} = \mathbf{g}_{ij} - g_{n,ij}\mathbf{n}_{ij} \tag{2.2.18}$$

is unchanged: $\mathbf{g}_{t,ij}^+ = \mathbf{g}_{t,ij}^-$. Thus, for $e > 0$, the collision reverses and rescales the approaching normal relative velocity while leaving the tangential component unchanged. For $e = 0$, the post-impact normal relative velocity vanishes. No tangential contact impulse, surface friction, or particle rotation is included.

The equal and opposite impulses conserve the linear momentum of the colliding pair:

$$m_i \mathbf{v}_i^+ + m_j \mathbf{v}_j^+ = m_i \mathbf{v}_i^- + m_j \mathbf{v}_j^-. \tag{2.2.19}$$

The corresponding change in pair translational kinetic energy is

$$E_{\mathrm{kin}}^+ - E_{\mathrm{kin}}^- = -\frac{1}{2}\mu_{ij}\left(1 - e^2\right)\left(g_{n,ij}^-\right)^2 \le 0. \tag{2.2.20}$$

Hence $e = 1$ gives an elastic collision, whereas $0 \leq e < 1$ dissipates normal relative kinetic energy while preserving pair momentum. For equal masses, $m_i = m_j = m$, the velocity update reduces to

$$\begin{aligned} \mathbf{v}_i^+ &= \mathbf{v}_i^- - \frac{1+e}{2} g_{n,ij}^- \mathbf{n}_{ij}, \\ \mathbf{v}_j^+ &= \mathbf{v}_j^- + \frac{1+e}{2} g_{n,ij}^- \mathbf{n}_{ij}. \end{aligned} \tag{2.2.21}$$

In the radial-relative-velocity notation used later,

$$w_r = \mathbf{g}_{ij} \cdot \mathbf{n}_{ij} = g_{n,ij}, \tag{2.2.22}$$

so that

$$w_r^+ = -e\, w_r^-. \tag{2.2.23}$$

For the positive restitution coefficients considered in the collisional particle calculations, contact therefore produces an immediate radial rebound. This statement concerns the instantaneous collision response only. The collision law above describes an isolated binary event. Simultaneous multiparticle contacts and rapidly accumulating collision sequences require additional numerical treatment and are therefore discussed with the collision-resolution algorithms rather than included in the physical binary-contact law. In the entire analysis, the particle contacts are modelled as instantaneous, frictionless binary impacts. The contact duration is assumed negligible relative to the resolved between-collision time scales. Particle positions therefore remain continuous through impact, whereas their velocities may undergo finite jumps.

2.2.4 Boundary Conditions

The particle model admits periodic boundaries and reflecting rigid walls. Periodic boundaries are used in the particle-laden-flow calculations developed later in the thesis. Reflecting walls are retained as part of the general hard-particle model.

Periodic boundaries. For a rectangular periodic domain, $\Omega_{\text{per}} = [0, L) \times [0, W)$, the particle coordinates are mapped into the fundamental cell according to

$$x_i \leftarrow x_i - L \left\lfloor \frac{x_i}{L} \right\rfloor, \qquad y_i \leftarrow y_i - W \left\lfloor \frac{y_i}{W} \right\rfloor. \tag{2.2.24}$$

A periodic boundary crossing changes only the coordinate representation of the particle. It does not produce an impulse, and the particle velocity is unchanged. Wrapped particle coordinates alone do not determine the shortest displacement between a pair separated across a periodic boundary. Define the coordinate differences

$$\Delta x_{ij} = x_i - x_j, \qquad \Delta y_{ij} = y_i - y_j. \tag{2.2.25}$$

The corresponding minimum-image components [14] are

$$\Delta x_{ij}^{\text{mi}} = \Delta x_{ij} - L\ \text{round}\left(\frac{\Delta x_{ij}}{L}\right), \Delta y_{ij}^{\text{mi}} = \Delta y_{ij} - W\ \text{round}\left(\frac{\Delta y_{ij}}{W}\right), \tag{2.2.26}$$

where round returns the nearest integer. The displacement from particle j to particle i is therefore

$$\mathbf{r}_{ij}^{\text{mi}} = \begin{pmatrix} \Delta x_{ij}^{\text{mi}} \\ \Delta y_{ij}^{\text{mi}} \end{pmatrix}. \tag{2.2.27}$$

In a periodic domain, $\mathbf{r}_{ij}^{\text{mi}}$ is used in every instantaneous pair quantity defined in Section 2.2.1:

$$\mathbf{r}_{ij} = \mathbf{r}_{ij}^{\text{mi}}, \qquad r_{ij} = \left\| \mathbf{r}_{ij}^{\text{mi}} \right\|. \tag{2.2.28}$$

The surface gap, contact normal, approach condition, and collision impulse are then evaluated from this minimum-image geometry.

The nearest periodic image is unique except when a displacement component is exactly one half of the corresponding box length. This degenerate case does not alter the physical boundary convention. The additional periodic-image rules required to predict future contacts over a finite ballistic interval are stated with the corresponding contact-search algorithms in Sections 2.3–2.5.

Reflecting rigid walls. For reflecting boundaries, the physical rectangle is $\Omega_{\mathrm{wall}} = [0, L] \times [0, W]$. Because the particles have finite radii, the centre of particle i must remain in the admissible region $a_i \le x_i \le L - a_i$, $a_i \le y_i \le W - a_i$.

Let $\mathbf{n}_w$ denote the outward unit normal of the wall reached by the particle, and let $v_{n,i}^- = \mathbf{v}_i^- \cdot \mathbf{n}_w$ denote the outward normal component of the pre-impact velocity. A particle approaching the wall from inside the domain therefore satisfies $v_{n,i}^- > 0$. The particle position is continuous at impact. For a wall restitution coefficient $e_w \in [0, 1]$, the post-impact velocity is

$$\mathbf{v}_i^+ = \mathbf{v}_i^- - (1 + e_w)\big(\mathbf{v}_i^- \cdot \mathbf{n}_w\big)\mathbf{n}_w, \tag{2.2.29}$$

so that $v_{n,i}^+ = -e_w v_{n,i}^-$, while the tangential velocity remains unchanged. Thus, $e_w = 1$ gives an elastic reflection, whereas $0 \le e_w < 1$ dissipates wall-normal kinetic energy.

Under periodic boundaries and force-free motion, particle–particle collisions leave the total particle momentum unchanged, as follows from the pair-momentum balance in Eq. (2.2.19). A reflecting wall instead supplies an external impulse to the particle population, so the particle momentum alone is not conserved in a wall-bounded calculation. This reflecting-wall model is distinct from the near-wall turbulent configuration of Chapter 4. There, particle motion is restricted to a periodic wall-parallel plane at a prescribed wall-normal position. The physical channel wall determines the sampled carrier field and the imposed particle height, but it does not act as a reflecting boundary for the reduced in-plane particle dynamics.

2.3 Contact Prediction, Transport, and Collision Response

Section 2.2 defines the admissible particle configurations and the velocity change associated with a binary impact. The present section specifies how these laws are executed during a force-free interval. Both numerical solvers use the same three operations:

1. predict the first admissible particle–particle or particle–wall contact;
2. transport the relevant particles ballistically to that time;
3. apply the corresponding collision response.

The event-driven solver uses these operations throughout the force-free simulation. The forced solver reuses them within each ballistic drift interval. The operations therefore form a common mechanical layer; the two solvers differ only in how candidate pairs are generated, how events are scheduled, and how ballistic motion is combined with continuous forcing.

2.3.1 First-Contact Prediction During Ballistic Motion

Consider particles i and j at a common reference time t_0. During a force-free interval, their velocities are constant, and their relative displacement after a non-negative lag τ is

$$\mathbf{r}_{ij}(\tau) = \mathbf{r}_{ij}^0 + \tau \mathbf{g}_{ij}, \qquad \mathbf{g}_{ij} = \mathbf{v}_i - \mathbf{v}_j, \tag{2.3.1}$$

where $\mathbf{r}_{ij}^0$ is the boundary-consistent pair displacement at t_0.

A geometric contact occurs when

$$\left\|\mathbf{r}_{ij}(\tau)\right\| = \sigma_{ij}. \tag{2.3.2}$$

Using $\mathbf{r}_{ij}(\tau) = \mathbf{r}_{ij}^0 + \tau \mathbf{g}_{ij}$, the contact condition becomes

$$A_{ij}\tau^2 + 2B_{ij}\tau + C_{ij} = 0, \tag{2.3.3}$$

with

$$A_{ij} = \|\mathbf{g}_{ij}\|^2, \qquad B_{ij} = \mathbf{r}_{ij}^0 \cdot \mathbf{g}_{ij}, \qquad C_{ij} = \|\mathbf{r}_{ij}^0\|^2 - \sigma_{ij}^2. \tag{2.3.4}$$

For an initially separated pair, $C_{ij} > 0$. A future contact requires $A_{ij} > 0$, $B_{ij} < 0$, and

$$\mathcal{D}_{ij} = B_{ij}^2 - A_{ij}C_{ij} \geq 0. \tag{2.3.5}$$

The first contact is given by the smaller root,

$$\tau_{ij}^\star = \frac{-B_{ij} - \sqrt{\mathcal{D}_{ij}}}{A_{ij}}. \tag{2.3.6}$$

For $\mathcal{D}_{ij} > 0$,

$$g_{n,ij}^{\star,-} = -\frac{\sqrt{\mathcal{D}_{ij}}}{\sigma_{ij}} < 0, \tag{2.3.7}$$

so the first root corresponds to an approaching contact. When $\mathcal{D}_{ij} = 0$, the trajectories are tangent and no collision impulse is applied.

If the pair is already at contact, $C_{ij} = 0$, an immediate collision is accepted only when $\mathbf{g}_{ij} \cdot \mathbf{n}_{ij} < 0$, in which case $\tau_{ij}^\star = 0$. Separating contacts receive no further impulse, while $C_{ij} < 0$ indicates an overlap and is handled separately by the numerical safeguards. In a time-driven solver with a finite drift interval Δt_{rem}, the pair contact is retained only when $0 \leq \tau_{ij}^\star \leq \Delta t_{\mathrm{rem}}$. In the event-driven solver, no finite drift horizon is imposed; the earliest admissible future event among the scheduled candidates is selected instead.

Periodic and wall contacts. In a periodic domain, contact prediction is carried out using the periodic images supplied by the corresponding solver. For each admissible image, the contact quadratic in Eq. (2.3.3) is evaluated, and the smallest admissible root is retained. The associated periodic image then fixes the pair displacement and contact normal at impact. The completeness of the image search is established separately for the event-driven and forced solvers. For reflecting walls, the contact time follows directly from ballistic motion. Along the x-direction,

$$\tau_{x,i} = \begin{cases} (L - a_i - x_i)/v_{x,i}, & v_{x,i} > 0, \\ (a_i - x_i)/v_{x,i}, & v_{x,i} < 0, \\ +\infty, & v_{x,i} = 0, \end{cases} \tag{2.3.8}$$

with the corresponding expression in y obtained by replacing $L, x_i, v_{x,i}$ with $W, y_i, v_{y,i}$. The earliest non-negative admissible value gives the next wall-contact time.

2.3.2 Ballistic Transport

Once an admissible event lag $\tau^\star$ has been identified, the system is advanced under the force-free dynamics:

$$\mathbf{x}_i(t_0 + \tau^\star) = \mathbf{x}_i(t_0) + \tau^\star \mathbf{v}_i(t_0), \qquad \mathbf{v}_i(t_0 + \tau^{\star,-}) = \mathbf{v}_i(t_0), \tag{2.3.9}$$

where the superscript $(-)$ denotes the state immediately before any collision response at the event time.

Under periodic boundaries, transported coordinates may be mapped back into the fundamental cell without changing the physical trajectory or velocity. For a predicted pair contact, the periodic image retained during contact prediction determines the corresponding contact geometry. For a reflecting-wall event, ballistic transport terminates at the predicted wall contact, after which the velocity is updated using Eq. (2.2.29).

2.3.3 Collision Response at Contact

At the predicted event time, the pair geometry and normal relative velocity are recomputed from the transported state. A particle–particle impulse is applied only if the pair is at contact within the prescribed numerical tolerance and remains dynamically approaching:

$$r_{ij}^{\star} \simeq \sigma_{ij}, \qquad g_{n,ij}^{\star,-} < 0. \tag{2.3.10}$$

The numerical tolerance used in Eq. (2.3.10) is $\varepsilon_{\mathrm{code}} = 10^{-12}$; it does not modify the particle diameter or collision law. Accepted particle–particle contacts are resolved using the impulse law of Section 2.2.3. Particle–wall contacts use the reflection law of Eq. (2.2.29). Separating or geometrically inconsistent candidates receive no collision impulse.

For $e > 0$, an accepted binary collision satisfies $g_{n,ij}^{\star,+} = -e\, g_{n,ij}^{\star,-} > 0$, so the immediate post-impact normal relative motion is separating. For $e = 0$, the post-impact normal relative velocity vanishes. Each accepted pair collision is assigned an event time t_e and particle indices i and j. The minimal event record is therefore $\mathcal{E}_e = (t_e, i, j)$. The collision time is determined by the contact-resolution procedure and is therefore distinct from quantities reconstructed from particle states stored at a finite output cadence. Any additional event-level quantities retained in a particular simulation are stated explicitly with that simulation's output protocol.

2.3.4 Shared Mechanical Operations

Table 2.1: **Mechanical operations shared by the event-driven and kick–drift solvers.** Each operation is defined by the physical model of Section 2.2; the solver sections determine how candidates are generated and how the operations are ordered.

Operation	Input	Output	Acceptance condition
First-contact prediction (`interaction_time`)	Pair or wall geometry, velocities, radii, boundary data, and optional time horizon	First admissible contact lag and associated periodic image or wall	Future approaching contact within the permitted horizon
Ballistic transport (`advance`)	Particle state and non-negative time increment	Position at the new time; unchanged pre-event velocity	Transport terminates no later than the selected event
Collision response (`jump`)	Transported contact geometry and pre-impact velocities	Post-impact velocities at unchanged positions	Geometric contact and inward normal relative velocity

The common mechanical layer of the two solvers is summarised in Table 2.1. Both use the same definitions of geometric contact and approaching motion; their differences lie in candidate generation, event scheduling, and the coupling to continuous forcing.

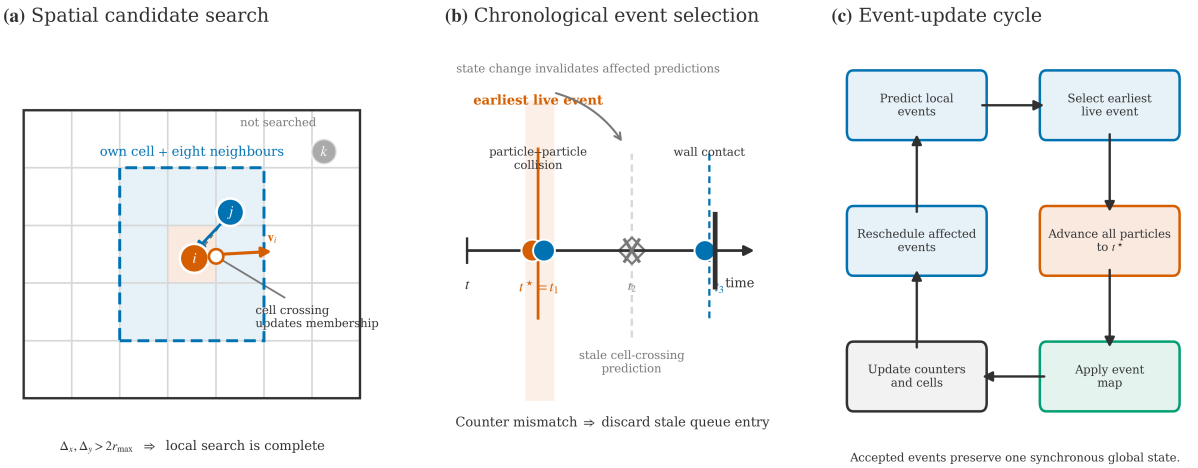

Figure 2.2: **Architecture of the event-driven solver.** **(a)** Local candidate generation from the current cell and its eight neighbours. **(b)** Chronological heap selection with counter-based lazy invalidation of stale predictions. **(c)** Synchronous advancement to the next live event, followed by the corresponding state update and local rescheduling.

2.4 Event-Driven Solver for the Force-Free Problem

I developed a force-free event-driven solver by adapting the serial billiard framework of Lubachevsky [13]. It shares the hard-particle mechanics of Sections 2.2 and 2.3 with the forced kick–drift solver: particles move ballistically between events, future contact times are predicted analytically, and particle–particle or particle–wall impacts produce instantaneous velocity jumps. The event-driven formulation is not used for the results of Chapters 3 and 4, which require continuous forcing.

Unlike a time-stepping method, the solver does not advance the particle state through increasing time steps. Instead, admissible future events are scheduled and the simulation clock advances directly to the earliest live event. Spatial sectoring restricts the generation of candidate pairs, while cell-boundary crossings are themselves treated as events so that the search neighbourhood remains consistent with particle motion.

2.4.1 Adaptation of the Lubachevsky Framework

Lubachevsky's original algorithm uses a component-local event representation: each particle retains a previously processed state and a newly scheduled state, and particles not involved in the selected event are not generally advanced to the current simulation time. The complete particle configuration is therefore reconstructed only when it is explicitly required.

The implementation developed here retains the same event-driven principles but uses a different data organisation. All particle positions are advanced synchronously to every selected event time. Future particle–particle, particle–wall, and cell-crossing predictions are stored as separate entries in a global priority queue implemented as a binary min-heap, while obsolete predictions are invalidated lazily through per-particle state counters. This synchronous heap-and-counter formulation provides a directly inspectable particle configuration after each processed event and gives the event-driven and forced solvers the same state representation. These features simplify configurational diagnostics and event logging, while providing a common state representation for the two numerical formulations. The trade-off is computational: every live event contains an $O(N)$ global operation. Consequently, the $O(\log N)$-per-event asymptotic performance derived for Lubachevsky's component-local implementation is not claimed for the algorithm mentioned in this chapter. Figure 2.2 summarises the organisation of the force-free solver into local candidate generation, chronological event selection, and synchronous state advancement.

2.4.2 Synchronous State and Spatial Sectoring

Let t denote the current simulation time and $t^\star > t$ the time of the earliest event. All particles are transported ballistically to $t^\star$,

$$\mathbf{x}_i(t^\star) = \mathbf{x}_i(t) + \left(t^\star - t\right)\mathbf{v}_i(t), \qquad \mathbf{v}_i(t^{\star,-}) = \mathbf{v}_i(t), \qquad i = 1, \ldots, N, \tag{2.4.1}$$

while their velocities remain unchanged until the event response is applied. For periodic boundaries, the transported coordinates are subsequently mapped into the fundamental cell. To restrict pair searches, the domain is partitioned into $n_x \times n_y$ rectangular cells with widths

$$\Delta_x = \frac{L}{n_x}, \qquad \Delta_y = \frac{W}{n_y}. \tag{2.4.2}$$

Let $a_{\max} = \max_{1 \le i \le N} a_i$,. The geometric completeness of the local nine-cell search requires $\Delta_x > 2a_{\max}$, $\Delta_y > 2a_{\max}$. The realised cell widths are therefore checked as run-level acceptance conditions for every calculation used as a force-free reference.

Under Eq. (2.4.2), particles that can reach contact must occupy either the same cell or neighbouring cells immediately before contact. Candidate partners of particle i are therefore sought in its own cell and the eight adjacent cells. Under periodic boundary conditions, cell indices are reduced modulo n_x and n_y, so the neighbourhood search continues consistently across the domain boundaries. A particle may leave its current search neighbourhood before its next physical collision. Cell-boundary crossings are therefore predicted and processed as events. Such an event carries no mechanical impulse: it updates the recorded cell membership and regenerates the local contact predictions associated with the particle's new neighbourhood.

2.4.3 Contact Prediction and Event Data Organisation

For each local candidate pair, the contact calculation of Section 2.3.1 is applied to the current ballistic state. Under periodic boundaries, the implementation explicitly tests the nine image displacements

$$\mathbf{r}_{ij}^{mn} = \mathbf{x}_i - \mathbf{x}_j - \begin{pmatrix} mL \\ nW \end{pmatrix}, \qquad m, n \in \{-1, 0, 1\}. \tag{2.4.3}$$

Roots that do not correspond to a future approaching contact are rejected. Among the admissible roots, the earliest is retained together with the periodic-image displacement at contact. This stored displacement is then used to compute the contact normal when that collision event is processed.

The event priority queue contains three classes of prediction: 1) particle–particle contacts; 2) particle–wall contacts, for reflecting boundaries; 3) cell-boundary crossings. Each prediction is stored as one entry in a binary min-heap. The first two fields are the absolute event time t_e and a monotonically increasing sequence number s_e; the remaining fields depend on the event type:

$$\begin{aligned} Q_e^{\text{pair}} &= \left(t_e, s_e, \text{pair}, i, j, \kappa_i^{\text{rec}}, \kappa_j^{\text{rec}}, \mathbf{r}_{ij}^\star\right), \\ Q_e^{\text{wall}} &= \left(t_e, s_e, \text{wall}, i, \mathcal{W}_i, \kappa_i^{\text{rec}}\right), \\ Q_e^{\text{cell}} &= \left(t_e, s_e, \text{cell}, i, \kappa_i^{\text{rec}}\right). \end{aligned} \tag{2.4.4}$$

Here κ_i^{rec} is the value of particle i's state counter when the prediction is generated, $\mathbf{r}_{ij}^\star$ is the retained periodic contact displacement, and $\mathcal{W}_i$ identifies the wall or walls reached by particle i. The heap is ordered lexicographically by (t_e, s_e). It therefore returns the prediction with the smallest stored time, while s_e provides a deterministic insertion-order tie break when two predictions have equal times. Heap insertion and removal require $O(\log M)$ operations for a heap containing M entries.

Each prediction stores the state counters of its participating particles and remains *counter-live* only while those counters are unchanged. Any velocity or cell-membership update increments the affected counter, thereby invalidating obsolete predictions. Stale heap entries are discarded lazily, with occasional cleanup to limit their accumulation. Further implementation details are given in Appendix A.2.

2.4.4 Chronological Event Cycle

Algorithm 1 summarises the chronological event cycle implemented in the force-free event-driven solver. For a periodic pair prediction, the stored image-consistent contact displacement is passed directly to the collision kernel. This displacement defines the contact normal, after which the normal relative velocity is evaluated and the impulse law of Section 2.2.3 is applied.

Algorithm 1: Synchronous event-driven hard-particle solver

Require: admissible initial state, final time T_{f}, live-event budget $N_{\mathrm{ev}}^{\max}$
construct the cell list and initialise particle state counters
generate the initial pair, wall, and cell-crossing predictions
set $t \leftarrow 0$
for $n_{\mathrm{ev}} = 1, \ldots, N_{\mathrm{ev}}^{\max}$ **do**
    discard stale heap entries until a counter-live entry is found
    if no counter-live entry remains **then**
        return the partial trajectory with terminal time t
    end if
    read the earliest counter-live event time $t^{\star}$
    if $t^{\star} \geq T_{\mathrm{f}}$ **then**
        advance all particles ballistically to T_{f}
        set $t \leftarrow T_{\mathrm{f}}$
        break
    end if
    advance all particles ballistically to $t^{\star}$
    wrap periodic coordinates where required
    set $t \leftarrow t^{\star}$
    reconcile geometric and recorded cell memberships
    increment counters and regenerate predictions for particles whose cell membership was repaired
    if the selected entry is a cell-crossing event **then**
        update or regenerate its cell-crossing prediction
        continue
    end if
    if the selected entry is a wall-contact prediction **then**
        apply the wall-restitution map
        increment the affected counter and reschedule the particle
    else
        recover the stored image-consistent contact displacement
        apply the pair impulse if the normal relative motion is approaching
        increment both particle counters and reschedule both particles
    end if
    record the processed wall or pair event
    perform heap cleanup when triggered by the implementation
end for
if $t < T_{\mathrm{f}}$ **then**
    classify the trajectory as incomplete
end if

2.4.5 Bookkeeping Safeguards and Event Accumulation

Because all particles are transported synchronously to each selected live-event time, the cell list is audited after every such transport. If the cell implied by a particle's current position differs from its stored membership, the stored membership is corrected, the particle's state counter is incremented, and the affected local predictions are recomputed. This operation changes neither $\mathbf{x}_i$ nor $\mathbf{v}_i$. It updates only the spatial-search bookkeeping. Since the cell-list audit is performed after transport to the selected event time, it does not by itself establish that no earlier contact was omitted. The corresponding implementation and acceptance conditions are reported in Appendix A.2.1.

For $e < 1$, inter-collision intervals may decrease rapidly and the dynamics may approach inelastic collapse. Each run is assigned a finite live-event budget $N_{\mathrm{ev}}^{\max}$. If this budget is exhausted before T_{f}, the trajectory is classified as incomplete and excluded from the force-free reference set. Budget exhaustion therefore indicates unresolved event accumulation; it does not, by itself, establish physical inelastic collapse. The solver further assumes isolated binary impacts.

2.4.6 Computational Cost and Numerical Role

Let M denote the current heap size, n_e^{stale} the number of stale entries discarded before the next counter-live event is obtained, and n_e^{new} the number of new predictions inserted during the subsequent rescheduling. Between heap cleanups, the work associated with one live event may be represented schematically as

$$C_e = O(N) + O\left[\left(1 + n_e^{\mathrm{stale}} + n_e^{\mathrm{new}}\right) \log M\right]. \tag{2.4.5}$$

The $O(N)$ contribution arises from the synchronous global transport and membership audit. The second term represents heap retrieval, lazy removal of stale predictions, and insertion of regenerated predictions. An occasional heap cleanup additionally requires an $O(M)$ scan and re-heapification.

Spatial sectoring restricts pair prediction to the current cell and its eight neighbours, so the rescheduling cost depends on the realised local occupancy rather than on an all-pairs search. The synchronous global update nevertheless retains an $O(N)$ operation at every live event. Consequently, the present implementation does not inherit the $O(\log N)$-per-event scaling of Lubachevsky's component-local formulation, and no universal asymptotic complexity is claimed for the complete solver.

2.4.7 Admissible Scope of the Event-Driven Reference

The event-driven solver is applicable to non-overlapping initial states without unresolved zero-time contacts. Initial configurations are therefore generated without overlaps or particles exactly at contact. The realised cell widths must satisfy Eq. (2.4.2), periodic contacts must be represented by the nine-image construction of Eq. (2.4.3), and the dynamics must remain within the isolated-binary-impact model.

Event records are interpreted according to the contact-time, approach-direction, periodic-image, and finite-precision conventions defined above. These conditions delimit the numerical scope of the event-driven implementation.

2.5 Forced Kick–Drift Solver

The event-driven method of Section 2.4 relies on ballistic motion between successive contacts. Continuous forcing removes that ballistic free-flight structure because particle velocities change between impacts. The forced solver therefore separates the two parts of the dynamics: continuous acceleration is applied to the particle velocities during ***kick*** stages, whereas transport and hard-particle contacts are resolved during a finite force-free ***drift*** interval. Therefore, the method is consequently time driven with respect to the forcing and event resolved within each drift interval.

For numerical validation of the solver, the continuous acceleration considered by the numerical engine is

$$\mathbf{f}_i^{\text{cont}} = \frac{\mathbf{u}(\mathbf{x}_i, t) - \mathbf{v}_i}{\tau_{p,i}} - \mu g \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}, \qquad \mathbf{v}_i \neq \mathbf{0}, \tag{2.5.1}$$

where $\mathbf{u}(\mathbf{x}, t)$ is the prescribed carrier flow velocity determined at the particle centre. $\tau_{p,i} > 0$ is the Stokes response time, and the friction constant μg determines the magnitude of an optional dry-friction acceleration . Setting $\mu = 0$ recovers the Stokes-drag-only formulation. At zero particle speed, the friction contribution is represented numerically by the clamped subflow defined in 2.5.9. It acts only during the continuous between-contact dynamics and must not be confused with tangential friction at an instantaneous particle–particle contact, which is absent from the smooth-disk collision law of Section 2.2.3. Equation (2.5.1) is introduced here only to define the numerical subproblems. The carrier fields, parameter regimes, coupling assumptions, and the decision whether dry friction is present are specified in the corresponding physical chapters.

2.5.1 Kick and Drift Maps

The forced particle dynamics combine two qualitatively different mechanisms: continuous evolution between contacts and instantaneous velocity changes at hard-particle impacts. Rather than integrating these components simultaneously, the kick–drift method separates them into two elementary evolution maps, each matched to the structure of the corresponding subproblem.

Let

$$\mathbf{Z} = (\mathbf{X}, \mathbf{V}) \tag{2.5.2}$$

denote the complete phase-space state, where $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$ and $\mathbf{V} = (\mathbf{v}_1, \ldots, \mathbf{v}_N)$.

The *kick* map

$$\mathcal{K}_h : (\mathbf{X}, \mathbf{V}) \longmapsto (\mathbf{X}, \mathbf{V}^\star) \tag{2.5.3}$$

advances the continuous velocity dynamics over a time-interval h while holding the particle positions fixed. In the present implementation this stage contains the Stokes-drag update and, where required, the dry-friction update introduced in Section 2.5. No particle transport or hard-particle collision is performed during the kick step.

The complementary *drift* map

$$\mathcal{D}_h : (\mathbf{X}, \mathbf{V}) \longmapsto (\mathbf{X}^+, \mathbf{V}^+) \tag{2.5.4}$$

advances the entire system for a finite time interval h with the continuous forcing switched off. As a result, between contacts, particle velocities are therefore constant and the trajectories are ballistic. This force-free nature of the drift stage is crucial. In the absence of continuous forcing, particle trajectories are piecewise linear and the time-to-contact problem admits the exact algebraic treatment described in Subsec. 2.3.1. Consequently, the drift operator preserves the exact hard-disk collision dynamics, while the kick operator accounts for the continuous evolution generated by drag and friction. This operator splitting therefore separates the continuous and discontinuous components of the dynamics, allowing each to be treated by the numerical method most appropriate to its structure.

The global timestep h controls how frequently continuous forcing is sampled and recombined with particle transport; it does *not* prescribe the instant at which a collision occurs. Contact times inside $\mathcal{D}_h$ are obtained independently from the ballistic first-contact problem, subject to candidate inclusion and the numerical tolerances specified below.

In our algorithm, the numerical method uses the kick–drift–kick arrangement[20]

$$\Phi_h = \mathcal{K}_{h/2} \circ \mathcal{D}_h \circ \mathcal{K}_{h/2}, \tag{2.5.5}$$

where maps act from right to left. A complete ballistic drift is therefore placed between two continuous

half-kicks. The numerical sequence is

$$\boxed{\text{continuous half-kick} \longrightarrow \text{ballistic, event-resolved drift} \longrightarrow \text{continuous half-kick}}.$$

For Stokes-only dynamics, and with carrier-field sampling arranged consistently across the two half-kicks, Eq. (2.5.5) has the usual Strang kick–drift–kick structure. This structural symmetry is not taken by itself as evidence of second-order accuracy for the complete hard-particle calculation. Carrier sampling, candidate generation, changes in collision ordering, finite event tolerances, and the treatment of additional non-smooth forcing can affect the observed convergence. The accuracy of the implemented scheme is therefore established through the timestep-refinement tests of Section 2.6. The following subsection specifies the continuous kick. The drift candidate set and chronological resolution of contacts are then developed in Sections 2.5.3 and 2.5.4, respectively.

2.5.2 Continuous Kick

During a kick, particle positions are held fixed and only the continuous velocity dynamics are advanced. For particle i, let $\mathbf{u}_i^k = \mathbf{u}(\mathbf{x}_i, t_k)$ denote the carrier velocity sampled at its fixed position at the time t_k associated with the kick. The Stokes-drag and dry-friction contributions of Eq. (2.5.1) are treated as separate velocity subflows.

Stokes-drag subflow. With $\mathbf{x}_i$ and $\mathbf{u}_i^k$ fixed [19] during a kick of duration h, the drag equation reduces to

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = \frac{\mathbf{u}_i^k - \mathbf{v}_i}{\tau_{p,i}}. \tag{2.5.6}$$

This linear relaxation can be integrated analytically:

$$\mathcal{K}_h^{\mathrm{drag}}: \qquad \mathbf{v}_i^+ = \mathbf{u}_i^k + \left(\mathbf{v}_i^- - \mathbf{u}_i^k\right) \exp\left(-\frac{h}{\tau_{p,i}}\right). \tag{2.5.7}$$

Equation (2.5.7) represents the exact flow of the *frozen-carrier* subproblem. It is not, in general, the exact solution of the complete between-contact dynamics, because $\mathbf{u}(\mathbf{x}, t)$ may vary both along the particle trajectory and explicitly in time. Carrier-field sampling and the separation of forcing from transport therefore remain sources of temporal discretisation error.

The exponential update adopted above avoids the stability restriction associated with an explicit discretisation of the isolated linear relaxation. In particular, the slip $\mathbf{v}_i - \mathbf{u}_i^k$ is multiplied by $\exp(-h/\tau_{p,i}) \in (0, 1]$, so the drag subflow relaxes monotonically toward the frozen carrier velocity for every $h \geq 0$.

Dry-friction subflow. The isolated dry-friction dynamics are

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\mu g \, \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}, \qquad \mathbf{v}_i \neq \mathbf{0}. \tag{2.5.8}$$

Hence the direction of motion is preserved while the speed decreases at the constant rate μg. Extending the dynamics by the convention that $\mathbf{v}_i = \mathbf{0}$ is a rest state gives the closed-form update

$$\mathcal{K}_h^{\mathrm{fric}}: \qquad \mathbf{v}_i^+ = \begin{cases} \mathbf{0}, & \|\mathbf{v}_i^-\| \leq \mu g \, h, \\ \left(1 - \dfrac{\mu g \, h}{\|\mathbf{v}_i^-\|}\right) \mathbf{v}_i^-, & \|\mathbf{v}_i^-\| > \mu g \, h. \end{cases} \tag{2.5.9}$$

The map therefore permits finite-time arrest without allowing dry friction alone to reverse the direction of motion.

Equation (2.5.9) defines the dry-friction subflow used in the numerical model. No separate static-friction complementarity condition or independently prescribed breakaway threshold is introduced, and static and kinetic friction coefficients are not distinguished.

When drag and dry friction are both active, each half-kick applies the drag subflow first and the dry-friction subflow second,

$$\mathcal{K}_{h/2} = \mathcal{K}^{\mathrm{fric}}_{h/2} \circ \mathcal{K}^{\mathrm{drag}}_{h/2}, \tag{2.5.10}$$

where the composition is read from right to left. This ordering is adopted here because the two isolated subproblems admit simple closed-form maps, whereas their simultaneous action does not, in general, retain a comparably simple closed-form vector update.

The chosen ordering also induces a numerical sticking criterion. For a particle initially at rest, the drag half-kick gives

$$\mathbf{v}_i^{\mathrm{drag}} = \mathbf{u}_i^k \left(1 - \exp\left[-\frac{h}{2\tau_{p,i}}\right]\right). \tag{2.5.11}$$

The subsequent dry-friction half-kick returns the particle to rest whenever

$$\left\|\mathbf{u}_i^k\right\| \left(1 - \exp\left[-\frac{h}{2\tau_{p,i}}\right]\right) \le \frac{\mu g\, h}{2}. \tag{2.5.12}$$

For $h/\tau_{p,i} \ll 1$, this condition approaches

$$\frac{\|\mathbf{u}_i^k\|}{\tau_{p,i}} \le \mu g, \tag{2.5.13}$$

with a relative correction of order $O(h/\tau_{p,i})$. This limiting relation is a consequence of the adopted discrete composition and is not introduced as an independent static-friction law. Reversing the subflow ordering would instead apply drag after the friction map and would therefore generate a non-zero velocity from a resting particle whenever $\mathbf{u}_i^k \neq \mathbf{0}$.

2.5.3 Candidate Generation for the Ballistic Drift

A naive contact search will examine all $\frac{N(N-1)}{2}$ unordered particle pairs during every drift. Since only sufficiently nearby particles can reach contact over a finite interval, the search can be restricted to a drift-start candidate set $\mathcal{P}_h$. At the beginning of each drift, a k-d tree is constructed from the particle centres. In the periodic implementation, the domain lengths are also supplied to the tree so that range queries use wrapped distances consistent with the minimum-image convention. The tree returns every unordered pair whose separation at the drift-start does not exceed a prescribed query radius r_{q}. This reduces the overall complexity as compared to the naive search. The actual computational cost depends on the number of candidate pairs fetched by the `scipy.spatial.cKDTree.query_pairs` range search.

Let

$$v^{\mathrm{d}}_{\max} = \max_{1\le i\le N} \left\|\mathbf{v}_i^{\mathrm{d}}\right\|, \qquad a_{\max} = \max_{1\le i\le N} a_i, \tag{2.5.14}$$

where $\mathbf{v}_i^{\mathrm{d}}$ denotes the velocity of particle i at the beginning of the drift, after the first continuous half-kick. The largest possible contact distance is $\sigma_{ij} = a_i + a_j \le 2a_{\max}$. If particle speeds remain bounded by $v^{\mathrm{d}}_{\max}$ throughout a drift of duration h, two particles can reduce their separation by at most $2v^{\mathrm{d}}_{\max}h$. This motivates the chosen query radius

$$r_{\mathrm{q}} = 2a_{\max} + 2v^{\mathrm{d}}_{\max}h. \tag{2.5.15}$$

For the nearest-image periodic construction used here, the search is further restricted to

$$r_{\mathrm{q}} < \frac{1}{2}\min(L, W). \tag{2.5.16}$$

This keeps the candidate horizon within the range for which the relevant periodic separation can be represented by a unique nearest image. If Eq. (2.5.16) is violated, the timestep must be reduced or a more general periodic-image treatment must be used.

2.5.4 Chronological Contact Resolution within a Drift

Let Δt_{rem} denote the unresolved duration of the current drift and $N_{\mathrm{drift}}^{\max}$ the maximum number of collision events allowed within one drift step. Algorithm 2 summarises one complete kick–drift timestep.

Algorithm 2: Kick–drift timestep with chronological within-drift contacts

Require: particle state at t_n, timestep h, event budget $N_{\mathrm{drift}}^{\max}$
Ensure: particle state at $t_n + h$
apply the first continuous half-kick
construct the drift-start k-d tree
determine r_{q} from Eq. (2.5.15)
generate the fixed candidate set $\mathcal{P}_h$
set $\Delta t_{\mathrm{rem}} \leftarrow h$
set $n_{\mathrm{ev}} \leftarrow 0$
while $\Delta t_{\mathrm{rem}} > \varepsilon_{\mathrm{ev}}$ and $n_{\mathrm{ev}} < N_{\mathrm{drift}}^{\max}$ **do**
    determine the earliest admissible pair contact in $\mathcal{P}_h$ and, where applicable, the earliest wall contact from the current ballistic state
    if no admissible event remains **then**
        advance all particles ballistically through Δt_{rem}
        wrap periodic coordinates where required
        set $\Delta t_{\mathrm{rem}} \leftarrow 0$
    else
        select the earliest event according to the implemented tie convention
        denote its lag by $\tau^\star$
        advance all particles ballistically through $\tau^\star$
        wrap periodic coordinates where required
        if the selected event is a particle–particle contact **then**
            reconstruct the boundary-consistent pair displacement
            apply the binary collision map
        else
            apply the wall-restitution map
        end if
        set $\Delta t_{\mathrm{rem}} \leftarrow \Delta t_{\mathrm{rem}} - \tau^\star$
        set $n_{\mathrm{ev}} \leftarrow n_{\mathrm{ev}} + 1$
    end if
end while
if $\Delta t_{\mathrm{rem}} > \varepsilon_{\mathrm{ev}}$ and $n_{\mathrm{ev}} \geq N_{\mathrm{drift}}^{\max}$ **then**
    record a cap activation
    advance ballistically through the residual drift duration
    apply the positional overlap-removal safeguard to candidate pairs in $\mathcal{P}_h$
    wrap periodic coordinates where required
    set $\Delta t_{\mathrm{rem}} \leftarrow 0$
end if
apply the second continuous half-kick

For a separated pair, an admissible first-contact lag should satisfy

$$\varepsilon_{\mathrm{ev}} < \tau^\star \leq \Delta t_{\mathrm{rem}}, \tag{2.5.17}$$

and the relative motion between collision partners must be approaching. A pair that is already at or inside the contact distance may instead generate an immediate event with $\tau^\star = 0$.

After each processed event, the predicted pair and wall contact times are then recomputed from the updated particle positions and velocities. The candidate pairs returned at the drift-start $\mathcal{P}_h$, however, remains fixed throughout the drift step. If no admissible events occur in a drift step, all particles are transported ballistically through the unresolved portion of the drift, after which the drift terminates. S

The within-drift event budget provides a termination safeguard for rapidly accumulating contact sequences. If the budget is exhausted before the drift is complete, the remaining interval is advanced ballistically and the positional overlap-removal safeguard is applied to candidate pairs in $\mathcal{P}_h$. This operation is a numerical fallback rather than part of the hard-particle dynamics, because it modifies particle positions without resolving the corresponding sequence of collision impulses. Its activation is therefore recorded and assessed through the sensitivity checks described in Section 2.5.5.

Contacts are thus not inferred from overlaps observed only at the end of a timestep. Within each drift, admissible contact times are obtained from the ballistic first-contact problem and processed chronologically, subject to the fixed candidate set, the event-time tolerance, the isolated-binary-impact assumption, and the within-drift event budget. The timestep h determines how frequently continuous forcing is coupled to particle transport and how often the candidate set is reconstructed; it does not prescribe the individual contact times resolved within a drift.

2.5.5 Accuracy, Safeguards, and Computational Cost

For smooth forcing, the kick–drift–kick composition has the usual formal second-order structure [20] only when the constituent maps, including carrier-field sampling, preserve the required symmetry. Hard-particle events, finite event tolerances, and changes in collision ordering can modify the observed convergence (see Fig. 2.3). When dry friction is active, the analytical drag and clamped friction subflows do not generally commute, while arrest at $\mathbf{v}_i = \mathbf{0}$ also introduces an additional non-smooth point. Consequently, no global convergence order is assigned without direct timestep-refinement evidence.

A finite event budget prevents a drift from stalling during unresolved rapid collision sequences, including regimes approaching *inelastic collapse* where a subpopulation of particles can keep on colliding indefinitely [15]. If the budget is exhausted, residual ballistic transport and overlap projection are used only as numerical safeguards, not as physical dynamics. Exhaustion therefore signals unresolved event accumulation rather than proving inelastic collapse. All activations are recorded, and cap-affected calculations require explicit sensitivity evidence. Simultaneous multiparticle contacts are outside the model scope and are processed serially

Let $P_h = |\mathcal{P}_h|$ denote the number of candidate pairs and E_h the number of events processed during one drift. The per-step cost can be represented schematically by

$$C_h = C_{\text{search}}(N, P_h) + C_{\text{events}}(P_h, E_h) + O(N). \tag{2.5.18}$$

The k-d tree avoids an unconditional $N(N-1)/2$ pair search, but the actual realised cost depends on particle density, query radius, timestep, and event activity. No universal $O(N \log N)$ complexity is therefore claimed for the complete solver before implementation.

2.6 Diagnostics, Verification, and Numerical Scope

The numerical evidence is organised according to the claim it supports. Diagnostic histories and conservative-limit checks first assess mechanical consistency of the solver and its numerical admissibility. Exact single-particle solutions then test the continuous forced update, while timestep and safeguard sensitivity are used to establish numerical resolution in the regimes for which results are reported.

Physical benchmarking against independently established collective behaviour is treated separately in Chapter 3 and is not used as a substitute for numerical verification.

No single test establishes the accuracy of the complete framework under arbitrary conditions. Numerical conclusions are therefore restricted to the solver, parameter regime, and observable addressed by the corresponding test. Detailed implementation audits and secondary acceptance checks are deferred to Appendix A.

2.6.1 Numerical Diagnostics and Conservative-Limit Checks

The calculations are monitored through complementary mechanical, event-based, and configurational diagnostics. These quantities serve different purposes: conservative-limit invariants test the accumulated particle mechanics, overlap and safeguard records monitor numerical admissibility, and configurational statistics characterise the spatial state used in the subsequent physical analysis.

Mechanical consistency. At the stored diagnostic times $t \in \mathcal{T}_{\mathrm{diag}}$, the total kinetic energy and linear momentum are

$$E(t) = \frac{1}{2}\sum_{i=1}^{N} m_i \|\mathbf{v}_i(t)\|^2, \qquad \mathbf{P}(t) = \sum_{i=1}^{N} m_i \mathbf{v}_i(t). \tag{2.6.1}$$

Under periodic boundaries in the force-free elastic limit $\mathbf{f}_i^{\mathrm{cont}} = \mathbf{0}$, $e = 1$, Both quantities are conserved by physical laws. Their maximum recorded residuals are defined as

$$\epsilon_E = \max_{t\in\mathcal{T}_{\mathrm{diag}}} \frac{|E(t) - E(0)|}{E(0)}, \qquad \epsilon_P = \max_{t\in\mathcal{T}_{\mathrm{diag}}} \frac{\|\mathbf{P}(t) - \mathbf{P}(0)\|}{P_{\mathrm{sc}}}, \tag{2.6.2}$$

where

$$P_{\mathrm{sc}} = \sum_i m_i \|\mathbf{v}_i(0)\| \tag{2.6.3}$$

These residuals test the accumulated force-free collision and transport sequence; they are not conservation criteria for forced or inelastic calculations. Excluded-volume consistency is assessed independently through the maximum audited penetration

$$\delta_{\max} = \max_{t\in\mathcal{T}_{\mathrm{aud}}} \max_{i<j} \left[\sigma_{ij} - r_{ij}(t)\right]_+, \qquad [q]_+ = \max(q, 0), \tag{2.6.4}$$

where r_{ij} is evaluated using the minimum-image separation under periodic boundaries.

The event record from the code output also provides the cumulative number of processed particle–particle pair events, denoted N_{coll}. For the kick–drift solver, N_{cap} denotes the number of drifts in which the within-drift event budget is reached and the numerical fallback is activated.

Configurational diagnostics. For spatial statistics, the domain is partitioned into M_ℓ equal cells of characteristic size ℓ, with $N_c(t)$ particles in cell c. The variance-to-mean ratio is

$$\widehat{\mathrm{VMR}}(t;\ell) = \frac{\dfrac{1}{M_\ell - 1}\displaystyle\sum_{c=1}^{M_\ell}\left[N_c(t) - \frac{N}{M_\ell}\right]^2}{N/M_\ell}. \tag{2.6.5}$$

With this Bessel-corrected variance, independent uniformly distributed particles at fixed N satisfy $\mathbb{E}[\widehat{\mathrm{VMR}}] = 1$. Values above unity indicate enhanced count fluctuations, consistent with clustering at scale ℓ, whereas values below unity indicate under-dispersion. VMR characterises count fluctuations at a specified scale and does not, by itself, measure cluster connectivity or characteristic cluster size.

The complete empirical occupancy distribution is given by:

$$P_M(m,t) = \frac{1}{M}\sum_{b=1}^{M} \mathbf{1}_{\{n_b(t)=m\}}, \tag{2.6.6}$$

where $P_M(m,t)$ is the fraction of boxes containing exactly m particles at time t. From this distribution, the negative Shannon entropy is defined as

$$H(t) = \sum_{m:P_M(m,t)>0} P_M(m,t)\ln P_M(m,t). \tag{2.6.7}$$

Unlike the VMR, H is not interpreted as a monotonic measure of spatial clustering. It also provides complementary information on the shape and temporal evolution of the complete occupancy distribution.

Where a statistically stationary analysis interval is required, stationarity is assessed from sustained behaviour of the relevant diagnostics over a finite time-period rather than from a single snapshot. The precise criterion and analysis window are specified in the subsequent chapters in which they are applied.

The conservative-limit diagnostics provide numerical checks of the hard-particle mechanics, whereas the accuracy of the continuous forced update is assessed independently against analytical single-particle solutions in Section 2.6.2.

2.6.2 Analytical Verification of the Continuous Update

The conservative-limit diagnostics of Section 2.6.1 examine the validity of accumulated hard-particle mechanics but do not, by themselves, determine the accuracy of the continuous forcing update. The latter is therefore isolated from collision dynamics and tested against closed-form single-particle solutions with dry friction switched-off. A uniform carrier tests the Stokes-relaxation update without spatial sampling error, while a centred linear shear additionally tests the composition of particle transport with sampling of a spatially varying carrier field.

Uniform carrier. For a constant carrier velocity $\mathbf{u}(\mathbf{x},t) = \mathbf{U}_0$, the frictionless particle equation

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathbf{U}_0 - \mathbf{v}}{\tau_p}$$

has the exact solution

$$\mathbf{v}(t) = \mathbf{U}_0 + (\mathbf{v}_0 - \mathbf{U}_0)\, e^{-t/\tau_p}, \tag{2.6.8}$$

$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{U}_0 t + \tau_p\,(\mathbf{v}_0 - \mathbf{U}_0)\left(1 - e^{-t/\tau_p}\right). \tag{2.6.9}$$

The quiescent case follows by setting $\mathbf{U}_0 = \mathbf{0}$. Figure 2.3(a) compares the numerical solution directly with Eq. (2.6.8). The numerical markers are indistinguishable from the analytical curves at the plotting scale, while the inset reports the corresponding position and velocity errors along the trajectory.

Linear shear. A more stringent test is provided by the centred frictionless shear

$$\mathbf{u}(\mathbf{x},t) = (\dot{\gamma}(y - y_c),\, 0)\,, \tag{2.6.10}$$

for which the streamwise carrier velocity sampled by the particle varies as its cross-stream position evolves. Define

$$Y_0 = y_0 - y_c, \qquad A = \dot{\gamma}\left(Y_0 + \tau_p v_{y0}\right). \tag{2.6.11}$$

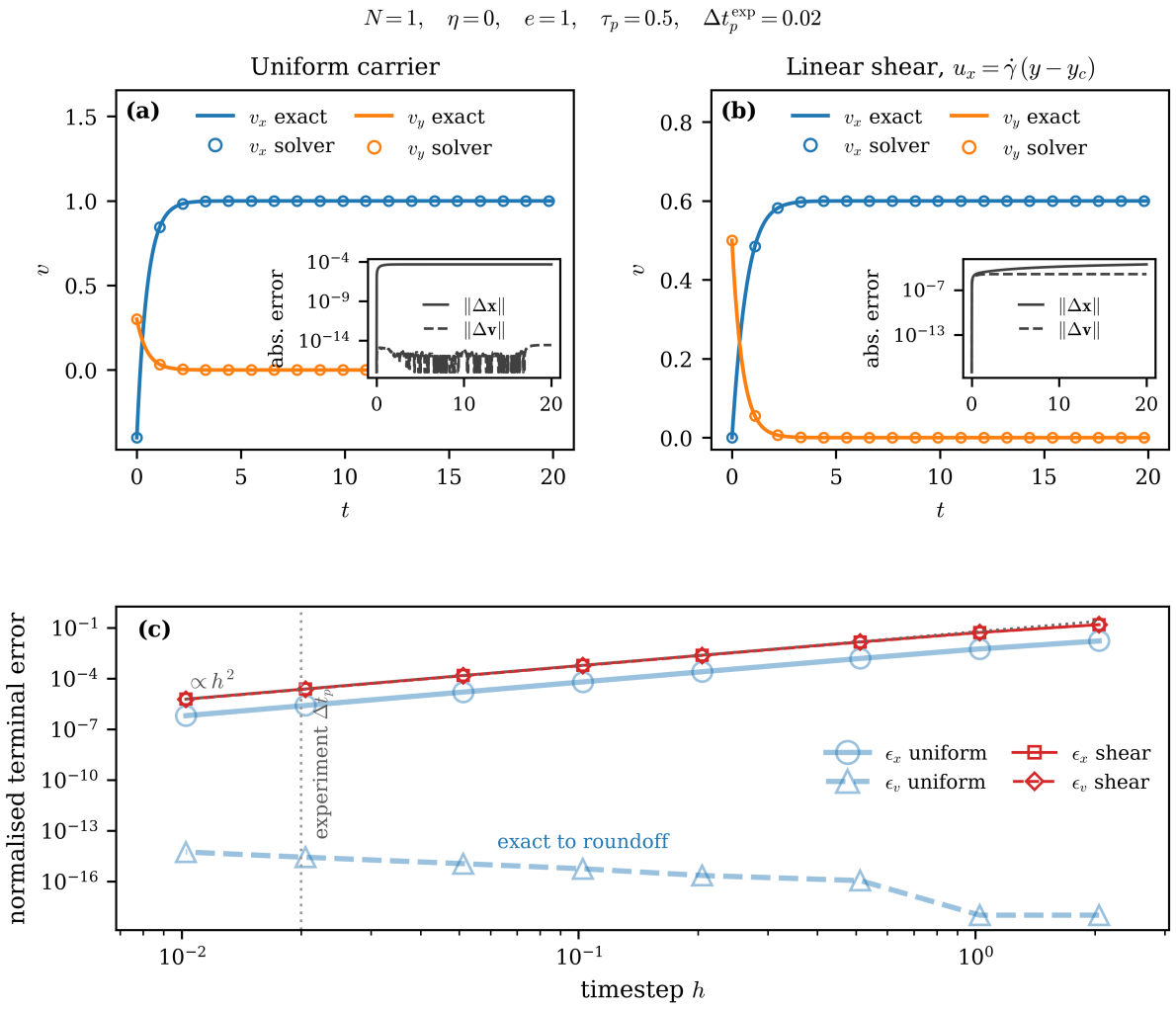

Figure 2.3: **Analytical verification of the continuous particle update. (a)** Velocity histories in a uniform carrier: analytical solutions are shown by solid lines and numerical solutions by markers; the inset reports the corresponding absolute trajectory errors. **(b)** Equivalent comparison for the centred linear shear $u_x = \dot{\gamma}(y - y_c)$, which additionally tests spatial carrier sampling along the particle trajectory. **(c)** Normalised terminal position and velocity errors under timestep refinement. The dotted guide is proportional to h^2; the uniform-carrier velocity error remains at floating-point level. The vertical line marks the timestep used for simulations in Chapter 4 and is included only for reference; timestep convergence of the complete turbulent-flow simulations is assessed separately in Chapter 4.

The cross-stream dynamics are

$$v_y(t) = v_{y0} e^{-t/\tau_p}, \qquad y(t) = y_0 + \tau_p v_{y0} \left(1 - e^{-t/\tau_p}\right), \tag{2.6.12}$$

while the streamwise velocity is

$$v_x(t) = A + (v_{x0} - A)\, e^{-t/\tau_p} - \dot{\gamma} v_{y0}\, t\, e^{-t/\tau_p}. \tag{2.6.13}$$

Integrating once more gives

$$\begin{aligned} x(t) = x_0 + At + \tau_p\, (v_{x0} - A) \left(1 - e^{-t/\tau_p}\right) \\ - \dot{\gamma} v_{y0} \tau_p^2 \left[1 - \left(1 + \frac{t}{\tau_p}\right) e^{-t/\tau_p}\right]. \end{aligned} \tag{2.6.14}$$

Figure 2.3(b) shows that both velocity components follow their analytical solutions throughout the test interval. Unlike the uniform-carrier problem, this test exercises the coupling between particle transport and spatial carrier sampling.

Timestep refinement. In the above analysis, for each timestep h, the numerical state is compared with the exact solution at a common terminal time T. Error analysis from Figure 2.3(c) shows that the uniform-carrier position error and the position and velocity errors in linear shear decrease approximately parallel to the h^2 reference over the tested range. The uniform-carrier velocity error instead remains at floating-point level, consistent with the analytical exponential update of the frozen uniform-drag subproblem; no convergence slope is therefore assigned to that quantity.

These tests verify the expected refinement of the smooth, frictionless continuous update for the two reference problems considered here. They do not establish global second-order accuracy of the complete hard-particle solver. Collision sequencing, fixed within-drift candidate selection, finite-precision event tolerances, non-smooth dry-friction dynamics, and safeguard activation introduce additional sources of numerical error whose influence must be assessed in the corresponding production regime.

2.6.3 Numerical Scope and Connection to Subsequent Chapters

The preceding tests establish the numerical validity of the developed solvers. The conservative-limit diagnostics of Section 2.6.1 examines the accumulated hard-particle mechanics through energy and momentum conservation, excluded-volume consistency, and safeguard records. The analytical comparisons of Section 2.6.2 independently test the smooth, frictionless continuous update in uniform and spatially varying carrier fields. Together, these checks establish the numerical basis on which the subsequent physical analyses are built.

The numerical evidence is therefore interpreted on a claim-by-claim basis. Conservative-limit diagnostics establish the consistency of the hard-particle transport and collision mechanics, while the analytical comparisons verify the continuous Stokes update in the reference problems considered. Resolution of a complete forced many-particle calculation must additionally be established through timestep and safeguard-sensitivity checks performed in its production regime. Solver-specific assumptions, numerical edge cases, and run-acceptance conditions are documented in Appendix A.

This requirement determines how numerical evidence is transferred to the physical chapters. Chapter 3 uses the common mechanics, solvers, and diagnostics established here to isolate collisional effects in controlled carrier flows. Comparison with the López–Puglisi benchmark problems then provides an independent qualitative test of the resulting many-particle behaviour; these comparisons are treated as physical benchmarks rather than as substitutes for numerical verification. Chapter 4 applies the same framework to the near-wall turbulent configuration and assesses numerical resolution through the three-level particle timestep refinement

$$\Delta t_p = 0.02,\ 0.01,\ 0.005.$$

That refinement tests the collision and pair statistics used in the turbulent analysis, and its conclusions are therefore attached specifically to the Chapter 4 regime and observables rather than extrapolated to unrelated parameter sets.

The contribution of the present chapter is methodological. It establishes a common hard-particle mechanics, a timestep-free event-driven formulation for the force-free problem, a forced kick–drift formulation with chronological contact resolution within each ballistic drift, and a common set of diagnostics and acceptance criteria. The subsequent physical chapters can therefore change the carrier flow and dynamical regime without changing the underlying definitions of contact and collision response.

These developments from Chapter 2 directly answer TQ1 (Section 1.2). Continuous forcing and

instantaneous hard-particle collisions are treated according to their distinct mathematical structure: forcing is advanced during kick stages, whereas transport and contact are resolved during ballistic intervals in which first-contact times can be determined algebraically and processed chronologically. The force-free solver applies this principle globally without a timestep, while the forced solver applies it within each drift of the kick–drift composition. The timestep therefore controls the coupling between continuous forcing and particle transport, not the individual collision times. Within the stated candidate-search, finite-precision, isolated-binary-impact, and safeguard assumptions, the resulting framework preserves explicit chronological resolution of first-contact events and thereby provides the numerical representation required by TQ1.

Chapter 3

Collective Dynamics in Controlled Carrier Flows

3.1 Introduction and Role of the Chapter

Chapter 2 established the hard-particle mechanics, numerical solvers, and diagnostic procedures used throughout the thesis. The present chapter moves from numerical verification to physical interpretation and qualitative validation. Its purpose is to determine whether the particle model recovers established mechanisms of inertial organisation and collision-modified collective behaviour in controlled carrier flows before addressing the substantially more complex near-wall turbulent environment studied in Chapter 4.

This intermediate step is important because collisions act on a particle population that has already been organised by the carrier flow and by finite particle response. Even without particle–particle interactions, inertial particles need not follow fluid trajectories exactly and can develop strongly non-uniform spatial distributions [4, 17, 18]. Collisions subsequently modify this pre-existing organisation through excluded volume, impulsive momentum exchange and, for $e < 1$, collisional dissipation. Their collective effect therefore depends not only on the collision law, but also on particle response, size and packing fraction, encounter frequency, and the structure imposed by the carrier.

The prescribed carrier flow considered in this chapter is the alternating sinusoidal shear flow. Its analytical form provides a controlled setting in which the competition among particle response, collision frequency, restitution, and carrier-induced separation can be examined without the multiscale complexity of turbulence. The alternating shear produces non-trivial Lagrangian transport [21, 22] and therefore provides a suitable framework for examining how finite-size collisions modify a particle population organised by the carrier flow. Collision and collisionless legs are compared across contrasting dynamical regimes.

Comparison with the results of López and Puglisi [8, 9] is intentionally qualitative. Their calculations employ a Direct Simulation Monte Carlo (DSMC) treatment, in which collisions are represented statistically through sampled pair interactions [24], whereas the present simulations track deterministic finite-size particles and resolve hard-particle contacts geometrically (Section 2.2). Particle number, particle size, and realised collision frequency also differ between the two formulations. A pointwise numerical reproduction is therefore neither expected nor claimed. Instead, the comparison is used to assess whether the deterministic hard-particle model recovers the qualitative changes between homogeneous and spatially inhomogeneous particle populations reported in the stochastic collision model.

A central feature of the López–Puglisi results is that collisions do not have a universal clustering or dispersive effect [8, 9]. Their influence depends on the dynamical regime established by the competition

between particle response, carrier transport, collisional interactions, and restitution. Collisions may therefore suppress an existing concentration pattern in one regime while promoting or maintaining spatial inhomogeneity in another. The *reversed-clustering* phenomenology is interpreted in this regime-dependent sense.

The chapter therefore addresses the question

> *How do finite-size collisions modify collective particle organisation across contrasting regimes of particle response, restitution, collision frequency, and carrier-induced separation, and to what extent does the deterministic hard-particle model recover the qualitative collision-modified regimes reported in stochastic collision simulations?*

The analysis consequently focuses on collisions as a mechanism that modifies, rather than independently determines, the collective organisation established by the carrier flow, with the sign and magnitude of that modification depending on the dynamical regime in which the analysis is done.

3.2 Microscopic Model for Particle Dynamics

The general hard-particle mechanics and numerical implementation were established in Chapter 2. The present chapter adopts the controlled-flow specialisation of López and Puglisi [8]: N identical finite-size particles are transported by a prescribed incompressible carrier field and interact through instantaneous binary collisions.

3.2.1 Equations of motion

Let $\mathbf{u} = \mathbf{u}(\mathbf{r}, t)$ be a prescribed carrier-fluid velocity field defined on a spatial domain $\Omega \subseteq \mathbb{R}^2$. The flow is assumed incompressible:

$$\nabla \cdot \mathbf{u} = 0. \tag{3.2.1}$$

In the two-dimensional setting considered in this study, the particles are hard disks; the same notation can be interpreted as describing hard spheres in a 3D domain. The position of particle i evolves according to

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i(t), \qquad i = 1, \ldots, N. \tag{3.2.2}$$

Between collisions, its velocity satisfies

$$\frac{d\mathbf{v}_i}{dt} = -\frac{1}{\tau_p}\left[\mathbf{v}_i(t) - \mathbf{u}\big(\mathbf{r}_i(t), t\big)\right]. \tag{3.2.3}$$

where τ_p is the particle response time. This is the heavy-particle approximation used in the reference model, in which Stokes drag is retained as the particle–fluid coupling while the additional hydrodynamic forces are neglected.

Let t_{ij}^k denote the time of the k-th collision between particles i and j, defined by the contact condition

$$\left\|\mathbf{r}_i\big(t_{ij}^k\big) - \mathbf{r}_j\big(t_{ij}^k\big)\right\| = \sigma, \tag{3.2.4}$$

where σ is the particle diameter. At contact, $\widehat{\mathbf{n}}_{ij}(t_{ij}^k)$ denotes a unit vector along the line joining the two particle centres. The collision parameter is $\gamma = \frac{1+e}{2}$, $0 \le e \le 1$, where e is the normal coefficient of restitution. Denoting by $\mathbf{v}_i^-$ and $\mathbf{v}_i^+$ the velocity of particle i immediately before and after the impact, respectively, the collision produces the jump

$$\mathbf{v}_i^+ - \mathbf{v}_i^- = -\gamma\left\{\left[\mathbf{v}_i^- - \mathbf{v}_j^-\right] \cdot \widehat{\mathbf{n}}_{ij}(t_{ij}^k)\right\}\widehat{\mathbf{n}}_{ij}(t_{ij}^k). \tag{3.2.5}$$

Equivalently, the normal relative velocity satisfies $g_n^+ = -e\, g_n^-$, while the tangential relative velocity is unchanged. The complete mechanical properties of this collision map are already discussed in Section 2.3. This is the equal-mass restitution law introduced mechanically in Section 2.3. The smooth drag dynamics and the discrete collision jumps can be written compactly in distributional form:

$$\frac{d\mathbf{v}_i(t)}{dt} = -\frac{1}{\tau_p}\left[\mathbf{v}_i(t) - \mathbf{u}(\mathbf{r}_i(t), t)\right] - \gamma \sum_{\substack{j=1 \\ j\neq i}}^{N} \sum_k \left\{\left[\mathbf{v}_i(t) - \mathbf{v}_j(t)\right] \cdot \hat{\mathbf{n}}_{ij}(t)\right\} \hat{\mathbf{n}}_{ij}(t)\, \delta\left(t - t_{ij}^k\right). \tag{3.2.6}$$

Together with

$$\frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} = \mathbf{v}_i, \tag{3.2.7}$$

this defines the deterministic hybrid microscopic system. The Dirac term is only a distributional representation of the instantaneous velocity jump; it is not a finite-duration collision force. This implies that the collision contributes to the particle dynamics only at the contact-times. Integrating Eq. (3.2.6) across an isolated collision recovers Eq. (3.2.5).

Rearranging Eq. (3.2.6) gives the fixed-point form given in [9] as:

$$\begin{aligned}\mathbf{v}_i = \mathbf{u}(\mathbf{r}_i, t) - \tau_p \frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} \\ - \gamma\tau_p \sum_{\substack{j=1 \\ j\neq i}}^{N} \sum_k \left[\left(\mathbf{v}_i^- - \mathbf{v}_j^-\right) \cdot \widehat{\mathbf{n}}_{ij}^{(k)}\right] \widehat{\mathbf{n}}_{ij}^{(k)}\, \delta\left(t - t_{ij}^{(k)}\right),\end{aligned} \tag{3.2.8}$$

3.2.2 Stokes relaxation and particle response

Between collisions, each particle velocity relaxes towards the local carrier velocity through Stokes drag, where τ_p is the particle response, or Stokes-relaxation, time. Small τ_p corresponds to rapid adjustment to changes in the carrier velocity, whereas larger τ_p produces stronger velocity memory and allows particle trajectories to depart more substantially from the corresponding fluid trajectories. In the limit $\tau_p \to 0$, away from collision events, $\mathbf{v}_i \to \mathbf{u}(\mathbf{r}_i, t)$.

3.2.3 Modelling Scope and Assumptions

The model inherits the assumptions established in Chapter 2. The particles are identical rigid disks of diameter σ, transported by a prescribed incompressible carrier field and interacting through instantaneous binary collisions. Following López and Puglisi [8], the particles are assumed to be much denser than the carrier fluid [16, 17]. Particle motion between collisions is therefore represented using Stokes drag as the retained contribution from the Maxey–Riley equation.

Under this heavy-particle approximation, the fluid-acceleration (pressure-gradient), added-mass, and Basset–Boussinesq history terms are neglected [8, 16, 17]. Faxén corrections are neglected because the particle diameter σ is small compared with the characteristic carrier-flow length scale $L/(2\pi)$. Gravity and Brownian forcing are also absent from the idealised model considered here

The particles are finite-sized mechanically but hydrodynamically unresolved. Their diameter determines contact, excluded volume, collision geometry, and the finite separation of particle centres at impact, but the disturbance flow generated by each particle is not accounted for. Hydrodynamic particle–particle interactions, lubrication, particle deformation and rotation, tangential contact friction, and enduring contact are consequently outside the scope of the model. The carrier field is prescribed independently of the particle population, so the coupling is one-way: particles respond to the flow, but particle motion and collision impulses do not modify it.

The model should therefore be seen as an idealised framework designed to isolate the combined effects of carrier transport, inertial relaxation, finite-size exclusion, and binary collisional momentum exchange. Its conclusions apply within this heavy-particle, one-way-coupled hard-disk description (see section 2.2) and are not transferred without qualification to regimes in which the neglected hydrodynamic or multi-particle interactions become dynamically important.

3.3 Alternating Sinusoidal Shear Flow

The particle calculations in this chapter use the prescribed alternating sinusoidal shear introduced below. The carrier field is analytical and one-way coupled: it drives the particle dynamics, whereas the particles do not modify the carrier flow.

The alternating sinusoidal shear considered by López and Puglisi [8] is defined on the periodic square $\Omega = [0, L]^2$ by

$$\mathbf{u}_{\mathrm{AS}}(\mathbf{r}, t) = \begin{cases} \left(\dfrac{2\pi L\alpha}{T} \sin\left(\dfrac{2\pi y}{L} \right), 0 \right), & t \bmod T < \dfrac{T}{2}, \\ \left(0, -\dfrac{2\pi L\alpha}{T} \sin\left(\dfrac{2\pi x}{L} \right) \right), & t \bmod T \geq \dfrac{T}{2}, \end{cases} \tag{3.3.1}$$

Here T is the forcing period and α is a dimensionless stirring amplitude. Throughout the analysis in this chapter, the domain size is chosen as

$$L = \sqrt{N}, \tag{3.3.2}$$

so that the domain area satisfies $L^2 = N$, therefore implying a unit mean particle number density. During the first half-period, the carrier flow velocity (3.3.1) is an x-directed sinusoidal shear varying only with y; during the second half-period, the shear direction is exchanged and the y-velocity varies only with x. The carrier flow is therefore piecewise steady, with its shear orientation switching every half-period.

Each phase is incompressible because its non-zero velocity component depends only on the transverse coordinate:

$$\nabla \cdot \mathbf{u}_{\mathrm{AS}} = \frac{\partial u_{\mathrm{AS},x}}{\partial x} + \frac{\partial u_{\mathrm{AS},y}}{\partial y} = 0. \tag{3.3.3}$$

The field is also compatible with the periodic boundaries because the sinusoidal factors are L-periodic. The scalar out-of-plane vorticity of the alternating-shear carrier is

$$\omega_{\mathrm{AS}}(\mathbf{r}, t) = \frac{\partial u_{\mathrm{AS},y}}{\partial x} - \frac{\partial u_{\mathrm{AS},x}}{\partial y} = -\frac{4\pi^2\alpha}{T} \begin{cases} \cos\left(\dfrac{2\pi y}{L} \right), & t \bmod T < \dfrac{T}{2}, \\ \cos\left(\dfrac{2\pi x}{L} \right), & t \bmod T \geq \dfrac{T}{2}. \end{cases} \tag{3.3.4}$$

Its maximum magnitude over space and time is therefore

$$\max_{\mathbf{r},t} |\omega_{\mathrm{AS}}(\mathbf{r}, t)| = \frac{4\pi^2\alpha}{T}, \tag{3.3.5}$$

for the positive values of α considered here.

The strongest velocity gradients occur along horizontal bands during the first half-period and then along vertical bands during the second half. Their repeated reorientation produces alternating stretching and folding of material elements and, for sufficiently strong forcing, gives rise to chaotic Lagrangian transport [21, 22]. In the present work, this carrier flow provides a tractable setting in which to examine how particle response, collision frequency, restitution, and carrier-induced transport combine to modify the collective particle distribution.

For this carrier flow, López and Puglisi [8], following Berthier *et al.* [23], report that for $\alpha > 0.4$ the passive-tracer dynamics contain a single connected chaotic region, with no numerically distinguishable Kolmogorov–Arnold–Moser (KAM) tori that partition the domain. KAM tori are invariant structures associated with quasi-periodic motion that can persist under sufficiently weak perturbations of an integrable Hamiltonian system [21, 22, 25]. Passive-tracer advection in a two-dimensional incompressible flow admits a Hamiltonian representation, with the streamfunction playing the role of the Hamiltonian. For a time-periodic carrier flow, surviving KAM tori therefore appear as invariant curves in the stroboscopic Poincaré map [21, 23]. Non-contractible invariant curves that wind around the periodic domain act as global barriers to Lagrangian transport, whereas contractible invariant curves bound local islands of regular motion. The absence of the former removes the class of barriers that would otherwise partition the accessible domain and is therefore consistent with the development of a connected chaotic region. Within the chaotic region, nearby passive-tracer trajectories separate exponentially. The corresponding Lyapunov exponent and carrier-induced separation time are quantified below.

The exponential separation rate of nearby passive tracers [8, 23] is estimated as

$$\lambda \simeq \frac{1.96\,\ln(3.35\alpha)}{T}, \qquad \tau_f = \lambda^{-1}, \tag{3.3.6}$$

where λ is the Lyapunov exponent characterising the exponential separation rate and τ_f is the corresponding characteristic carrier-induced separation time. This description applies specifically to the passive-tracer dynamics of the prescribed carrier flow.

3.3.1 Competing time scales

The dynamical regimes considered in this chapter are characterised by four time scales: the particle response time τ_p, the carrier-flow period T, the measured mean collision time τ_c, and the carrier-induced separation time τ_f [8, 9]. Their relative magnitudes provide a framework for classifying regimes by comparing inertial relaxation, collisional redistribution, and chaotic carrier transport.

The ratio τ_p/T compares particle relaxation with the temporal variation of the carrier flow. When $\tau_p \ll T$, particle velocity adjusts rapidly to changes in the local carrier velocity and exhibits a tracer-like behaviour in the collisionless case. As τ_p/T increases, the particle velocity becomes more reluctant to adjust to changes in the carrier velocity and their trajectories begin to depart from the fluid pathlines. The ratio τ_p/τ_c compares inertial relaxation with the frequency of collisions: increasing τ_p/τ_c corresponds to more frequent collisional redistribution relative to particle relaxation. The ratio τ_c/τ_f compares collisional and carrier-induced separation time scales. Thus, $\tau_c < \tau_f$ means that collisions occur, on average, more rapidly than chaotic separation, whereas $\tau_f < \tau_c$ means that carrier-induced separation acts on the shorter characteristic time scale. These inequalities establish only the relative rates of the competing processes; neither ordering implies, by itself, clustering or dispersal.

3.4 Clustering diagnostic and steady-state detection

The phenomenon under study is a property of the *steady* spatial distribution, so its measurement requires two ingredients: a scalar that quantifies the degree of spatial clustering, and a criterion for deciding when the simulation has reached a statistically stationary state at which that scalar is to be evaluated. Both follow the construction of the benchmark [8], so that the comparison is made on equal terms.

Particle-occupancy distribution. Spatial inhomogeneity is characterised using the particle-in-cell occupancy distribution [8, 9]. The periodic domain is partitioned into M equal boxes, and the number

n_ℓ of particles in each box ℓ is recorded. The empirical occupancy distribution is

$$P_M(m) = \frac{1}{M}\sum_{\ell=1}^{M} \mathbf{1}_{\{n_\ell = m\}}, \tag{3.4.1}$$

so that $P_M(m)$ is the fraction of boxes containing exactly m particles. Following the benchmark, the partition is chosen such that $\frac{N}{M} = 5$, giving a mean occupancy of five particles per box. For a spatially homogeneous population, the Poisson distribution is used as the benchmark reference,

$$P_M^{\mathrm{Pois}}(m) = e^{-\mu_{\mathrm{occ}}}\frac{\mu_{\mathrm{occ}}^m}{m!}, \qquad \mu_{\mathrm{occ}} = \frac{N}{M} = 5, \tag{3.4.2}$$

where μ_{occ} is the mean box occupancy. For finite fixed N, the exact uniform occupancy statistics are multinomial rather than strictly Poisson; the Poisson distribution is retained here for consistency with the benchmark comparison.

A nearly homogeneous particle distribution follows this reference closely, whereas aggregation produces a broader occupancy distribution, with increased weight in sparsely populated and highly occupied boxes. The full $P_M(m)$ curve therefore provides the distribution-level measure used to distinguish homogeneous and clustered states in the alternating-shear calculations. The complementary VMR, kinetic-energy, and occupancy-entropy diagnostics were defined previously in section 2.6.

Statistical stationarity detection. When a statistically stationary regime can be identified, spatial statistics are evaluated only after the transient dynamics have decayed. Stationarity is monitored using the total particle kinetic energy $E(t)$ defined in Eq. (2.6.1), together with the VMR and the negative occupancy entropy $H(t)$ defined in Eq. (2.6.7). The kinetic energy provides the primary dynamical criterion, while VMR and $H(t)$ provide complementary checks that the spatial occupancy statistics have also ceased to evolve systematically.

For each diagnostic, its candidate onset time is the earliest sampled time after which the quantity remains within ±5% of its mean over the final 15% of recorded frames. A statistically stationary regime is accepted only when $E(t)$ satisfies this criterion and at least one of VMR or $H(t)$ does so as well. The stationarity onset t_{ss} is then taken as the latest onset among the diagnostics entering the acceptance criterion.

The stationarity diagnostics are used to qualify the interpretation of the reported late-time statistics rather than to determine the configurations or sampling windows shown in the results. In the individual case descriptions of Sec. 3.5, the displayed particle configuration and quoted VMR refer to the final recorded state of the corresponding simulation. For the distribution-level comparison in Fig. 3.3, $P_M(m)$ and the VMR values shown in the legend are averaged over the final 30 recorded frames. The same final-30-frame VMR averages are used in Sec. 3.5.3. When the stationarity criterion is satisfied, a final-window statistic may be interpreted as representative of a statistically stationary regime provided that the averaging window lies after the detected onset t_{ss}. When the criterion is not satisfied, the corresponding terminal-snapshot and final-window quantities are interpreted only as late-time diagnostics

3.5 Results

Particle organisation is assessed primarily through the box-occupancy distribution $P_M(m)$, which provides the main observable for qualitative comparison with the López–Puglisi benchmark. For a spatially homogeneous population, $P_M(m)$ remains close to the Poisson reference, whereas systematic broadening towards sparsely and highly occupied boxes indicates spatial inhomogeneity. Representative particle configurations are used to identify the associated morphology.

The variance-to-mean ratio (VMR) is used only as a compact scalar summary of the occupancy fluctuations and is not interpreted independently as a definitive measure of clustering. In the individual case descriptions below, the quoted VMR refers to the final recorded particle configuration and is therefore a terminal-snapshot diagnostic. The full occupancy distribution $P_M(m)$ provides the primary evidence for the corresponding regime classification. A separate temporal averaging convention is introduced explicitly in Sec. 3.5.3 for the direct comparison of the elastic low- and high-inertia regimes. There, VMR is averaged over the final 30 recorded frames in order to reduce sensitivity to a single terminal configuration. These final-window averages should therefore not be confused with the terminal-snapshot VMR values reported in the individual case descriptions.

3.5.1 High-Inertia Regimes in the Alternating Sinusoidal Shear

The first three high-inertia simulations Case (a) - Case (c) use the same carrier parameters, particle number $N = 980$, and particle response time $\tau_p = 10$, allowing the effects of collisions and restitution to be compared within a common dynamical setting. Their differences therefore isolate the population-level consequences of enabling collisions and varying the restitution coefficient. A fourth simulation (Case (d)) uses $N = 320$, a different carrier configuration, and $\tau_p = 2$ to realise the complementary ordering $\tau_f < \tau_c$; it should therefore be interpreted as a regime comparison rather than as a one-parameter continuation of the preceding cases.

Case (a): Collisionless inertial organisation. In the absence of collisions, the particles develop a strongly inhomogeneous filamentary distribution [Fig. 3.1 (Case a)]. The corresponding occupancy distribution is substantially broader than the Poisson reference, with enhanced probability of both sparsely and highly occupied boxes [Fig. 3.3 (Case a)]. At the final recorded configuration,

$$\mathrm{VMR}^{(a)}_{\mathrm{final}} = 6.31. \tag{3.5.1}$$

No statistically stationary onset is detected within the simulated interval, so this terminal-snapshot value characterises the late-time state rather than a stationary-state mean. The terminal configuration and the broad $P_M(m)$ distribution consistently demonstrate strong collisionless spatial inhomogeneity. Finite particle response therefore establishes a structured population before any particle–particle interaction is introduced.

Case (b): Suppression by elastic collisions. Enabling elastic collisions, $e = 1$, produces a significantly different high-inertia state [Fig. 3.1 (Case b)]. The collision leg becomes approximately homogeneous, whereas the corresponding collisionless leg develops a strongly inhomogeneous filamentary population (as in Case (a)). The collision leg reaches $\mathrm{VMR}^{(b)}_{\mathrm{final}} = 0.81$, whereas the collisionless leg has $\mathrm{VMR}_{\mathrm{nocoll,final}} = 6.31$ at the final recorded configuration. These terminal-snapshot values provide a compact summary of the contrasting late-time states but are not interpreted as stationary-state ensemble means, since no statistically stationary onset is detected for the collisionless high-inertia case.

The full $P_M(m)$ distributions provide the primary evidence. The collision-on population lies close to the Poisson reference, whereas the collisionless population is substantially broader and strongly non-Poissonian [Fig. 3.3 (case b)]. The value $\mathrm{VMR}^{(b)}_{\mathrm{final}} < 1$ indicates weak sub-Poissonian occupancy fluctuations at the adopted box scale rather than a distinct clustering regime.

For the collision leg, $\widehat{\tau}_c \simeq 0.101$, $\tau_f \simeq 0.989$, $\tau_p = 10$, so that $\widehat{\tau}_c < \tau_f < \tau_p$. Collisions therefore occur repeatedly before either carrier-induced separation or Stokes relaxation acts over its characteristic time. Because $e = 1$, each impact conserves kinetic energy; the suppression of spatial organisation therefore cannot be attributed to collisional dissipation. Instead, repeated elastic encounters redistribute particle relative motion, producing a net dispersive effect at the population level. This qualitatively reproduces the elastic high-inertia regime reported in [8], despite the different microscopic collision treatments adopted.

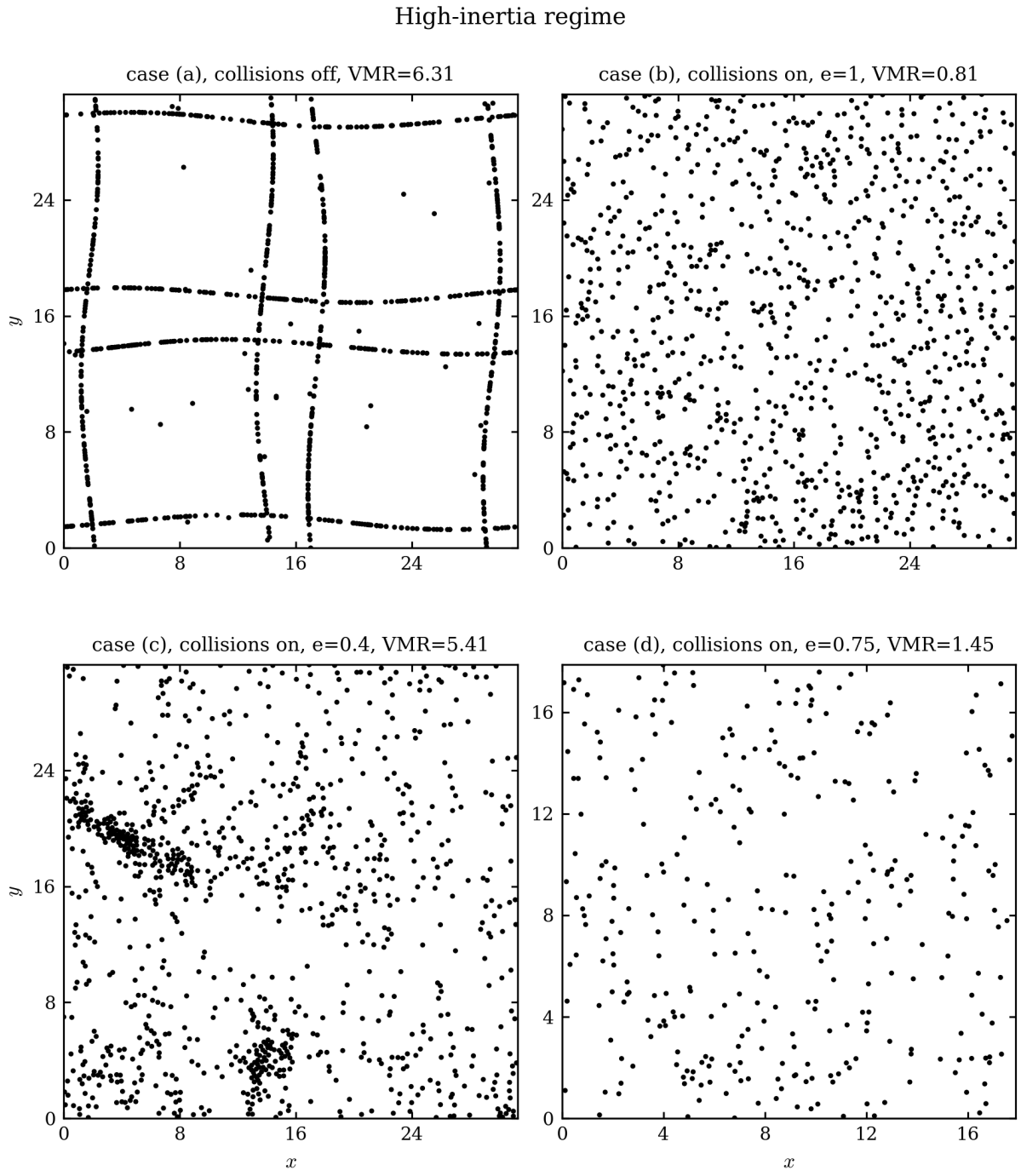

Figure 3.1: **High-inertia regimes in the alternating sinusoidal shear.** Spatial configurations for **(a)** collisionless particles, **(b)** elastic collisions, $e = 1$, **(c)** inelastic collisions, $e = 0.40$, and **(d)** $e = 0.75$ in the stronger carrier-separation regime. Elastic collisions suppress the high-inertia collisionless organisation, stronger dissipation recovers aggregation, and the $\tau_f < \widehat{\tau}_c$ regime remains approximately homogeneous.

Case (c): Recovery of aggregation with stronger dissipation. The collision response changes significantly when the restitution coefficient is reduced to $e = 0.40$. Dense particle-rich regions coexist with a broader inhomogeneous background [Fig. 3.1 (Case c)], and the occupancy distribution develops the longest high-m tail among the high-inertia cases. At the final recorded configuration,

$$\mathrm{VMR}^{(c)}_{\mathrm{final}} = 5.41. \tag{3.5.2}$$

No statistically stationary state is detected over the simulated interval. Spatial concentration and collision activity continue to increase, which is consistent with progressive inelastic clustering that may approach inelastic collapse [15]. The present simulation does not establish inelastic collapse itself.

Unlike the elastic case, each collision dissipates normal relative kinetic energy. The resulting organisation nevertheless reflects the coupled evolution of collisional dissipation, collision frequency, carrier transport, and spatial concentration, and is therefore not attributed to restitution alone. The terminal configuration and broad $P_M(m)$ distribution [Fig. 3.3 (case c)] show that the strong homogenisation observed in the high-inertia elastic case does not persist in this strongly dissipative regime.

Case (d): Near-Poisson state under stronger carrier-induced separation. The final high-inertia calculation examines the complementary regime in which carrier-induced separation acts faster than repeated collisions. For the collision leg,

$$\tau_f \simeq 0.084, \qquad \widehat{\tau}_c \simeq 0.171, \qquad \tau_f < \widehat{\tau}_c. \tag{3.5.3}$$

Despite the inelastic restitution coefficient $e = 0.75$, the particle occupancy distribution remains comparatively close to the Poisson reference [Fig. 3.3 (case d)]. At the final recorded configuration,

$$\mathrm{VMR}^{(d)}_{\mathrm{final}} = 1.45. \tag{3.5.4}$$

This terminal-snapshot value indicates a modest excess of occupancy fluctuations relative to the Poisson reference at the adopted box scale, rather than exact spatial homogeneity. The ordering $\tau_f < \widehat{\tau}_c$ implies that nearby carrier trajectories separate on a shorter characteristic time than successive collisions occur. Carrier-induced separation can therefore disperse nearby particles before repeated impacts sustain persistent aggregation. This behaviour is qualitatively consistent with the corresponding regime reported by López and Puglisi [8]. Because Case (d) uses a different carrier configuration and particle parameters, and no matched collisionless counterpart is reported, this result is interpreted only as a regime-level observation. It does not by itself establish a causal suppression of clustering by carrier-induced separation.

3.5.2 Low-Inertia Collision-Induced Organisation

The two low-inertia simulations **Case (e) - Case(f)** use the same carrier parameters, particle number $N = 980$, and particle response time $\tau_p = 0.1$, so their difference isolates the population-level effect of enabling exactly elastic collisions within the same dynamical regime. The collisionless case provides the reference particle organisation, while the collisional case uses $e = 1$. The low-inertia regime exhibits the opposite population-level response to elastic collisions as compared to the high-inertia regime. For $\tau_p = 0.1$, the collisionless leg remains approximately homogeneous [Fig. 3.2 (Case e)], with

$$\mathrm{VMR}^{(e)}_{\mathrm{final}} = 0.97. \tag{3.5.5}$$

Consistently, its occupancy distribution $P_M(m)$ remains close to the Poisson reference [Fig. 3.3 case (e)]. When exactly elastic collisions are enabled, $e = 1$, the collision leg develops pronounced particle-rich bands [Fig. 3.2 (Case f)] and

$$\mathrm{VMR}^{(f)}_{\mathrm{final}} = 3.58. \tag{3.5.6}$$

The corresponding $P_M(m)$ is substantially broader than the Poisson reference, with enhanced probability of both sparsely and highly occupied boxes. The full occupancy statistics therefore provide the primary evidence for collision-induced spatial inhomogeneity, while the VMR supplies a compact scalar summary of the same redistribution. The observed morphology is predominantly banded and is not interpreted as a population of compact clusters.

For the collision leg, $\widehat{\tau}_c \simeq 0.102$, $\tau_p = 0.1$, $\tau_f \simeq 1.45$. Thus, collisions occur on a time scale comparable with the particle relaxation time and substantially shorter than the characteristic carrier-induced separation

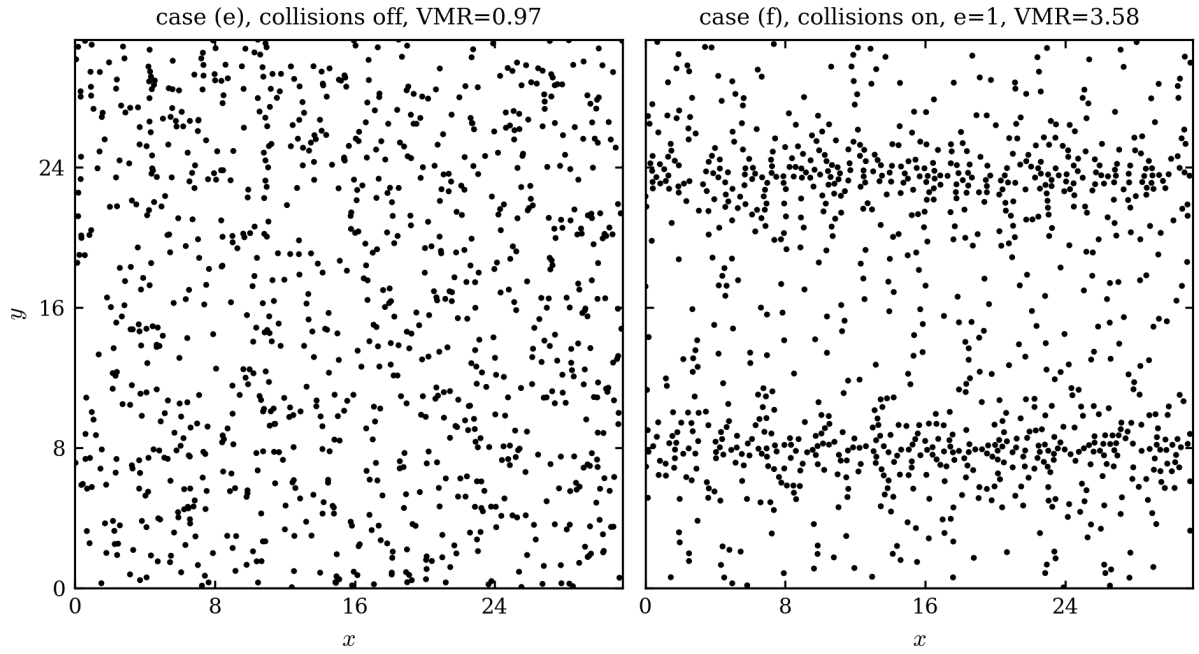

Figure 3.2: **Low-inertia collision-induced organisation.** Spatial configurations **(e)** without collisions and **(f)** with elastic collisions at $e = 1$. The collisionless population remains approximately homogeneous, whereas the collisional population develops pronounced banded spatial inhomogeneity.

time. Relative to the high-inertia cases, the smaller response time allows particle velocities to adjust more rapidly to changes in the prescribed carrier field, but the collisionless trajectories are not assumed to coincide with passive-tracer paths.

When elastic collisions are enabled, repeated impacts redistribute particle relative motion before carrier-induced separation acts over its characteristic time scale. Because $e = 1$, the resulting increase in spatial inhomogeneity cannot be attributed to collisional energy dissipation. Instead, it is associated with collision-induced redistribution within this low-inertia regime. The resulting broadening of the occupancy distribution is qualitatively consistent with the low-inertia elastic regime reported by López and Puglisi [8].

3.5.3 Reversed-Clustering Phenomenology

The low- and high-inertia parameter sets exhibit opposite population-level responses to the same exactly elastic collision law. Because the two regimes also differ in their carrier-flow parameters, the comparison is interpreted as a regime-level reversal rather than as an isolated effect of particle inertia. For this direct comparison, VMR is averaged over the final 30 recorded frames of each of the four elastic collision-on/collision-off simulations. This temporal averaging is used specifically in the present subsection to reduce sensitivity to the terminal snapshot and differs from the terminal-snapshot VMR values quoted in the preceding individual case descriptions.

In the low-inertia regime,

$$\overline{\mathrm{VMR}}^{\mathrm{low}}_{30,\mathrm{nocoll}} = 1.02, \qquad \overline{\mathrm{VMR}}^{\mathrm{low}}_{30,\mathrm{coll}} = 3.505. \tag{3.5.7}$$

Thus, the collisionless population remains close to the Poisson reference, whereas enabling elastic collisions produces substantially larger occupancy fluctuations and a more spatially inhomogeneous particle distribution.

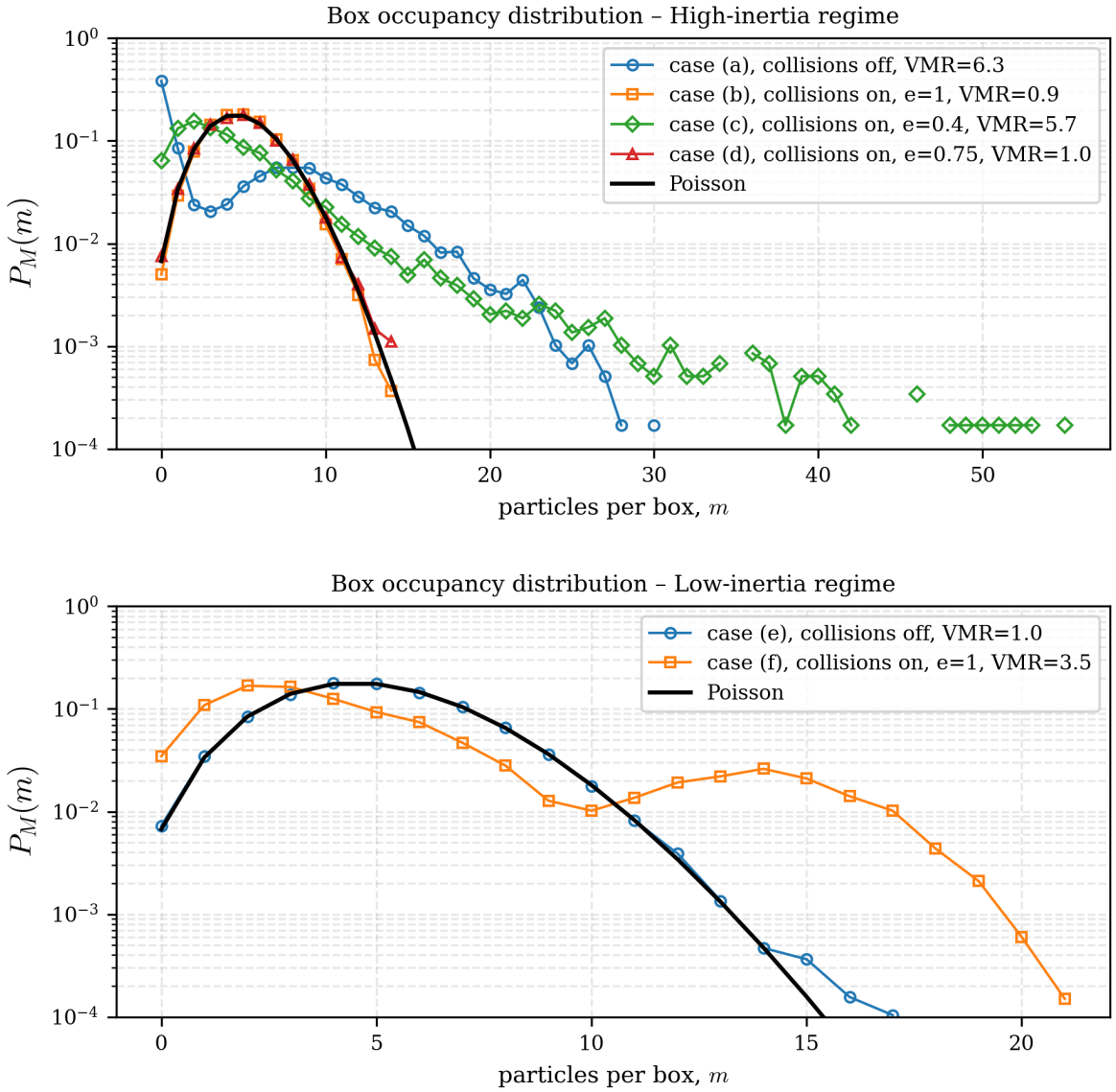

Figure 3.3: **Box-occupancy distributions and reversed clustering in the alternating sinusoidal shear.** The distributions $P_M(m)$ and the VMR values are averaged over the final 30 recorded frames; the Poisson reference has mean occupancy $N/M = 5$. **Top: high-inertia regimes.** The collisionless population is strongly non-Poissonian, whereas elastic collisions ($e = 1$) substantially narrow the occupancy distribution towards the Poisson reference. Stronger collisional dissipation ($e = 0.40$) produces a broad high-occupancy tail, while the distinct $e = 0.75$ regime, in which carrier-induced separation acts on the shorter characteristic time scale, remains comparatively close to the Poisson reference. **Bottom: low-inertia regime.** The collisionless population remains close to the Poisson reference, whereas enabling elastic collisions ($e = 1$) substantially broadens $P_M(m)$, indicating increased spatial inhomogeneity. The full occupancy distributions provide the primary distribution-level evidence for the regime classifications and for the reversal of the elastic collision response.

In the high-inertia regime, the ordering is reversed:

$$\overline{\mathrm{VMR}}^{\mathrm{high}}_{30,\mathrm{nocoll}} = 6.323, \qquad \overline{\mathrm{VMR}}^{\mathrm{high}}_{30,\mathrm{coll}} = 0.929. \tag{3.5.8}$$

No statistically stationary onset is detected for the high-inertia collisionless run. Its value 6.323 is therefore interpreted as a late-time final-window average rather than as a stationary-state mean. The full occupancy distributions provide the primary distribution-level evidence for this reversal [Fig. 3.3]. At low inertia, elastic collisions broaden $P_M(m)$ relative to the collisionless population, increasing the weight of both sparsely and highly occupied boxes. At high inertia, the collision-on population instead

has a substantially narrower occupancy distribution than the collisionless population and lies close to the Poisson reference. The value $\overline{\mathrm{VMR}}^{\mathrm{high}}_{30,\mathrm{coll}} = 0.929$ indicates weak under-dispersion at the adopted box scale rather than a separate clustering regime.

Elastic collisions therefore enhance spatial inhomogeneity in the low-inertia regime but suppress it in the high-inertia regime. Because $e = 1$, this reversal cannot be attributed to collisional energy dissipation; rather, it reflects a change in the population-level effect of collision-induced redistribution between the two dynamical regimes. López and Puglisi [8] referred to the corresponding inversion as *reversed clustering*: with collisions active, a lower-inertia population can become more strongly inhomogeneous than a higher-inertia one. A comparable reversal of the collisional effect was also reproduced for the Solomon–Gollub cellular flow [26] and was compared with the benchmark results [9]. Because of the page constraints of the present thesis, those additional simulation results are not documented in detail here.

3.5.4 Physical Synthesis, Chapter Contribution, and Scope

This chapter provides the controlled-flow bridge between the numerical framework of Chapter 2 and the turbulent analysis of Chapter 4. The alternating sinusoidal shear offers an analytically prescribed environment in which the effects of particle response, collision frequency, restitution, and carrier-induced separation can be clearly distinguished and analysed without the multiscale heterogeneity of turbulence. The objective is therefore not merely to reproduce particle patterns, but to identify how the same hard-particle interaction modifies populations belonging to different dynamical regimes.

The simulation results clearly demonstrate that finite-size collisions have no universal clustering or dispersive effect on particle spatial configurations. At high inertia, the collisionless leg develops a broad, strongly non-Poissonian occupancy distribution, whereas frequent elastic collisions produce an approximately homogeneous collision leg. A strongly inelastic case instead develops pronounced late-time aggregation, although no statistically stationary state is detected. When carrier-induced separation acts faster than successive collisions, $\tau_f < \widehat{\tau}_c$, the population remains approximately homogeneous despite inelastic impacts. At low inertia, the same exactly elastic collision law produces the opposite response: the collisionless leg remains close to the Poisson reference, while the collision leg develops substantial banded spatial inhomogeneity.

The principal physical result is therefore the reversal of the elastic collision response between low and high inertia regimes. Because this reversal occurs at $e = 1$, collisional energy dissipation is not required. Elastic impacts alone can enhance or suppress collective organisation by redistributing particle relative motion within populations whose collisionless dynamics are already different. Restitution modifies this redistribution further, while the competition between collision frequency and carrier-induced separation controls whether collision-associated organisation can persist. The final state must consequently be interpreted through the combined dynamical regime, rather than considering inertia, restitution or the collision law in isolation.

The contribution of this chapter is to establish this regime-dependent population response using representative configurations and, principally, the full box-occupancy distributions $P_M(m)$. The observed ordering is qualitatively consistent with the published López–Puglisi regimes, which are used only as external physical benchmarks. Since the low- and high-inertia calculations use different carrier parameters, the result is a regime-level reversal rather than a one-parameter determination of a critical inertia.

The central conclusion is that the effect of collisions depends on the carrier- and inertia-conditioned organisation of the particle population, leading to regime-dependent redistribution rather than to a universal clustering or dispersive response. These results also provide the answer to TQ2 (Section 1.2). The population-level consequence of enabling collisions cannot be inferred from the contact law alone; it depends on the organisation and time-scale balance of the population on which those collisions act.

Chapter 4

Collective Dynamics of Colliding Inertial Particles in Near-Wall Turbulence

4.1 Motivation: Collision-induced dynamics of inertial particles in turbulence

The preceding chapters examined in detail how finite-size interparticle collisions modify the dynamics and collective organisation of inertial particles in idealised carrier flows. A central outcome of that analysis was that collisions can play a dual dynamical role: depending on the dynamical regime, they may suppress an existing spatial organisation through repeated redistribution of particle relative motion or promote spatial inhomogeneity within an otherwise approximately homogeneous population [8, 9]. The controlled carrier flows considered in this work, including the Solomon–Gollub cellular flow and the time-periodic alternating sinusoidal shear flow, were intentionally chosen for their relatively simple and well-controlled kinematics, allowing inertia-driven and collision-induced effects to be distinguished. In addition, their velocity-gradient structure can be characterised directly, enabling particle organisation and collision statistics to be related to strain- and vorticity-dominated regions and to coherent transport structures of the carrier flow. Such controlled systems are therefore well suited to identifying the mechanisms underlying collision-modified collective behaviour, but they leave open the question of whether those mechanisms remain robust in more realistic flow environments.

In many natural and industrial particulate flows, particles are transported by ***turbulence***, whose velocity field is broadband, strongly time dependent and spatially heterogeneous [10, 17]. Extending the analysis to a turbulent carrier flow is therefore a natural next step. The central question of the present chapter is whether the mechanisms identified in the idealised settings persist in a turbulent carrier flow and, where they do not, which features of the turbulent environment modify them.

Unlike the analytically prescribed carrier flows considered previously, turbulence cannot in general be represented by a small number of coherent modes or simple closed-form velocity fields. Despite this broadband variability, coherent flow structures exist within the fluctuating background and significantly contribute to momentum transport, mixing, and particle dynamics. This organisation is particularly evident in wall-bounded turbulence, where the near-wall region contains elongated streamwise velocity *streaks*, intense shear, vortical motions, and strong velocity gradients [27, 29, 30].

Near-wall velocity streaks are elongated regions in which the instantaneous streamwise velocity is persistently lower or higher than the local mean streamwise velocity. Regions with negative streamwise fluctuation, $u' < 0$, are conventionally associated with low-speed streaks, whereas regions with $u' > 0$ are associated with high-speed streaks [31, 32]. In the quantitative analysis below, however, the carrier-speed classes are defined more restrictively using the standardised fluctuation ξ and the fixed thresholds

introduced in Eqs. (4.4.34) and (4.4.36). This near-wall organisation [28] creates a strongly anisotropic environment in which particle concentration [3], pair-relative motion, and collision occurrence [33] reflect the combined influence of particle inertia, turbulent fluctuations, and persistent carrier structure [34]. The temporal evolution of the sampled wall-parallel field is retained through reconstruction of the archived direct numerical simulation (DNS) sequence, so the particles experience an evolving carrier field rather than a frozen turbulent snapshot.

The present chapter examines particle motion driven by a time-resolved sequence of wall-parallel velocity fields extracted at a fixed wall-normal location from a turbulent channel-flow DNS. The carrier field is obtained from the fully three-dimensional incompressible Navier–Stokes solution, whereas the particle dynamics are deliberately restricted to a fixed wall-normal position $y = y_0$ near the lower wall. The particles are driven only by the local streamwise and spanwise carrier-velocity components. The resulting system is therefore not a two-dimensional turbulent flow, but a wall-parallel particle model driven by a time-dependent planar restriction of a fully three-dimensional turbulent carrier. This configuration provides an idealised setting, where finite-size particles are constrained to move along a surface while undergoing mutual collisions.

As in the preceding chapters, the collisional simulation contains finite-size inertial particles whose binary impacts are governed by the same restitution law introduced in Section 2.2. The collisionless control uses a *ghost-particle* representation [35–37] in which the same nominal particle radius is retained for geometric diagnostics, while excluded-volume interactions are disabled. Particles in this control are therefore permitted to overlap, and no collision impulse is applied during contact. Since the two legs share the same initial particle positions and velocities, particle response time, and prescribed carrier-flow realisation, their difference provides a matched collision-on/collision-off comparison in which the interparticle collision dynamics constitute the only deliberate change.

4.2 Research questions and structure of the analysis

The objective of this chapter is to determine how finite-size interparticle collisions modify inertial-particle organisation in a strongly anisotropic, time-dependent near-wall turbulent flow. The four research questions introduced below (RQ1–RQ4) decompose TQ3 (Section 1.2) into complementary questions concerning carrier-organised particle structure, collision selection, event-scale response, and same-pair recurrence with cumulative redistribution. The detailed analysis presented below focuses on the case $\tau_p = 0.5$.

RQ1 — How does the carrier flow organise particle structure and velocity coherence, and how do collisions modify this organisation across scale? The analysis first characterises the organisation of the near-wall carrier field, including low- and high-speed streamwise-velocity regions, vorticity, and the wall-parallel balance between strain and rotation, and relates these features to the spatial and kinematic organisation of the inertial-particle population [29, 30]. The multi-scale particle organisation is then quantified using concentration and pair statistics, scale-dependent number fluctuations, pair-axis nematic order, connected-structure morphology, density coherence, and particle-velocity correlations.

RQ1 therefore asks whether collisions modify only local pair dynamics or redistribute particle organisation across larger spatial scales. Complementary structural diagnostics are used to distinguish changes in particle-number fluctuations, pair alignment, connected morphology, density coherence, and velocity organisation. Their combined behaviour identifies which properties and spatial scales of the carrier-organised particle field are modified by collisions.

RQ2 — Which available pair configurations and carrier environments are preferentially selected for collision? Collision events occur within a pair population in an already spatially inhomogeneous and strongly anisotropic environment set by the carrier flow. Raw angular or environmental event counts

can therefore be large simply because a class contains more susceptible near-contact pairs or supplies greater inward radial motion; they do not, by themselves, demonstrate preferential collision selection. The analysis consequently distinguishes

$$\boxed{\text{particle occupancy} \neq \text{pair availability} \neq \text{inward-motion supply} \neq \text{collision conversion.}}$$

Each conditional statistic is normalised by the reference population appropriate to the quantity being measured. Angular selection is evaluated relative to the accumulated near-contact pair population and interpreted together with the corresponding inward radial kinematics. Environmental conditioning then separates differences in collision occurrence associated with pair availability, inward-motion supply, and residual event yield. Throughout, unity denotes proportional representation relative to the relevant exposure; these quantities are therefore relative selection statistics rather than absolute collision probabilities or hazards.

RQ2 (4.2) therefore asks: *after accounting for encounter availability and inward-motion supply, which pair orientations and carrier environments are over-represented among realised collisions?*

RQ3 — How does event-centred pair dynamics depart from isolated collision–drag evolution? Collision selection identifies which configurations reach contact, but not how the selected pairs evolve through and after impact. Collision events are therefore aligned at their contact time t_e and expressed in the event-centred lag

$$\tau = t - t_e.$$

The radial relative velocity, separation from contact, and pair-axis memory are then evaluated on the pre- and post-collision branches. The pre-collision branch characterises the selected incoming population, whereas the post-collision branch records the subsequent full-system evolution of the same pairs.

The measured trajectories are compared with isolated-pair reference constructions containing the hard-particle collision rule alone and the same collision followed by Stokes relaxation. These references identify the additional effects resulting from incorporating collision kinematics and Stokes drag to the system. Any systematic departure of the measured trajectories from them therefore quantifies dynamics absent from the isolated-pair description, including the time-dependent carrier forcing, interactions with neighbouring particles, and subsequent collisions.

RQ4 — How do repeated pair collisions accumulate into population-level redistribution? The final question connects the event-scale collision response to same-pair recurrence and longer-time reorganisation of the particle phase. Collision histories are grouped by unordered pair so that first and subsequent impacts, collision multiplicity, and inter-collision waiting times can be distinguished. The spatial character of recurrence is then quantified using the maximum resolved separation between consecutive impacts and geometrically defined proximity episodes, which determine whether repeated collisions occur while the pair remains within a prescribed local neighbourhood.

Repeated impacts do not, by themselves, imply collision-created confinement. A recurrent pair may be selected from a carrier-organised configuration that already favours slow separation or subsequent re-approach, while the wider particle neighbourhood continues to change. Pair-local recurrence is therefore examined together with initial-neighbour retention in the matched collisional and ghost-particle populations, thereby distinguishing repeated contact of the same pair from persistent membership of a wider many-particle neighbourhood. These recurrence and neighbourhood-turnover diagnostics are then interpreted jointly with the multiscale structural differences established under RQ1 (4.2).

RQ4 (4.2) therefore asks: *do repeated impacts remain predominantly local to selected pairs, and how does this pair-local recurrence coexist with wider neighbourhood persistence and the cumulative redistribution of particle organisation?*

Organisation of the chapter. The four questions form a mechanistic and statistical hierarchy,

$$\underbrace{\text{flow organisation}}_{\text{where particles are}} \rightarrow \underbrace{\text{pair availability}}_{\text{what can collide}} \rightarrow \underbrace{\text{collision selection}}_{\text{what collides}} \rightarrow \underbrace{\text{event response and redistribution}}_{\text{what changes afterwards}}, \tag{4.2.1}$$

Each level is examined across particle response time τ_p and treated as a distinct statistical object: the underlying particle organisation, the available encounter population, the realised collision-event population, and the subsequent post-collision dynamics are analysed separately.

The analysis builds on previous work on preferential concentration and near-wall particle organisation [29, 30], anisotropic pair geometry [34], turbulent collision rates and relative-velocity effects [35, 36, 38], collision-conditioned geometry and localisation [38, 39],and repeated encounters [35, 37]. This literature is based almost entirely on fully developed three-dimensional turbulence or on homogeneous isotropic turbulence. The model in this chapter is an idealised one, so these diagnostics are not carried over unchanged: each observable is adapted to the assumptions of the present model. Therefore, the cited studies serve only as a source of inspiration for the observables and statistics used; rather than as definitions applied directly. The contribution of the present chapter is to combine these levels of evidence within the same prescribed anisotropic carrier field, thereby separating the organisation supplied by the turbulent flow, the configurations selected for collision, the local response surrounding collision events, and the cumulative redistribution produced by the collision sequence.

4.3 Numerical experiment

4.3.1 Wall-bounded turbulent carrier-field

The **DNS** was performed at the Institut de Physique de Nice (InPhyNi) by Dr. Jérémie Bec, who provided the time-resolved velocity dataset used in the present study.Direct numerical simulation (DNS) resolves the Navier–Stokes equations without introducing a turbulence model, with the spatial and temporal resolution chosen to capture the dynamically relevant turbulent scales [40, 41].

In the present work, I reconstructed the supplied velocity record at the wall-normal position required by the reduced particle model (see 4.3.4) and used the resulting time-dependent carrier field to drive both the collisional and collisionless particle simulations.

The three-dimensional fluid velocity $\mathbf{u}(\mathbf{x}, t)$ and kinematic pressure $p(\mathbf{x}, t)$ satisfy

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \qquad \nabla \cdot \mathbf{u} = 0. \tag{4.3.1}$$

The stationary no-slip walls are located at $y = \pm h = \pm 1$, and periodic boundary conditions are imposed in the streamwise and spanwise directions, denoted by x and z, respectively. Distance from the lower wall is written as

$$\tilde{y} \equiv y + h = y + 1, \qquad 0 \leq \tilde{y} \leq 2. \tag{4.3.2}$$

All quantities are reported in the native nondimensional units of the DNS: lengths are scaled by the channel half-height $h = 1$, velocities by the solver's convective reference velocity, and times by the corresponding advective time scale. In these units,

$$\nu = 1.25 \times 10^{-4} = \frac{1}{8000}, \qquad U_b = 0.835. \tag{4.3.3}$$

The DNS time-resolved record covers $0 \leq t \leq 200$ and is stored at regular intervals $\Delta T = 0.2$, providing 1001 snapshots including both endpoints.

Wall units and temporal coverage. Let $\langle u \rangle$ denote the mean streamwise velocity. The friction velocity is diagnosed from the mean wall shear according to

$$u_\tau = \left[\nu \left| \frac{\partial \langle u \rangle}{\partial \tilde{y}} \right|_w \right]^{1/2}, \qquad \delta_\nu = \frac{\nu}{u_\tau}, \qquad t_\nu = \frac{\nu}{u_\tau^2}, \tag{4.3.4}$$

where δ_ν and t_ν are the viscous length and time scales, respectively. A superscript + denotes normalisation by the corresponding viscous scale; for example,

$$\tilde{y}^+ = \frac{\tilde{y}}{\delta_\nu}, \qquad u^+ = \frac{u}{u_\tau}, \qquad t^+ = \frac{t}{t_\nu}.$$

The measured wall gradient gives

$$Re_\tau = \frac{u_\tau h}{\nu} \simeq 360, \qquad u_\tau \simeq 0.045, \qquad \delta_\nu \simeq 2.78 \times 10^{-3}, \qquad t_\nu \simeq 6.17 \times 10^{-2}. \tag{4.3.5}$$

The carrier record therefore spans approximately

$$T^+_{\mathrm{rec}} \simeq 3240, \qquad \Delta T^+ \simeq 3.24.$$

Because ν is prescribed by the DNS solver whereas u_τ is diagnosed from the converged mean wall gradient, Re_τ is a measured property of the archived flow state rather than an independently prescribed parameter. All wall-unit conversions used throughout this chapter employ the values in Eq. (4.3.5). The complete carrier-flow and particle parameters are summarised in Appendix B.1.

4.3.2 Reduced wall-parallel particle model

The principal modelling assumption is that all particle centres lie on a single wall-parallel plane near the lower wall. Unless stated otherwise, all modelling assumptions of the microscopic model discussed in Chapter 3 (see 3.2.3) remain in force throughout the present chapter. Their wall-normal positions are imposed kinematically as

$$\tilde{y}_i(t) = \tilde{y}_0 = a = \frac{\sigma}{2} = 5.217 \times 10^{-3}, \qquad i = 1, \ldots, N, \tag{4.3.6}$$

where a and $\sigma = 2a$ are the particle radius and diameter, respectively. Geometrically, $\tilde{y}_0 = a$ is the center height of a sphere of radius a in contact with the lower wall. With the wall units scaling of Eq. (4.3.4),

$$\tilde{y}_0^+ \simeq 1.88, \qquad \sigma^+ \simeq 3.76, \tag{4.3.7}$$

so the particle motion is confined deep within the viscous sublayer. The viscous sublayer is the immediate near-wall region, conventionally $\tilde{y}^+ \lesssim 5$, in which molecular viscous forces dominates momentum transfer [27, 29] and the mean streamwise velocity is approximately linear, $u^+ \simeq \tilde{y}^+$ **(see Fig. 4.2).**

Because every particle center has the same wall-normal coordinate, the near-wall particle dynamics are restricted to the two-dimensional (x, z) plane. Their equatorial cross-sections form the hard-disk system used in the two-dimensional collision model developed in Chapter 2. The resulting configuration is therefore a two-dimensional particle system driven by a wall-parallel restriction of a fully three-dimensional turbulent carrier field.

The imposed wall-normal constraint represents an idealised sedimented monolayer in which the particles are assumed to remain in contact with the lower wall throughout the simulation. This approximation is motivated in part by the greater material density of the particles relative to the carrier fluid [16] and reduces the particle dynamics to motion within the wall-parallel surface. The wall is treated as smooth

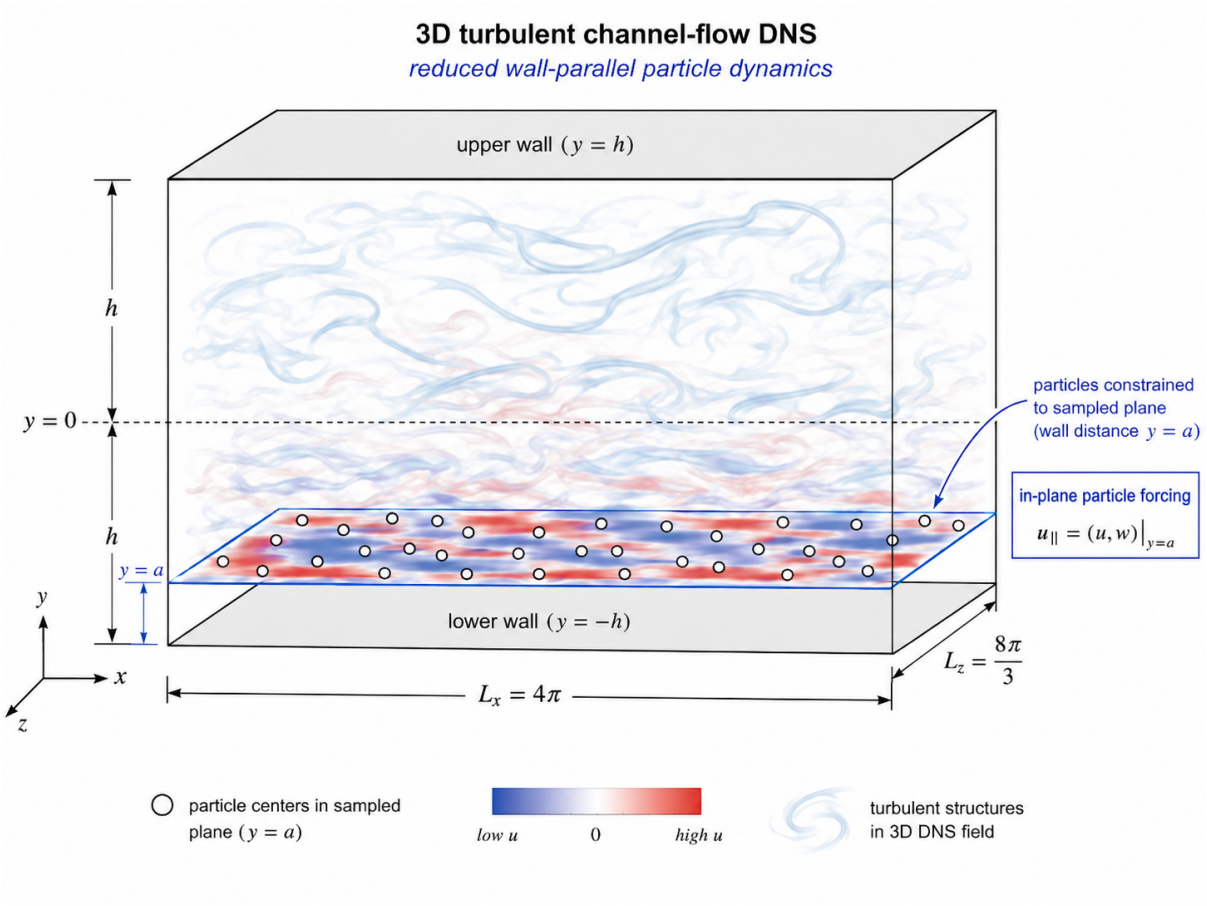

Figure 4.1: **Reduced wall-parallel particle experiment.** A single wall-parallel plane at wall distance $y_0 = a$ is extracted from a time-resolved turbulent channel-flow DNS at $\mathrm{Re}_\tau = u_\tau h/\nu \simeq 360$; the wall-normal height is exaggerated in the schematic for clarity. A population of N inertial particles evolves within this plane under Stokes drag driven by the local in-plane carrier velocity $(u, w)(x, \tilde{y}_0, z, t)$ and interacts, in the collisional system, through instantaneous hard-disk collisions. The plane is periodic in both wall-parallel directions and spans $L_x \times L_z = 4\pi \times 8\pi/3 \simeq 12.566 \times 8.378$. The particle dimensions, wall-normal offset, and streak morphology are **schematic** and are not drawn to scale.

and frictionless, so no tangential particle–wall force acts on the in-plane motion. Wall-normal dynamics, including detachment from the surface and the corresponding balance of gravity and wall-normal forces, are outside the scope of the present model.

Each particle is driven only by the wall-parallel carrier velocity evaluated at its centre:

$$\mathbf{u}_{\parallel,i}(t) = \begin{pmatrix} u \\ w \end{pmatrix} \big(x_i(t), \tilde{y}_0, z_i(t), t\big). \tag{4.3.8}$$

These components provide the forcing in the in-plane Stokes-drag equation. The response time τ_p is therefore interpreted as the prescribed effective relaxation time of the reduced wall-parallel particle model.

The plane spans the complete wall-parallel extent of the DNS domain,

$$L_x \times L_z = 4\pi \times \frac{8\pi}{3} \simeq 12.566 \times 8.378, \tag{4.3.9}$$

and is periodic in both x and z.

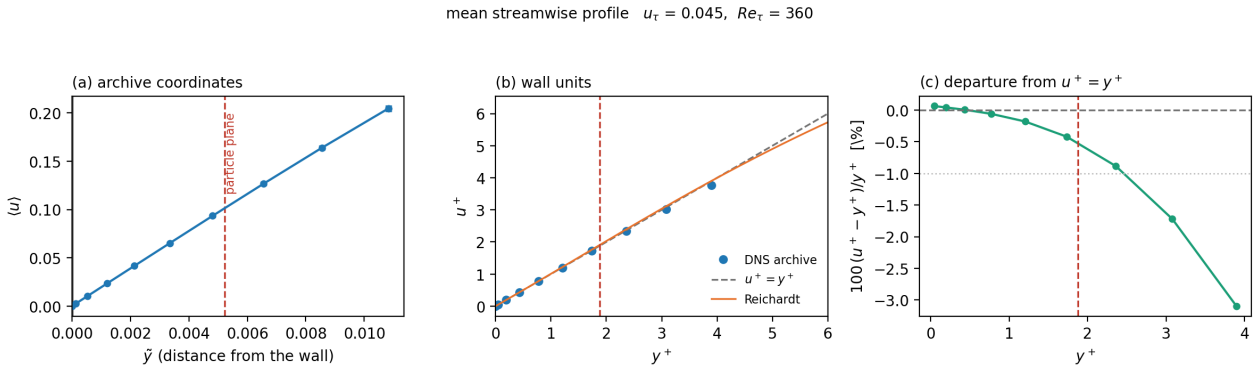

Figure 4.2: **Mean near-wall streamwise velocity profile of the archived DNS. (a)** Mean streamwise velocity $\langle u \rangle$ against wall distance $\tilde{y}$; the dashed line marks the prescribed particle plane $\tilde{y}_0 = a$. **(b)** The same profile in viscous units, compared with the viscous-sublayer relation $u^+ = \tilde{y}^+$ and the Reichardt composite profile [42]. **(c)** Relative deviation from $u^+ = \tilde{y}^+$. All retained planes lie within the viscous sublayer, and the particle centres are located at $\tilde{y}_0^+ \simeq 1.88$.

4.3.3 Wall-normal sampling of the carrier field

The restriction of the particle motion to $\tilde{y} = \tilde{y}_0$ is a modelling choice. The finite wall-normal sampling described here is a separate limitation of the available DNS dataset. Only the ten wall-normal collocation planes nearest to the lower wall are retained. These planes lie within the viscous sublayer; Therefore, the available record does not extend into the buffer layer ($\tilde{y}^+ \sim 5$–30) or the logarithmic region at larger wall distances [27] (see Fig. 4.2). The scope of the present particle study is therefore restricted to the immediate near-wall region.

In this analysis, the prescribed particle height lies between the retained DNS planes at approximately

$$\tilde{y}^+_{\text{lower}} \simeq 1.73, \qquad \tilde{y}^+_0 \simeq 1.88, \qquad \tilde{y}^+_{\text{upper}} \simeq 2.36. \tag{4.3.10}$$

The carrier velocity at $\tilde{y}_0$ is therefore not taken directly from a stored plane but reconstructed by wall-normal interpolation between the neighbouring DNS planes. The complete spatial and temporal reconstruction procedure is described in Sec. 4.3.4.

4.3.4 Carrier-field reconstruction and flow diagnostics

The turbulent carrier flow is supplied as a precomputed, time-resolved DNS dataset; the Navier–Stokes equations are therefore not solved as part of the present work. The numerical task is instead to reconstruct the recorded velocity field at the continuously varying particle positions and times required by the solver. For a particle with centre $\mathbf{x}_i = (x_i, \tilde{y}_0, z_i)$, the in-plane carrier velocity entering the drag law is

$$\mathbf{u}_{\parallel,i}(t) = (u, w)\big(x_i(t), \tilde{y}_0, z_i(t), t\big). \tag{4.3.11}$$

The DNS velocity field is available only at discrete spatial grid points and archived times, whereas the particle solver requires $\mathbf{u}_\parallel(x_i(t), \tilde{y}_0, z_i(t), t)$ at every integration step. Therefore, we need to reconstruct the carrier flow velocities at arbitrary positions and time-points sampled by the particles.

The reconstruction combines three interpolation operations. First, periodic cubic B-splines are used in the wall-parallel coordinates (x, z) [43, 44]. The periodic construction ensures that particles crossing a domain boundary sample the same continuous carrier field from either side. Second, the velocity at the prescribed particle height $\tilde{y}_0 = a$ is obtained using a local quadratic Lagrange interpolant constructed from the three nearest archived wall-normal planes. This step is necessary because $\tilde{y}_0$ does not coincide with a stored DNS plane. Finally, Catmull–Rom cubic interpolation is applied in time to reconstruct the velocity between successive archived snapshots [45]. The resulting space–time interpolant supplies the instantaneous wall-parallel velocity components (u, w) at every particle centre throughout the simula-

tion window. The three interpolation stages are implemented with the `scipy.interpolate` module: `RectBivariateSpline` for the wall-parallel directions; `lagrange` for the n_y-point wall-normal reconstruction on the stretched grid; and `CubicHermiteSpline` in time, supplied with the centred-difference tangents $m_i = (f_{i+1} - f_{i-1})/(t_{i+1} - t_{i-1})$, which makes it equivalent to Catmull–Rom interpolation [44].

For the flow-topology diagnostics used later in the chapter, the spatial derivatives in the periodic wall-parallel directions were evaluated spectrally [46]. For a periodic field f, differentiation in physical space corresponds to multiplication of its Fourier coefficients by the appropriate wavenumber in Fourier space. Thus, for example,

$$\frac{\partial f}{\partial x} = \mathcal{F}^{-1}\left[ik_x \widehat{f}\right], \tag{4.3.12}$$

where $\widehat{f} = \mathcal{F}[f]$, k_x is the streamwise wavenumber, and $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the forward and inverse Fourier transforms, respectively. The spectral differentiation was implemented using NumPy's `numpy.fft` module [47]. I also tested central finite-difference approximations for these derivatives, but they were less accurate at the present grid resolution; I therefore retained spectral differentiation in the periodic wall-parallel directions. The wall-normal derivative $\partial_{\tilde{y}} v$ is evaluated separately from the derivative of the local quadratic Lagrange interpolant constructed through the three nearest Chebyshev planes.

These gradients define the in-plane divergence $\nabla_\parallel \cdot \mathbf{u}_\parallel = \frac{\partial u}{\partial x} + \frac{\partial w}{\partial z}$, and the wall-normal vorticity $\omega_y = \frac{\partial w}{\partial x} - \frac{\partial u}{\partial z}$. The symmetric part of the in-plane velocity-gradient tensor provides the strain-rate quantities used later to distinguish locally strain-dominated and rotation-dominated regions and to condition particle and collision statistics. Although the underlying three-dimensional carrier flow obtained DNS is incompressible, its 2D wall-parallel restriction need not have **zero divergence**. Instead,

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = -\frac{\partial v}{\partial \tilde{y}}.$$

It also reveals an important feature of the reduced model: local convergence or divergence within the (x, z) plane can occur because the compensating flux lies in the omitted wall-normal direction. See Appendix B.2.3 for the carrier-flow topology.

I have also verified the interpolation procedure through grid-node recovery, boundary and periodicity checks, polynomial reproduction, and manufactured space–time fields. Relative RMS errors were approximately 10^{-8} for the known-answer fields and 4×10^{-4} for the time-dependent sinusoidal field, with no measurable loss of accuracy in the endpoint intervals. The complete validation is reported in Appendix B.2.2.

4.3.5 Microscopic dynamics, collision protocol, and control

The model considers a system of N identical inertial particles transported by the carrier field and constrained to the wall-parallel plane $\tilde{y} = \tilde{y}_0$. Each particle i is characterised by its microscopic description $(x_i(t), v_i(t))$, where $x_i(t)$ and $v_i(t)$ represents the position (centre) and velocity at time t respectively. The particles have a fixed finite diameter σ for interparticle contact. The hydrodynamic forcing (Stokes drag) acting on a particle is evaluated from the instantaneous carrier velocity $u(x_i, y_0 = a, z, t)$ computed at its centre. Between collisions, particle i evolves according to

$$\dot{\mathbf{x}}_i = \mathbf{v}_i, \qquad \dot{\mathbf{v}}_i = \frac{\mathbf{u}_\parallel(\mathbf{x}_i, t) - \mathbf{v}_i}{\tau_p}, \qquad \mathbf{u}_\parallel = (u, w)\big|_{\tilde{y}_0}, \tag{4.3.13}$$

where $\mathbf{x}_i = (x_i, z_i)$ and $\mathbf{v}_i = (v_{x,i}, v_{z,i})$ are the in-plane particle position and velocity, respectively, and τ_p is the particle response time. Refer to section 3.2 for detailed analysis of the microscopic model. The restriction $\mathbf{u}_\parallel(\mathbf{x}_i, t)$ is obtained from the reconstructed DNS field at the instantaneous particle centre, as described in the preceding subsection.

The microscopic dynamics are advanced using the same kick–drift integration scheme developed in Chapter 2. During each timestep, the Stokes-drag dynamics are treated in the kick stage and particle positions are advanced ballistically during the drift stage. Binary contacts occurring within a drift interval are localised and resolved using the same hard-disk collision and restitution law introduced in the preceding chapters. Thus, the present calculation replaces the idealised carrier flow by a time-resolved turbulent forcing while retaining the same microscopic particle integrator and collision mechanics.

Let L and W be the length and width respectively of the rectangular periodic domain. The nominal mean interparticle spacing associated with such a domain is given by

$$s = \sqrt{\frac{LW}{N}}$$

and the corresponding two-dimensional geometric area fraction is

$$\phi_A = \frac{N\pi(\sigma/2)^2}{LW}$$

For $N = 15000$, we get $s = 0.08378$ and $\phi_A = 0.012$. In our analysis, three particle response times are considered,

$$\tau_p = 0.5,\ 8,\ 16, \tag{4.3.14}$$

corresponding, under the wall scaling adopted in the simulations, to

$$\tau_p^+ = \frac{\tau_p u_\tau^2}{\nu} \simeq 8,\ 130,\ 260.$$

These cases span progressively increasing particle inertia, from dynamics responsive to near-wall viscous time scales to substantially more inertia-dominated motion. The detailed single-case analysis presented below focuses on $\tau_p = 0.5$ with $N = 15000$ and diameter $\sigma = 1.0434 \times 10^{-2}$. The larger-response-time simulations were performed as additional parameter cases and are not documented in this chapter. Particle states are stored every

$$\Delta t_{\text{save}} = 0.02 = \frac{\Delta T}{10}. \tag{4.3.15}$$

resulting in 10000 snapshots for the entire simulation period $T = 200$. Refer **appendix B.2.4** to storage and temporal localisation of collision geometry around snapshot frames.

If τ_p and σ are prescribed independently, then the material density ratio $\frac{\rho_p}{\rho_f}$ can be obtained from the relation 4.3.16

$$\tau_p = \frac{2}{9}\frac{\rho_p}{\rho_f}\frac{a^2}{\nu}, \tag{4.3.16}$$

which gives the nominal density ratios

$$\frac{\rho_p}{\rho_f} \simeq 10.3,\ 165,\ 331$$

for $\tau_p = 0.5$, 8, and 16, respectively. The resulting variation in density ratio is relevant to the validity of the drag-only approximation and is therefore retained explicitly in assessing the model assumptions. Throughout the present study, the fluid–particle coupling is taken to be ***one way*** by construction. It isolates the consequences of hard-particle collisions within a fixed turbulent carrier field but does not include the two-way momentum feedback that may arise at this packing fraction. The collision-on and collision-off legs nevertheless remain directly comparable because they use identical initial particle states, parameters, and carrier-flow realisations.

Hard-particle contact and event logging To isolate the cumulative population-level effects these collision impulses, a matched collisionless simulation is launched from the same initial particle state and driven by the same prescribed time-dependent carrier-flow. In this *ghost-particle* approximation (GPA), the particle radius serves as a geometrical contact scale, but excluded-volume interactions are completely ignored. As a result, particles may overlap each other and no collision impulse is applied at the point of contact [35].

Collisions are represented as instantaneous frictionless hard-disk contacts. Let $\widehat{\mathbf{n}}_{ij}$ be the contact normal from particle j to i pre-impact normal relative velocity and let $w_r^- = (\mathbf{v}_i^- - \mathbf{v}_j^-) \cdot \widehat{\mathbf{n}}_{ij} < 0$ be the pre-impact normal relative velocity. For equal masses and normal restitution coefficient e,

$$w_r^+ = -e\, w_r^-, \tag{4.3.17}$$

while the tangential relative component is unchanged by the frictionless impulse (Refer Subsec. 2.2.3).

Every collision is logged at least as (t_e, i, j), where t_e is the event time and i, j are particle indices involved in the event. Particle states used for diagnostics are saved every $\Delta t_{\text{save}} = 0.02$. Therefore, an event-conditioned quantity reconstructed from saved positions is not necessarily evaluated at the exact contact time.

Paired collision-on/off design The collisional and collisionless simulations are launched from the same particle initial state and are driven by the same prescribed carrier-flow record. For a distribution-level observable Y evaluated over a common time-period, the collision effect can be obtained as:

$$\Delta Y = Y_{\text{coll}} - Y_{\text{nocoll}}. \tag{4.3.18}$$

Paired temporal resampling is used so that common carrier-flow excursions are not counted as independent variability. Also, the particle-by-particle differencing requires additional care. Define the minimum-image counterpart displacement

$$\delta \mathbf{x}_i(t) = \mathbf{x}_i^{\text{coll}}(t) - \mathbf{x}_i^{\text{nocoll}}(t). \tag{4.3.19}$$

At sufficiently early time this compares nearby trajectories associated with the same initial particle label. Once collision-generated perturbations are increased to a substantial fraction of the mean spacing, the labels still correspond mathematically but they no longer represent the neighbouring local microstates; the complete propagation diagnostic is reported in Appendix B.2.7.

4.4 Observables, conditioning populations, and statistical estimators

The analysis studies in detail the full particle population, available pairs, approaching near-contact pairs, collision events, recolliding pairs, and the particle field at later times. Each observable is therefore defined with respect to a specific quantity of interest, conditioning population, and reference. This separation ensures that collision occurrence is not interpreted interchangeably with particle occupancy, pair availability, inward-motion exposure, or conversion into contact. Event-centred changes are likewise distinguished from the subsequent evolution produced by carrier forcing, neighbouring-particle interactions, and later collisions.

The diagnostics are organised according to four levels of the mechanistic analysis:

- **Carrier-organised particle population:** multiscale structural and kinematic observables characterise particle concentration, anisotropy, density coherence, connected morphology, and velocity organisation in each simulation leg;
- **Cumulative collision-associated redistribution:** matched comparisons between the collision and collisionless legs assess the accumulated redistribution of particle organisation through changes in particle-number fluctuations, pair-axis order, density coherence, and connected morphology.

- **Available and approaching pairs:** pair geometry and radial relative kinematics quantify the near-contact encounter population, the fraction of pairs that are approaching, and the inward motion available to bring them towards contact;
- **Contact and repeated encounters:** collision-selection statistics identify the pair configurations and carrier environments that are over- or under-represented among realised contacts, while event-centred and recurrence statistics characterise the immediate pair response and subsequent same-pair encounters;

Most quantities are established standard observables and are restated only to fix their definitions and normalisations in the present context. Specialised conditionings and reference constructions are defined where introduced.

4.4.1 Particle structure, anisotropy, and velocity coherence

Before conditioning on individual collision events, the particle phase is characterised using complementary structural and kinematic observables. Number fluctuations, radial pair concentration, density coherence, orientational order, connected morphology, and velocity coherence describe distinct properties and probe different spatial scales. Collision-on and collision-off statistics are evaluated using identical estimators, analysis parameters, and matched saved times.

Under the minimum-image convention, radial and angular statistics employing circular-shell normalisation are restricted to

$$r < r_{\mathrm{geom}} \equiv \frac{1}{2}\min(L_x, L_z), \tag{4.4.1}$$

for which all angular directions have equal geometric support.

Multiscale concentration. To quantify particle-number fluctuations across spatial scales, the wall-parallel periodic domain is partitioned into M_ℓ equal square cells of side length ℓ. Let $N_{\ell,c}(t)$ denote the number of particle centres contained in cell c at time t. Since the total particle number N is fixed, the mean occupancy at scale ℓ is $\overline{N}_\ell = \frac{N}{M_\ell}$.

For each snapshot, the Bessel-corrected variance of the cell occupancies is

$$S_N^2(t;\ell) = \frac{1}{M_\ell - 1}\sum_{c=1}^{M_\ell}\left[N_{\ell,c}(t) - \overline{N}_\ell\right]^2. \tag{4.4.2}$$

The corresponding snapshot variance-to-mean ratio is

$$\mathrm{VMR}(t;\ell) = \frac{S_N^2(t;\ell)}{\overline{N}_\ell}, \tag{4.4.3}$$

and the scale-dependent statistic reported below is obtained by averaging over the retained snapshots,

$$\mathrm{VMR}(\ell) = \langle \mathrm{VMR}(t;\ell)\rangle_t = \left\langle \frac{S_N^2(t;\ell)}{\overline{N}_\ell}\right\rangle_t. \tag{4.4.4}$$

The appropriate reference for the present fixed-N system is a population of N independently and uniformly distributed particle centres. For a partition into M_ℓ equal cells, the corresponding occupancy vector satisfies

$$\left(N_{\ell,1},\ldots,N_{\ell,M_\ell}\right) \sim \text{Multinomial}\left(N;\frac{1}{M_\ell},\ldots,\frac{1}{M_\ell}\right). \tag{4.4.5}$$

With the variance convention above, this independent-uniform fixed-N reference satisfies

$$\boxed{\mathbb{E}_{\text{ref}}\left[\text{VMR}(t;\ell)\right] = 1.} \tag{4.4.6}$$

Values above unity therefore indicate excess particle-number fluctuations at the observation scale ℓ, consistent with enhanced spatial concentration, whereas values below unity indicate suppressed fluctuations relative to the independent-uniform fixed-N reference. VMR quantifies the amplitude of number fluctuations at a prescribed spatial scale; it does not, by itself, measure cluster connectivity or define a characteristic cluster size.

For the numerical reference used in the results, 500 independent-uniform fixed-N configurations are processed using the same estimator, providing the reference mean and its corresponding standard deviation. The retained analysis scans the observation range $2 \le \ell/s \le 10$, where s is the nominal mean interparticle spacing.

Radial pair distribution and correlation dimension. Radial pair concentration is quantified by the radial distribution function [14, 36, 48]. For a separation bin $B_k = [r_k^-, r_k^+)$, let $n_k(t)$ be the number of unordered pairs in that bin and

$$A_k = \pi\left[(r_k^+)^2 - (r_k^-)^2\right] \tag{4.4.7}$$

its annular area. The estimator is

$$g_k = \left\langle \frac{2L_xL_z\, n_k(t)}{N(N-1)A_k} \right\rangle_t, \tag{4.4.8}$$

so that $g_k = 1$ for a spatially uniform pair population at the same mean number density, $g_k > 1$ indicates pair excess, and $g_k < 1$ pair depletion. For hard particles, $g(r) = 0$ for $r < \sigma$ by excluded volume; this geometric constraint is not interpreted as dynamical depletion.

The cumulative pair abundance is described by the correlation integral

$$C(r) = \left\langle \frac{2}{N(N-1)} \sum_{i<j} \mathbb{I}\left(r_{ij} < r\right) \right\rangle_t, \tag{4.4.9}$$

whose local logarithmic slope defines the scale-dependent correlation dimension,

$$D_2(r) = \frac{\mathrm{d}\log C(r)}{\mathrm{d}\log r}. \tag{4.4.10}$$

A homogeneous two-dimensional distribution satisfies $C(r) \propto r^2$ and hence $D_2 = 2$, whereas $D_2 < 2$ indicates enhanced concentration over the corresponding scale range [49]. For hard particles, no dimension is inferred inside the excluded region or from the immediate crossover near $r = \sigma$.

Pair-axis nematic order. An unordered particle pair defines a head–tail-symmetric axis rather than a directed vector. Its orientation is therefore represented by an angle $\theta_{ij} \in [0, \pi)$ relative to the streamwise direction. For the set $\mathcal{P}_r$ of M_r pairs in a separation shell centred on r, the complex nematic order parameter is [50, 51]

$$Q_2(r) = \frac{1}{M_r} \sum_{(i,j)\in\mathcal{P}_r} \exp\left(2\mathrm{i}\theta_{ij}\right), \tag{4.4.11}$$

where the factor of two accounts for the head–tail symmetry and ensures invariance under $\theta_{ij} \mapsto \theta_{ij} + \pi$. The corresponding nematic magnitude and director are

$$S(r) = |Q_2(r)|, \qquad \Theta(r) = \frac{1}{2} \arg Q_2(r), \tag{4.4.12}$$

with Θ defined modulo π. The magnitude satisfies $0 \le S \le 1$, with $S = 1$ denoting perfect alignment of the sampled pair axes. For an isotropic angular population, the second harmonic vanishes in the infinite-sample limit, although finite sampling generally produces a small nonzero value of S. In the present analysis, $S(r)$ measures the strength of head–tail-symmetric pair-axis alignment, while $\Theta(r)$ identifies its preferred orientation relative to the streamwise direction.

Cluster orientation provides a direct diagnostic of the anisotropy imposed on the particle population by the near-wall streak architecture[52], as examined in Subsecs. 4.5.1.1 and 4.5.1.2. Clusters are the connected components identified by periodic DBSCAN [53], and within each component c the particle coordinates are first unwrapped across the periodic boundaries. The gyration tensor of component c is

$$\mathbf{G}_c = \frac{1}{N_c} \sum_{i \in c} (\mathbf{x}_i - \mathbf{x}_c)\,(\mathbf{x}_i - \mathbf{x}_c)^{\mathsf{T}}, \tag{4.4.13}$$

where N_c is the component size and $\mathbf{x}_c$ its centroid in the unwrapped coordinates. The eigenvector of the largest eigenvalue defines the cluster major axis, whose angle Θ_c is, like θ_{ij}, defined modulo π. Cluster-axis order is measured with the same doubled-angle estimator,

$$Q_{2,\mathrm{cl}} = \frac{\sum_c w_c \exp(2\mathrm{i}\Theta_c)}{\sum_c w_c}, \tag{4.4.14}$$

with $w_c = 1$ for equal-cluster weighting and $w_c = N_c$ for particle-number weighting; the weighting is stated with each result.

Particle-density coherence. Following the count-based analysis above, the spatial coherence of the particle field is obtained from the periodic two-point correlation of the unsmoothed cell-count field. The single-particle self contribution and the mean-density mode are removed in Fourier space following the standard structure-factor decomposition. Each frame is then normalised at zero separation before averaging over the common frame support of the matched collisional and collisionless cases. The streamwise and spanwise coherence lengths, (ℓ_x) and (ℓ_z), are defined by the first positive-lag $(1/e)$ crossings of the corresponding axial correlations [54]. They measure axis-aligned density coherence rather than individual cluster dimensions. The calculation was implemented in Python, using NumPy [47] (`numpy.histogram2d` for the cell-count fields and `numpy.fft` for the discrete Fourier transforms) and Matplotlib for visualisation.

Particle-velocity organisation. For pairs whose minimum-image separation lies in a bin centred on r, the total particle-velocity correlation is defined as

$$C_v(r) = \frac{\langle \mathbf{v}_i \cdot \mathbf{v}_j \rangle_r}{\langle |\mathbf{v}|^2 \rangle}. \tag{4.4.15}$$

The numerator is averaged over all supported pairs in the separation bin, while the denominator is the instantaneous one-particle mean-square speed. The estimator is evaluated separately in each frame and subsequently averaged over the matched frame set. Because the instantaneous population-mean velocity is not removed in Eq. (4.4.15), $C_v(r)$ includes the common downstream advection of the particle population. It therefore measures total velocity coherence rather than coherence of the velocity fluctuations.

The angular organisation of the velocity fluctuations is evaluated separately after removing the instantaneous population mean,

$$\mathbf{v}'_i(t_m) = \mathbf{v}_i(t_m) - \frac{1}{N}\sum_{j=1}^{N}\mathbf{v}_j(t_m). \tag{4.4.16}$$

The corresponding fluctuation-velocity direction is

$$\phi'_i(t_m) = \operatorname{atan2}\Big(v'_{z,i}(t_m), v'_{x,i}(t_m)\Big), \tag{4.4.17}$$

and its polar order is

$$P'(t_m) = \left|\frac{1}{N'_m}\sum_{i\in\mathcal{I}'_m}\exp\big(\mathrm{i}\phi'_i(t_m)\big)\right|. \tag{4.4.18}$$

Here, $\mathcal{I}'_m$ excludes particles whose fluctuation speed is below 10^{-9} times the framewise RMS fluctuation speed, and $N'_m = |\mathcal{I}'_m|$. The magnitude satisfies $0 \le P' \le 1$: $P' = 1$ denotes perfect directional alignment, whereas an isotropic fluctuation-direction population has vanishing polar order in the infinite-sample limit. The corresponding angular density $g_{\phi'}(\phi)$ is compared with the isotropic reference $1/(2\pi)$.

4.4.2 Collision-selection statistics

Collision events are realised from a pair population that is already spatially inhomogeneous, streamwise anisotropic, and characterised by non-uniform relative-motion statistics. Raw collision counts alone can therefore be large either because more susceptible pairs are available or because the available encounters are more efficiently converted into contact. Analysing raw event counts alone consequently conflates pair availability with collision conversion. The collision statistics are therefore normalised by the population of *available encounters* appropriate to the quantity being measured [36, 37]. The central distinction is

$$\boxed{\text{pair availability} \;\neq\; \text{approaching-pair exposure} \;\neq\; \text{collision conversion.}} \tag{4.4.19}$$

Angular selection relative to pair availability. Let $p_{\mathrm{coll}}(\theta)$ denote the normalised angular density of collision events and $p_{\mathrm{near}}(\theta)$ the corresponding angular density of near-contact pair observations in the shell

$$\sigma \le r_{ij} \le 1.5\sigma, \tag{4.4.20}$$

with the unoriented pair axis folded onto $\theta \in [0, \pi)$. The angular collision-selection function is defined by

$$R(\theta) = \frac{p_{\mathrm{coll}}(\theta)}{p_{\mathrm{near}}(\theta)}. \tag{4.4.21}$$

For angular bins k,

$$p_k = \frac{N_{\mathrm{coll},k}}{N_{\mathrm{coll}}}, \qquad q_k = \frac{N_{\mathrm{near},k}}{N_{\mathrm{near}}}, \qquad R_k = \frac{p_k}{q_k}, \tag{4.4.22}$$

so that

$$\sum_k q_k R_k = 1. \tag{4.4.23}$$

The proportional-sampling null is therefore

$$\boxed{R(\theta) = 1.} \tag{4.4.24}$$

Values $R > 1$ identify orientations that are over-represented (preferentially selected) among collisions relative to their availability in the near-contact population, whereas $R < 1$ denotes under-representation (under-selection). Note that, $R(\theta)$ is a *relative selection statistic*; it is neither an absolute collision probability nor an absolute collision hazard. Because recurrent pairs may contribute several events to

p_{coll}, the event-weighted estimate is supplemented by a first-collision calculation in which each unordered pair contributes its first recorded collision.

Inward pair kinematics. The angular collision selection is interpreted together with the radial kinematics of the available pair population. The radial relative velocity is defined as

$$w_r = (\mathbf{v}_i - \mathbf{v}_j) \cdot \widehat{\mathbf{r}}_{ij}, \qquad \widehat{\mathbf{r}}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}}, \tag{4.4.25}$$

so that $w_r < 0$ denotes an approaching pair and $w_r > 0$ a separating pair. For the available pair population at orientation θ, the probability of inward radial motion is

$$P_-(\theta) = P(w_r < 0 \mid \theta), \tag{4.4.26}$$

and the mean inward speed conditional on approach is

$$S_-(\theta) = \langle -w_r \mid w_r < 0, \theta \rangle \, . \tag{4.4.27}$$

Their product,

$$\begin{aligned} A_-(\theta) &= P_-(\theta) S_-(\theta) \\ &= \langle [-w_r]_+ \mid \theta \rangle \, , \qquad [-w_r]_+ \equiv \max(-w_r, 0), \end{aligned} \tag{4.4.28}$$

measures the mean inward radial-motion supply per available pair at orientation θ. By construction,

$$A_-(\theta) \geq 0.$$

The decomposition separates the probability that an available pair is approaching from the magnitude of its inward motion conditional on approach [36]; pair abundance is treated separately through the angular availability distribution.

Exposure-normalised environmental conversion. Spatial and environmental collision counts are interpreted relative to the time for which approaching near-contact pairs are available[39]. For a class C, the accumulated approaching pair-time is

$$T_{\mathrm{app}}(C) = \sum_m \Delta t_m \sum_{i<j} \mathbf{1}_C(i, j, t_m) \, \mathbf{1}_{\sigma \leq r_{ij}(t_m) \leq 1.5\sigma} \, \mathbf{1}_{w_{r,ij}(t_m) < 0}, \tag{4.4.29}$$

where $\mathbf{1}_C$ assigns each pair to the environmental class under consideration. If $N_{\mathrm{coll}}(C)$ is the corresponding collision count, the collision rate per unit approaching near-contact pair-time is

$$\lambda(C) = \frac{N_{\mathrm{coll}}(C)}{T_{\mathrm{app}}(C)}. \tag{4.4.30}$$

Relative to the pooled rate over all classes,

$$\boxed{\mathcal{E}(C) = \frac{N_{\mathrm{coll}}(C)/T_{\mathrm{app}}(C)}{N_{\mathrm{coll}}/T_{\mathrm{app}}} = \frac{N_{\mathrm{coll}}(C)/N_{\mathrm{coll}}}{T_{\mathrm{app}}(C)/T_{\mathrm{app}}}.} \tag{4.4.31}$$

The statistic is reported only for classes with positive exposure and sufficient denominator support. The proportional-exposure reference is

$$\boxed{\mathcal{E}(C) = 1.} \tag{4.4.32}$$

Thus, $\mathcal{E}(C) > 1$ and $\mathcal{E}(C) < 1$ indicate collision rates per unit approaching-pair time above and below

the pooled rate, respectively. Whether a departure from unity is statistically resolved is determined from the corresponding uncertainty estimates.

$\mathcal{E}(C)$ is a relative event-rate statistic, not an absolute collision probability, hazard, or causal effect of the environment. Because T_{app} accounts for the duration of approach but not the magnitude of inward relative velocity, this statistic is distinct from the inward-flux-normalised collision yield.

In particular,

$$\boxed{\mathcal{E}(C) > 1 \;\Rightarrow\; \text{most collisions occur in } C.} \tag{4.4.33}$$

A class with little approaching-pair exposure can therefore have an enhanced relative collision rate while contributing only a small fraction of all collision events.

Carrier-speed conditioning. The local carrier state is classified using the standardised streamwise velocity fluctuation

$$\xi(x,z,t) = \frac{u'(x,z,t)}{\sigma_{u'}}, \qquad u'(x,z,t) = u(x,\tilde{y}_0,z,t) - \overline{U}(\tilde{y}_0), \qquad \sigma_{u'} = \left\langle u'^2 \right\rangle_{x,z,t}^{1/2}, \tag{4.4.34}$$

where

$$\overline{U}(\tilde{y}) = \langle u(x,\tilde{y},z,t)\rangle_{x,z,t}, \qquad \tilde{y}_0 = a, \tag{4.4.35}$$

so that $\overline{U}(\tilde{y}_0)$ is the Reynolds-mean streamwise velocity at the fixed analysis height. The carrier-speed classes are

$$\mathrm{LS} : \xi \le -1, \qquad \mathrm{I} : |\xi| < 1, \qquad \mathrm{HS} : \xi \ge 1. \tag{4.4.36}$$

Here LS, I, and HS denote low-, intermediate-, and high-speed carrier environments, respectively. These are instantaneous threshold-defined classes rather than geometrically identified or temporally tracked streak objects. Their use in the environmental availability, inward-motion, and collision-yield decomposition is described in Sec. 4.5.4.

Flow-topology conditioning. Carrier-flow topology is studied using the same exposure-normalised conversion statistic $\mathcal{E}(C)$. For each approaching near-contact pair and collision event, the topology is evaluated at the minimum-image pair midpoint. The local balance between in-plane strain and rotation is described by the Okubo–Weiss-type indicator [55, 56]

$$W = s_n^2 + s_s^2 - \omega_y^2, \tag{4.4.37}$$

where

$$s_n = \frac{\partial u}{\partial x} - \frac{\partial w}{\partial z}, \qquad s_s = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}, \qquad \omega_y = \frac{\partial w}{\partial x} - \frac{\partial u}{\partial z}. \tag{4.4.38}$$

Here, s_n and s_s are the normal and shear components of the in-plane strain, while ω_y is the wall-normal vorticity. With this sign convention, $W > 0$ denotes strain-dominated in-plane kinematics [55, 56].

Because the sampled wall-parallel velocity field is a restriction of a three-dimensional incompressible flow and can have non-zero in-plane divergence , W is interpreted as a planar diagnostic rather than as the full three-dimensional velocity-gradient invariant. For conditioned estimation, W is divided into equal-exposure quantile bins using the accumulated approaching near-contact pair-time, and the same bin boundaries are applied to the collision events. Thus, $\mathcal{E}(W) = 1$ denotes collision occurrence proportional to the available exposure at that topology, whereas values above or below unity denote enhanced or suppressed conversion, respectively.

4.4.3 Event-centred observables and isolated-pair references

Event-centred observables characterise the evolution of a particle pair before and after a recorded collision. For event e, let $g_e = \{i_e, j_e\}$ denote the unordered focal pair and t_e its event-resolved collision time. The event-centred lag is

$$\tau = t - t_e, \tag{4.4.39}$$

so that $\tau < 0$ and $\tau > 0$ denote the pre- and post-collision branches, respectively, while $\tau = 0$ denotes the exact contact time.

For the focal pair, the minimum-image separation vector and distance are

$$\mathbf{r}_e(\tau) = \mathbf{x}_{i_e}(t_e + \tau) - \mathbf{x}_{j_e}(t_e + \tau), \qquad r_e(\tau) = |\mathbf{r}_e(\tau)|, \tag{4.4.40}$$

and the radial relative velocity is

$$w_{r,e}(\tau) = \left[\mathbf{v}_{i_e}(t_e + \tau) - \mathbf{v}_{j_e}(t_e + \tau)\right] \cdot \widehat{\mathbf{r}}_e(\tau). \tag{4.4.41}$$

where, $w_{r,e} < 0$ denotes approach and $w_{r,e} > 0$ separation.

For any event-centred quantity $X_e(\tau)$, the ensemble average is

$$\langle X(\tau)\rangle_e = \frac{1}{M(\tau)} \sum_{e \in \mathcal{E}(\tau)} X_e(\tau), \tag{4.4.42}$$

where $\mathcal{E}(\tau)$ is the set of events possessing a valid state at lag τ, and $M(\tau) = |\mathcal{E}(\tau)|$. Because the supported event set can vary with lag, $M(\tau)$ and the corresponding number of distinct unordered pairs are reported with the curves. The first observable measures the mean approach or recession of the event-selected pairs, while the second measures their mean separation from contact.

The primary radial observables are

$$\langle w_r(\tau)\rangle_e\,, \qquad \left\langle \frac{r(\tau)}{\sigma} - 1 \right\rangle_e\,, \tag{4.4.43}$$

which measure the mean radial approach or recession and the mean diameter-normalised separation from contact of the event-selected population.

Event-axis memory and independent-draw baseline. Let

$$\theta_e(\tau) = \operatorname{atan2}\big(r_{e,z}(\tau), r_{e,x}(\tau)\big) \bmod \pi \tag{4.4.44}$$

denote the folded pair-axis orientation, appropriate to the head–tail symmetry of an unordered pair [51].Because particle positions are continuous through an instantaneous hard-particle collision, the reference axis $\theta_{e,0}$ is taken from the saved state nearest t_e, without a pre- or post-collision side constraint. The raw event-axis memory is

$$C_2^e(\tau) = \frac{1}{M(\tau)} \sum_{e \in \mathcal{V}(\tau)} \cos\big(2[\theta_e(\tau) - \theta_{e,0}]\big)\,. \tag{4.4.45}$$

The factor of two accounts for the head–tail symmetry of the line of centres.

Because the carrier flow imposes persistent streamwise anisotropy, statistically independent axes need not have zero raw correlation. The appropriate baseline is therefore the correlation expected from independent draws from the measured angular marginals. Using the same lag-supported event population in every

term, define

$$Q_{2,\tau}(\tau) = \frac{1}{M(\tau)} \sum_{e \in \mathcal{V}(\tau)} \exp[2\mathrm{i}\theta_e(\tau)]\,, \tag{4.4.46}$$

$$Q_{2,0}(\tau) = \frac{1}{M(\tau)} \sum_{e \in \mathcal{V}(\tau)} \exp\left[2\mathrm{i}\theta_{e,0}\right], \tag{4.4.47}$$

so that

$$C^e_{2,\mathrm{ind}}(\tau) = \mathrm{Re}\left[Q_{2,\tau}(\tau)\overline{Q_{2,0}(\tau)}\right]. \tag{4.4.48}$$

The baseline-corrected excess memory is

$$C^e_{2,\mathrm{exc}}(\tau) = C^e_2(\tau) - C^e_{2,\mathrm{ind}}(\tau). \tag{4.4.49}$$

For the same lag-supported population, the corresponding contact-axis excess is

$$D_2(\tau) = 1 - \left|Q_{2,0}(\tau)\right|^2, \tag{4.4.50}$$

and the normalised event-axis memory is

$$\widetilde{C}^e_2(\tau) = \frac{C^e_{2,\mathrm{exc}}(\tau)}{D_2(\tau)}. \tag{4.4.51}$$

The normalisation is reported only when $D_2(\tau)$ is sufficiently different from zero for numerical stability. Thus $\widetilde{C}^e_2 = 0$ corresponds to axes that are independent subject to the same measured angular marginals, while positive values indicate excess same-pair orientational memory beyond that anisotropic baseline. An isotropic zero baseline would retain correlation caused solely by the carrier-imposed angular preference and could therefore be mistaken for same-pair memory.

Collision times are resolved independently of the saved-state cadence. Particle states are stored every 0.02 time units. For each non-zero requested lag, the nearest saved state on the corresponding temporal side of the focal event is used. No measured radial velocity or separation is assigned exactly at $\tau = 0$; the closest reported measured states are at $\tau = -0.02$ and $+0.02$. Further implementation and uncertainty details are given in Appendix B.3.5.

Collision-only reference. The isolated-pair references are constructed from a flux-weighted proxy for the measured collision population. Approaching near-contact pairs satisfying

$$\sigma \le r \le 1.5\sigma, \qquad w_r < 0, \tag{4.4.52}$$

provide samples of the incoming radial and tangential relative-velocity components (w_r^-, w_t^-). To account for the greater contact flux associated with faster approach, these observations are sampled with weight proportional to $-w_r^-$, consistent with the inward-relative-motion weighting used in particle-collision statistics [36–38].

The sampled relative velocity is imposed at synthetic contact,

$$\mathbf{r}_0 = \sigma\widehat{\mathbf{n}}, \qquad \Delta\mathbf{v}^- = w_r^-\widehat{\mathbf{n}} + w_t^-\widehat{\mathbf{t}}, \tag{4.4.53}$$

where $\widehat{\mathbf{n}}$ and $\widehat{\mathbf{t}}$ are the normal and tangential unit vectors at contact. For a smooth frictionless collision with normal restitution coefficient ε_n,

$$w_r^+ = -\varepsilon_n w_r^-, \qquad w_t^+ = w_t^-, \tag{4.4.54}$$

so that

$$\Delta\mathbf{v}^{+} = -\varepsilon_n w_r^{-}\widehat{\mathbf{n}} + w_t^{-}\widehat{\mathbf{t}}. \tag{4.4.55}$$

This is the same hard-particle restitution law used in the particle simulations; see Eq. (4.3.17).

For the collision-only reference, carrier forcing, Stokes relaxation, neighbouring-particle interactions, and subsequent collisions are omitted. The relative velocity is therefore constant on each side of contact, giving the piecewise-ballistic trajectory

$$\mathbf{r}_{\mathrm{bal}}(\tau) = \begin{cases} \mathbf{r}_0 + \Delta\mathbf{v}^{-}\tau, & \tau < 0, \\ \mathbf{r}_0, & \tau = 0, \\ \mathbf{r}_0 + \Delta\mathbf{v}^{+}\tau, & \tau > 0. \end{cases} \tag{4.4.56}$$

This reference therefore represents the evolution implied by the collision rule and ballistic continuation alone.

Collision–drag reference. The collision–drag reference uses the same proxy contact population and collision rule but additionally retains isolated homogeneous Stokes relaxation of the relative velocity,

$$\frac{\mathrm{d}\Delta\mathbf{v}}{\mathrm{d}\tau} = -\frac{\Delta\mathbf{v}}{\tau_p}. \tag{4.4.57}$$

It describes two isolated particles whose relative velocity relaxes towards zero in a common background velocity, while differential carrier forcing, neighbouring-particle interactions, and subsequent collisions are omitted. For $\tau > 0$,

$$\Delta\mathbf{v}^{+}_{\mathrm{drag}}(\tau) = \Delta\mathbf{v}^{+}\exp\!\left(-\frac{\tau}{\tau_p}\right). \tag{4.4.58}$$

The negative-lag branch is obtained by backward continuation of the same reduced Stokes equation from the incoming synthetic contact state. Because this continuation amplifies the relative velocity as $\exp(|\tau|/\tau_p)$, it is retained only over the validated backward-lag range. The analytical trajectories and precise lag restrictions are given in Appendix B.3.5.

Reference hierarchy and interpretive boundary. The two isolated-pair references form the hierarchy

collision-only	:	collision rule + ballistic continuation,
collision–drag	:	collision rule + isolated Stokes relaxation.

(4.4.59)

Both references are reduced diagnostic models constructed from the same flux-weighted proxy contact population. Both omit the spatially and temporally varying carrier forcing, interactions with neighbouring particles, and subsequent collisions; the collision-only reference additionally omits Stokes relaxation.

They are therefore not event-level counterfactuals for the measured collisions. Differences between the measured event-centred statistics and the isolated references quantify departure from these reduced proxy descriptions. Such departures can reflect both dynamics absent from the reference and the approximation involved in representing the measured collision population, and cannot be identified with the isolated causal effect of the focal collision.

For example,

$$\left\langle \frac{r(\tau)}{\sigma} - 1 \right\rangle_{\mathrm{meas}} < \left\langle \frac{r(\tau)}{\sigma} - 1 \right\rangle_{\mathrm{drag}} \tag{4.4.60}$$

indicates that the measured event population remains closer, separates more slowly, or re-approaches sooner than predicted by isolated collision–drag evolution. It does not by itself establish collision-induced

binding, an attractive collision impulse, or trapping by the carrier flow.

4.4.4 Residence, recurrence, and neighbourhood persistence

Repeated collisions of the same unordered pair may arise from persistent local co-residence, repeated nearby motion, or reunion after a large excursion. Geometric residence is compared between the matched collision-on and collision-off legs. Multiplicity, inter-collision excursion and waiting time are defined only for the collisional leg because they require recorded collision events.

Geometric proximity episodes. Pairwise proximity is defined using the hysteretic thresholds

$$r_{\mathrm{in}} = 1.5\sigma, \qquad r_{\mathrm{esc}} = 2\sigma. \tag{4.4.61}$$

The entry radius coincides with the near-contact shell used in the collision-selection analysis, while the larger escape radius prevents small fluctuations about r_{in} from repeatedly terminating and restarting an episode.

For unordered pair g, proximity episode m begins at the first resolved time $a_{g,m}$ for which $\sigma \le r_g(a_{g,m}) \le r_{\mathrm{in}}$, and ends at the first subsequent resolved time $b_{g,m}$ satisfying $r_g(b_{g,m}) > r_{\mathrm{esc}}$. Its resolved duration is

$$T_{\mathrm{res},g,m} = b_{g,m} - a_{g,m}. \tag{4.4.62}$$

Once the episode boundaries are fixed, the observed number of collisions within the episode is

$$N^{\mathrm{obs}}_{g,m} = \sum_{e \in \mathcal{E}_g} \mathbf{1}_{a_{g,m} \le t_e < b_{g,m}}, \tag{4.4.63}$$

where $\mathcal{E}_g$ contains the event-resolved collision times of pair g. Episodes already active at the beginning of the retained record are excluded because their entry times are unknown. Episodes still active at the record end are retained as right-censored observations [57]; their recorded collision counts are therefore lower bounds on their complete-episode counts.

Collision multiplicity and waiting times. For unordered pair g, let

$$t_{g,1} < \cdots < t_{g,M_g} \tag{4.4.64}$$

denote its retained collision times. The observed collision multiplicity is

$$M_g = \#\{t_{g,k}\}, \tag{4.4.65}$$

and is a finite-record quantity rather than an intrinsic property of the pair. For $k = 1, \ldots, M_g - 1$, the inter-collision waiting time [58, Ch. 4], is

$$T_{g,k} = t_{g,k+1} - t_{g,k}. \tag{4.4.66}$$

The first-to-second gap, $T_{g,1} = t_{g,2} - t_{g,1}$, provides a pair-balanced summary because each recurrent pair contributes at most one such interval.

Resolved inter-collision excursion and local recurrence. For consecutive collisions $t_{g,k}$ and $t_{g,k+1}$, the maximum resolved separation is

$$\widehat{R}_{\mathrm{max},gk} = \max_{t_m \in (t_{g,k}, t_{g,k+1})} \frac{r_g(t_m)}{\sigma}, \tag{4.4.67}$$

where the maximum is taken over saved states strictly inside the interval. Thus $\widehat{R}_{\max,gk}$ is a sampled lower bound on the true continuous-time maximum. Intervals containing no interior saved state have undefined $\widehat{R}_{\max,gk}$ and are excluded from excursion-conditioned statistics rather than assigned the contact value.

Define the supported intervals of pair g by

$$\mathcal{K}_g^{\sup} = \left\{ k \in \{1, \dots, M_g - 1\} : \widehat{R}_{\max,gk} \text{ is defined} \right\}, \qquad n_g^{\sup} = |\mathcal{K}_g^{\sup}|. \tag{4.4.68}$$

Its within-pair local-recurrence fraction at operational threshold R_c is

$$c_g(R_c) = \frac{1}{n_g^{\sup}} \sum_{k \in \mathcal{K}_g^{\sup}} \mathbf{1}_{\widehat{R}_{\max,gk} \le R_c}. \tag{4.4.69}$$

For any multiplicity class C, define

$$\mathcal{G}_C^{\sup} = \left\{ g : M_g \in C,\ n_g^{\sup} > 0 \right\}. \tag{4.4.70}$$

The pair-balanced local-recurrence estimator is

$$f_{\text{local}}^{\text{pair}}(C; R_c) = \frac{1}{|\mathcal{G}_C^{\sup}|} \sum_{g \in \mathcal{G}_C^{\sup}} c_g(R_c), \tag{4.4.71}$$

so every recurrent pair receives equal weight irrespective of how many supported intervals it contributes. The corresponding interval-weighted estimator is

$$f_{\text{local}}^{\text{int}}(C; R_c) = \frac{\displaystyle\sum_{g \in \mathcal{G}_C^{\sup}} \sum_{k \in \mathcal{K}_g^{\sup}} \mathbf{1}_{\widehat{R}_{\max,gk} \le R_c}}{\displaystyle\sum_{g \in \mathcal{G}_C^{\sup}} n_g^{\sup}}. \tag{4.4.72}$$

This estimator gives greater weight to highly recurrent pairs because they supply more intervals. Pair-balanced and interval-weighted statistics are therefore reported separately.

The primary analysis uses $R_c = 2$, corresponding to the escape radius in Eq. (4.4.61). Hence $\widehat{R}_{\max,gk} \le 2$ means that no separation beyond 2σ is resolved at the available saved states before the next collision. This is an operational definition of pair-local recurrence; it does not define a bound state, attractive interaction, mechanically closed cage, or dynamically invariant region. Sensitivity is also examined at the neighbouring thresholds $R_c = 1.5$ and $R_c = 2.5$, showing that the qualitative conclusion does not depend on the particular choice $R_c = 2$.

Neighbourhood persistence. To distinguish pair-local recurrence from persistence of the wider many-particle environment, define the neighbours of particle i within radius r_N at time t as

$$\mathcal{N}_i(t; r_N) = \left\{ j \ne i : r_{ij}(t) \le r_N \right\}, \tag{4.4.73}$$

where r_{ij} is the minimum-image pair distance [14].

For reference time t_0, the retained fraction of the initial neighbourhood after lag τ is

$$Q_N(\tau; t_0) = \frac{1}{N_{\text{elig}}(t_0)} \sum_{i \in \mathcal{I}(t_0)} \frac{|\mathcal{N}_i(t_0; r_N) \cap \mathcal{N}_i(t_0 + \tau; r_N)|}{|\mathcal{N}_i(t_0; r_N)|}, \tag{4.4.74}$$

where

$$\mathcal{I}(t_0) = \{i : |\mathcal{N}_i(t_0; r_N)| > 0\}, \qquad N_{\text{elig}}(t_0) = |\mathcal{I}(t_0)|. \tag{4.4.75}$$

Particles without neighbours at t_0 are therefore excluded from that reference-time average. Equation (4.4.74) is a particle-averaged analogue of the neighbour-list correlation function defined by Rabani, Gezelter and Berne [59]: their ratio of averages, $\langle|\mathcal{N}_i(t_0; r_N) \cap \mathcal{N}_i(t_0+\tau; r_N)|\rangle_i / \langle|\mathcal{N}_i(t_0; r_N)|\rangle_i$, weights each particle by its number of initial neighbours, whereas here each eligible particle receives equal weight.

The reported neighbourhood persistence is

$$Q_N(\tau) = \frac{1}{N_{t_0}(\tau)} \sum_{t_0 \in \mathcal{T}_0(\tau)} Q_N(\tau; t_0), \tag{4.4.76}$$

where $\mathcal{T}_0(\tau)$ contains the supported reference times and $N_{t_0}(\tau) = |\mathcal{T}_0(\tau)|$. Thus $Q_N(0) = 1$, while its decay measures replacement of particles belonging to the initial neighbourhood.

This statistic compares neighbourhood membership only at t_0 and $t_0 + \tau$; it does not require a neighbour to remain continuously within r_N over the intervening interval. In the reported analysis, $r_N = 1.5s$, where s is the mean interparticle spacing.

4.5 Results

The results are organised from the pre-existing organisation of the particle population to the cumulative consequences of collisions. All comparisons between matched collision-on and collision-off systems are based on simulations sharing the same carrier-flow record and identical initial state. We first characterise the spatial and kinematic organisation of particles in the near-wall flow (Secs. 4.5.1.1–4.5.2). We then determine which available near-contact pairs collide and where these events occur, distinguishing pair availability from conversion into collisions (Secs. 4.5.3 and 4.5.4). Finally, we study how event-centred and recurrence statistics subsequently describe the immediate response to an impact and repeated encounters of the same pair (Secs. 4.5.5 and 4.5.6).

4.5.1 Carrier architecture and multiscale particle organisation

The detailed analysis focuses on the case $\tau_p = 0.5$. The collision-on and collision-off simulations are launched from the same labelled particle configuration and are driven by the same prescribed carrier-flow realisation. All geometric, material and numerical parameters are therefore identical; the only controlled difference is whether particle–particle collisions are resolved. Within this matched one-way-coupled system, differences between the two legs can therefore be attributed to the inclusion of collision effects. At later times, however, these differences represent the accumulated dynamical response initiated by collisions rather than the effect of an individual collision.

4.5.1.1 Near-wall carrier-flow streak architecture

Figure 4.3(a) shows the instantaneous streamwise velocity fluctuation at $t = 200$, evaluated on the reconstructed wall-parallel field at the prescribed particle height $\tilde{y}_0 = a = 5.217 \times 10^{-3}$. The carrier-speed classes at each point are defined from the standardised streamwise fluctuation introduced in Eqs. (4.4.34)–(4.4.36). The field is organised into narrow, elongated regions of negative and positive u', aligned predominantly with the streamwise direction. The thresholded representation in Fig. 4.3(b) retains the same directional organisation, showing that the observed streak structure is not confined to weak fluctuations around $u' = 0$.

For the displayed frame, the carrier-speed classes are defined by $u' < -\sigma_{u'}$ for low-speed regions and $u' > \sigma_{u'}$ for high-speed regions, with $\sigma_{u'} = 0.0414$; the remaining points form the intermediate class.

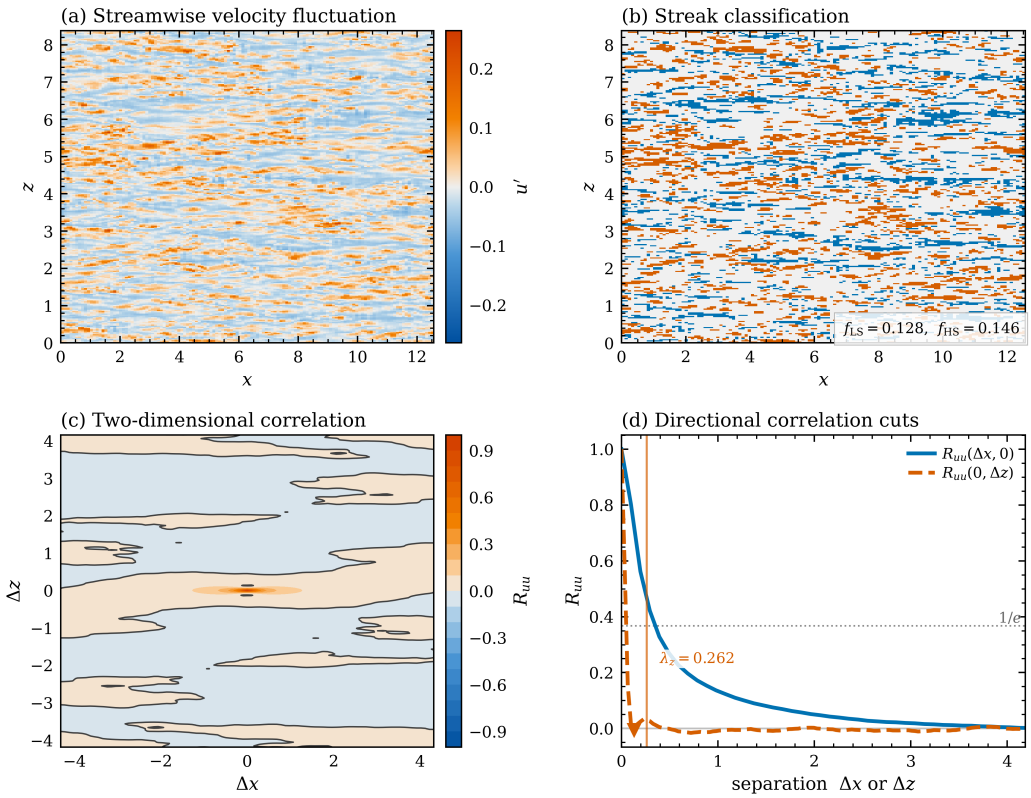

Figure 4.3: **Carrier-flow streak architecture at** $y = a = 0.005217$**. (a)** Streamwise velocity fluctuation $u'(x, z)$ at $t = 200$. **(b)** Low-, intermediate-, and high-speed regions defined by thresholds $u' = \pm\sigma_{u'}$, with mean area fractions 0.128, 0.726, and 0.146, respectively. **(c)** Two-dimensional autocorrelation averaged over 100 snapshots; the contour marks $R_{uu} = 0$. **(d)** Streamwise and spanwise correlation cuts. The $1/e$ scales are 0.351 and 0.049 and the spanwise streak spacing is $\lambda_z = 0.262$. Brackets reported in the text denote moving-block-bootstrap 95% intervals.

Across the analysed record, the low-speed, intermediate, and high-speed regions occupy mean area fractions

$$f_{\mathrm{LS}} = 0.128\ [0.127, 0.129], \qquad f_{\mathrm{I}} = 0.726\ [0.725, 0.727], \qquad f_{\mathrm{HS}} = 0.146\ [0.145, 0.147],$$

respectively, where *square brackets* denote moving-block-bootstrap 95% intervals [60, 61].

These fractions quantify the spatial availability of each streak class and are required when interpreting the conditioned collision statistics: a class occupying a larger area can contain more events even if its local collision yield is unchanged.

The record-averaged two-dimensional autocorrelation in Fig. 4.3(c) is strongly elongated in the streamwise direction, showing that the anisotropy visible in the instantaneous field persists statistically. Defining the streamwise and spanwise $1/e$ correlation scales as $\ell^u_{x,1/e}$ and $\ell^u_{z,1/e}$, respectively, gives

$$\ell^u_{x,1/e} = 0.351\ [0.345, 0.358], \qquad \ell^u_{z,1/e} = 0.049\ [0.049, 0.050],$$

where the brackets denote moving-block-bootstrap 95% intervals. The streamwise coherence scale is therefore approximately 7.2 times the spanwise scale. The first minimum of the mean spanwise autocorrelation occurs at $\Delta z_{\min} \simeq 0.131$, corresponding to the characteristic separation between neighbouring opposite-sign streak regions. Following the standard interpretation of the first spanwise autocorrelation minimum, the associated same-sign streak spacing is estimated as twice this lag [62],

$$\lambda_z = 2\,\Delta z_{\min} \simeq 0.262, \qquad \lambda_z^+ \simeq 94. \tag{4.5.1}$$

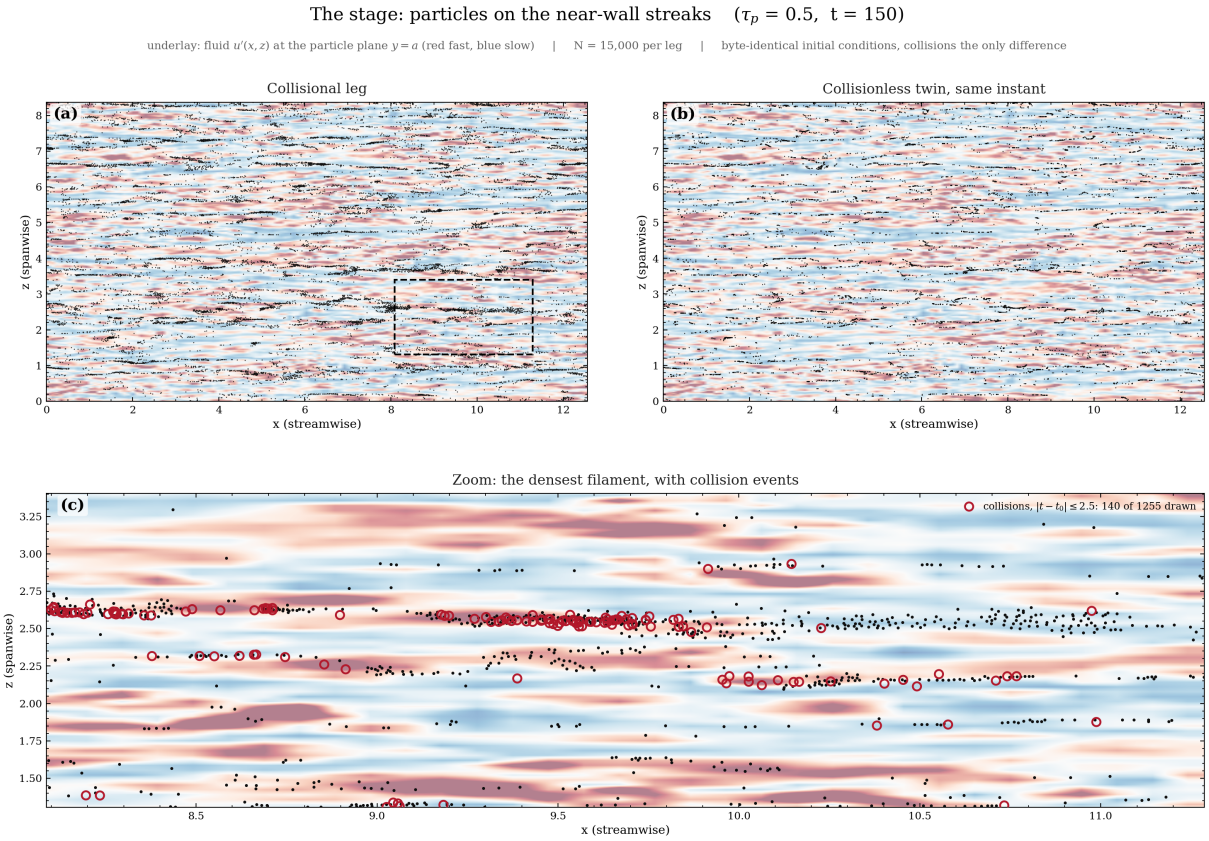

Figure 4.4: **Particles embedded within the near-wall carrier streaks for $\tau_p = 0.5$ at $t = 150$. (a,b)** Collision-on and collision-off particle configurations superimposed on the same instantaneous streamwise carrier-velocity fluctuation $u'(x, z)$ at $y = a$. Blue and red denote negative and positive u', respectively, and black points denote particle centres. Both legs contain $N = 15\,000$ particles, originate from the same labelled initial state, and share the same carrier-flow record. **(c)** Enlargement of the dashed region in panel (a). Black points show particle positions at $t = 150$, while open red circles mark recorded collision locations within $|t - 150| \le 2.5$; 140 of the 1 255 events are displayed for legibility. The event overlay is illustrative and is not normalised by approaching-pair exposure; collision-location selection is quantified separately in Sec. 4.5.4.

Because the carrier flow is prescribed and identical in the matched collision-on and collision-off simulations, particles do not modify its architecture under the one-way-coupling assumption. The correlations reported here therefore characterise a common anisotropic environment rather than a collision-induced difference. This carrier-imposed streamwise axis provides the reference direction for the particle-pair, cluster-alignment, and collision-angle statistics considered below.

4.5.1.2 Multiscale organisation of the particle population

Starting from an approximately homogeneous spatial configuration, both particle populations develop narrow, streamwise-elongated structures (Fig. 4.5). This organisation becomes apparent by approximately $t = 40$ and persists throughout the remainder of the simulation. Comparison with the carrier architecture established in Sec. 4.5.1.1 shows that the preferred particle orientation is aligned with the near-wall streamwise streaks. The dominant direction of the particle organisation is therefore imposed by the prescribed carrier field rather than created by particle–particle collisions.

Figure 4.4 illustrates this relationship at $t = 150$. In both simulation legs, the elongated particle-rich structures are embedded within the same instantaneous carrier-streak field. The collision locations shown in the magnified collision-on region lie within an already concentrated particle structure and are included only to provide spatial context. Their locations are not normalised by encounter availability or inward-motion exposure and therefore cannot, by themselves, be interpreted as evidence of preferential collision selection; that question is addressed quantitatively in Sec. 4.5.4.

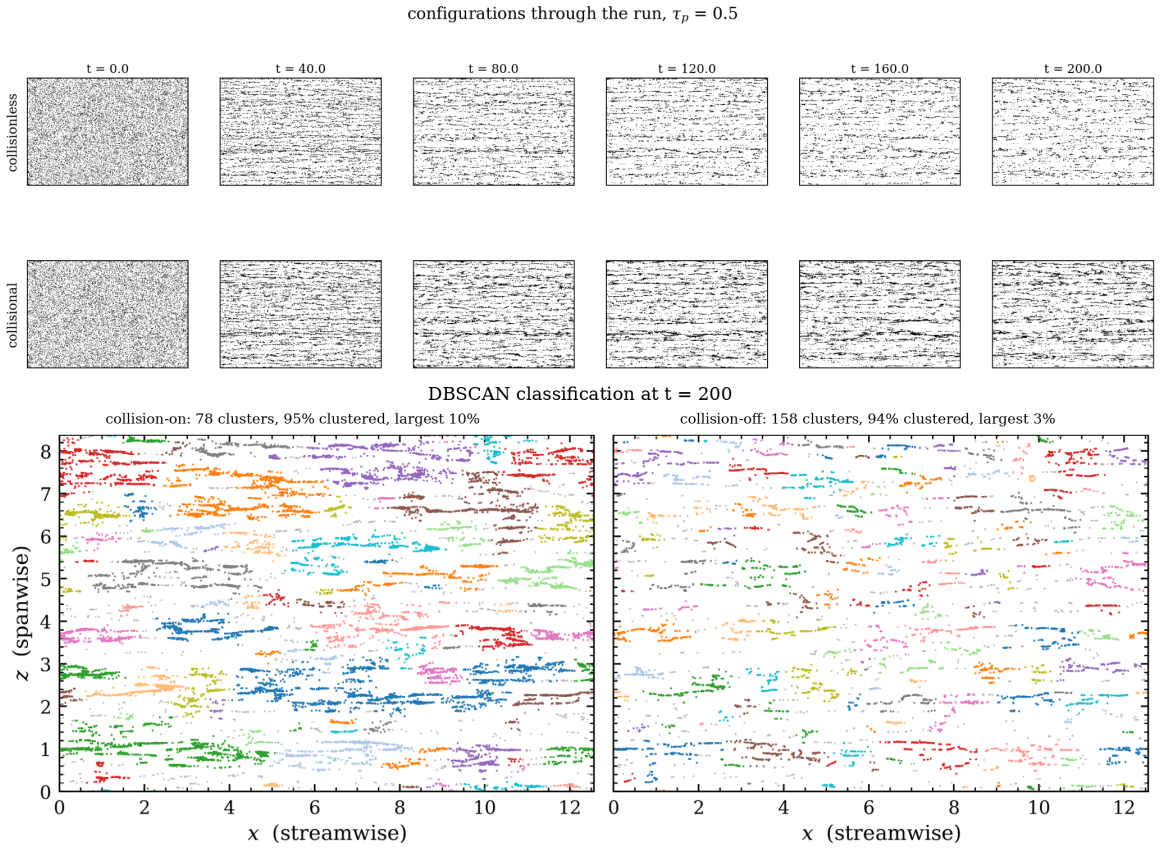

Figure 4.5: **Development and endpoint morphology of particle organisation for $\tau_p = 0.5$.** *Top:* Particle configurations at $t = 0, 40, 80, 120, 160$, and 200 in the collision-off and collision-on simulations. The two legs begin from the same labelled configuration and share the same prescribed carrier-flow realisation. *Bottom:* DBSCAN classifications at $t = 200$, obtained with $\varepsilon = 2.2s = 0.1843$, mean particle spacing $s = 0.08378$, and $N_{\min} = 12$.

The principal difference between the two simulation legs is morphological. At later times, the collision-on population exhibits broader and more spatially connected particle-rich structures, whereas the collision-off population retains a finer and more fragmented organisation. The configurations therefore suggest that collisions redistribute the carrier-aligned particle structure across spatial scales without introducing a different preferred orientation. This qualitative observation is quantified below using connected-structure, number-fluctuation, pair-orientation, and density-correlation diagnostics.

The endpoint classification quantifies the late-time morphological difference at the prescribed DBSCAN scale (see Fig 4.5). At $t = 200$, the collision-on population is partitioned into 78 clusters, compared with 158 in the collision-off population. By contrast, the fractions assigned to clusters are nearly identical, 95% and 94%, respectively. Enabling collisions therefore reduces the number of identified structures by approximately by a half without substantially modifying the fraction of population belonging to clusters.

The largest-cluster fraction provides a complementary measure of this reorganisation. The largest collision-on cluster contains 10% of the particle population, compared with 3% in the collision-off case. Under the same classification rule, the collision-on population therefore contains approximately 3.3 times the largest-cluster fraction while being divided into only half as many clusters. This distinction demonstrates collisions scatter particles resulting in repartitions of close-structures into fewer and larger structures with a longer major axis along streamwise direction.

A cluster-level orientation check also gives the same conclusion (Fig. B.6). Over 25 matched frames, the size-weighted cluster-axis order is $\langle S_{\mathrm{cl}} \rangle_t = 0.829 \pm 0.027$ in the collision-off leg and 0.907 ± 0.029 for the collision-on leg, where the uncertainties denote one standard deviation across frames. Both values

exceed the 95th percentile, $S_{\mathrm{cl}} = 0.588$, obtained from uniformly distributed particles processed using the same DBSCAN estimator. The larger collision-on point estimate is consistent with the repartition of particles into fewer, larger streamwise structures. However, no statistically resolved collision-induced increase is claimed without a paired confidence interval for the difference.

Together, the temporal configurations and endpoint classification support a picture in which the carrier flow establishes the streamwise organisation, while collisions shift its morphology towards coarser, more connected structures. Because DBSCAN is sensitive to classification parameters and provides only an operational classification of the final configuration [53], these results do not establish the merger history of individual clusters or demonstrate scale-independent coarsening. The temporal persistence and scale dependence of the reorganisation are therefore examined next by using grid-based density fluctuations, pair statistics and density correlations.

4.5.1.3 Scale-dependent number fluctuations (VMR)

While the endpoint DBSCAN classification characterises connectivity at one prescribed neighbourhood scale, the variance-to-mean ratio (VMR) measures the amplitude of particle-number fluctuations at a given observation scale. Figure 4.6 shows that both populations remain strongly inhomogeneous: at $t = 40$, 120, and 199, their VMR values lie well above the uniform fixed-N reference throughout the retained range, $2 \leq \ell/s \leq 10$. At the finest retained scale, $\ell/s = 2$, the paired mean difference, $\Delta\mathrm{VMR} = \mathrm{VMR}_{\mathrm{coll}} - \mathrm{VMR}_{\mathrm{nocoll}}$, is approximately -0.8, -6.8, and -12.4 at the three observation times, respectively. Its magnitude generally decreases as ℓ/s increases and is relatively stronger in the two later windows. Collisions reduce the VMR over nearly the entire retained range. The collision-on curve remains below the collision-off curve at every retained scale at $t = 40, 120$, and 199.

Collisions therefore do not homogenise the particle field: the collision-on VMR remains far above its uniform reference at every retained scale. Instead, their accumulated effect predominantly suppresses fine- and intermediate-scale count fluctuations. Read together with the nearly unchanged DBSCAN-classified fraction but fewer and larger endpoint clusters [Fig. 4.5], this result separates fluctuation amplitude from connected morphology: weaker number variance can coexist with organisation into larger structures. VMR does not itself measure connectivity or define a characteristic length; accordingly, the proposed collision-associated coarsening is tested next using pair-axis geometry analysis and density correlations. The estimator construction, scale-support criteria, fixed-N reference, and temporal resampling are documented in Appendix B.3.1.1.

4.5.1.4 Scale-dependent pair-axis geometry

Section 4.5.1.1 established that the near-wall carrier field has a persistent streamwise orientation, while Sec. 4.5.1.2 showed that both particle populations inherit this direction but develop different late-time morphologies. The pair-axis order parameters defined in Eqs. (4.4.11)–(4.4.12) now separates these two effects by measuring the strength and direction of particle-pair alignment as functions of pair separation. The angular normalisation, frame-averaging conventions, finite-sample isotropic reference, and paired bootstrap procedure are detailed in Appendix B.3.1.3.

Figure 4.7(a) shows the frame-averaged angular distribution over the prescribed local shell

$$r \in [s, 5s] = [0.0838, 0.419].$$

Both simulation exhibit larger lobes along $\theta = 0°$ and 180° while depleted at $\theta = 90°$ and 270°. Because the separation axis is nematic, 0° and 180° represent the same streamwise orientation. This result confirms, at pair level, the streamwise organisation observed in Sec. 4.5.1.2. Over this broad shell, the nematic magnitude averaged equally over the 25 analysed frames is

$$S_{\mathrm{off}} = 0.206\,[0.199, 0.213], \qquad S_{\mathrm{on}} = 0.214\,[0.209, 0.220].$$

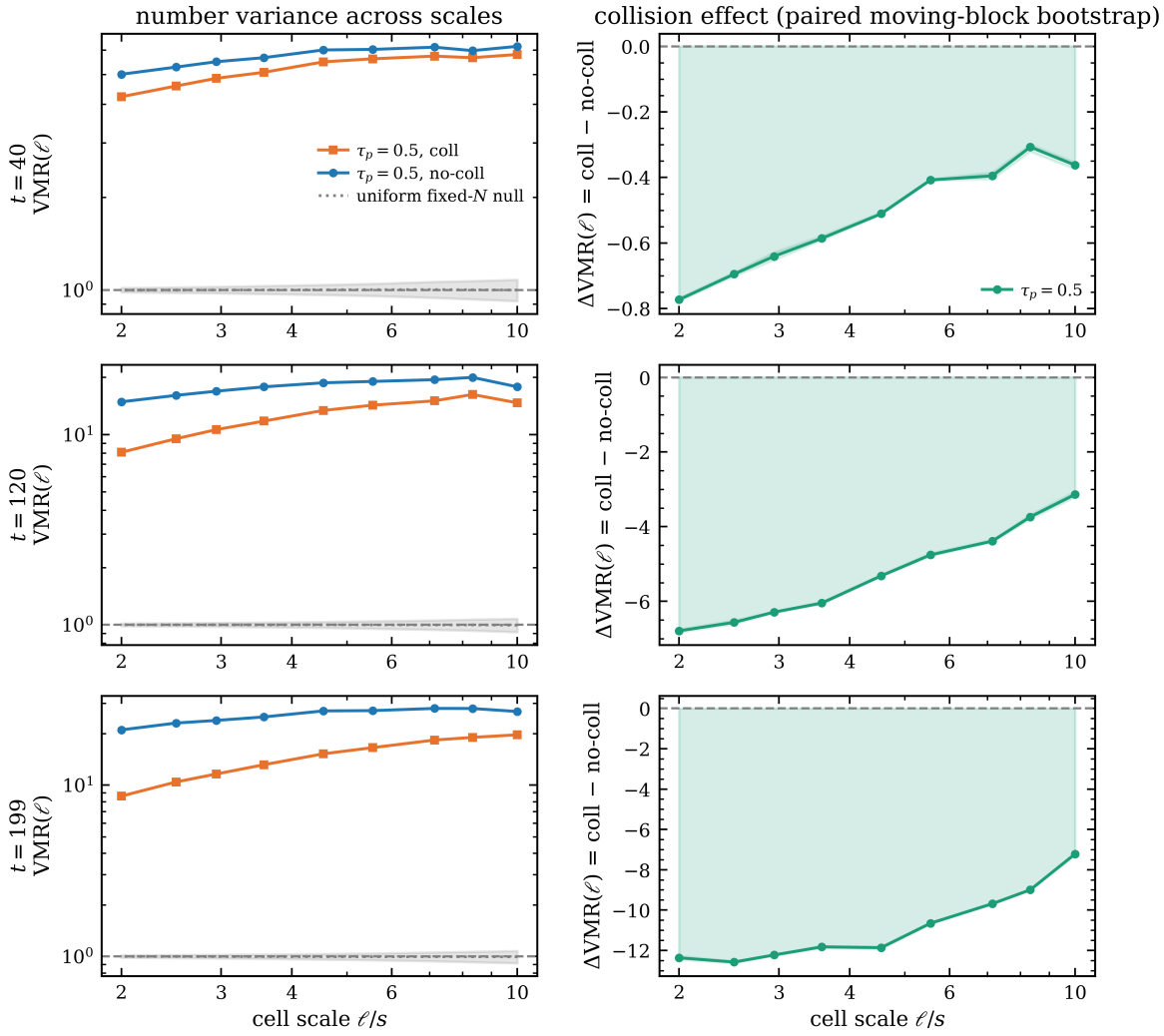

Figure 4.6: **Scale-dependent particle-number fluctuations, $\tau_p = 0.5$.** Rows are $t = 40$, 120 and 199; each symbol averages the 11 consecutive matched frames centred on that time. *Left:* VMR against square-cell scale ℓ/s for both legs. Grey shows the mean and standard deviation of 500 uniform fixed-N configurations under the same estimator. *Right:* the paired difference $\Delta\mathrm{VMR} = \mathrm{VMR}_{\mathrm{coll}} - \mathrm{VMR}_{\mathrm{nocoll}}$, with the 2.5–97.5 percentile envelope of 2000 moving-block bootstrap replicates, three-frame blocks, identical block indices in both legs. Nine supported grids are retained, 15×10 ($\ell/s = 10$) to 75×50 ($\ell/s = 2$); coarser and finer grids lack support and are omitted.

where square brackets denote moving-block-bootstrap 95% confidence intervals. The paired difference is obtained as:

$$\Delta S = S_{\mathrm{on}} - S_{\mathrm{off}} = 0.008$$

Paired confidence intervals for $\Delta S(r)$ are reported in Appendix B.3.1.4; A reduction can be observed in the two finest shells with a small increase at intermediate separations.

The scale-conditioned statistic reveals a substantially richer response [Fig. 4.7(b)]. The calculation uses contiguous shells $[r_h/2, r_h]$, plotted at their geometric centres $r_m = r_h/\sqrt{2}$. Pair-axis order is strongest at the smallest resolved separations and decreases rapidly with increasing r_m/s in both simulations. In the smallest shell,

$$r \in [s/2, s],$$

the mean order decreases from approximately 0.62 in the collision-off case to 0.46 in the collision-on case, a reduction of about 26%. The two mean curves subsequently converge, and the collision-on

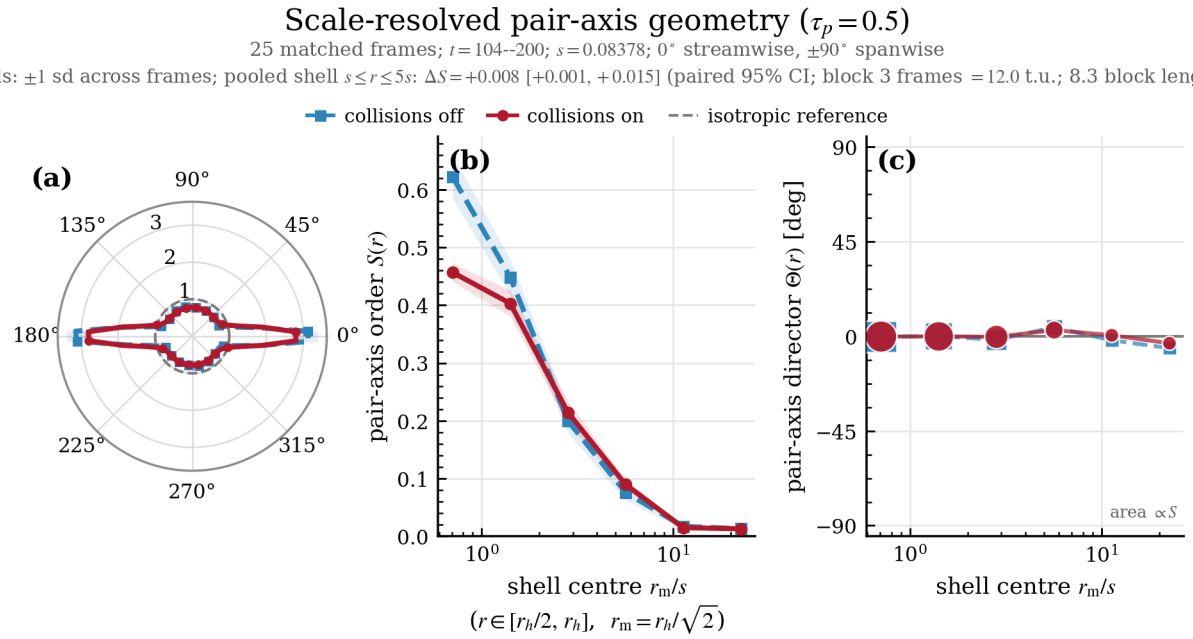

Figure 4.7: **Scale-dependent particle-pair orientation for $\tau_p = 0.5$.** Statistics are evaluated over 25 matched frames spanning $t = 104$–200. **(a)** Frame-averaged angular pair distribution over $r \in [s, 5s] = [0.0838, 0.419]$; the grey circle denotes the isotropic reference $g_\theta = 1$. **(b)** Frame-averaged nematic magnitude in shells $[r_h/2, r_h]$, plotted at the geometric shell centre $r_m = r_h/\sqrt{2}$. Shading denotes ±1 standard deviation across frames **(c)** Pair-axis director obtained by pair-count-weighted pooling of the complex nematic moments across frames. Marker size represents the corresponding order magnitude. Brackets in the text denote moving-block-bootstrap 95% intervals; the collision-on/off difference uses identical temporal blocks in both legs.

average becomes slightly larger than that of collisionless over intermediate range. Subsequently, both curves decay to very weak but still resolved values at the largest separations shown. Therefore, collisions weakens the local-pair axis anisotropy by scattering the small streak particle structures in the collisionless, then responds immediately to the strong streamwise orientation due to carrier forcing.

Despite the scale-dependent change in alignment strength, the director remains close to $\Theta = 0°$ in both simulations as observed in [Fig. 4.7(c)]. The director is obtained by combining the complex shell-wise moments across frames with their pair counts before extracting the phase. Its agreement with the streamwise axis independently identified from the carrier correlation in Sec. 4.5.1.1 shows that collisions do not rotate the preferred particle-pair orientation or establish a competing direction. Instead, they just redistribute how strongly the carrier-selected direction is expressed across separation scales. The corresponding density maps for spatial organisation and collision-induced redistribution are shown in Fig. B.3.

4.5.1.5 Collisions extend the streamwise coherence of the particle-density field

The VMR reflects the strength of particle-number fluctuations within the spatial distribution, but does not determine whether those fluctuations form many compact structures or fewer spatially extended ones. We therefore examine the self-term-corrected particle-density autocorrelation. The calculation uses the raw particle-count field on a common 840×560 grid. Although the mean occupancy is only $N/(840 \times 560) = 0.032$ particles per cell, subtraction of the exact single-particle self-term removes the corresponding zero-lag discreteness contribution before normalisation. The grid spacing, $\Delta x \simeq \Delta z \simeq 0.0150$, therefore provides the relevant resolution grid. A coherence length is retained only when the excess density variance is resolved and the first positive-lag $1/e$ crossing lies beyond the first grid cell.

At $t = 120$, the collisionless correlation is confined to a small compact neighbourhood of zero lag, whereas the collision-on field exhibits a noticeably longer streamwise core [Fig. 4.8(a,b)]. The maps are averages of the 21 individually normalised correlations from the matched frames centred on $t = 120$.

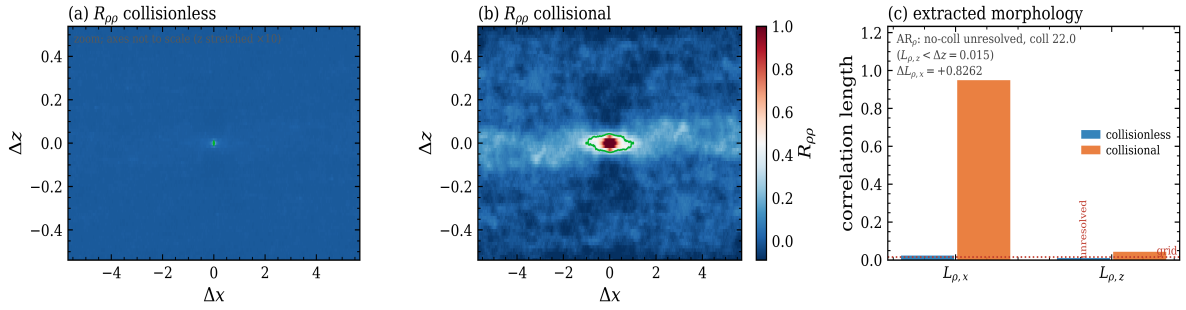

Figure 4.8: **Collision-enhanced particle-density coherence,** $\tau_p = 0.5$**. (a,b)** Autocorrelation maps, collisions off and on, averaging the 21 individually normalised frames centred on $t = 120$; identical 840×560 grid, frame list, lag coordinates and colour scale. Green marks $R_{\rho\rho} = 1/\mathrm{e}$; the spanwise axis is stretched sevenfold. **(c)** Correlation lengths from those maps. The mean paired per-frame difference is $\overline{\Delta L}_{\rho,x}^{(120)} = 0.826$ and the collision-on aspect ratio 22; the collisionless ratio is unresolved, its spanwise crossing lying inside the first cell.

For the same common frame set $\mathcal{W}_{120}$, the paired mean streamwise difference is

$$\overline{\Delta L}_{\rho,x}^{(120)} = \frac{1}{|\mathcal{W}_{120}|} \sum_{m \in \mathcal{W}_{120}} \left[L_{\rho,x}^{\mathrm{coll}}(t_m) - L_{\rho,x}^{\mathrm{nocoll}}(t_m) \right] = 0.826. \tag{4.5.2}$$

For each frame, the coherence length is obtained by linear interpolation of the first positive-lag $1/\mathrm{e}$ crossing between adjacent lag samples. The spanwise length of the window-averaged collision-on correlation is also resolved, giving the aspect ratio

$$\mathrm{AR}_{\rho}^{\mathrm{coll}} = \frac{L_{\rho,x}^{\mathrm{coll}}}{L_{\rho,z}^{\mathrm{coll}}} = 22.$$

For the collisionless leg the spanwise $1/\mathrm{e}$ crossing occurs within the first grid cell, $L_{\rho,z}^{\mathrm{nocoll}} < \Delta z = 0.0150$, so its aspect ratio is resolution-dependent and is not assigned a numerical value [Fig. 4.8(c)]. The robust comparison is consequently the resolved increase in streamwise length, not a collision-on/off difference between two aspect ratios.

The instantaneous directional cuts provide a complementary check at three well-separated times [Fig. 4.9]. At $t = 40$, 120, and 199, the collision-on streamwise correlation remains above $1/\mathrm{e}$ over a substantially greater separation than its collisionless counterpart. The spanwise correlations decay very rapidly in both legs, confirming that the collision-induced particle re-arrangement occurs mostly within the streamwise structure established by the carrier flow.

The VMR and autocorrelation therefore describe complementary aspects of the same redistribution. Collisions reduce the amplitude of local number fluctuations while increasing the distance over which the remaining density fluctuations are correlated. Together with the nearly unchanged DBSCAN-classified fraction but fewer and larger endpoint structures (Fig. 4.5), this supports a consistent interpretation: *the carrier flow selects the streamwise orientation, whereas collisions reorganise the particle population towards a larger characteristic spatial scale.*

This agreement among independent diagnostics constitutes evidence of collision-associated coarsening. However, it does not provide a clear view on the merger history of individual clusters and the mechanism by which density variance transfers between scales.

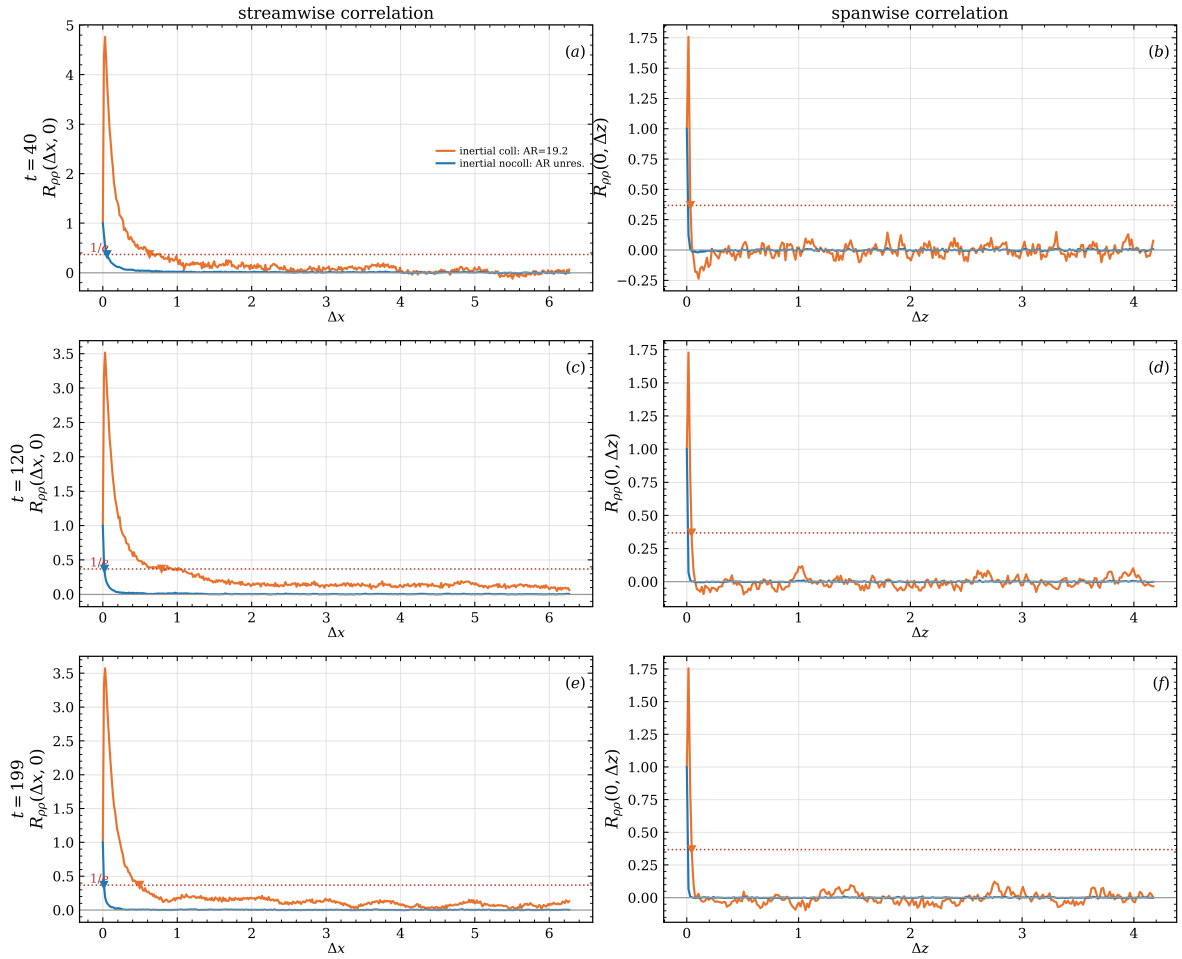

Figure 4.9: **Single-frame density correlation cuts, $\tau_p = 0.5$.** Streamwise **(a,c,e)** and spanwise **(b,d,f)** cuts of $R_{\rho\rho}$ at $t = 40$ **(a,b)**, $t = 120$ **(c,d)** and $t = 199$ **(e,f)**; collision-on in orange, collisionless in blue. Dotted lines mark $1/e$ and markers the resolved crossings. Removing the single-particle self-term before normalising allows $R_{\rho\rho} > 1$ away from zero lag. Same 840×560 grid and lag coordinates as Fig. 4.8.

4.5.2 Pair-relative kinematics and particle-velocity organisation

The preceding structural analysis showed that the prescribed carrier field sets the dominant streamwise organisation of the particle phase, while collisions modify that organisation in a scale-dependent manner. We now examine the associated particle kinematics through three complementary diagnostics. The radial relative velocity w_r characterises pairwise approach and separation; the uncentred velocity correlation $C_v(r)$ measures total particle-velocity coherence, including the common downstream motion; and the angular distribution of the mean-subtracted particle velocities characterises the directional organisation of the velocity fluctuation.

Collisions modify radial pair motion primarily near contact. The mean absolute radial relative speed increases with separation in both simulation legs [Fig. 4.10(a)]. Immediately outside contact, the collision-on frame-mean curve lies above the collision-off curve, and the difference persists over the adjacent local range. The curves subsequently converge and approach $\langle |w_r| \rangle \simeq 2.6 \times 10^{-2}$ at the largest separations shown. Collisions therefore modify the amplitude of radial relative motion most clearly at local pair scales, while their influence becomes weak once the separation is large compared with the particle diameter.

Because $\langle |w_r| \rangle$ combines approaching and separating pairs, its increase cannot be interpreted as enhanced

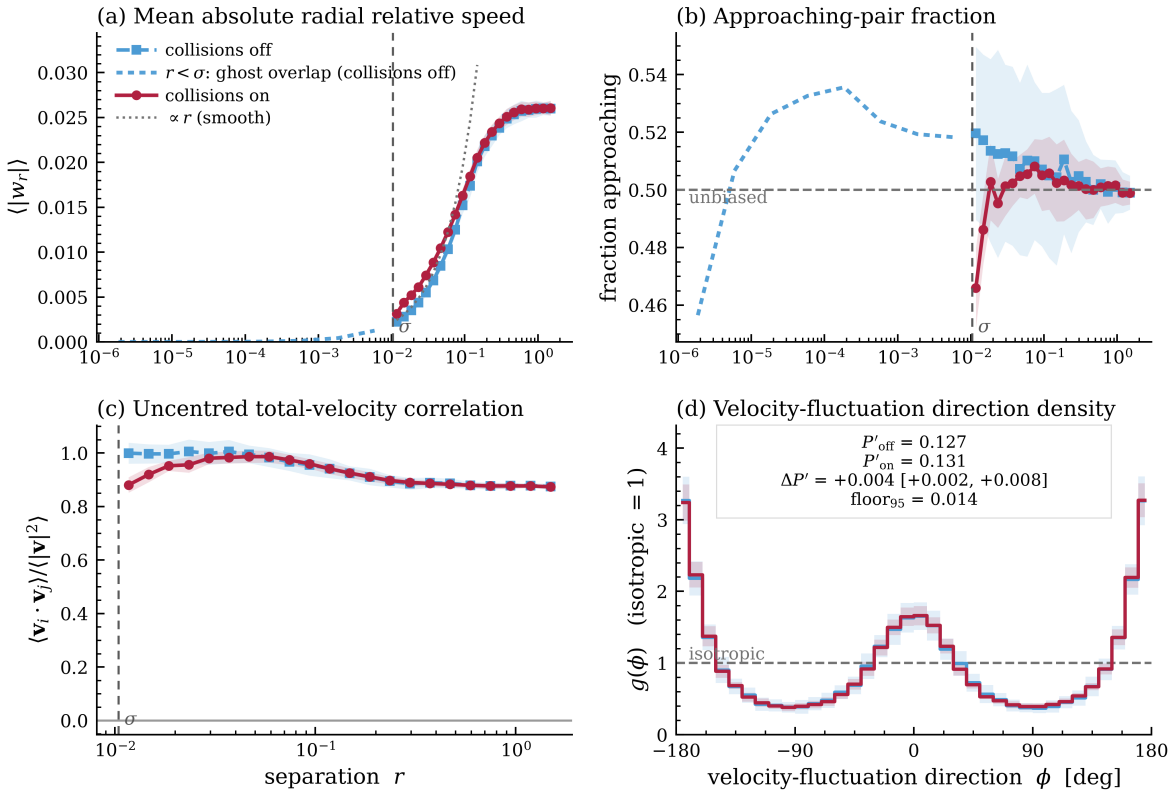

Figure 4.10: **Pair-relative kinematics and particle-velocity organisation for $\tau_p = 0.5$.** Statistics are evaluated over 25 matched frames covering $104 \leq t \leq 200$. **(a)** Mean absolute radial relative speed $\langle |w_r| \rangle$; the grey dotted line shows a linear $\propto r$ scaling guide corresponding to locally smooth relative motion. **(b)** Fraction of pairs with $w_r < 0$; marks the equal approach–separation reference $1/2$. **(c)** Uncentred total-velocity correlation $C_v(r) = \langle \mathbf{v}_i \cdot \mathbf{v}_j \rangle_r / \langle |\mathbf{v}|^2 \rangle$. **(d)** Normalised angular density $g_\phi(\phi)$ of the instantaneous mean-subtracted particle velocities; $g_\phi = 1$ is the isotropic reference. The displayed $P'_{\mathrm{iso},95} \simeq 0.014$ is the finite-sample isotropic reference (see B.3.2). Shading denotes ±1 standard deviation across frames. Both legs use identical radial bins for $r \geq \sigma$. Dotted collision-off curves below σ correspond to ghost-particle overlap and are excluded from collision-on/off comparisons.

inward transport. The sign-resolved statistic gives the opposite near-contact ordering [Fig. 4.10(b)]. In the first resolved bin outside contact, approximately 47% of collision-on pairs are approaching, compared with approximately 52% in the collision-off population. Both fractions tend towards $1/2$ as the separation increases. The near-contact depletion of approaching collision-on pairs is consistent with the restitution law in Eq. (4.3.17), which converts approaching normal relative motion into separating motion at contact.

Total particle-velocity coherence remains carrier dominated. The uncentred total-velocity correlation $C_v(r)$, defined in Sec. 4.4.1, remains close to unity in both simulation legs [Fig. 4.10(c)]. At the smallest common separation, the collision-on frame-mean value is approximately 0.88, compared with approximately 1.00 in the collision-off case. This difference decreases rapidly with separation, and the two curves nearly coincide at larger r. The local reduction is consistent with collision impulses redistributing the velocities of contacting and recently separated particles.

Because the instantaneous population mean is retained in C_v, this statistic is dominated by the common downstream motion imposed by the carrier flow. Panel (c) therefore shows that collisions produce a predominantly local perturbation to an otherwise strongly coherent total particle-velocity field.

The velocity-fluctuation direction remains streamwise organised. After subtracting the instantaneous population-mean velocity, the angular densities of the fluctuation directions remain strongly non-uniform [Fig. 4.10(d)]. Both legs show enhanced probability near $\phi = 0°$ and $\phi = \pm 180°$, corresponding to the streamwise axis, and depletion near $\phi = \pm 90°$, corresponding to the spanwise axis. The two angular densities are nearly coincident. Collisions therefore neither create a competing velocity direction nor substantially reshape the global fluctuation-direction distribution.

The frame-averaged first-harmonic polar orders are

$$P'_{\mathrm{nocoll}} = 0.127, \qquad P'_{\mathrm{coll}} = 0.131, \qquad \Delta P' = +0.004\,[+0.002, +0.008]. \tag{4.5.3}$$

where the square brackets denote the paired moving-block-bootstrap 95% confidence interval for $\Delta P'$. Although the interval excludes zero under the adopted resampling procedure, the estimated change is only about 3% of the collisionless value and is therefore small in magnitude.

Answer to RQ1 Together with the structural results of Sec. 4.5.1.2, these findings provide a direct answer to RQ1 for the focused case, $\tau_p = 0.5$. The prescribed near-wall carrier determines the directional organisation of the particle phase: both simulation legs develop streamwise-elongated bands, their pair and cluster axes remain preferentially streamwise, and nearby particles exhibit strong positive total-velocity correlation. Hence, it is observed that collisions do not create this orientation or introduce a competing director. Instead, their influence is scale dependent. Collision-induced effects weaken the exceptionally strong pair-axis order formed by the carrier flow and then modify radial relative motion at the smallest separations and suppress particle-number fluctuations. The initial highly clustered population is reorganised into fewer, larger structures with longer streamwise density coherence. The collision-induced difference in total-velocity coherence is comparatively modest over the common physical support $r \geq \sigma$. Thus, the carrier selects the preferred direction and organises particle motion, whereas collisions primarily alter the strength, characteristic scale, and morphology of the resulting particle structures.

4.5.3 Angular collision selection and inward pair kinematics

The preceding sections established that the available nearby-pair population is already strongly anisotropic: particle pairs preferentially align with the streamwise streaks. A histogram of collision-axis angles alone cannot therefore establish preferential collision selection, because an orientation may contribute many events simply because it contains more susceptible pairs. To separate availability from selection, the angular distribution of all 298,029 collision events recorded over the time period $t = 104$–200 is compared with the accumulated near-contact pair-time and its inward radial motion. All quantities are evaluated in the common shell $\sigma \leq r \leq 1.5\sigma$, where, for the detailed $\tau_p = 0.5$ case, $\sigma = 1.0434 \times 10^{-2}$. The corresponding discrete estimators and temporal-resampling procedure are documented in Appendix B.3.3.

Figure 4.11(a) shows that near-contact pair-time is concentrated along the streamwise axis, consistent with the pair-axis order reported in subsection 4.5.1.4. The collision distribution remains streamwise biased in absolute terms but has noticeably broader lobes. Hence, the most abundant pair orientations are not converted proportionally into collision events.

For angular bin k, let T_k **denote the accumulated near-contact pair-time** and C_k **the collision-event count**. With $T = \sum_k T_k$ and $C = \sum_k C_k$, define as in subsection 4.4.2:

$$R_k = \frac{p_{\mathrm{coll},k}}{p_{\mathrm{near},k}} = \frac{C_k/T_k}{C/T}. \tag{4.5.4}$$

Thus, R_k **is the collision rate per unit near-contact pair-time in bin** k, normalised by the rate pooled over all orientations. It is a relative event-rate statistic, not an absolute collision probability. Values above and below unity denote over- and under-representation among collision events after correcting for pair availability.

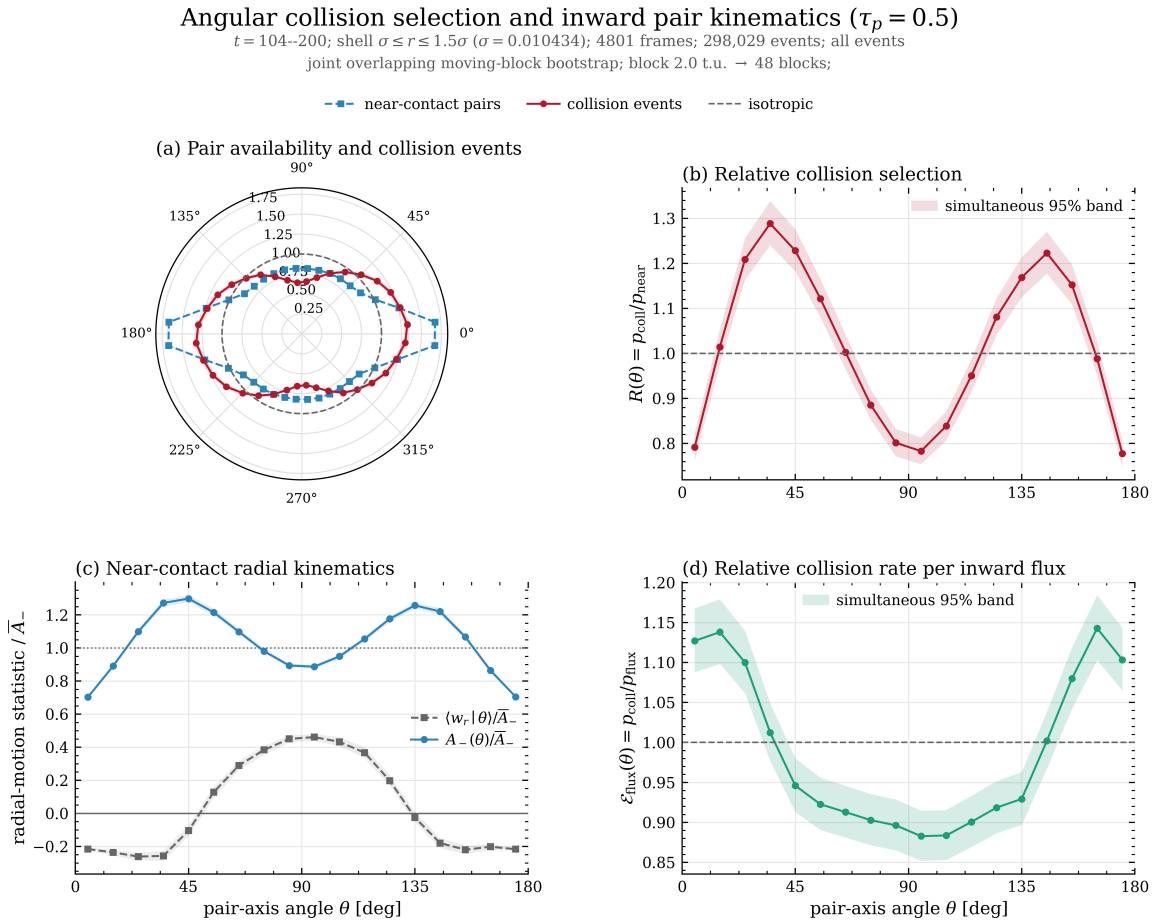

Figure 4.11: **Angular collision selection and inward pair kinematics for $\tau_p = 0.5$.** Statistics use 4,801 frames over $t = 104$–200, all 298,029 collision events, and the shell $\sigma \le r \le 1.5\sigma$. **(a)** Relative-to-isotropic angular distributions of accumulated near-contact pair-time and collision events. **(b)** Relative collision selection $R = p_{\mathrm{coll}}/p_{\mathrm{near}}$. **(c)** Pair-time-weighted signed radial velocity and positive-part inward motion, both normalised by the global mean inward motion $\overline{A}_-$. **(d)** Relative collision yield per unit inward flux, $\mathcal{E}_{\mathrm{flux}} = p_{\mathrm{coll}}/p_{\mathrm{flux}}$. Unity denotes proportional selection relative to the corresponding denominator. Shading gives pointwise 95% intervals in panels **(a,c)** and simultaneous 95% bands in panels **(b,d)**, obtained by joint overlapping moving-block resampling in time.

The availability correction reveals preferential oblique selection [Fig. 4.11(b)]. The relative selection reaches approximately 1.29 near $\theta = 35°$ and 1.22 near $\theta = 145°$, whereas it decreases to approximately 0.78–0.80 near the streamwise and spanwise axes. These departures from $R = 1$ are resolved by the simultaneous confidence band. Streamwise orientations account for a large fraction of the available near-contact pair population and therefore contribute many collision events in absolute terms. After normalisation by pair availability, however, their relative collision rate is lower than that of the preferentially selected oblique orientations.

To determine the kinematic origin of this preferential selection, define the mean inward-motion supply per unit near-contact pair-time in angular bin B_k as

$$A_{-,k} = \frac{F_k}{T_k} = \langle [-w_r]_+ \,|\, \theta \in B_k \rangle_T = P_{-,k} S_{-,k}, \qquad [-w_r]_+ = \max(-w_r, 0), \tag{4.5.5}$$

where F_k is the accumulated inward radial-motion exposure in bin k. For conciseness, F_k is referred to as the *inward-flux exposure*. Here, $P_{-,k}$ is the fraction of near-contact pair-time spent approaching, while $S_{-,k}$ is the mean inward speed conditional on approach. Separating pair states contribute zero to

F_k. The pair-time-weighted mean inward-motion supply over all orientations is

$$\overline{A}_- = \frac{\sum_k F_k}{\sum_k T_k} = \sum_k p_{\mathrm{near},k} A_{-,k}. \tag{4.5.6}$$

The relative inward supply $A_{-,k}/\overline{A}_-$ reaches approximately 1.3 near the selected oblique directions but falls to approximately 0.7 near the streamwise axis [Fig. 4.11(c)]. Its close correspondence with R_k shows that the oblique collision maxima are associated primarily with greater inward radial motion per unit available pair-time.

The signed mean radial velocity provides different information. It is negative near the streamwise axis, indicating net contraction, and positive towards the spanwise axis, indicating net separation. This does not conflict with the smaller streamwise value of $A_{-,k}/\overline{A}_-$: the signed mean compares approaching and separating motion within the same angular bin, whereas $A_{-,k}/\overline{A}_-$ compares the positive inward contribution with the population-average inward contribution. Streamwise pair states can therefore contract weakly on average while supplying less inward motion per unit pair-time than oblique states.

The **normalised angular distribution of inward radial-motion exposure** is

$$p_{\mathrm{flux},k} = \frac{F_k}{\sum_j F_j} = p_{\mathrm{near},k} \frac{A_{-,k}}{\overline{A}_-}. \tag{4.5.7}$$

The residual event yield after accounting for this supply gives r**elative collision yield per unit inward-flux exposure**, defined as:

$$\mathcal{E}_{\mathrm{flux},k} = \frac{p_{\mathrm{coll},k}}{p_{\mathrm{flux},k}} = \frac{C_k/F_k}{C/\sum_j F_j}. \tag{4.5.8}$$

Hence $\mathcal{E}_{\mathrm{flux},k} = 1$ means that bin k contributes the same fraction of collision events as of inward radial-motion exposure. Combining Eqs. (4.5.4)– (4.5.8) gives the exact factorisation

$$\boxed{R_k = \underbrace{\mathcal{E}_{\mathrm{flux},k}}_{\text{relative event yield}} \underbrace{\frac{A_{-,k}}{\overline{A}_-}}_{\text{relative inward supply}}}. \tag{4.5.9}$$

Figure 4.11(d) shows a smaller but resolved angular variation in the residual yield. It reaches approximately 1.13 near the streamwise axis and decreases to approximately 0.88 near the spanwise axis. The enhanced streamwise yield does not compensate for the weak streamwise inward supply:

$$R_{\mathrm{str}} \simeq \underbrace{1.13}_{\text{relative yield}} \times \underbrace{0.70}_{\text{relative inward supply}} \simeq 0.79. \tag{4.5.10}$$

At the preferentially selected oblique orientations, $A_{-,k}/\overline{A}_- \simeq 1.3$, while $\mathcal{E}_{\mathrm{flux},k}$ remains close to unity. The enhanced collision selection at these orientations is therefore driven primarily by the greater inward-motion supply rather than by an unusually large residual event yield once that supply is available.

The resulting physical picture has three distinct layers. The carrier-organised particle field supplies a strongly streamwise-biased population of available near-contact pairs; pair-relative kinematics redistribute the inward-motion supply towards oblique orientations; and the residual conversion from inward exposure to realised collision events provides a smaller angular correction. Thus, the angular distribution of collision events reflects a hierarchy of pair availability, inward-motion supply, and residual collision conversion rather than pair abundance alone.

Physical interpretation of the angular collision selection To interpret the origin of this oblique inward-motion enhancement, we compare the measured angular dependence with the local radial motion generated by a smooth carrier-velocity gradient.

A local smooth-carrier approximation, corresponding to the Saffman–Turner shear mechanism in the tracer limit [63], provides a kinematic explanation for this angular dependence. Let $\mathbf{r} = r\widehat{\mathbf{r}}$ with $\widehat{\mathbf{r}} = (\cos\theta, \sin\theta)$ denote a wall-parallel pair separation, and define the in-plane velocity gradient $(\mathbf{A}_\parallel)_{\alpha\beta} = \partial_\beta u_\alpha$ with $\mathbf{u}_\parallel = (u, w)$. For separations small compared to the local carrier length scale, the radial velocity difference satisfies

$$\frac{\delta u_r}{r} = \frac{[\mathbf{u}_\parallel(\mathbf{x}+\mathbf{r}) - \mathbf{u}_\parallel(\mathbf{x})]\cdot\widehat{\mathbf{r}}}{r} \simeq \widehat{\mathbf{r}}^{\mathsf{T}}\mathbf{A}_\parallel\widehat{\mathbf{r}} = \widehat{\mathbf{r}}^{\mathsf{T}}\mathbf{S}_\parallel\widehat{\mathbf{r}}, \tag{4.5.11}$$

where $\mathbf{S}_\parallel = (\mathbf{A}_\parallel + \mathbf{A}_\parallel^{\mathsf{T}})/2$ is the in-plane rate-of-strain tensor. The antisymmetric part $\mathbf{\Omega}_\parallel = (\mathbf{A}_\parallel - \mathbf{A}_\parallel^{\mathsf{T}})/2$ contributes no radial component ($\widehat{\mathbf{r}}^{\mathsf{T}}\mathbf{\Omega}_\parallel\widehat{\mathbf{r}} = 0$, Batchelor [65]). Writing

$$\mathbf{S}_\parallel = \begin{pmatrix} \partial_x u & \frac{1}{2}(\partial_z u + \partial_x w) \\ \frac{1}{2}(\partial_z u + \partial_x w) & \partial_z w \end{pmatrix} \tag{4.5.12}$$

gives

$$\frac{\delta u_r}{r} \simeq (\partial_x u)\cos^2\theta + (\partial_z w)\sin^2\theta + \tfrac{1}{2}(\partial_z u + \partial_x w)\sin(2\theta). \tag{4.5.13}$$

The measured carrier architecture is consistent with a large off-diagonal contribution. The streamwise and spanwise $1/e$ correlation lengths, $\ell^u_{x,1/e} = 0.351$ and $\ell^u_{z,1/e} = 0.049$ (Fig. 4.3), differ by a factor of about 7.2, indicating much stronger spanwise variation of u than streamwise. This suggests $\partial_z u$ may be dominant, but the coherence lengths alone do not fix the exact gradient ratio. If $\partial_z u$ dominates $\partial_x w$ and the diagonal terms, then Eq. (4.5.13) reduces to $\delta u_r/r \simeq \frac{1}{2}\partial_z u \sin(2\theta)$, which is maximal at $\theta = 45°, 135°$ and zero along the axes. A spanwise pair experiences no radial approach from $\partial_z u$, nor does a purely streamwise pair; but oblique pairs project this shear onto their line of centres.

Near-wall streak flanks are associated with strong wall-parallel velocity gradients and oppositely signed wall-normal vorticity [64]. Consequently, local shear configurations of both signs occur within the sampled near-wall field. The positive-part statistic $A_- = \langle[-w_r]_+\rangle$ retains only inward radial motion [37], so opposite-signed local shears can contribute inward motion in the two oblique angular sectors. The measured maxima near 35° and 145° are displaced by approximately 10° from the idealised 45°/135° orientations predicted by a pure $\partial_z u$ shear. Such a displacement is qualitatively consistent with the contribution of the remaining components of the local strain tensor. For a two-dimensional symmetric strain tensor, a principal-axis orientation θ_s satisfies $\tan(2\theta_s) = \frac{\partial_z u + \partial_x w}{\partial_x u - \partial_z w}$, [65]. Thus, non-zero diagonal strain and the $\partial_x w$ contribution can rotate the directions of strongest instantaneous radial compression away from 45° and 135°.

This comparison is intended only as a kinematic interpretation. The relevant velocity-gradient components were not conditionally measured at the collision events, and, because $\tau_p^+ \simeq 8.1$, the inertial-particle relative velocity w_r need not follow the instantaneous carrier-velocity difference δu_r exactly. The smooth-carrier argument therefore explains why oblique inward motion is plausible, but it does not quantitatively predict the measured $A_-(\theta)$ or establish a causal origin for the observed collision-angle maxima.

4.5.4 Where collisions convert: carrier-speed and flow-topology conditioning

The angular analysis established which pair orientations are preferentially converted into collisions. The environmental component of RQ2 asks whether this conversion also depends on the instantaneous carrier environment sampled by a near-contact encounter. We first examine this question for the low-,

intermediate-, and high-speed carrier classes using the same pair-availability and inward-motion decomposition introduced in Sec. 4.5.3. We then examine the broader carrier environments sampled by the particle population, including wall-parallel divergence, strain, rotation, and the local topology indicator. Raw collision counts cannot establish preferential environmental selection. A carrier class may contain more collision events simply because it occupies more of the domain, contains more near-contact pair-time, or supplies greater inward radial motion. The carrier-speed analysis therefore separates near-contact pair availability, inward-motion supply, and residual collision yield before interpreting differences between classes.

Environmental rate decomposition. Let q denote either a carrier-speed class or a flow-topology bin, and let $\mathcal{T}_f$ be the set of carrier save times. See subsec. 4.4.2 for the classification criterion. For any $(\mathbf{x}, t)$, write $t^\star = \arg\min_{t_f \in \mathcal{T}_f} |t_f - t|$ (so $|t^\star - t| \le 0.1$ time units for the 0.2-unit carrier cadence) and define the class indicator

$$\chi_q(\mathbf{x}, t) = \begin{cases} 1, & \text{the carrier state at } (\mathbf{x}, t^\star) \text{ belongs to class } q, \\ 0, & \text{otherwise,} \end{cases} \tag{4.5.14}$$

where the standardised streamwise fluctuation ξ is defined in Eq. (4.4.34), and the low-, intermediate-, and high-speed thresholds are given in Eq. (4.4.36). For the retained exposure times t_m, define the accumulated near-contact pair-time and inward-flux exposure as

$$T_q = \sum_m \Delta t_m \sum_{i<j} \chi_q\left(\mathbf{x}_{ij}^{\mathrm{mid}}(t_m), t_m\right) \mathbf{1}\left(\sigma \le r_{ij}(t_m) \le 1.5\sigma\right), \tag{4.5.15}$$

$$F_q = \sum_m \Delta t_m \sum_{i<j} \chi_q\left(\mathbf{x}_{ij}^{\mathrm{mid}}(t_m), t_m\right) \mathbf{1}\left(\sigma \le r_{ij}(t_m) \le 1.5\sigma\right) [-w_{r,ij}(t_m)]_+, \tag{4.5.16}$$

where $[s]_+ = \max(s, 0)$, $\mathbf{x}_{ij}^{\mathrm{mid}}$ is the periodic pair midpoint, and $\chi_q(\mathbf{x}, t)$ is the indicator of environmental class q.

Collision events use the same indicator, evaluated at their own time: with $t_e^\star$ the carrier snapshot nearest to the event time t_e as above, the class-specific collision count is

$$C_q = \sum_e \chi_q(\mathbf{x}_e, t_e), \tag{4.5.17}$$

where $\mathbf{x}_e$ is the logged contact position. Exposure is evaluated every 25 stored particle frames, giving $\Delta t_{\mathrm{row}} = 0.5$. Each collision is assigned to its nearest exposure row only for moving-block resampling, so the maximum assignment offset is 0.25; environmental classification itself uses the carrier snapshot nearest to the actual event time.

With notations as in subsection 4.5.3, we can define the corresponding class-based relative statistics as follows:

$$J_q = \frac{C_q/T_q}{C/T} = \frac{p_{C,q}}{p_{T,q}}, \qquad \Phi_q = \frac{F_q/T_q}{F/T} = \frac{p_{F,q}}{p_{T,q}}, \qquad \mathcal{E}_{\mathrm{flux},q} = \frac{C_q/F_q}{C/F} = \frac{p_{C,q}}{p_{F,q}}. \tag{4.5.18}$$

Here, J_q is the relative collision rate per unit near-contact pair-time, Φ_q is the relative inward-flux exposure supplied per unit pair-time, and $\mathcal{E}_{\mathrm{flux},q}$ is the relative collision yield per unit inward-flux exposure. These definitions give the exact factorisation

$$\boxed{J_q = \Phi_q \, \mathcal{E}_{\mathrm{flux},q}.} \tag{4.5.19}$$

Thus, a large collision rate per unit pair-time may arise from a greater supply of inward radial motion, a greater residual event yield after that supply is available, or both. Each statistic is referenced to unity,

Table 4.1: **Carrier-speed collision and inward-flux totals for $\tau_p = 0.5$.** The event and exposure shares are $p_{C,q} = C_q/C$ and $p_{F,q} = F_q/F$, respectively. The final column reports $\mathcal{E}_{\mathrm{flux},q} = p_{C,q}/p_{F,q}$, with simultaneous 95% moving-block-bootstrap intervals.

Class	C_q	$p_{C,q}$	F_q	$p_{F,q}$	$\mathcal{E}_{\mathrm{flux},q}$
Low speed	54,263	0.182	164.877	0.165	1.102 [1.077, 1.127]
Intermediate speed	199,569	0.670	680.759	0.682	0.981 [0.959, 1.004]
High speed	44,197	0.148	151.856	0.152	0.974 [0.952, 0.996]
Total	298,029	1.000	997.492	1.000	1.000

but under a different weighting.

The quantity F_q is an accumulated finite-shell measure of inward radial motion. It is analogous to the inward pair flux entering flux formulations of turbulent collision kernels [36, 37], but it is evaluated throughout $\sigma \le r \le 1.5\sigma$, rather than on the contact surface.

Low-speed regions have the largest residual collision yield. The low-, intermediate-, and high-speed environments are defined by the Reynolds fluctuation and fixed thresholds in Eqs. (4.4.34) and (4.4.36). Their mean and RMS references are pooled over the 193 retained exposure times in $t \in [104, 200]$, each classified from its nearest stored carrier field. These classes represent instantaneous carrier-speed environments and are not geometrically identified or temporally tracked streak objects.

Table 4.1 reports the raw collision and inward-flux-exposure totals underlying the normalised yields shown in Fig. 4.12. Here, C_q is an integer event count, whereas F_q is the accumulated inward-flux. Low-speed environments contain 54,263 collision events, corresponding to 18.2% of the total 298,029 events, while contributing $F_{\mathrm{LS}} = 164.877$, or 16.5%, of the total inward-motion exposure $F = 997.492$. Their residual collision yield is therefore

$$\mathcal{E}_{\mathrm{flux,LS}} = \frac{p_{C,\mathrm{LS}}}{p_{F,\mathrm{LS}}} = \frac{0.182}{0.165} = 1.102. \tag{4.5.20}$$

Thus, after normalisation by inward-motion exposure, the low-speed class exhibits a residual collision yield approximately 10% above proportional conversion [Fig. 4.12]. The intermediate-speed value, $\mathcal{E}_{\mathrm{flux,I}} = 0.981\ [0.959, 1.004]$, is not resolved from unity because its simultaneous 95% interval contains the proportional-conversion reference. The high-speed value, $\mathcal{E}_{\mathrm{flux,HS}} = 0.974\ [0.952, 0.996]$, shows a small but resolved suppression, with its simultaneous upper bound slightly below unity. The largest resolved residual departure is therefore the enhanced collision yield in the low-speed class.

The pair-time decomposition explains why the flux-normalised result differs from the raw collision-rate ordering. The measured pair-time shares are $p_{T,q} = 0.182,\ 0.706,\ 0.112$ for the low-, intermediate-, and high-speed classes, giving

$$J_q \simeq 1.00,\ 0.95,\ 1.32, \qquad \Phi_q \simeq 0.91,\ 0.97,\ 1.36.$$

High-speed environments therefore exhibit an elevated collision rate per unit near-contact pair-time, $J_{\mathrm{HS}} \simeq 1.32$, primarily because they provide a substantially greater inward-motion supply, $\Phi_{\mathrm{HS}} \simeq 1.36$. After normalisation by that supply, their residual collision yield is slightly below proportional conversion, $\mathcal{E}_{\mathrm{flux,HS}} = 0.974$. Low-speed environments show the complementary behaviour. Their collision rate per unit near-contact pair-time is close to the pooled value, $J_{\mathrm{LS}} \simeq 1.00$, despite a weaker inward-motion supply, $\Phi_{\mathrm{LS}} \simeq 0.91$, because their residual collision yield is enhanced, $\mathcal{E}_{\mathrm{flux,LS}} = 1.102$. The complete three-factor decomposition and its uncertainty are reported in Appendix Fig. B.8.

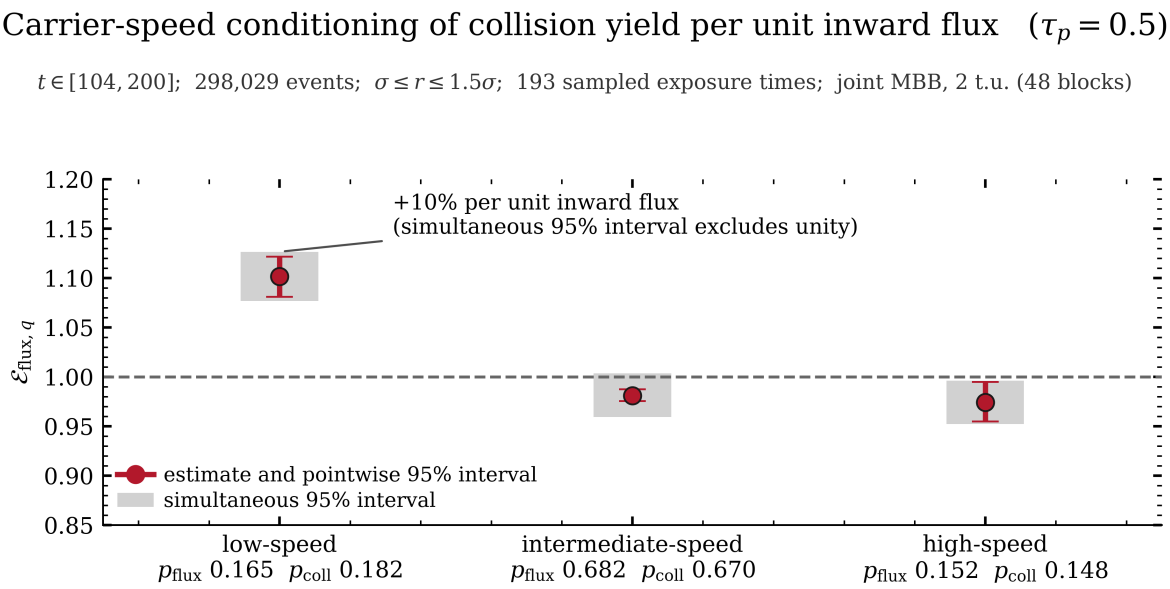

Figure 4.12: **Carrier-speed-conditioned collision yield for $\tau_p = 0.5$.** Statistics use 298,029 events and 193 exposure times over $t = 104$–200, with $\sigma \le r \le 1.5\sigma$. Markers show $\mathcal{E}_{\mathrm{flux},q}$; unity denotes collision occurrence proportional to the inward-flux exposure supplied by the class, and is the exposure-weighted mean of $\mathcal{E}_{\mathrm{flux},q}$ by Eq. (4.5.18). Red error bars and grey rectangles denote pointwise and simultaneous 95% intervals, respectively, obtained by jointly resampling event counts and exposure in 2.0-time-unit moving blocks. Class labels give $p_{F,q}$ and $p_{C,q}$.

Environmental availability is biased towards low-speed regions but is not measurably changed by collisions. For a carrier observable X, define the standardised particle-sampling bias

$$B_X = \frac{\langle X \rangle_P - \langle X \rangle_E}{s_E(X)}, \tag{4.5.21}$$

where $\langle X \rangle_P$ is the mean carrier value sampled at particle positions, while $\langle X \rangle_E$ and $s_E(X)$ are the instantaneous Eulerian mean and spatial standard deviation on the same wall-parallel plane. The complete grid-level estimator, temporal averaging, periodic-shift null, and paired uncertainty calculation are given in Appendix B.3.4.2. B_X compares carrier statistics sampled at particle positions with their Eulerian counterparts, the diagnostic through which preferential concentration was first identified [18].

Figure 4.13 shows a negative streamwise-velocity sampling bias in both windows centred at $t_c = 120$ and 199. In each window, $B_{u'}$ is of order -0.15, indicating that particles preferentially sample lower streamwise velocities relative to a spatially uniform reference. A weaker negative bias is also observed for the in-plane velocity divergence, $B_{\nabla_\parallel \cdot \mathbf{u}_\parallel} \simeq -0.05$, indicating preferential sampling of inertial particles in regions where the carrier is mildly compressed. The agreement between the two displayed windows shows that these sampling preferences are not restricted to a single observation time.

The same environmental preference is evident in the carrier-speed classes introduced above. Using the record-averaged area fractions

$$p_{A,q} = 0.128,\ 0.726,\ 0.146,$$

together with the corresponding near-contact pair-time fractions gives $\frac{p_{T,q}}{p_{A,q}} \simeq 1.42,\ 0.97,\ 0.77$. A value above unity means that a class contains a larger share of near-contact pair-time than would be expected from its spatial area alone, whereas a value below unity indicates relative depletion. Low-speed environments are therefore enriched in near-contact pair-time by about 42% relative to their area share, while high-speed environments are depleted by about 23%. The intermediate-speed class is close to proportional representation.

The sampling biases for the Okubo–Weiss-type indicator W, the in-plane strain magnitude $|S|$, and the

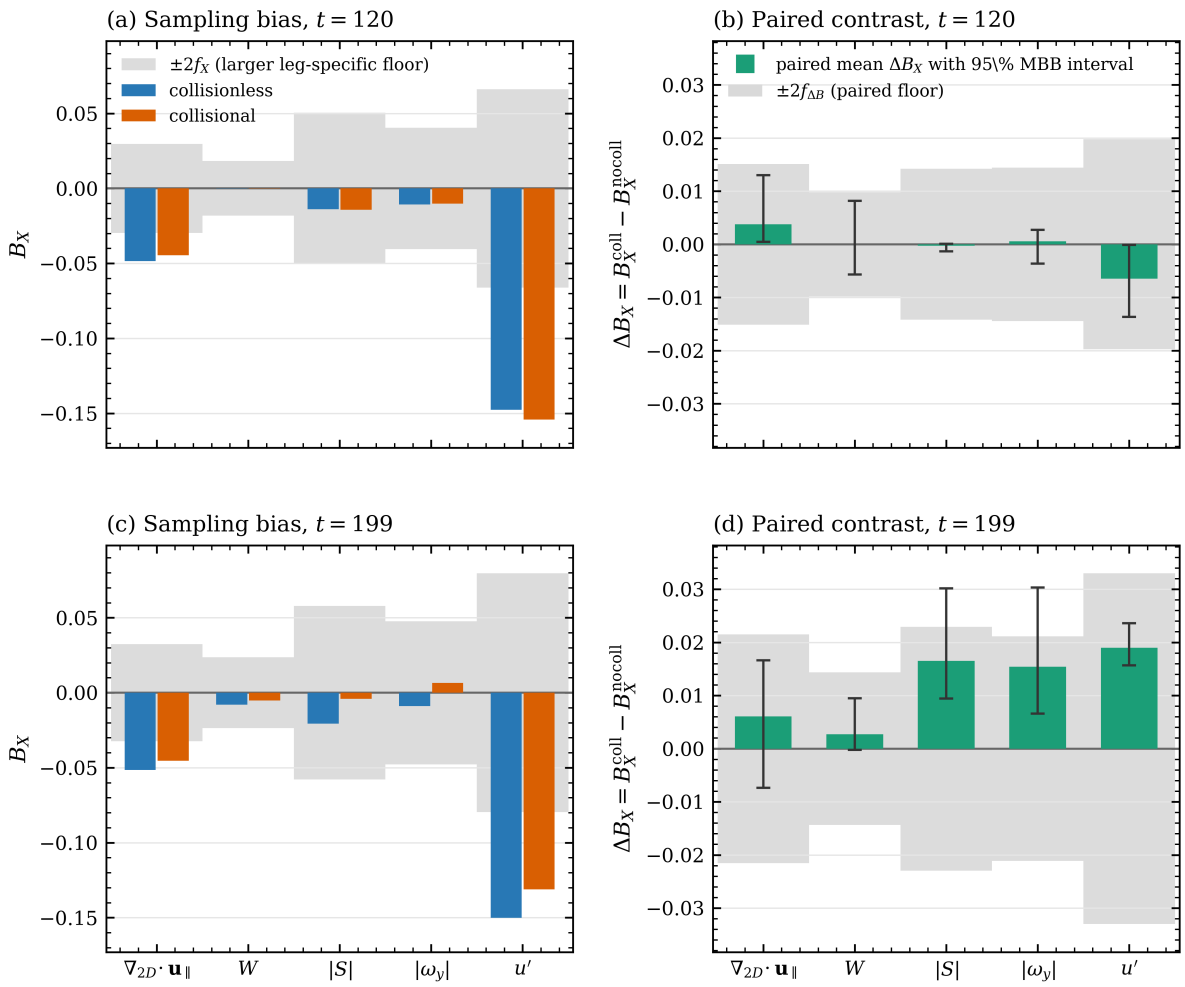

Figure 4.13: **Environmental sampling and paired collision effect for $\tau_p = 0.5$ at $t = 120$ and 199. (a,c)** Standardised one-particle sampling biases, B_X, in the collisionless and collisional systems at $t = 120$ and 199, respectively. Grey bands denote ±2 times the larger of the two leg-specific random-translation detection floors. **(b,d)** Matched differences, $\Delta B_X = B_X^{\mathrm{coll}} - B_X^{\mathrm{nocoll}}$, at the corresponding times. Error bars denote pointwise 95% paired non-circular moving-block-bootstrap intervals, and grey bands denote ±2 times the paired random-translation detection floor.

wall-normal vorticity magnitude $|\omega_y|$ remain within the corresponding displayed random-translation detection bands in both sampling windows [Fig. 4.13]. Under the adopted criterion, there is therefore no resolved displacement of their particle-sampled means relative to the spatially translated reference populations.

For every displayed observable, $\Delta B_X = B_X^{\mathrm{coll}} - B_X^{\mathrm{nocoll}}$ remains within the paired numerical-null floor at t =120 and 199. Some pointwise bootstrap intervals exclude zero, but the corresponding contrasts remain smaller than the empirical alignment floor and therefore do not satisfy the adopted detection criterion. The carrier flow organises particle sampling towards lower-velocity and mildly compressive regions, but the matched comparison provides no strong evidence that collisions modify this environmental availability significantly.

Answer to the environmental component of RQ2 The environmental dependence of the collision rate is explained primarily by the organisation of near-contact pair availability and inward-motion supply. Low-speed regions contain more near-contact pair-time per unit area but provide weaker inward radial motion. After this reduced supply is accounted for, a modest but resolved enhancement remains in the collision yield per unit inward-flux exposure. High-speed regions contain less near-contact pair-time per unit area but provide substantially stronger inward motion, producing an elevated collision rate per unit

pair-time despite a slightly suppressed residual yield.

The matched collision-on/collision-off comparison resolves no corresponding collision-induced change in environmental sampling under the adopted detection criterion. These results are conditional on the near-contact shell $\sigma \leq r \leq 1.5\sigma$ and establish statistical association rather than the isolated causal action of any single carrier variable, since carrier state, concentration, pair orientation, relative motion, residence, and recurrence covary. Having established how the carrier environment conditions collision occurrence, the next section examines the pair dynamics through and after impact.

4.5.5 Event-conditioned pair kinematics through collision

Section 4.5.4 established how pair availability and inward-motion supply determine which carrier environments convert near-contact pairs into collisions. Having identified where collisions occur, the analysis now turns from event selection to event-centred evolution. RQ3 asks how the measured event-conditioned pair dynamics compare with reduced isolated-pair descriptions containing the collision rule alone or the same collision rule together with isolated Stokes relaxation.

For each retained collision event at time t_e, the dynamics are expressed in the event-centred lag

$$\tau = t - t_e,$$

where $\tau < 0$ and $\tau > 0$ denote the pre- and post-collision branches, respectively. The analysis uses events satisfying $t_e \in [104, 200]$. Because finite-lag particle states must remain within the available trajectory record, the number of events contributing to an ensemble statistic can decrease with increasing $|\tau|$.

The event-centred observables and ensemble averages are defined in Sec. 4.4.3. The measured statistics are compared with two reduced isolated-pair references constructed from a flux-weighted proxy for the incoming collision population. These references are diagnostic reduced models rather than event-level counterfactual trajectories of the same realised collisions. Finite-cadence treatment, pair-clustered uncertainty, reference-model construction, and validation checks are documented in Appendix B.3.5.

To distinguish the direct collision response from the subsequent full-system evolution, Fig. 4.14 compares the measured trajectories with two isolated-pair references. The collision-only reference applies the frictionless elastic impulse and then propagates the pair ballistically. The collision–drag reference applies the same impulse and also includes isolated Stokes relaxation with $\tau_p = 0.5$. Both references omit the unsteady carrier field, neighbouring-particle interactions, and subsequent collisions.

Contact reverses radial approach directly. Figure 4.14(a) shows negative radial relative velocity immediately before contact and positive radial relative velocity immediately afterwards. This sign reversal is a direct consequence of the frictionless elastic binary collision rule. The impulse therefore converts inward normal motion into outward normal motion while leaving the tangential relative velocity unchanged.

The first recorded collision of each unordered pair approaches more rapidly and rebounds more strongly than the all-event population. The all-event mean assigns equal weight to individual collision events and therefore contains repeated contributions from recurrent pairs, whereas each unordered pair can contribute at most once to the first-recorded-collision subset [66]. The observed difference then establishes that the selected first-collision population has stronger near-contact radial approach and rebound than the event-weighted population. A corresponding distinction between the first contact of a pair and its later contacts has been reported for inertial particles in turbulence, where the contact-time distribution of first collisions differs from that of repeated collisions because first contacts are preferentially caustic, high-relative-velocity events [35].

Terminal recorded collisions shows the strongest sustained outward radial motion over the reported positive lags. A terminal event is the last recorded collision of its unordered pair before $t = 200$.

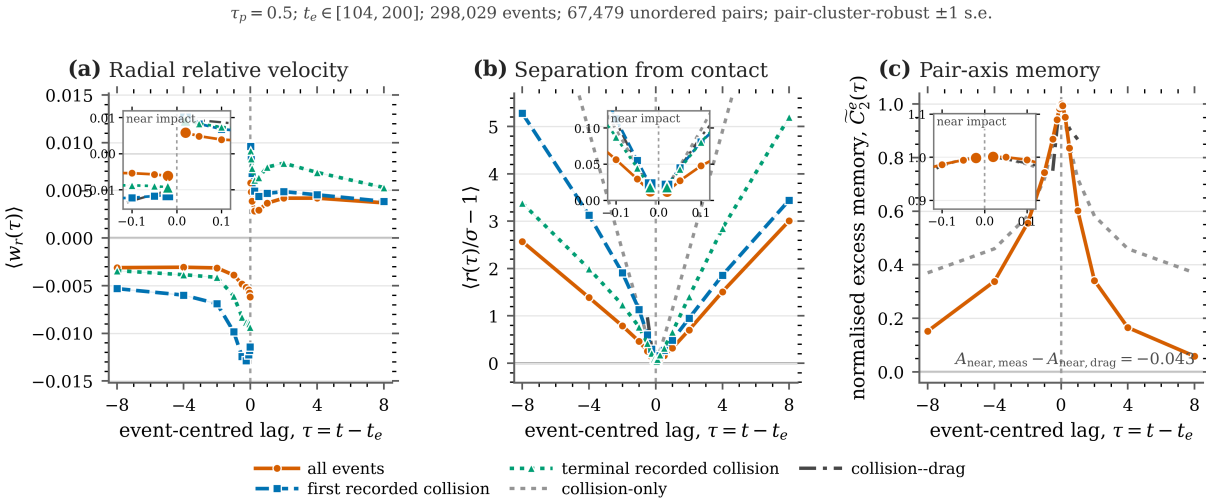

Figure 4.14: **Event-conditioned pair kinematics through collision for $\tau_p = 0.5$ and $\varepsilon_n = 1$. (a)** Mean radial relative velocity and **(b)** mean normalised separation from contact for all retained collisions and for the first-ever and terminal recorded collision of each unordered pair. **(c)** Independent-baseline-corrected, normalised pair-axis memory for the all-event population, compared with isolated collision-only and collision–drag references. Curves use saved states on the requested side of contact and are not joined through $\tau = 0$; the vertical dotted line marks the exact collision time. Insets magnify $|\tau| \leq 0.1$. Shading denotes pair-cluster-robust ±1 standard error. Statistics comprise 298,029 collisions from 67,479 unordered pairs over $t_e \in [104, 200]$.

Sustained $w_r > 0$ carries the pair away from contact and is therefore consistent with the absence of a later collision within the retained record. Because the terminal class is defined using that subsequent record, the result establishes an association between sustained recession and observed terminality; it does not show that the final collision causes permanent escape.

Impact produces a local rebound, whereas finite-lag separation reflects the complete system. For each pair,

$$\frac{\mathrm{d}r}{\mathrm{d}t} = w_r. \tag{4.5.22}$$

The reversal of w_r at contact therefore produces the minimum in mean separation around $\tau = 0$ [Fig. 4.14(b)]. The pair approaches before impact and recedes immediately afterwards: the direct collision response is locally dispersive, not attractive. The small positive separation at the closest plotted lag reflects the saved-state cadence; the exact event occurs at $r = \sigma$.

First-ever collisions approach more rapidly and rebound more strongly than the all-event population. Nevertheless, their post-impact separation does not increase in proportion to the stronger rebound and approaches the pooled curve at finite lag. This behaviour is consistent with the surrounding carrier and particle dynamics reducing or reversing the outward relative motion. It does not, by itself, establish co-residence, caging, or binding.

Terminal recorded collisions exhibit the strongest sustained outward motion and the largest post-impact separation. The follow-up audit gives a median terminal-event headroom of 41.1, while only 10.4% of terminal events occur within five time units of the simulation time-record boundary. The observed recession is therefore not concentrated immediately before $t = 200$ (see Fig. B.9).

The collision–drag reference retains only the collision impulse and isolated Stokes relaxation. The measured finite-lag evolution additionally contains the effects of the unsteady and spatially varying carrier field, neighbouring-particle interactions, and subsequent collisions. A measured separation below the collision–drag reference therefore indicates that the full-system dynamics keep the pair closer, or return it to proximity sooner, than the isolated collision–drag evolution predicts. This difference cannot be attributed to an attractive collision impulse or, by itself, to collision-induced trapping. Finally, the

minimum-image separation remains well below the periodic half-domain limits throughout $-8 \le \tau \le 8$, excluding periodic-image ambiguity over the reported lag range.

Pair-axis memory shows no clear additional post-collision retention beyond isolated collision–drag dynamics. Having established that the immediate collision response is radial, Fig. 4.14(c) examines the persistence of the line-of-centres orientation through contact. The raw nematic event-axis memory $C_2^e(\tau)$ measures the correlation between the pair axis at lag τ and the saved reference axis nearest the collision time.

For the present frictionless elastic collision rule, the particle positions, and hence the line of centres, are continuous through contact. The tangential relative velocity is unchanged, while the normal component reverses sign. The collision-only ballistic reference is therefore exactly symmetric between equal positive and negative lag magnitudes. To quantify any pre–post asymmetry, let

$$\mathcal{L} = \{0.02,\ 0.05,\ 0.10,\ 0.25,\ 0.50\}$$

denote the lag magnitudes supported by the measured, collision-only, and collision–drag curves on both sides of contact, and define

$$A_X = \frac{1}{|\mathcal{L}|} \sum_{s \in \mathcal{L}} \left[C^e_{2,X}(+s) - C^e_{2,X}(-s) \right], \qquad X \in \{\text{meas}, \text{bal}, \text{drag}\}. \tag{4.5.23}$$

Thus, $A_X > 0$ denotes greater mean pair-axis memory on the post-collision branch, $A_X < 0$ denotes lower post-collision memory, and $A_X = 0$ denotes matched-lag pre–post symmetry.

The measured curve is nearly symmetric, $A_{\text{meas}} = -0.001$, whereas the isolated collision–drag reference exhibits a modest positive post–pre asymmetry, $A_{\text{drag}} = +0.042$. The collision-only reference gives $A_{\text{bal}} \simeq -6\times10^{-7}$, numerically indistinguishable from its exact zero-asymmetry value. The corresponding measured-minus-drag contrast is

$$A_{\text{meas}} - A_{\text{drag}} = -0.043.$$

No separate confidence interval is constructed for this aggregate contrast, so it is interpreted descriptively rather than as a formal significance test. The measured pair-axis memory therefore provides no clear evidence of an additional post-collision orientational-retention effect beyond that represented by the isolated collision–drag reference. The most direct instantaneous mechanical signature of contact remains the reversal of radial relative motion rather than an enhancement of pair-axis memory. The reference construction and exact ballistic-symmetry derivation are documented in Appendix B.3.5.

Answer to RQ3 and transition to RQ4 The event-conditioned analysis separates the immediate mechanical response to contact from the subsequent evolution of the particle pair. At impact, the hard-particle collision rule reverses the inward normal component of the relative velocity and initiates radial separation, while the line of centres and tangential relative velocity remain continuous. The direct collision response is therefore locally dispersive rather than attractive.

At finite lag, the measured radial velocity, separation, and pair-axis memory reflect the complete many-particle evolution under the prescribed time-dependent carrier field. Differences from the isolated collision-only and collision–drag references cannot be attributed uniquely to the focal collision, because those reduced references omit the spatially and temporally varying carrier forcing, interactions with neighbouring particles, and subsequent collisions. Moreover, they are constructed from a flux-weighted proxy for the contact population rather than from event-by-event counterfactual realisations of the same measured collisions.

Over the near-impact lag magnitudes supported on both sides of contact, the raw event-axis memory shows no clear additional post-collision retention relative to the collision–drag reference. The matched-

lag aggregate contrast is interpreted descriptively because no separate confidence interval is constructed for it.

RQ3 is therefore answered most directly by the radial dynamics: contact produces an immediate collision-driven rebound, whereas the subsequent pair trajectory reflects full-system evolution rather than the isolated continuation of the collision impulse. The supported sequence is

$$\boxed{\text{carrier-organised approach} \longrightarrow \text{impact-driven radial rebound} \longrightarrow \text{full-system pair evolution}}. \tag{4.5.24}$$

Section 4.5.6 extends this event-centred picture to RQ4 by determining whether the same unordered pairs subsequently return to contact and whether those re-collisions occur without a large intervening spatial excursion.

4.5.6 Collision multiplicity and spatial locality of same-pair recurrence

The preceding event-centred analysis established that contact reverses inward radial motion and produces an immediate rebound. The collision record nevertheless contains many unordered pairs that collide more than once. Repeated contact alone does not determine the intervening pair history: re-collision may follow rapid local re-approach, prolonged association within the same neighbourhood, or a larger excursion followed by reunion. Figure 4.15 therefore characterises completed same-pair recurrence intervals jointly in time and space.

For pair $g = \{i, j\}$, let $t_{g,1} < \cdots < t_{g,M_g}$ denote its retained collision times. Its observed multiplicity, inter-collision waiting time, and sampled maximum excursion are

$$M_g = \#\{t_{g,k}\}, \qquad T_{g,k} = t_{g,k+1} - t_{g,k}, \qquad \widehat{R}_{\max,gk} = \max\left\{1, \max_{t_{g,k}<t_n<t_{g,k+1}} \frac{r_g(t_n)}{d}\right\}, \qquad d \equiv \sigma. \tag{4.5.25}$$

Here, $T_{g,k}$ is obtained from the event log, whereas $\widehat{R}_{\max,gk}$ is evaluated from the saved particle states strictly between the two impacts. Intervals containing no interior saved state are excluded from the excursion-conditioned estimators. Multiplicity remains conditional on the finite observation window, and $\widehat{R}_{\max}$ is a sampled lower bound on the continuous-time maximum. These limitations and the cadence sensitivity are quantified in Appendix B.3.6.

The colliding-pair population contains a broad multiplicity spectrum. Panel 4.15(a) shows the probability mass $P(M = m)$ and complementary cumulative probability $P(M \geq m)$ of the observed multiplicity among unordered pairs with at least one retained collision in $t \in [104, 200]$. The 298,029 retained collision events are generated by 67,479 distinct unordered pairs. Their observed multiplicity has mean 4.42, median 2, and maximum 94. Approximately 66.8% of colliding pairs have $M_g \geq 2$, while 11.5% have $M_g \geq 10$. The distribution is therefore strongly right skewed: most colliding pairs contribute relatively few events, whereas a smaller recurrent subset returns to contact many times.

Same-pair recurrence also affects the weighting of collision-conditioned statistics. Of the 298,029 retained events, 230,550 (77.4%) occur after an earlier retained collision of the same unordered pair. Consequently, an event-weighted statistic describes the distribution over collision events and gives greater representation to highly recurrent pairs. It is therefore distinct from a pair-balanced statistic, in which each unordered pair receives equal weight, and the retained events should not be interpreted as independent first encounters.

Most completed same-pair repeat intervals remain spatially local. A completed repeat interval is classified as *supported* when at least one saved particle state lies strictly between its two bounding collision times. Of the 230,550 completed same-pair repeat intervals, 228,437 (99.1%) are supported in this sense. The remaining 2,113 intervals (0.9%) complete within one saved-state interval and contain no interior

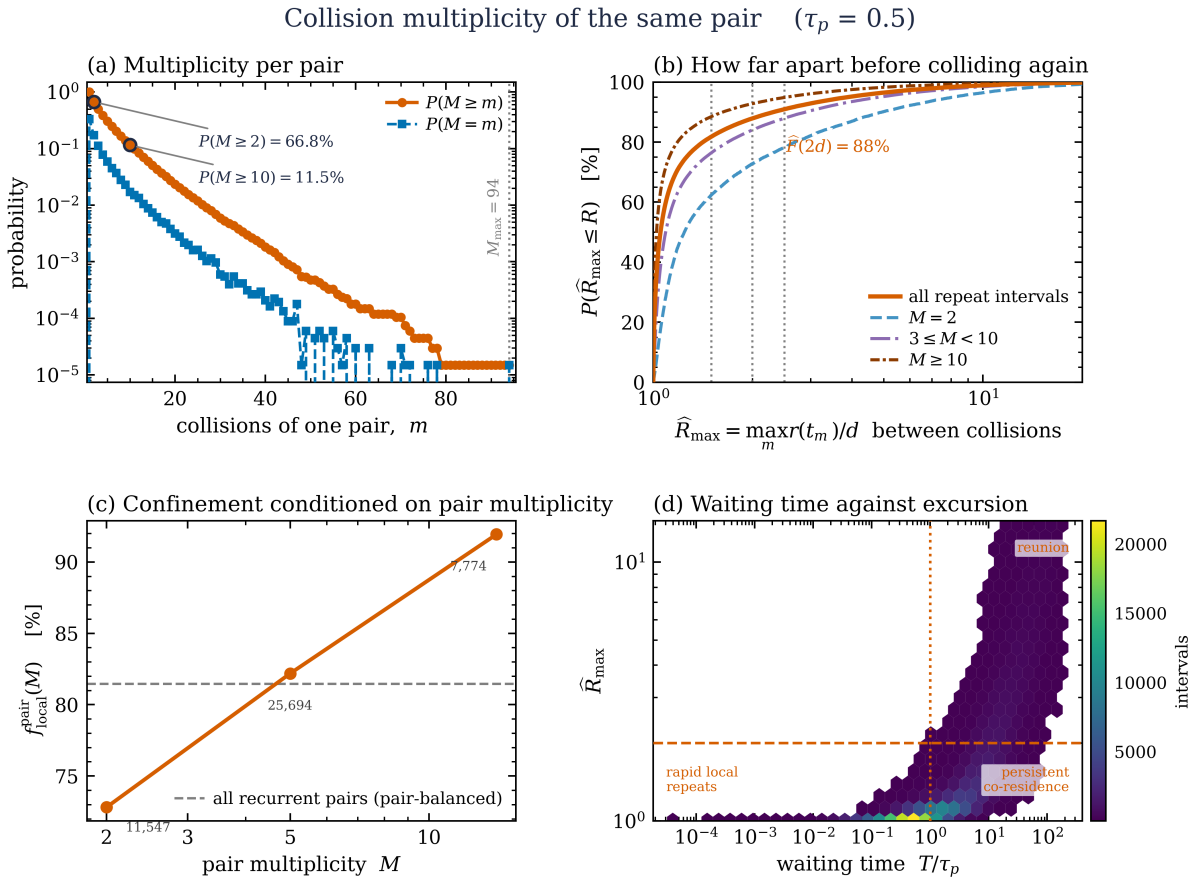

Figure 4.15: **Same-pair collision multiplicity and spatial recurrence for $\tau_p = 0.5$. (a)** Probability mass $P(M = m)$ and complementary cumulative probability $P(M \geq m)$ of the observed collision multiplicity across unordered pairs with at least one retained collision. **(b)** Cumulative distribution of the maximum resolved centre-to-centre separation, $\widehat{R}_{max} = r_{max}/\sigma$, over completed repeat intervals with at least one interior saved state, shown for the pooled interval-weighted population and by pair-multiplicity class. Approximately 87.9% of supported intervals satisfy $\widehat{R}_{max} \leq 2$. **(c)** Pair-balanced local-recurrence fraction ($\widehat{R}_{max} \leq 2$) by pair-multiplicity class. **(d)** Joint distribution of the exact normalised inter-collision waiting time T/τ_p and the corresponding resolved maximum excursion $\widehat{R}_{max}$ for supported completed repeat intervals. The reference lines $T/\tau_p = 1$ and $\widehat{R}_{max} = 2$ partition the plane into operational regions used to distinguish rapid local recurrence, longer local co-residence, and large-excursion reunion. These thresholds are descriptive classification scales, not dynamical phase boundaries and not evidence of collision-induced trapping.

stored state; $\widehat{R}_{max}$ is therefore undefined for them, and they are excluded from excursion-conditioned statistics.

An interval completed within one save step provides insufficient time for a substantial excursion at the measured near-contact velocity scale: $w\,\Delta t_{save} \sim 10^{-3}$, corresponding to approximately $0.1d$. This displacement is far smaller than that required to cross the operational confinement threshold $R_{max} = 2$.

Panel 4.15(b) reports the interval-conditioned cumulative distribution

$$F_R(R) = P\left(R_{max} \leq R \,|\, \text{a completed supported repeat interval}\right). \tag{4.5.26}$$

Thus, $F_R(R)$ is the fraction of observed and temporally supported same-pair repeat intervals whose largest resolved separation before the next collision does not exceed R particle diameters.

For the pooled population of completed, temporally supported repeat intervals,

$$\widehat{F}_R(1.5) = 0.818, \qquad \widehat{F}_R(2) = 0.879, \qquad \widehat{F}_R(2.5) = 0.909. \tag{4.5.27}$$

Thus, approximately 88% of completed supported repeat intervals do not exceed a maximum separation of two particle diameters before the next impact. The conclusion also holds for neighboring locality thresholds: 81.8% remain within $1.5d$, while 90.9% remain within $2.5d$. Among the observed completed intervals, same-pair recurrence is therefore predominantly local rather than dominated by wide separation followed by reunion.

This result also supports the event-centered rebound inference obtained from resolving the apparent tension with the event-conditioned rebound obtained from (Fig. 4.14, panel a). The collision impulse results in ($w_r > 0$) and separates the particles. For another collision to occur while $\widehat{R}_{\max} \leq 2$, the outward relative motion must subsequently decay or reverse before the pair leaves the operational $2d$ neighbourhood. In the present one-way-coupled model we study, this return cannot result from an attractive collision impulse or from a collision-induced modification of the carrier flow. It occurs during the subsequent full-system particle evolution, which contains Stokes relaxation towards the local carrier velocity, the unsteady and spatially varying carrier field, interactions with neighbouring particles, and any later collisions.

Note that, in this analysis, I have deliberately chosen ($\widehat{R}_{\max} = 2$) as an operational definition of spatial locality. It clearly distinguishes recurrence that remains within a prescribed neighbourhood from an recollision producing larger pair-separation. However, it is not sufficient to define a mechanically closed cage, an attractive interaction, permanent binding, or a dynamically invariant region.

The multiplicity-conditioned distributions show a strong association between recurrence frequency and spatial locality. At $\widehat{R}_{\max} = 2$,

$$\begin{aligned} \widehat{F}_R(2 \mid M = 2) &= 0.728, \\ \widehat{F}_R(2 \mid 3 \leq M < 10) &= 0.840, \\ \widehat{F}_R(2 \mid M \geq 10) &= 0.928. \end{aligned} \tag{4.5.28}$$

Completed intervals belonging to high-multiplicity pairs are therefore more strongly concentrated at small excursions less than twice the diameter separation between particle centers, whereas pairs that collide only twice are more likely to separate farther before returning to contact. A physically consistent interpretation is selection: pair histories that remain in local carrier-organised and many-particle configurations provide more opportunities for renewed approach and are consequently enriched in the high-multiplicity classes. The next analysis tests this interpretation using geometrically defined residence episodes constructed independently of collision count.

The multiplicity–locality association persists under equal-pair weighting. The pooled recurrence distribution in Fig. 4.15(b) is interval weighted. A recurrent pair with observed multiplicity M_g can contribute as many as $M_g - 1$ completed repeat intervals, whereas a pair with $M_g = 2$ contributes only one. Consequently, highly recurrent pairs receive greater weight in the pooled interval population.

Panel 4.15(c) removes this direct interval-count weighting using the pair-balanced estimator defined in Eqs. (4.4.68)–(4.4.71). For each recurrent unordered pair, the within-pair quantity $c_g(2)$ measures the fraction of its supported repeat intervals for which no saved state resolves a separation beyond 2σ. These within-pair fractions are then averaged across contributing pairs, so that every pair receives equal total weight irrespective of the number of supported intervals it contains.

The pair-balanced local-recurrence fraction increases systematically with observed multiplicity:

$$f_{\text{local}}^{\text{pair}}(C;2) = \begin{cases} 0.728, & M_g = 2, \\ 0.822, & 3 \leq M_g < 10, \\ 0.919, & M_g \geq 10. \end{cases} \tag{4.5.29}$$

The three classes contain 11,547, 25,694, and 7,774 contributing unordered pairs, respectively. Across

all 45,015 recurrent pairs with at least one supported interval,

$$f_{\mathrm{local}}^{\mathrm{pair}}(C_{\mathrm{all}};2) = 0.815, \qquad C_{\mathrm{all}} = \{M_g \geq 2\}, \tag{4.5.30}$$

compared with the pooled interval-weighted value $\widehat{F}_R(2) = 0.879$.

The difference between the interval-weighted and pair-balanced estimators has a direct statistical interpretation. Higher-multiplicity pairs contribute more repeat intervals and also exhibit larger within-pair fractions of local recurrence. Nevertheless, the monotonic ordering with multiplicity persists when every contributing pair is assigned equal total weight. The association between high observed multiplicity and small resolved inter-collision excursions is therefore not solely an artefact of interval weighting. Since multiplicity is itself a finite-record observable rather than a controlled parameter, this result establishes association rather than causation.

This behaviour is consistent with *local co-residence*, used here in an operational sense analogous to the stable-states definition of residence times [67]: successive collisions of the same pair occur without any intermediate saved state showing a separation beyond 2σ. The particles are neither bound nor maintained at contact; after impact they can separate, rotate, and move with the surrounding population before re-approaching, as in the recollision sequences of inertial pairs in mixing flows [66]. Because $\widehat{R}_{\max}$ is evaluated only at saved states, this classification does not imply uninterrupted confinement between outputs. The carrier-organised, streamwise particle-rich structures identified earlier, of the kind in which near-wall particles are known to accumulate [29, 30] and which govern collision occurrence [39], provide a setting in which repeated encounters are plausible, but the present statistics do not identify a specific carrier structure or many-particle configuration as the mechanism maintaining a recurrent pair.

For elastic collisions,

$$w_r^- < 0 \quad \longrightarrow \quad w_r^+ = -w_r^- > 0,$$

so contact produces an immediate radial rebound. Re-collision therefore requires the relative motion subsequently to return to inward approach during the full-system evolution, which includes Stokes relaxation, carrier forcing, neighbour interactions, and any intervening collisions. The observed sequence is consequently

collision $\longrightarrow$ radial rebound $\longrightarrow$ full-system evolution $\longrightarrow$ renewed approach $\longrightarrow$ local re-collision.

Thus, same-pair recurrence is common and predominantly local in resolved separation, with higher observed multiplicity associated with stronger pair-local recurrence. This does not constitute evidence for attractive binding, collision-induced trapping, or a persistent many-particle cage.

Waiting time and excursion distinguish rapid local recurrence, persistent co-residence, and large-excursion reunion. Panel 4.15(d) combines the exact normalised waiting time T/τ_p with the corresponding resolved maximum excursion $\widehat{R}_{\max}$. The reference lines $\frac{T}{\tau_p} = 1$, , $\widehat{R}_{\max} = 2$ provide an operational classification of completed supported intervals though they may not represent the actual dynamical phase boundaries.

Intervals satisfying

$$\frac{T}{\tau_p} < 1, \qquad \widehat{R}_{\max} \leq 2 \tag{4.5.31}$$

can be attributed with rapid local repeating events. These pairs return to contact within one particle-response time, with no resolved pair-separation beyond the local $2d$ neighbourhood. Following the collision-induced rebound established in the preceding section, such rapid recurrence requires the outward relative motion to decay or reverse before a large excursion develops. This reversal occurs during the subsequent action of Stokes relaxation, the time-dependent carrier field, neighbouring-particle interactions, and any intervening collisions; the present statistic does not isolate their individual contributions.

Intervals satisfying

$$\frac{T}{\tau_p} \geq 1, \qquad \widehat{R}_{\max} \leq 2 \tag{4.5.32}$$

can be associated with persistent local co-residence. Their resolved trajectories remain within two diameters for at least one particle-response time before returning to contact. Over one response time, an unforced relative-velocity difference subject to Stokes drag would decay by a factor e^{-1}. Recurrence in this region therefore occurs after substantial relaxation of the velocity imparted at the preceding impact. A consistent physical interpretation is that the particles remain within the same local carrier- and particle-organised configuration while their relative motion is reorganised and inward approach is restored. The figure does not exactly determine whether this restoration is governed primarily by carrier forcing, neighbouring-particle interactions, the velocity state inherited from earlier impacts, or their combined action.

Intervals with

$$\widehat{R}_{\max} > 2 \tag{4.5.33}$$

contain a resolved maximum excursion beyond the local 2d neighborhood from the prescribed before the pair returns again to contact. These can be classified as large-excursion reunions. They occur predominantly at longer waiting times and occupy the comparatively sparse upper region of the distribution. Since $\widehat{F}_R(2) = 0.8788$, approximately 12.1% of completed supported repeat intervals follow this route. Large-excursion reunion is therefore an observed but secondary pathway to same-pair recurrence.

The resolved range of $\widehat{R}_{\max}$ increases with waiting time partly because longer intervals permit larger excursions and contain more saved states at which the maximum can be sampled. Very short intervals provide less temporal support, and those without an interior saved state are excluded rather than assigned $\widehat{R}_{\max} = 1$. The joint distribution classifies completed recurrence intervals in time and space; it does not imply that longer waiting times cause larger excursions.

4.5.6.1 Neighbour retention and collision counts within proximity episodes

The preceding recurrence analysis showed that most completed same-pair repeat intervals are local in resolved separation and that higher observed multiplicity is associated with smaller inter-collision excursions. These pair-level statistics do not, however, determine whether several impacts can occur during a single geometrically defined period of local proximity, nor do they determine whether repeated pair contact is accompanied by persistence of the wider particle neighbourhood. Figure 4.16 addresses these two questions separately.

The complete analysis interval is $104 \leq t \leq 200$. Panel 4.16(a) uses all 4,801 saved particle states together with the exact collision times recorded within this interval. Panel 4.16(b) uses 25 reference times,

$$t_0 = 104, 106, \ldots, 152,$$

and follows the initial neighbourhood over $0 \leq \tau \leq 48$, so that every reference time has complete temporal support to $t = 200$.

Repeated collisions occur within a single period of local proximity. Proximity episodes are defined as in Sec. 4.4.4, using the hysteretic thresholds $r_{\text{in}} = 1.5\sigma$ and $r_{\text{esc}} = 2\sigma$. For the present analysis, this definition is applied over the retained interval $104 \leq t \leq 200$. Episodes already active at $t = 104$ are excluded because their entry times are not observed, while episodes still active at $t = 200$ are retained as right-censored observations. The observed collision count $N_{g,m}^{\text{obs}}$ is evaluated according to Eq. (4.5.34).

At the saved-state interval $\Delta t_s = 0.02$, the unresolved displacement is only a few hundredths of σ at the measured near-contact velocity scale. Crossing from $r = \sigma$ to $r = r_{\text{esc}} = 2\sigma$ within one interval would require an average radial relative speed of approximately 0.52, far above the observed scale. The

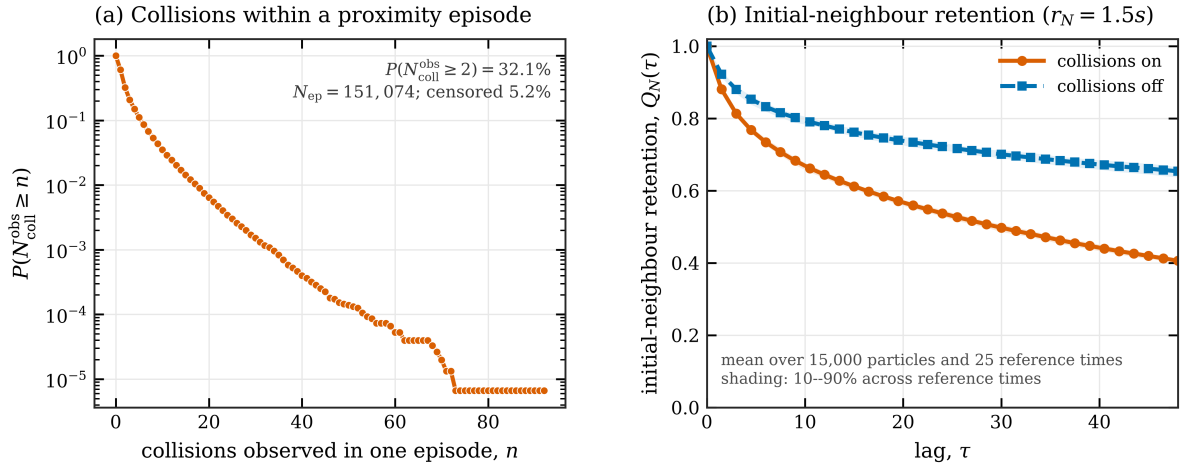

Figure 4.16: **Collision counts within proximity episodes and initial-neighbour retention for $\tau_p = 0.5$. (a)** Complementary cumulative distribution of the observed number of collisions within each geometrically defined proximity episode. Episodes active at $t = 200$ are right-censored, so their displayed counts are lower bounds on their complete-episode counts. **(b)** Mean fraction $Q_N(\tau)$ of the neighbours initially located within $r_N = 1.5s$ that remain within this radius after lag τ, where s is the mean interparticle spacing. The matched collisional and collisionless legs are processed identically. Shading denotes the descriptive 10–90% range across the 25 reference times and is not a confidence interval. Panel (a) uses all 4,801 saved states over $t = 104$–200; panel (b) uses $t_0 = 104, 106, \ldots, 152$ and $0 \leq \tau \leq 48$.

saved-state cadence may therefore shift the resolved escape time slightly but cannot plausibly hide a complete σ-wide excursion.

Once the episode boundaries have been fixed, the number of logged collisions within the episode is

$$N_{g,m}^{\mathrm{obs}} = \sum_{e \in \mathcal{E}_g} \mathbf{1}_{a_{g,m} \leq t_e < b_{g,m}}, \tag{4.5.34}$$

where $\mathcal{E}_g$ contains the exact collision times of pair g. Episodes already active at $t = 104$ are excluded because their entry times are not observed. Episodes remaining within $2d$ at $t = 200$ are retained as right-censored observations, for which $N_{g,m}^{\mathrm{obs}}$ is a lower bound on the complete-episode collision count.

Multiple collisions occur during single periods of uninterrupted local proximity. Figure 4.16(a) contains 151,074 proximity episodes. Of these, 60.9% contain at least one observed collision and 32.1% contain at least two. Only 5.2% remain active at $t = 200$. Additional collisions may occur after the available record, so the observed count of a censored episode is a lower bound. The reported multiple-collision fraction is therefore conservative.

Among the 143,289 episodes observed to terminate before $t = 200$, the median collision count is one. Conditional on containing at least one collision, the median is two, and 31.7% of completed episodes contain at least two collisions. The close agreement between the completed-episode fraction and the corresponding all-episode value shows that the multiple-collision result is not generated by right censoring at the end of the record. Thus, repeated collisions of the same pair can occur without any saved state resolving an intervening excursion beyond the operational 2σ neighbourhood.

Neighbourhood persistence is stronger in the collisionless control. For particle i, let $\mathcal{N}_i(t_0)$ denote the particles located within $r_N = 1.5s$ at reference time t_0. Initial-neighbour retention is defined as

$$Q_N(\tau) = \left\langle \frac{|\mathcal{N}_i(t_0) \cap \mathcal{N}_i(t_0 + \tau)|}{|\mathcal{N}_i(t_0)|} \right\rangle, \tag{4.5.35}$$

where the average includes particles with at least one initial neighbour and all 25 reference times. The statistic measures whether the same particles belong to the neighbourhood at t_0 and $t_0 + \tau$; it does not require them to remain continuously within r_N during the intervening interval.

Both simulations retain substantial neighbourhood memory, but the collisionless curve lies above the collisional curve at every positive sampled lag. At $\tau = 48$, approximately 65.4% of the initial neighbours remain in the collisionless leg, compared with 40.6% in the collisional leg. This comparison therefore provides no evidence that collisions create neighbourhood persistence. The lower retention in the collisional leg can be read together with the effect of collision impulses increasing relative displacement and accelerating the replacement of nearby particles.

The collisionless reference permits ghost-particle overlap because excluded volume is removed. So its neighbourhood persistence therefore cannot be interpreted as mechanical confinement or caging. However, the persistence of neighbours in the absence of contact forces shows that excluded-volume interactions are not required to maintain local particle association. Consistent with the pair-velocity results, carrier-organised co-transport provides a plausible explanation: particles experiencing similar streamwise forcing separate slowly and can remain members of the same local neighbourhood. Persistent neighbourhood membership from Fig. 4.16 should therefore be interpreted as a kinematic consequence of carrier-organised transport, not as evidence of attractive binding, collision-induced trapping, or a mechanical caging.

Answer to RQ4 : pair-local recurrence without a collision-stabilised many-particle cage. The recurrence diagnostics consistently show that repeated collisions are predominantly local at the pair level. Among completed repeat intervals, approximately 88% satisfy $\widehat{R}_{\max} \leq 2$. After assigning equal weight to each recurrent unordered pair, the local-recurrence fraction increases from approximately 73% for $M = 2$ to approximately 92% for $M \geq 10$. Also, 32.1% of all geometrically defined proximity episodes contain at least two observed collisions. Because episodes still active at $t = 200$ are right-censored, this final fraction is based on observed collision counts during the simulation windows. Together, these results show that same-pair re-collisions usually occur before the pair undergoes a large spatial excursion, and that this proximity becomes stronger with increasing observed multiplicity.

This pair-local recurrence does not imply persistent membership of the wider particle neighbourhood. Initial-neighbour retention is substantially lower in the collisional system than in the collisionless control, for example $Q_N(48) = 0.406$ compared with 0.654. Collisions are therefore associated with faster turnover of the surrounding neighbours even while repeated encounters of the same pair remain predominantly local. The combined evidence supports a sequence of collision-induced rebound, followed by full-system evolution and, for a selected subset of pairs, local re-approach. It does not support collision-created confinement, permanent binding, or a stable many-particle cage: repeated collisions occur within a reorganising local environment whose broader membership is continually changing.

Chapter 5

Conclusions

The **central aim** of this thesis was to investigate the collective dynamics and spatial organisation of inertial particles transported by prescribed carrier flows. One of the core questions addressed in this thesis was **how finite-size interparticle collisions modify the spatial organisation already established by inertial transport and the carrier flow**.

The thesis addressed this question through three connected stages. First, it established a numerical framework for collisional inertial-particle dynamics. Second, it examined collision-modified particle organisation in controlled, prescribed carrier flows and compared the resulting regimes qualitatively with published benchmark behaviour. Finally, it extended the analysis to a reduced wall-parallel particle model driven by a time-resolved turbulent-channel carrier field. This progression and its unifying physical picture are summarised in Fig. 5.1. The principal findings from these three stages are discussed below.

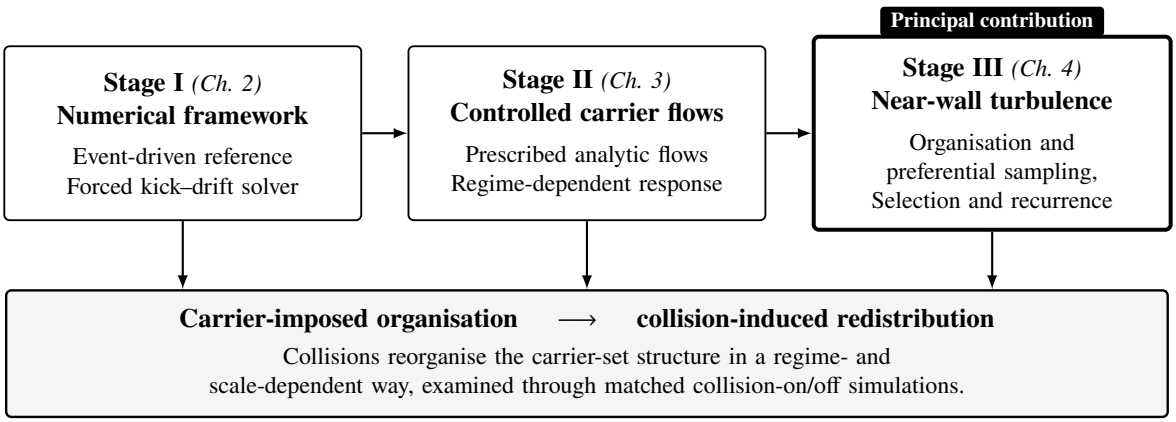

Figure 5.1: **Progression and unifying physical picture of the thesis.** The carrier flow sets the dominant particle organisation; hard-particle collisions redistribute it in a regime- and scale-dependent manner. Stage III (Chapter 4) is the principal original contribution.

Chapter 2 established the numerical foundation for the physical analyses that followed. A timestep-free, event-scheduled hard-particle solver provided an independent force-free reference under its stated assumptions, while the forced kick–drift formulation used operator splitting to couple continuous Stokes-drag dynamics to chronologically resolved hard-particle collisions within each ballistic drift. Analytical tests, cross-solver comparisons, conservation diagnostics, and timestep-refinement studies supported the numerical reliability of the solvers within the tested regimes and observables, and delineated the numerical scope within which collision-associated differences could be interpreted physically.

In Chapter 3, the numerical framework developed in Chapter 2 was applied to a controlled physical setting,

and the resulting many-particle behaviour was assessed qualitatively against collision-modified regimes reported in the literature. For this purpose, simulations were performed in the analytically prescribed alternating sinusoidal shear flow. The periodic reorientation of the shear produces stretching and folding of material elements and, for sufficiently strong forcing, chaotic advection and non-trivial Lagrangian transport. This provides a controlled setting in which carrier-induced separation and mixing compete with particle response and collisional redistribution, without the additional multiscale complexity of turbulence.

The controlled simulations demonstrated that exactly elastic collisions suppress the strong spatial organisation developed by the high-inertia collisionless population, while generating substantial spatial inhomogeneity at low inertia, thereby producing a reversed-clustering ordering without collisional energy dissipation. Stronger inelasticity permitted pronounced aggregation to re-emerge, whereas a regime in which carrier-induced separation acted faster than successive collisions remained approximately homogeneous. These results established that the population-level effect of collisions is regime dependent and is governed by the interplay among particle response, collision frequency, restitution, and carrier transport, rather than by the collision law alone.

The **principal original contribution** of this thesis was developed in Chapter 4, where the analysis was extended to inertial particles transported by a time-resolved wall-parallel carrier field reconstructed from a turbulent-channel DNS. The matched collisional and collisionless simulations were launched from the same labelled particle state and experienced the same prescribed carrier-flow record, so their comparison isolates the accumulated statistical consequence of enabling hard-particle interactions. The analysis connects carrier-imposed organisation, collision selection, event-scale response, same-pair recurrence, and population-level redistribution within a common multiscale framework.

The dominant orientation of the particle organisation is imposed by the carrier flow. Its persistent streamwise streak architecture [Sec. 4.5.1.1] is inherited by both the collisional and collisionless populations, which exhibit streamwise-elongated particle organisation, preferentially streamwise pair and connected-structure axes, and strong particle-velocity coherence [Sec. 4.5.1.2]. Collisions therefore do not create the preferred direction; instead, their effects are scale dependent. They weaken the strongest short-range pair-axis alignment and modify radial relative motion near contact [Sec. 4.5.2], while reducing particle-number fluctuations and increasing the streamwise coherence length of the particle-density field [Sec. 4.5.1.5]. Consistently, the late-time configurations and DBSCAN analysis identify fewer, larger connected particle-rich structures in the collisional system under the adopted clustering criterion. The cumulative effect of collisions is therefore best interpreted as a redistribution of particle organisation across spatial scales within an anisotropic architecture already imposed by the carrier, rather than as simple homogenisation or scattering.

The collision statistics further show that raw event counts cannot, by themselves, be interpreted as collision propensity. Near-contact pairs are already strongly biased towards the streamwise direction, whereas realised collisions are preferentially selected from particular approaching configurations [Sec. 4.5.3]. Separating pair availability, inward-motion supply, and realised collision events shows that the angular variation in collision occurrence is governed primarily by the available inward radial motion, with a smaller residual variation in event yield once that exposure is accounted for. Environmental conditioning gives the same general hierarchy [Sec. 4.5.4]: particles preferentially sample lower streamwise velocities and regions of mildly negative wall-parallel divergence, while the matched collisional and collisionless populations show no resolved collision-induced change in the sampled carrier environment under the adopted detection criterion. The carrier therefore organises the encounter population from which collisions are selected.

At the event scale, Sec. 4.5.5 characterises the immediate mechanical response to contact. For a frictionless elastic collision, the normal component of the relative velocity reverses sign while the tangential component remains unchanged, producing an instantaneous radial rebound. The direct contact response is therefore locally dispersive rather than attractive. At finite lag, however, the measured pair

evolution also contains the effects of time-dependent carrier forcing, neighbouring-particle interactions, and subsequent collisions. The collision-only and collision–Stokes-drag trajectories are therefore reduced diagnostic references rather than exact event-level counterfactuals. After accounting for the persistent carrier-imposed orientation anisotropy, no excess post-collision pair-axis retention is resolved relative to the isolated collision–drag reference.

Despite this immediate rebound, substantial same-pair recurrence is observed [Sec. 4.5.6]. Approximately 88% of completed repeat intervals with adequate temporal support show no resolved pair separation beyond two particle diameters before the next impact. Under equal-pair weighting, the local-recurrence fraction increases from approximately 73% for pairs with two recorded collisions to approximately 92% for pairs with at least ten. The proximity-episode analysis [Sec. 4.5.6.1] provides complementary evidence, with 32.1% of episodes containing at least two observed collisions. This pair-local recurrence does not imply persistent many-particle confinement: at $\tau = 48$, the retained fraction of initial neighbours is approximately 0.406 in the collisional system and 0.654 in the collisionless reference. Selected pairs can therefore return repeatedly to contact while the wider particle neighbourhood turns over more rapidly.

Taken together, these findings support a scale-dependent collision response within a particle field whose dominant orientation is imposed by the carrier flow. The carrier organises the population of available encounters; contact produces an immediate radial rebound; and subsequent full-system dynamics govern re-approach and recurrence. At the population level, collisions predominantly reduce particle-number fluctuations and short-range pair-axis alignment while increasing streamwise density coherence. Selected pairs can therefore undergo repeated local encounters even as the wider neighbourhood turns over more rapidly.

These near-wall conclusions concern the focused $\tau_p = 0.5$ case within the reduced, one-way-coupled wall-parallel model. Within this scope, the combined structural, collision-selection, and recurrence diagnostics provide a coherent physical interpretation of the observed redistribution: *the carrier flow establishes the dominant particle organisation, while finite-size collisions modify its strength, spatial scale, and subsequent evolution*

5.1 Limitations and Future Extensions

The present study deliberately adopts a reduced particle model in order to isolate how finite-size hard-particle collisions modify collective particle dynamics within a strongly anisotropic near-wall turbulent carrier field. This reduction makes it possible to distinguish carrier organisation, collision selection, event-scale response, same-pair recurrence, and population-level redistribution within a common controlled framework, while also defining the scope of the conclusions. Unless stated otherwise, the hard-particle, drag-only, and one-way-coupling assumptions introduced in Chapter 2 remain in force.

5.1.1 Limitations

Wall-parallel geometrical scope and three-dimensional extension. In the model defined in Sec. 4.3, particle centres are constrained to the single plane

$$\tilde{y} = \tilde{y}_0 = a, \qquad \tilde{y}_0^+ \simeq 1.88,$$

and only the streamwise and spanwise components of the carrier velocity enter their dynamics. Although the carrier field is reconstructed from a fully three-dimensional turbulent-channel DNS dataset, the particle dynamics form a wall-parallel hard-disk system rather than a fully three-dimensional suspension. Consequently, wall-normal transport mechanisms relevant to wall-bounded particle dynamics, including turbophoretic migration, detachment from the wall, and three-dimensional particle–particle collision geometry, are excluded by construction. In addition, the available DNS archive contains only the wall-

normal planes nearest the lower wall and therefore restricts the present analysis to the immediate near-wall region; an equivalent study cannot be performed here throughout the buffer and logarithmic layers.

The most direct physical extension is therefore to release the wall-normal constraint and evolve finite-size particles in three dimensions using a carrier record with sufficient wall-normal coverage. Such a formulation would allow wall-normal inertial transport, particle–wall interactions, and three-dimensional interparticle collisions to act simultaneously, thereby testing whether the collision-selection, recurrence, and structural-redistribution mechanisms identified in Chapter 4 persist as particles sample different wall-normal environments.

Hydrodynamic and contact-model scope. The particles are mechanically finite-sized but hydrodynamically unresolved: their diameter determines excluded volume and contact geometry, whereas carrier forcing is represented by centre-sampled Stokes relaxation. Particle–wall and particle–particle contacts are assumed to be non-adhesive. The supporting wall is additionally smooth and frictionless, so no tangential particle–wall force acts on the in-plane dynamics. Wall-corrected drag, lift, lubrication, history effects, and interparticle hydrodynamic interactions are neglected, as are particle rotation, surface roughness, and tangential particle–particle contact dynamics. One-way coupling further excludes momentum feedback from the particles to the carrier turbulence. These assumptions isolate the effects of excluded volume and instantaneous normal hard-particle impulses while restricting direct extrapolation to physical suspensions in which the neglected mechanisms become dynamically important.

Parameter and inferential scope. The detailed quantitative analysis presented in Chapter 4 focuses on the $\tau_p = 0.5$ case for one particle size, concentration, wall-normal location, and turbulent carrier record at $Re_\tau \simeq 360$. Additional simulations were performed for $\tau_p = 0.5,\ 8$ and 16, but these parameter cases are not reported or analysed in detail in the present thesis. The conclusions drawn here should therefore be interpreted as results for the focused $\tau_p = 0.5$ configuration rather than as scaling laws with particle response time, particle size, concentration, Reynolds number, or wall distance.

The matched collision-on/collision-off design isolates, within the adopted model, the accumulated population-level consequence of enabling hard-particle interactions from a common initial particle state under the same prescribed carrier-flow realisation. It does not provide a persistent particle-wise counterfactual. Once the two particle populations have dynamically diverged, late-time differences should therefore be interpreted statistically rather than as the consequence of a particular earlier collision. The collisionless ghost-particle reference also permits overlap, so separations below the physical contact distance have no direct hard-particle counterpart.

Collision-selection statistics are conditional on the finite near-contact shell

$$\sigma \le r \le 1.5\sigma.$$

They quantify relative pair availability, inward-motion supply, and collision occurrence within this shell rather than an exact contact-limit collision kernel, absolute collision probability, or causal environmental effect. Likewise, the collision-only and collision–Stokes-drag constructions are reduced diagnostic references rather than exact event-level counterfactuals. They omit differential time-dependent carrier forcing, neighbouring-particle interactions, and later collisions, so departures from these references represent the combined influence of dynamics absent from the isolated-pair description.

The recurrence diagnostics also inherit the finite saved-state cadence. The sampled maximum separation $\widehat{R}_{\max}$ is a lower bound on the true continuous-time maximum pair separation, so $\widehat{R}_{\max} \le 2$ denotes *resolved* local recurrence rather than uninterrupted confinement between saved states. Proximity episodes are similarly defined through resolved entry and escape states and may be right-censored at the end of the observation window. Initial-neighbour retention $Q_N(\tau)$ compares neighbourhood membership at two endpoints and does not require continuous residence during the intervening interval. Event-weighted

and pair-balanced recurrence statistics are therefore kept separate because they assign different statistical weight to highly recurrent pairs.

5.1.2 Ongoing Activities

Ongoing work extends the wall-confined particle model used in this thesis by progressively introducing rotational and frictional contact physics. The particles remain confined to the wall-parallel plane and are driven by the same prescribed one-way-coupled carrier flow, but the model hierarchy adds, in sequence, fluid-torque-driven particle rotation, tangential restitution during particle–particle collisions, Coulomb-limited particle–particle friction, and particle–wall friction with sliding and rolling. This staged construction is designed to isolate the effect of each additional mechanism relative to the smooth, frictionless collision model used as the baseline in the present thesis while retaining the same controlled modelling framework.

A complementary activity concerns the wall-normal component of the collision impulse that is excluded from the confined dynamics. Because wall-normal particle motion remains kinematically suppressed, only wall-parallel impulses are applied to the simulated trajectories. An auxiliary unconstrained three-dimensional contact calculation can nevertheless be evaluated diagnostically from each pre-collision state to estimate the hypothetical wall-normal impulse associated with the corresponding three-dimensional contact law. These diagnostic impulses are recorded but are never applied to the particle dynamics. Their distributions can therefore be used to quantify how frequently the unconstrained contact model predicts wall-normal impulses above a prescribed threshold and, once the confinement assumption is relaxed, to investigate whether such events may contribute to particle detachment or resuspension.

Research output. A first-author manuscript based on the research presented in this thesis is currently in preparation with Dr. Christophe Henry and Dr. Laetitia Giraldi.

5.1.3 Parallel Research Direction: Continuum Modelling for Wall-Bounded Turbulent Carrier Flows

In parallel with the particle-level work presented in this thesis, I am developing a continuum description of colliding inertial-particle populations derived from the underlying microscopic dynamics. The framework has already been implemented for analytically prescribed carrier flows, including the Solomon–Gollub cellular flow [26], and builds on the continuum approach of López and Puglisi [9]. Starting from the microscopic Stokes-drag and hard-particle collision model, the particle velocity is expanded in the particle response time. At higher order, the expansion contains unresolved collisional contributions involving derivatives of collision sums and terms associated with successive and repeated collision events. Retaining the complete sequence of collision times, collision partners, and contact orientations would preserve microscopic information that a reduced continuum description is intended to eliminate. The unresolved collisional remainder is therefore represented through a stochastic closure: its singular collision-dependent contribution is modelled as an effective sequence of collision-localised random kicks whose covariance is related to the local collision-rate field. The underlying microscopic dynamics remain deterministic; the stochasticity enters only through the reduced description of unresolved collision history. The resulting closed particle-velocity representation is then used to derive a reduced evolution equation for the particle-density field.

The current objective is to extend this continuum formulation from idealised analytically prescribed carrier flows to wall-bounded turbulent carrier flows. The near-wall analysis developed in Chapter 4 identifies additional statistical structure that such an extension must represent, including strong directional anisotropy, spatially non-uniform pair availability, orientation-dependent inward relative motion, preferential collision selection, and substantial same-pair recurrence. These particle-level measurements provide a basis for determining how the existing stochastic collisional closure should be generalised in

an anisotropic turbulent environment and whether a one-particle density description must be supplemented by separation- and orientation-resolved pair information. A subsequent extension beyond the fixed wall-parallel plane would additionally require the wall-normal dependence of these statistics to be represented. The aim of this parallel research direction is therefore to construct and assess a reduced continuum model that retains the tractability of the stochastic closure while incorporating the collision statistics and anisotropic organisation identified in the present near-wall turbulent setting.

5.1.4 Long-Term Research Direction: AI-Based Optimisation of Flexible Particle Shapes

A natural long-term research direction emerging from this thesis is to extend the present physics-based description from rigid finite-size particles to flexible particles whose geometry and mechanical properties influence their transport and preferential sampling of turbulent flows. Studies of non-spherical particles and deformable particle models show that particle geometry and flexibility can modify transport, orientation, and preferential sampling in turbulence [10, 68]. The present thesis provides a mechanistic foundation for this extension by examining how prescribed carrier flows, particle response, finite-size interactions, pair geometry, and collision dynamics contribute to particle organisation. The associated inverse-design problem is to determine how particle geometry and mechanical properties can be selected to produce a prescribed preferential-sampling behaviour.

This direction is directly aligned with the proposed PhD project *AI-Based Optimization of Flexible Particle Shapes in Turbulent Flows*, supervised by Prof. Christophe Eloy (IRPHE, Centrale Méditerranée) and Dr. Laetitia Giraldi (Inria Centre at Université Côte d'Azur). Within the proposed reduced Stokes-flow framework, flexible particles are represented as assemblies of rigid spheres, with elastic deformation projected onto a finite-dimensional mechanical space. Such a formulation combines a reduced representation of compliant-body mechanics with bead-based low-Reynolds-number hydrodynamics [11, 69]. Particle geometry and mechanical properties then become design variables in an inverse-design problem. Bayesian optimisation provides a natural approach for shape optimisation in such reduced hydrodynamic models, as demonstrated for parametrically optimised microswimmers by Palazzolo *et al.* [71], while reinforcement learning provides a complementary strategy for learning flow-dependent policies in complex carrier flows [70]. These studies provide closely related methodological precedents rather than direct models of the flexible-particle problem considered here.

The methodology can first be developed and assessed in controlled carrier flows, including cellular flows, before being extended to turbulent environments of increasing complexity. Possible objectives include enhancing or suppressing preferential sampling of regions characterised by prescribed strain, vorticity, or related flow quantities. In this progression, the present thesis supplies the mechanistic particle-transport foundation, while the proposed research introduces particle shape and flexibility as design variables and AI-assisted optimisation as a systematic means of identifying particle properties that produce targeted behaviour in complex flows.

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Declaration on the Use of Artificial Intelligence

Artificial-intelligence tools were used in a **very limited supporting capacity** during the preparation of this thesis, **with the full knowledge of the thesis supervisors.**

Their use was primarily restricted to language polishing of selected sentences to improve clarity, grammar, academic style, and publication-quality presentation, as well as to assistance with the LaTeX formatting of complex mathematical expressions. With regard to numerical implementation, AI assistance was limited to supporting the use of the Numba package and just-in-time (JIT) compilation to improve the computational efficiency of simulations involving large particle populations, typically exceeding 10,000 particles, and long simulation windows.

AI tools were also consulted regarding standard averaging procedures for selected time-dependent statistical quantities presented in Chapter 4 and to help identify appropriate standard Python library routines for numerical operations such as spectral differentiation, Fourier transforms, and inverse Fourier transforms. Where applicable, the corresponding tools and resources have been acknowledged or cited in the thesis.

All AI-assisted suggestions related to numerical implementation were independently reviewed and verified before use. **All mathematical and physical modelling, methodological choices, numerical experiments, analysis and interpretation of results, scientific conclusions, and final responsibility for the content of this thesis are entirely my own.**

Appendix A

Numerical Implementation and Audit Details of the Solver

This appendix supplements Chapter 2. The particle model, contact mechanics, event-driven architecture, kick–drift algorithm, diagnostics, and verification evidence are defined in the main chapter and are not repeated here. The purpose of this appendix is narrower: it records only those implementation conventions that affect the interpretation of an event record or delimit the numerical scope of a calculation. Values stored with an individual run take precedence over the interface defaults quoted below.

A.1 Numerical Controls and Event Classification

The numerical thresholds used by the solvers classify quantities close to zero in floating-point arithmetic; they do not alter the hard-particle geometry and collision law defined in Section 2.2, nor the ballistic contact condition of subsection 2.3.1. The event-driven implementation uses the default absolute tolerance $\varepsilon_{\mathrm{code}} = 10^{-12}$, a coincidence tolerance of 10^{-12} for wall-contact times, and a minimum cell-crossing lag of 10^{-9}. Its default live-event budget is 5×10^5. These are implementation-level absolute thresholds rather than a dimensionally normalised error model; their adequacy must therefore be reconsidered if the characteristic length, velocity, or time scales change substantially.

For event-driven pair prediction, the code treats $\|\Delta\mathbf{v}\|^2 < \varepsilon_{\mathrm{code}}$ as negligible relative motion. A discriminant smaller than $-\varepsilon_{\mathrm{code}}$ is rejected, whereas a value in $[-\varepsilon_{\mathrm{code}}, 0)$ is set to zero. Admissible roots are examined in increasing order; a future pair event must have a lag greater than $\varepsilon_{\mathrm{code}}$ and inward relative motion at the predicted contact configuration. Consequently, an exactly touching and approaching pair is not resolved through ordinary future-event prediction. Reference initial states must therefore contain neither material overlaps nor unresolved zero-time particle–particle or particle–wall contacts.

A.2 Event-Driven Solver: Acceptance Conditions and Event Records

A.2.1 Neighbour search and completion conditions

The nine-cell neighbour argument of Sec. 2.4.2 holds only when the realised cell widths satisfy

$$\Delta_x > 2a_{\max}, \qquad \Delta_y > 2a_{\max}, \qquad a_{\max} = \max_i a_i, \tag{A.2.1}$$

which every event-driven simulation must meet. Pair prediction examines the local 3×3 set of periodic images and, where several give the same earliest admissible lag, retains the first in the fixed search order; configurations for which that choice is physically ambiguous lie outside the admissible scope.

After transport to a live-event time the implementation recomputes every particle's cell index, repairs discrepant memberships, increments the corresponding counters and regenerates the affected predictions. This changes neither positions nor velocities and is not a physical event. Since repair counts are not retained and the selected queue entry is not re-tested after reconciliation, this is a search safeguard, not evidence of event completeness.

A trajectory is accepted as complete only when the returned terminal time agrees with the requested T_{f} within the stated tolerance. An earliest counter-live entry with $t_e \geq T_{\mathrm{f}}$ is not applied and the state is advanced to T_{f}, so events are processed on $0 \leq t < T_{\mathrm{f}}$. If no counter-live entry remains, the implementation returns at the last processed time rather than completing the final ballistic interval; exhausting the event budget before T_{f} raises an incomplete-run error.

A.2.2 Processed Pair Events and Physical Collisions

The internal queue contains particle–particle, particle–wall, and cell-crossing predictions. Cell crossings and stale entries are search operations and are not exported as physical collision rows. A further distinction is required within the pair branch: the implementation closes the particle free paths and increments the pair-event counters before the collision resolver is called. The resolver can subsequently return unchanged velocities when the supplied geometry is degenerate or the normal relative motion is separating. A code-level pair event is therefore not automatically equivalent to a resolved collision impulse.

For a processed pair event e, define

$$\delta_e = r_{ij}(t_e) - \sigma_{ij}, \qquad g_{n,e}^{-} = \left(\mathbf{v}_i^{-} - \mathbf{v}_j^{-}\right) \cdot \mathbf{n}_{ij}, \qquad J_e = \|\mathbf{v}_i^{+} - \mathbf{v}_i^{-}\| + \|\mathbf{v}_j^{+} - \mathbf{v}_j^{-}\|, \tag{A.2.2}$$

where $R_{ij} = r_i + r_j$ is the contact distance.

A processed pair row is counted as a resolved collision only if $|\delta_e| \leq \varepsilon_{\mathrm{code}}$, $g_{n,e}^{-} < 0$ and $J_e > 0$. Rows failing any of these tests are retained in the record but flagged as no-operation events. Any calculation that interprets the code-level pair-event count as a physical collision count must therefore report the corresponding tolerances and the fraction of no-operation pair rows. Without this audit, the precise description is a *processed counter-live pair event*.

A.3 Forced Solver: Candidate and Safeguard Scope

This section supplements the forced-particle integration and within-drift contact treatment used in the solver. Only the candidate-search and fallback conventions required to interpret the cross-validation and production-refinement evidence are recorded here.

A.3.1 Fixed Drift-Start Candidate Set

At the beginning of a drift of duration h, the implemented query radius is

$$r_{\mathrm{q}}^{\mathrm{raw}} = 2a_{\max} + 2v_{\max}^{\mathrm{d}} h, \qquad v_{\max}^{\mathrm{d}} = \max_i \|\mathbf{v}_i^{\mathrm{d}}\|, \tag{A.3.1}$$

where $\mathbf{v}_i^{\mathrm{d}}$ is the particle velocity immediately before the ballistic drift. Under periodic boundaries, the radius supplied to the spatial search is

$$r_{\mathrm{q}}^{\mathrm{use}} = \min\left[r_{\mathrm{q}}^{\mathrm{raw}}, 0.49 \min(L, W)\right]. \tag{A.3.2}$$

The standard output does not automatically retain the raw and used query radii, candidate count, or a clipping indicator. An un-clipped search may therefore be claimed only when $r_{\mathrm{q}}^{\mathrm{raw}} \leq 0.49 \min(L, W)$ has been checked for the run in question.

The spatial tree is constructed once at the beginning of each drift and returns a fixed unordered-pair set $\mathcal{P}_h$. After an impact, contact times are recomputed from the updated particle state, but only for pairs already contained in $\mathcal{P}_h$. Intermediate collisions can alter individual particle speeds, so Eq. (A.3.1) is not an unconditional completeness bound for arbitrary within-drift collision sequences.

The fixed-set construction is therefore treated as a numerical approximation whose adequacy is assessed for the reported regime through timestep refinement, overlap monitoring, and safeguard records.

A.3.2 Finite-Precision Event and Fallback Conventions

The forced solver uses $\varepsilon_{\mathrm{code}} = 10^{-12}$ by default. For a separated pair, the entry root is retained only when

$$\varepsilon_{\mathrm{code}} < \tau^{\star} \leq \Delta t_{\mathrm{rem}} \tag{A.3.3}$$

and the pair is approaching. A contacting or overlapping approaching pair can instead generate a zero-lag pair event, whereas an overlapping separating pair is ignored. Particle–wall events require a strictly positive lag greater than $\varepsilon_{\mathrm{code}}$.

When the earliest wall and pair lags are equal, the wall event is processed first. Equal pair-event lags follow the deterministic candidate-scan ordering. These rules define a serial implementation convention; simultaneous multiparticle contact is not solved as a coupled constraint problem.

The within-drift event budget is $N_{\mathrm{drift}}^{\max} = c_{\mathrm{cap}} N$, with $c_{\mathrm{cap}} = 2000$ for the alternating-shear runs of Chapter 3 and $c_{\mathrm{cap}} = 100$ for the near-wall runs of Chapter 4 (interface default $c_{\mathrm{cap}} = 10$). If this budget is reached, the implementation records a cap activation, advances the particles ballistically through the residual drift, applies the positional overlap-removal safeguard to the original candidate pairs in $\mathcal{P}_h$, and wraps the corrected configuration under periodic boundaries. This positional projection is a numerical fallback rather than part of the hard-particle collision law, because it changes the configuration without reconstructing the corresponding sequence of physical contact impulses.

A calculation in which this fallback contributes materially is therefore not accepted as an unqualified production calculation without explicit timestep, event-budget, and overlap-sensitivity evidence. The simulation of Chapter 4 is supported by the three-level timestep refinement, overlap record, and cap-hit record reported in Appendix B.2. That evidence establishes resolution only for the tested parameter regime and observables; it does not imply candidate completeness or safeguard inactivity for arbitrary timestep, particle density, restitution, forcing, or search radius.

A.4 Event-driven implementation

Queue and invalidation. Wall, pair and cell-crossing predictions share one priority queue ordered by predicted time, with a sequence number as tie-breaker so equal times order deterministically and payloads are never compared. Each entry records the collision counters of its participants, which an impulse increments, so every entry predicted from the superseded state becomes stale. Stale entries are discarded when popped, and the queue is rebuilt only once it is bloated.

Membership reconciliation. Advancing every particle to the event time lets one drift past a cell boundary before its own crossing event fires, leaving its new neighbours without pair predictions. Drifted particles are therefore located in one vectorised pass after each transport, their membership repaired, counters incremented and predictions regenerated; positions and velocities are unchanged, so this is not a physical event. It costs $O(N)$ per event, so this is not the $O(\log N)$ event-driven algorithm, and is retained because cell-crossing events alone were found to permit overlaps in finite precision.

Completion. Events are processed on $0 \leq t < T_{\mathrm{f}}$. An earliest live entry with $t_e \geq T_{\mathrm{f}}$ is not applied: the state is advanced to T_{f}, the distance covered in that interval being accumulated but not closed, since it is

an unfinished free path. If no live entry remains the implementation returns at the last processed time, so a trajectory is accepted as complete only when the returned terminal time agrees with T_{f} within the stated tolerance. Exhausting the event budget raises an incomplete-run error.

Omissions. The listing shows the event loop. The predictors, the impulse, the cell-list helpers, the heap rebuild and the state recorder appear in Sec. 2.4; snapshot interpolation, the collision table and progress reporting are omitted.

```
import heapq
import numpy as np


# Queue entries carry a sequence number, so that entries with equal predicted
# times are ordered deterministically and their payloads are never compared,
# and the collision counters of the particles involved. An entry whose
# counters no longer match is stale and is discarded when popped.
def next_seq():
    event_seq[0] += 1
    return event_seq[0]


def push_wall(i):
    t_w, walls = time_to_wall(pos[i], vel[i], L, W, radii[i], eps, wall_tol)
    if np.isfinite(t_w):
        heapq.heappush(heap, (current_time + t_w, next_seq(), 'w', i,
                              frozenset(walls), counters[i]))


def push_pair(i, j):
    if i > j:
        i, j = j, i
    t_p, dx = time_to_ball_collision_periodic(pos[i], vel[i], pos[j], vel[j],
                                              radii[i], radii[j], L, W, eps)
    if np.isfinite(t_p):
        dx_store = tuple(dx.tolist()) if dx is not None else None
        heapq.heappush(heap, (current_time + t_p, next_seq(), 'b', i, j,
                              counters[i], counters[j], dx_store))


def push_cell_crossing(i):
    cx, cy = int(particle_cell[i][0]), int(particle_cell[i][1])
    t_c = time_to_cell_crossing(pos[i, 0], pos[i, 1], vel[i, 0], vel[i, 1],
                                cx, cy, cell_size_x, cell_size_y, eps)
    # 1e-9 rather than eps, to avoid repeated tiny crossing events when a
    # particle lands very close to a cell boundary
    if np.isfinite(t_c) and t_c > 1e-9:
        heapq.heappush(heap, (current_time + t_c, next_seq(), 'c', i,
                              counters[i]))


def schedule_all_events(i):
    if boundary == 'wall':
        push_wall(i)
    push_cell_crossing(i)
    for j in get_neighbour_particles(i, particle_cell, cells, nx, ny,
                                     boundary):
        push_pair(i, j)


# Main loop. Events are processed on the half-open interval 0 <= t < T_final.
for _ in range(max_events):

    # earliest entry whose counters are still live; stale entries are dropped
    active_entry = None
    while heap:
        entry = heapq.heappop(heap)
        kind = entry[2]
        if kind == 'b':
            if counters[entry[3]] == entry[5] and \
               counters[entry[4]] == entry[6]:
                active_entry = entry
```

```
                break
        elif counters[entry[3]] == entry[-1]:
            active_entry = entry
            break

    # No live entry remains: return at the last processed time, without
    # completing the final ballistic interval.
    if active_entry is None:
        break

    t_ev = active_entry[0]
    kind = active_entry[2]

    # The earliest live event lies at or beyond T_final: it is not applied.
    # The state is advanced to T_final and the run ends. The distance covered
    # in this last interval is an unfinished path and is not accumulated.
    if t_ev >= T_final:
        dt = T_final - current_time
        lambda_current += np.linalg.norm(vel, axis=1) * dt
        pos += vel * dt
        if boundary == 'periodic':
            pos = wrap_positions(pos, L, W)
        current_time = T_final
        record_state()
        break

    # transport every particle to the event time
    dt = t_ev - current_time
    lambda_current += np.linalg.norm(vel, axis=1) * dt
    pos += vel * dt
    if boundary == 'periodic':
        pos = wrap_positions(pos, L, W)
    current_time = t_ev

    # Membership reconciliation. Advancing every particle lets one drift past
    # a cell boundary before its own crossing event fires, leaving its new
    # neighbours without pair events. Drifted particles are found in one
    # vectorised pass, their membership repaired, counters incremented and
    # predictions regenerated. Positions and velocities are unchanged, so
    # this is not a physical event.
    new_cx = np.minimum((pos[:, 0] / cell_size_x).astype(int), nx - 1)
    new_cy = np.minimum((pos[:, 1] / cell_size_y).astype(int), ny - 1)
    crossed = np.where((new_cx != particle_cell[:, 0]) |
                       (new_cy != particle_cell[:, 1]))[0]
    for k in crossed:
        old = (int(particle_cell[k, 0]), int(particle_cell[k, 1]))
        new = (int(new_cx[k]), int(new_cy[k]))
        cells.setdefault(old, set()).discard(k)
        cells.setdefault(new, set()).add(k)
        particle_cell[k] = [new[0], new[1]]
        counters[k] += 1
        schedule_all_events(k)

    # cell crossing: no velocity change. If the reconciliation above already
    # moved the particle, only a fresh crossing prediction is needed.
    if kind == 'c':
        i = active_entry[3]
        old = (int(particle_cell[i][0]), int(particle_cell[i][1]))
        new = get_cell_index(pos[i, 0], pos[i, 1], cell_size_x, cell_size_y,
                             nx, ny)
        if new != old:
            cells.setdefault(old, set()).discard(i)
            cells.setdefault(new, set()).add(i)
            particle_cell[i] = [new[0], new[1]]
            counters[i] += 1
            schedule_all_events(i)
        else:
            push_cell_crossing(i)
        continue

    # wall hit: reflect the normal component
    if kind == 'w':
        i, walls = active_entry[3], active_entry[4]
```

```
        vel[i] = reflect_from_wall(vel[i], walls, e_wall)
        counters[i] += 1
        wall_collision_count += 1
        update_particle_cell(i, pos, cells, particle_cell,
                             cell_size_x, cell_size_y, nx, ny)
        schedule_all_events(i)

    # ball-ball contact
    else:
        i, j, dx_store = active_entry[3], active_entry[4], active_entry[7]
        dx = np.array(dx_store) if dx_store is not None else None
        # close the free path of both particles. The segment before a
        # particle's first contact is discarded, not counted.
        for p in (i, j):
            if bb_count_particle[p] > 0:
                lambda_sum[p] += lambda_current[p]
                lambda_count[p] += 1
            lambda_current[p] = 0.0
            bb_count_particle[p] += 1
        vel[i], vel[j] = resolve_ball_collision(pos[i], vel[i], masses[i],
                                                pos[j], vel[j], masses[j],
                                                e_ball, eps, dx)
        counters[i] += 1
        counters[j] += 1
        ball_ball_collision_count += 1
        for p in (i, j):
            update_particle_cell(p, pos, cells, particle_cell,
                                 cell_size_x, cell_size_y, nx, ny)
            schedule_all_events(p)

    # stale entries accumulate; the heap is rebuilt once it is bloated
    if len(heap) > heap_cleanup_threshold:
        cleanup_heap()
    record_state()

else:
    raise RuntimeError("max_events reached before T_final.")
```

Listing A.1: Event-driven main loop: prediction and scheduling, lazy invalidation by collision counters, membership reconciliation, free-path closure, and the completion conditions.

A.5 Numerical implementation of Kick-Drift Solver

The production module is approximately 3,100 lines, most of which is input validation, diagnostic bookkeeping and figure generation. The listing below reproduces the parts that define the scheme: the carrier fields, candidate generation, contact-time prediction, impulsive resolution, the collision cap and its fallback, the drag kick, and the main loop.

The solver uses $\varepsilon_{\text{code}} = 10^{-12}$. For a separated pair the entry root is retained only when $\varepsilon_{\text{code}} < \tau^\star \leq \Delta t_{\text{rem}}$ and the pair is approaching; an overlapping approaching pair generates a zero-lag event, an overlapping separating pair is ignored, and wall events require a lag strictly greater than $\varepsilon_{\text{code}}$.

The number of impulses within one drift is bounded by $N^{\max}_{\text{ev,drift}}$. On reaching it the implementation records a cap activation, advances ballistically through the residual drift, and applies the positional overlap-removal safeguard to the candidate pairs. This changes the configuration without reconstructing the corresponding impulses, so it is a numerical fallback rather than part of the collision law, and a calculation in which it contributes materially requires timestep, event-budget and overlap-sensitivity evidence. For Chapter 4 that evidence is the three-level refinement, overlap record and cap-hit record of Appendix B.2, which establishes resolution for the tested regime and observables only.

```
import numpy as np
from scipy.spatial import cKDTree


# Carrier fields. Each returns a closure field(pos, t) giving U_f at every
```

```
# particle position, which is the only interface the drag kick requires.

# U_f = const: every particle relaxes towards a single translation.
def uniform_fluid(U0):
    U0 = np.asarray(U0, dtype=float)

    def field(pos, t):
        out = np.empty_like(pos)
        out[:] = U0
        return out
    return field


# U_f = (S y, 0): steady linear shear of uniform gradient dU/dy = S.
def shear_fluid(shear_rate):
    def field(pos, t):
        out = np.zeros_like(pos)
        out[:, 0] = shear_rate * pos[:, 1]
        return out
    return field


# Alternating shear: streamwise shear for the first half of each period,
# spanwise shear for the second. Each half is steady and integrable, but
# alternating the direction makes the trajectories chaotic. Incompressible,
# with amplitude amp = 2 pi L alpha / T.
def alternating_shear_flow(amp, L, W, period):
    kx = 2.0 * np.pi / L
    ky = 2.0 * np.pi / W

    def field(pos, t):
        out = np.zeros_like(pos)
        if (t % period) < 0.5 * period:
            out[:, 0] = amp * np.sin(ky * pos[:, 1])
        else:
            out[:, 1] = -amp * np.sin(kx * pos[:, 0])
        return out
    return field


# Candidate pairs are generated from drift-start positions and velocities.
# The search radius is based on the drift-start speed bound and therefore
# does not provide an unconditional completeness guarantee after intermediate
# collisions. It is capped below half the box size so that the periodic
# neighbour query remains single-valued.
def candidate_pairs(pos, vel, radii, L, W, dt_drift):
    r_max = radii.max()
    v_max = np.linalg.norm(vel, axis=1).max()
    r_query = min(2.0 * r_max + 2.0 * v_max * dt_drift, 0.49 * min(L, W))
    tree = cKDTree(pos, boxsize=[L, W])
    pairs = tree.query_pairs(r=r_query, output_type="ndarray")
    if len(pairs) == 0:
        return np.empty(0, dtype=np.int64), np.empty(0, dtype=np.int64)
    return pairs[:, 0].copy(), pairs[:, 1].copy()


# Earliest contact among the candidates. No force acts during the drift, so
# the velocity is constant there and the quadratic |r + w t| = R is exact.
def earliest_ball_collision(pi, pj, pos, vel, radii, L, W, t_max, eps):
    best_t = t_max
    best_k = -1
    for k in range(pi.shape[0]):
        i = pi[k]
        j = pj[k]
        dx0 = pos[i, 0] - pos[j, 0]
        dx1 = pos[i, 1] - pos[j, 1]
        dx0 -= L * np.floor(dx0 / L + 0.5)      # minimum image
        dx1 -= W * np.floor(dx1 / W + 0.5)
        dv0 = vel[i, 0] - vel[j, 0]
        dv1 = vel[i, 1] - vel[j, 1]
        R = radii[i] + radii[j]
        a = dv0 * dv0 + dv1 * dv1
        if a < eps:                             # no relative motion
```

```
            continue
        b = 2.0 * (dx0 * dv0 + dx1 * dv1)
        c = dx0 * dx0 + dx1 * dx1 - R * R
        if c <= 0.0:                            # currently overlapping
            if b < 0.0 and best_t > 0.0:        # and still approaching
                best_t = 0.0
                best_k = k
            continue
        disc = b * b - 4.0 * a * c
        if disc < 0.0:                          # the pair never touches
            continue
        t1 = (-b - disc ** 0.5) / (2.0 * a)     # entry root
        if eps < t1 <= best_t and b < 0.0:      # approaching guard
            best_t = t1
            best_k = k
    return best_t, best_k


# Impulse along the line of centres: w_r+ = -e w_r-, tangential unchanged.
def resolve_collision(vi0, vi1, mi, vj0, vj1, mj, e_ball, dx0, dx1, eps):
    dist = (dx0 * dx0 + dx1 * dx1) ** 0.5
    if dist < eps:
        return vi0, vi1, vj0, vj1
    n0 = dx0 / dist
    n1 = dx1 / dist
    rel = (vi0 - vj0) * n0 + (vi1 - vj1) * n1   # relative normal velocity
    if rel >= 0.0:                              # separating: no impulse
        return vi0, vi1, vj0, vj1
    f = (1.0 + e_ball) * rel / (mi + mj)
    return (vi0 - mj * f * n0, vi1 - mj * f * n1,
            vj0 + mi * f * n0, vj1 + mi * f * n1)


# Displacement fallback used only when the cap is reached: any residual
# overlap is removed along the line of centres, split by inverse mass so
# that the centre of mass of the pair does not move. Positions change,
# velocities do not.
def push_apart(pi, pj, pos, radii, inv_m, L, W, eps, iterations):
    for _ in range(iterations):
        for k in range(pi.shape[0]):
            i = pi[k]
            j = pj[k]
            dx0 = pos[i, 0] - pos[j, 0]
            dx1 = pos[i, 1] - pos[j, 1]
            dx0 -= L * np.floor(dx0 / L + 0.5)
            dx1 -= W * np.floor(dx1 / W + 0.5)
            d = (dx0 * dx0 + dx1 * dx1) ** 0.5
            R = radii[i] + radii[j]
            if d < R - eps:
                if d < eps:                     # coincident centres
                    n0, n1, d = 1.0, 0.0, eps
                else:
                    n0, n1 = dx0 / d, dx1 / d
                pen = R - d
                wi, wj = inv_m[i], inv_m[j]
                ws = wi + wj
                pos[i, 0] += n0 * pen * wi / ws
                pos[i, 1] += n1 * pen * wi / ws
                pos[j, 0] -= n0 * pen * wj / ws
                pos[j, 1] -= n1 * pen * wj / ws


# Drag: dv/dt = (U_f - v)/tau, solved exactly with U_f frozen over dt_k.
def apply_kick(dt_k, t_cur):
    Uf = fluid_velocity(pos, t_cur)
    vel[:] = Uf + (vel - Uf) * np.exp(-dt_k / tau)[:, None]


# Straight-line motion of every particle, folded back into the box.
def advance(dt_a):
    pos += vel * dt_a
    pos[:, 0] = np.mod(pos[:, 0], L)
    pos[:, 1] = np.mod(pos[:, 1], W)
```

```

# Free flight over dt, resolving every contact in chronological order.
# The cap bounds the number of impulses per step; it is a guard against a
# non-terminating event sequence, not part of the model.
def drift(dt_drift):
    global cap_hits
    if radii.max() <= 1e-6:                    # collisionless run
        advance(dt_drift)
        return 0
    pi, pj = candidate_pairs(pos, vel, radii, L, W, dt_drift)

    remaining = dt_drift
    n_events = 0
    while remaining > eps and n_events < cap:
        bt, bk = earliest_ball_collision(pi, pj, pos, vel, radii,
                                         L, W, remaining, eps)
        if bk == -1:                           # nothing left in this step
            advance(remaining)
            return n_events
        advance(bt)                            # move up to the contact
        i = pi[bk]
        j = pj[bk]
        dx0 = pos[i, 0] - pos[j, 0]
        dx1 = pos[i, 1] - pos[j, 1]
        dx0 -= L * np.floor(dx0 / L + 0.5)
        dx1 -= W * np.floor(dx1 / W + 0.5)
        vel[i, 0], vel[i, 1], vel[j, 0], vel[j, 1] = \
            resolve_collision(vel[i, 0], vel[i, 1], masses[i],
                              vel[j, 0], vel[j, 1], masses[j],
                              e_ball, dx0, dx1, eps)
        remaining -= bt
        n_events += 1

    # Cap reached: finish the step ballistically and remove any residual
    # overlap. Every occurrence is counted and reported with the run.
    cap_hits += 1
    advance(remaining)
    if len(pi):
        push_apart(pi, pj, pos, radii, 1.0 / masses, L, W, eps, 4)
    return n_events


# Main loop: splitting, half kick / drift / half kick.
cap = cap_factor * N                           # cap_factor = 2000
cap_hits = 0
for step in range(1, n_steps + 1):
    t_prev = (step - 1) * dt
    apply_kick(dt / 2.0, t_prev)
    total_collisions += drift(dt)
    apply_kick(dt / 2.0, t_prev + dt)
```

Listing A.2: Core of the drag–collision solver: carrier fields, candidate generation, contact-time prediction, impulsive resolution, the collision cap and its displacement fallback, the exponential drag kick, and the Strang main loop.

Appendix B

Numerical and Statistical Validation of the Near-Wall Turbulence Analysis

This appendix documents the parameter values, numerical conventions, reference populations, resolution tests, censoring rules, and dependence-aware uncertainty procedures supporting the observables defined in Chapter 4. These details are separated from the main chapter so that the physical interpretation is not distracted by the estimator implementation. Unless stated otherwise, all comparisons use identical particle records, temporal windows, geometric definitions, and estimator parameters in the collisional and collisionless simulations.

B.1 Parameters of the Reduced Turbulent-Flow Experiment

Table B.1: Carrier, sampling and particle parameters. Simulation units use the channel half-height $h = 1$ and the bulk velocity as reference scales; wall units use $\delta_\nu = \nu/u_\tau$ and $t_\nu = \nu/u_\tau^2$. The wall-normal coordinate $\tilde{y}$ is measured from the wall. Entries marked – are dimensionless or have no wall-unit form.

Quantity	Simulation units	Wall units
Carrier flow		
Viscosity ν	1.25×10^{-4} (= 1/8000)	–
Bulk velocity U_b	0.835	$U_b^+ \simeq 18.6$
Re_b; Re_τ	1.34×10^4; $\simeq 360$	–
Friction scales u_τ; δ_ν; t_ν	0.045; 2.78×10^{-3}; 6.2×10^{-2}	1 each, by definition
Wall-normal grid N_y; $\Delta\tilde{y}_{\min}$	193 (Chebyshev); 1.34×10^{-4}	$\Delta\tilde{y}^+_{\min} \simeq 0.048$
Domain $L \times W$	$4\pi \times 8\pi/3$	4520×3020
Carrier grid ($x \times z$)	128×256	$\Delta x^+ \simeq 35$; $\Delta z^+ \simeq 12$
Sampled plane and archive		
Archive: planes; ΔT; T	10 ($\tilde{y} \le 1.08 \times 10^{-2}$); 0.2; 200	$\tilde{y}^+ \le 3.9$; $3.2\,t_\nu$; $3.2 \times 10^3\,t_\nu$
Plane height $\tilde{y}_0 = a$	5.217×10^{-3}	1.9
Particles		
Number N	15 000	–
Diameter σ	1.0434×10^{-2}	3.8
Spacing $s = \sqrt{LW/N}$	8.378×10^{-2} (= $8.03\,\sigma$)	$\simeq 30$
Area fraction ϕ_A	1.22×10^{-2}	–
Response times τ_p	0.5; 8; 16	8; 130; 260
Density ratio ρ_p/ρ_f	10.3; 165; 331	–
Restitution e	1.0 (elastic)	–
Integration step Δt; save step Δt_{save}	0.02; 0.02 (= $\Delta T/10$)	$0.32\,t_\nu$ each

Table B.1 summarises the simulation parameters and their corresponding representations in viscous wall units used throughout the turbulent-flow chapter.

B.2 Numerical integration and interpolation validation

B.2.1 Particle-timestep convergence

Three dedicated simulations were used to test the production particle timestep $\Delta t_p = 0.02$ against the refinements $\Delta t_p/2 = 0.01$ and $\Delta t_p/4 = 0.005$. The simulations share the same $N = 15\,000$ initial particle states, carrier-flow realisation, physical parameters, analysis interval $t \in [10, 20]$, and output times. Only the solver time-step Δt_p is varied. Collision quantities are recorded from the recorded contact events, whereas $R(\theta)$ additionally uses the near-contact pair population in $\sigma \le r \le 1.5\sigma$. Pair axes are folded onto $[0, \pi)$ and divided into 36 equal 5° bins.

For a scalar observable Y, the reported medium-to-fine change is

$$\varepsilon_Y^{m\to f} = \frac{\left|Y_{\Delta t_p/4} - Y_{\Delta t_p/2}\right|}{\left|Y_{\Delta t_p/4}\right|}. \tag{B.2.1}$$

The prescribed tolerances are 2% for the integrated scalar statistics, 5% for the support-weighted RMS difference of the complete $R(\theta)$ curve, and 5° for its folded peak direction.

Between the two finest timesteps, the collision count, distinct-pair count, mean incoming speed, and recurrent-pair fraction change by at most 0.71%, below the prescribed 2% tolerance. Because the analysis window is fixed, the collision rate has the same relative change as the event count and is not listed separately. The single-bin maximum of $R(\theta)$ changes by 2.1%, while its folded direction moves by one angular bin. The support-weighted RMS differences of the complete $R(\theta)$ curve are 4.3% for the event-weighted estimator and 5.0% when only the first collision of each unordered pair is retained. The latter therefore meets the 5% criterion at its acceptance boundary.

The collision counts are not expected to be identical across timesteps. Small changes in impact time alter the subsequent trajectories and collision sequence, so timestep refinement does not preserve a one-to-one correspondence between later events. The relevant test is therefore the stability of the population-level estimators, rather than event-by-event identity. The stable scalar statistics and angular-selection structure, together with the absence of cap hits and unresolved overlaps, support the robustness of choice $\Delta t_p = 0.02$ used for the observables considered here.

Table B.2: **Particle-timestep convergence of the collision statistics.** The final column gives the change between the two finest timesteps; the peak-angle change is expressed in 5° bins.

Observable	0.020	0.010	0.005	Fine-step change
Collision events	9 838	9 827	9 758	0.71%
Distinct unordered pairs	4 485	4 485	4 472	0.29%
Mean pre-impact inward speed, $\langle -w_r^- \rangle$	0.006500	0.006529	0.006530	0.011%
Recurrent-pair fraction, $P(M_g \ge 2)$	0.4285	0.4355	0.4327	0.64%
Peak selection, $\max_\theta R(\theta)$	1.436	1.397	1.427	2.1%
Folded peak direction	47.5°	47.5°	42.5°	one bin
Cap hits	0	0	0	—
Unresolved overlaps	0	0	0	—

B.2.2 Validation of carrier-field interpolation

Carrier velocities at off-grid particle positions are reconstructed using periodic cubic B-spline interpolation in the wall-parallel (x, z) directions, local quadratic Lagrange interpolation in the wall-normal direction, and cubic Catmull–Rom interpolation in time. The carrier record contains $N_t = 1001$ snapshots over $t \in [0, 200]$, with nominal spacing $\Delta t_c = 0.2$. The maximum departure from uniform temporal spacing is 1.70×10^{-14}, confirming the uniform-spacing requirement of the temporal interpolant to floating-point accuracy.

Basic consistency checks recover the stored grid-node values with a maximum absolute error of 1.78×10^{-15}, the no-slip condition at the lower wall to 4.19×10^{-15}, and periodic translation by L_x and L_z to 1.05×10^{-14}. The wall-normal interpolant reproduces linear and quadratic manufactured profiles with maximum errors of 1.60×10^{-16} and 1.16×10^{-16}, respectively, consistent with its polynomial degree.

The complete space–time reconstruction was assessed using the same reader and interpolation procedure employed in the particle calculations. The periodic manufactured fields were

$$u(x, z, t) = \sin(k_x x)\cos(k_z z)\,\Theta(t), \qquad w(x, z, t) = \cos(k_x x)\sin(k_z z)\,\Theta(t), \qquad k_x = \frac{2\pi}{L_x}, \quad k_z = \frac{2\pi}{L_z}. \tag{B.2.2}$$

The fields were stored on the 128×256 wall-parallel grid used in the analysis, with $\Delta t_c = 0.2$, and each manufactured record contained 80 consecutive snapshots. Linear and quadratic choices of $\Theta(t)$ provide known-answer tests because the interior temporal stencil reproduces these temporal dependences exactly; the remaining error therefore reflects the spatial reconstruction and floating-point effects. A sinusoidal $\Theta(t)$, containing four periods over the manufactured record, was used to assess the combined space–time interpolation error.

For each temporal field, 2×10^4 random off-grid space–time queries were evaluated, comprising 3000 queries in each endpoint interval and 14 000 in the interior. For an interpolated quantity q_h and its exact value $q_\star$, the reported errors are

$$\epsilon_{\mathrm{rel,RMS}} = \frac{\|q_h - q_\star\|_2}{\|q_\star\|_2}, \qquad \epsilon_{\max} = \max_m \left| q_h^{(m)} - q_\star^{(m)} \right|. \tag{B.2.3}$$

The relative error is evaluated as a ratio of norms over each temporal region rather than pointwise, because the manufactured fields pass through zero.

Table B.3: **Manufactured-field validation of the carrier-velocity interpolation.** For each row, the largest relative RMS and maximum absolute errors over the two tested wall-parallel velocity components are reported. The first, interior, and final regions contain 3000, 14 000, and 3000 off-grid queries, respectively.

Temporal field	Region	$\epsilon_{\mathrm{rel,RMS}}$	$\epsilon_{\max}$
Linear	first	1.01×10^{-8}	2.81×10^{-10}
Linear	interior	1.02×10^{-8}	2.35×10^{-8}
Linear	final	1.02×10^{-8}	2.53×10^{-8}
Quadratic	first	1.01×10^{-8}	3.24×10^{-12}
Quadratic	interior	1.02×10^{-8}	2.21×10^{-8}
Quadratic	final	1.02×10^{-8}	2.51×10^{-8}
Sinusoidal	interior	4.09×10^{-4}	5.30×10^{-4}

For the linearly and quadratically varying fields, $\epsilon_{\mathrm{rel,RMS}} \simeq 10^{-8}$ throughout the tested temporal regions. Since these temporal dependences are reproduced exactly by the interior stencil, this residual is consistent

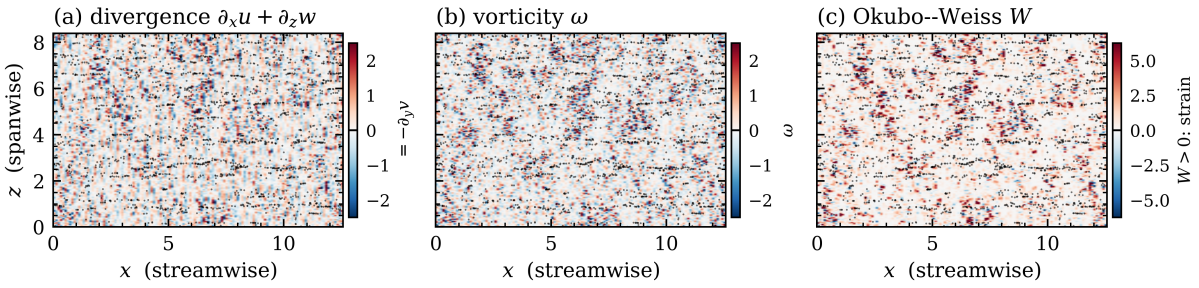

Figure B.1: **Carrier-flow topology on the sampled plane at** $t = 150$**. (a)** In-plane divergence $\partial_x u + \partial_z w = -\partial_y v$. **(b)** Vorticity $\omega = \partial_x w - \partial_z u$, shown on the same symmetric colour scale as the divergence. **(c)** Okubo–Weiss parameter $W = s_n^2 + s_s^2 - \omega^2$; positive and negative values identify strain- and rotation-dominated regions, respectively. Black markers denote collision-on particle centres for $\tau_p = 0.5$ at the same instant. The prescribed carrier field is identical in the two matched simulation legs.

with the spatial reconstruction and floating-point effects rather than temporal truncation. For the smooth sinusoidal field, the interior combined space–time relative RMS error is approximately 4×10^{-4}.

The complete four-point temporal stencil is available over $t \in [0.2, 199.8]$. The first and final intervals use the boundary treatment of the reader and were therefore tested separately. For the known-answer fields, the endpoint and interior relative RMS errors agree within 1.3%, while the largest endpoint-to-interior ratio of maximum absolute error is 1.08. The endpoint treatment therefore introduces no material loss of accuracy relative to the interior over the tested fields and resolution, and is retained in the carrier-field reconstruction used in the present analysis.

B.2.3 Carrier-flow topology

Figure B.1 illustrates the instantaneous carrier-flow quantities used for topology-conditioned statistics. On the sampled plane,

$$\nabla_{\parallel} \cdot \mathbf{u}_{\parallel} = \partial_x u + \partial_z w = -\partial_y v, \qquad \omega = \partial_x w - \partial_z u, \qquad W = s_n^2 + s_s^2 - \omega^2,$$

where $s_n = \partial_x u - \partial_z w$ and $s_s = \partial_x w + \partial_z u$. Thus, the restriction of the three-dimensional incompressible carrier flow can exhibit local in-plane convergence and divergence. With the convention used throughout this study, $W > 0$ denotes strain-dominated regions and $W < 0$ denotes rotation-dominated regions. The particle overlay provides spatial context only: no preferential sampling or collision propensity is inferred from this single snapshot.

B.2.4 Temporal localisation of collision geometry

Collision times t_e are resolved independently of the saved-state cadence. When event geometry is reconstructed from the nearest stored particle state, the temporal offset is

$$\delta t_e = t_{\text{sample}} - t_e, \tag{B.2.4}$$

with

$$|\delta t_e| \leq \frac{\Delta t_{\text{save}}}{2}. \tag{B.2.5}$$

Figure B.2 shows that these offsets are distributed approximately uniformly and symmetrically about zero, indicating no systematic alignment between collision events and output times. The influence of this finite reconstruction offset is assessed separately through neighbouring frame-selection sensitivity tests.

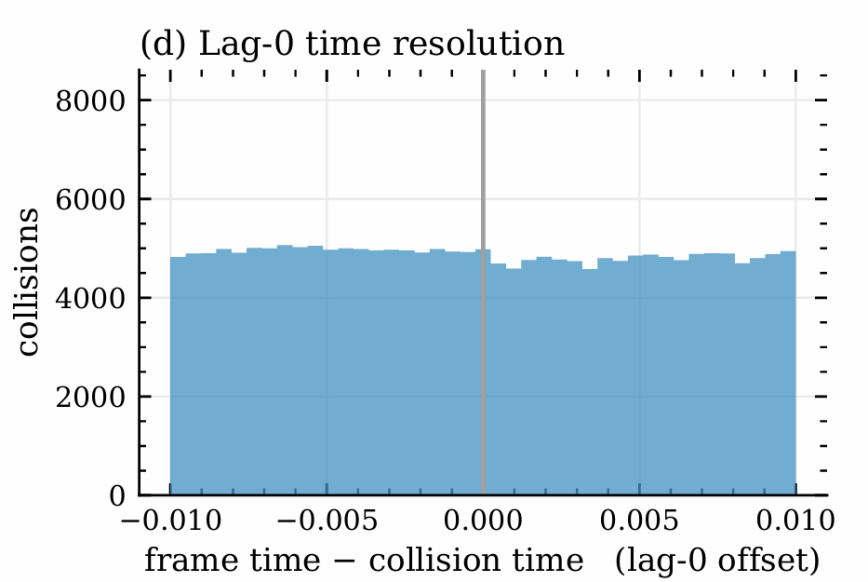

Figure B.2: **Temporal localisation of collision geometry.** Distribution of the offset $\delta t_e = t_{\text{sample}} - t_e$ between the exact collision time and the nearest saved particle state used for event reconstruction. The offsets span the admissible interval $|\delta t_e| \le \Delta t_{\text{save}}/2$ and are approximately uniform about zero, with no evident preferential event-to-frame alignment.

B.2.5 Pair geometry and angular-resolution checks

The minimum-image transformation is applied before evaluating pair separation, axis orientation, or radial velocity. In particular,

$$w_r = (\mathbf{v}_i - \mathbf{v}_j) \cdot \frac{\mathbf{r}_{ij}}{r_{ij}} \tag{B.2.6}$$

uses the same minimum-imaged separation vector as the distance and angle. This avoids assigning the distance to one periodic image and the radial velocity to another.

All angular directions are geometrically available without rectangular-domain truncation when

$$r < \frac{1}{2}\min(L, W). \tag{B.2.7}$$

Statistics extending beyond this range are either compared directly with the matched geometric reference or excluded from scalar orientation interpretation.

The principal near-contact interval is

$$\sigma \le r \le 1.5\sigma. \tag{B.2.8}$$

The outer shell radius is varied in the sensitivity analysis to verify that the qualitative collision-selection result is not produced by one particular choice of shell width.

B.2.6 Time-averaged particle-density redistribution

To visualise persistent particle-density structures, the wall-parallel domain is divided into a periodic 48×32 grid. For frame m, the normalised cell-count fluctuation is defined as

$$\eta_{\alpha\beta}^{(m)} = \frac{H_{\alpha\beta}^{(m)}}{\left\langle H^{(m)} \right\rangle_{\alpha\beta}} - 1, \tag{B.2.9}$$

where $H_{\alpha\beta}^{(m)}$ is the number of particle centres in cell (α, β), and $\langle\cdot\rangle_{\alpha\beta}$ denotes an average over all grid cells. The time-averaged field is

$$\overline{\eta}_{\alpha\beta} = \frac{1}{N_f} \sum_{m=1}^{N_f} \eta_{\alpha\beta}^{(m)}, \qquad N_f = 251, \qquad t_m \in [195, 200]. \tag{B.2.10}$$

The collision-induced redistribution is represented by the matched difference

$$\Delta\overline{\eta}_{\alpha\beta} = \overline{\eta}_{\alpha\beta}^{\text{coll}} - \overline{\eta}_{\alpha\beta}^{\text{nocoll}}, \tag{B.2.11}$$

where the collisional and collisionless simulations are sampled at identical times and use the same spatial grid.

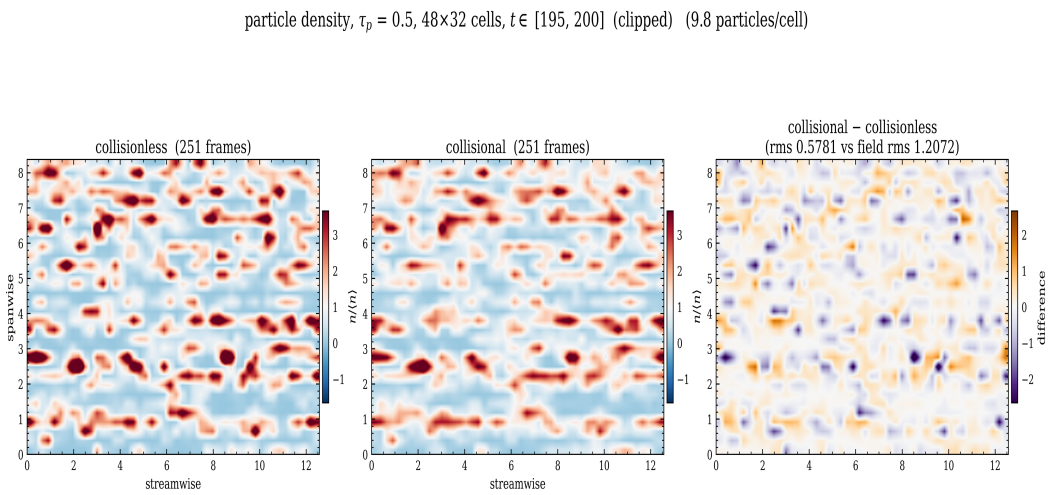

Figure B.3: Time-averaged particle-count fluctuations for $\tau_p = 0.5$, computed from 251 frames over $t \in [195, 200]$ on a periodic 48×32 grid: (a) collisionless case, (b) collisional case, and (c) matched difference $\Delta\overline{\eta} = \overline{\eta}^{\text{coll}} - \overline{\eta}^{\text{nocoll}}$. Panels (a) and (b) use identical colour limits, while panel (c) uses a symmetric diverging scale centred at zero. Colour clipping is applied only for visualisation; all reported statistics are evaluated from the unclipped fields.

Both cases retain predominantly streamwise-aligned density organisation, whereas the difference field reveals local redistribution between the matched simulations. Its root-mean-square magnitude is

$$\Delta\eta_{\text{rms}} = \left\langle (\Delta\overline{\eta})^2 \right\rangle_{xz}^{1/2} = 0.5781, \tag{B.2.12}$$

where $\langle\cdot\rangle_{xz}$ denotes an average over the wall-parallel grid.

Temporal averaging suppresses short-lived fluctuations and therefore emphasises structures that persist over the sampling interval. However, the difference field includes the accumulated nonlinear divergence of the two simulations and is not interpreted as an instantaneous causal response to an individual collision.

B.2.7 Validity of labelled-particle pairing

Section 4.3.5 details the implementation design used for collision and collision legs. Because the two legs begin from identical labeled states and share the same carrier record, the minimum-image displacement

$$\delta\mathbf{x}_i(t) = \text{MI}\left[\mathbf{x}_i^{\text{coll}}(t) - \mathbf{x}_i^{\text{nocoll}}(t)\right]$$

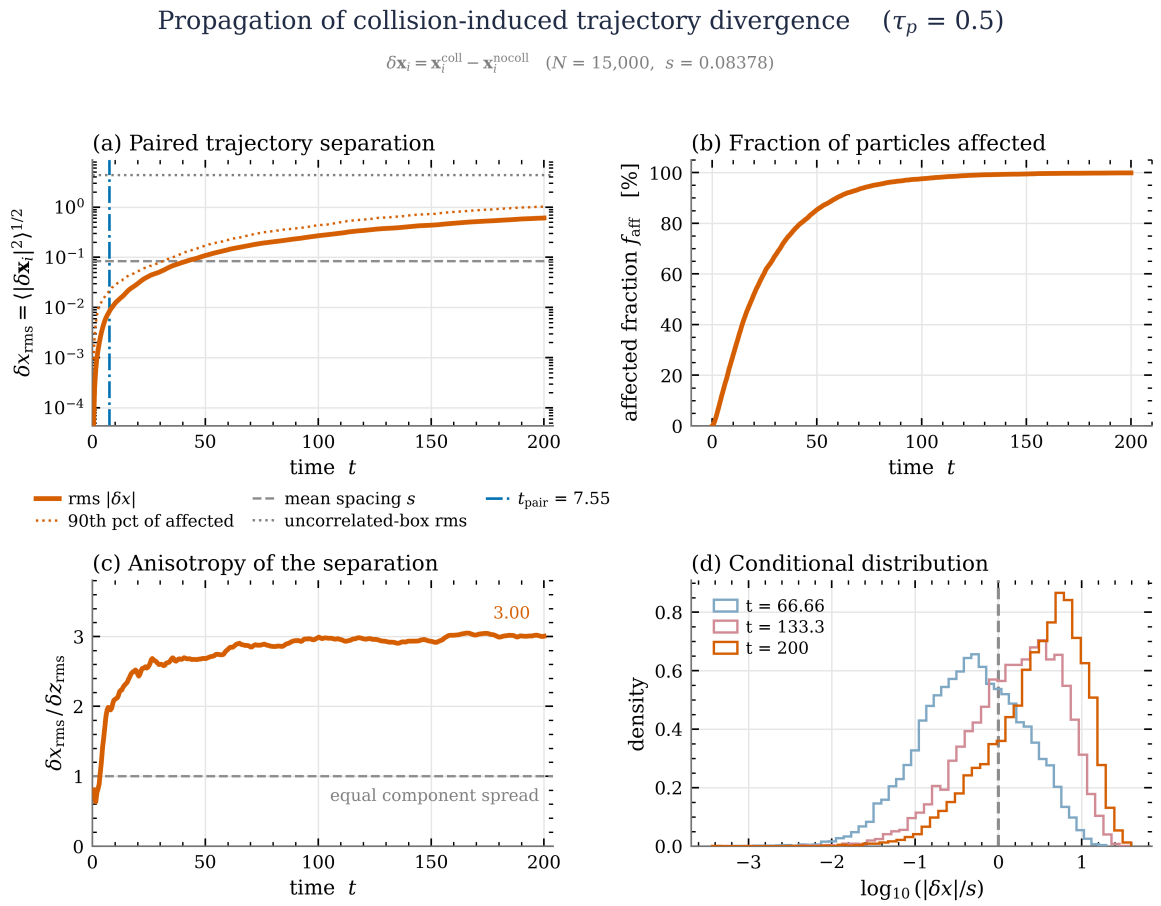

Figure B.4: **Propagation of collision-induced trajectory divergence for $\tau_p = 0.5$. (a)** RMS displacement and 90th percentile over affected particles; $t_{\mathrm{pair}} = 7.55$ marks the conservative pairing limit. **(b)** Affected fraction. **(c)** Streamwise and spanwise RMS components. **(d)** Distribution of $|\delta\mathbf{x}|/s$ over affected particles; the dashed line marks one mean spacing.

measures the accumulated trajectory divergence initiated by collisions. The pairing horizon is defined by the first crossing

$$t_{\mathrm{pair}} = \inf\left\{t > 0 : \left\langle|\delta\mathbf{x}_i(t)|^2\right\rangle_i^{1/2} \geq 0.1s\right\} = 7.55,$$

where $s = 0.08378$ is the mean particle spacing. A particle is classified as affected when $|\delta\mathbf{x}_i| > 10^{-12}s$; under one-way coupling, unreached particles remain identical in the two legs.

Collision-induced perturbations eventually reach almost the entire population and develop more strongly in the streamwise direction. Beyond t_{pair}, however, equal labels no longer represent nearby local counterparts. Late-time differences therefore measure accumulated trajectory divergence under continued carrier forcing and further collisions, not the displacement produced by an isolated collision or a Lyapunov exponent.

B.3 Validation, implementation, and sensitivity of the statistical diagnostics

This appendix supplements Section 4.4 and follows its six-subsection structure exactly. The main text defines the physical estimands, conditioning populations, denominators, null expectations, and interpretive limits; the present appendix records the numerical construction and robustness checks required to reproduce those estimators. To avoid duplicating the Methods, definitions already given in the main text are referenced rather than reintroduced unless an implementation detail requires an explicit

formula.

The correspondence is therefore

$$\boxed{\begin{array}{c}\text{particle structure} \longleftrightarrow \text{Appendix validation of structure}\\ \text{collision selection} \longleftrightarrow \text{Appendix validation of selection}\\ \text{event-centred dynamics} \longleftrightarrow \text{Appendix validation of event references}\\ \text{residence and recurrence} \longleftrightarrow \text{Appendix validation of recurrence}\end{array}} \tag{B.3.1}$$

B.3.1 Computation, validation, and uncertainty of particle-structure and velocity diagnostics

This subsection documents the numerical implementation and validation of the observables introduced in Section 4.4.1. Definitions and physical interpretations given in the main text are not repeated. All collision-on and collision-off quantities are calculated using identical analysis parameters, matched saved times, and the same carrier-flow record.

Common geometry and averaging conventions. All pair-based statistics use unordered pairs $i < j$ and componentwise minimum-image displacements in the periodic $L_x \times L_z$ domain. Circular shell normalisations are used only over the geometric support specified by Eq. (4.4.1). The implementation is verified by checking that every retained displacement satisfies

$$|\Delta x_{ij}^{\mathrm{MI}}| \le \frac{L_x}{2}, \qquad |\Delta z_{ij}^{\mathrm{MI}}| \le \frac{L_z}{2}. \tag{B.3.2}$$

A uniform periodic translation of the complete particle configuration leaves every structural statistic unchanged to numerical precision.

Unless stated otherwise, a statistic is first evaluated separately at every saved time and then averaged over the analysed interval. This equal-frame convention prevents snapshots containing more accepted pairs or clusters from receiving disproportionate weight. Bins failing their pre-specified support criterion are masked rather than interpolated or extrapolated.

B.3.1.1 VMR estimator and validation

For each frame, the periodic domain is divided into M_ℓ square cells of side ℓ. If $N_{c,q}(t_m)$ is the count in cell c for grid translation q, the framewise estimator is

$$\widehat{\mathrm{VMR}}_m(\ell) = \frac{1}{Q}\sum_{q=1}^{Q} \frac{\dfrac{1}{M_\ell - 1}\displaystyle\sum_{c=1}^{M_\ell}\left[N_{c,q}(t_m) - N/M_\ell\right]^2}{N/M_\ell}. \tag{B.3.3}$$

The production calculation uses $Q = 9$ deterministic periodic translations on a 3×3 subcell lattice. These reduce sensitivity to the arbitrary grid origin but are not treated as independent observations.

Exactly square cells are obtained using $(n_x, n_z) = (3m, 2m)$, consistent with $L_x/L_z = 3/2$. Nine grids are retained, from 15×10 to 75×50 cells, corresponding to

$$2 \le \ell/s \le 10, \qquad \ell/s = \sqrt{N/M_\ell}.$$

The finest grid has four particles per cell on average, while the coarsest has 150 cells. Scales outside these conservative support limits are omitted.

For independent uniform points at fixed N, the cell counts are multinomial. The Bessel-corrected variance in Eq. (B.3.3) gives the exact expectation $\mathbb{E}[\widehat{\mathrm{VMR}}] = 1$. The grey reference in Fig. 4.6 is obtained from 500 independent uniform fixed-N configurations processed with the same grids and translations; its envelope is one standard deviation across configurations.

Each reported curve is the mean over 11 consecutive matched frames. Collision effects are formed framewise,

$$\Delta\mathrm{VMR}_m(\ell) = \widehat{\mathrm{VMR}}_m^{\mathrm{coll}}(\ell) - \widehat{\mathrm{VMR}}_m^{\mathrm{nocoll}}(\ell), \tag{B.3.4}$$

before averaging. The displayed percentile range uses 2 000 paired moving-block-bootstrap replicates with three-frame blocks and identical block draws in both legs. Since the window contains only $11/3 \simeq 3.7$ nominal block lengths, this range is *indicative*, not a formal 95% confidence interval. Known-answer tests verify the uniform value VMR = 1, the single-cell limit VMR = N, exact square-cell geometry, deterministic translation averaging, and recovery of a prescribed paired difference. VMR quantifies scale-dependent count-fluctuation amplitude; it does not independently measure connectivity, characteristic length, or coarsening.

B.3.1.2 Density-field construction and autocorrelation

At each saved time t_m, the periodic wall-parallel domain is divided into an $N_x^\rho \times N_z^\rho$ grid of square cells. The unsmoothed particle-count field is

$$H_{\alpha\beta}(t_m) = \sum_{i=1}^{N} \mathbf{1}_{C_{\alpha\beta}}\big(\mathbf{x}_i(t_m)\big), \qquad \sum_{\alpha,\beta} H_{\alpha\beta}(t_m) = N. \tag{B.3.5}$$

Particle positions are reduced periodically before binning, and particle number is verified in every frame. No spatial smoothing or count capping is applied in the reported calculation.

Using the unnormalised forward-transform convention implemented by `numpy.fft.fft2`, the discrete transform is

$$\widehat{H}_{pq}(t_m) = \sum_{\alpha=0}^{N_x^\rho-1} \sum_{\beta=0}^{N_z^\rho-1} H_{\alpha\beta}(t_m) \exp\left[-2\pi\mathrm{i}\left(\frac{p\alpha}{N_x^\rho} + \frac{q\beta}{N_z^\rho}\right)\right]. \tag{B.3.6}$$

For this count field, $|\widehat{H}|^2$ contains the exact single-particle self contribution N. The pair-only spectrum is therefore defined as

$$P_\rho(p,q;t_m) = \begin{cases} |\widehat{H}_{pq}(t_m)|^2 - N, & (p,q) \neq (0,0), \\ 0, & (p,q) = (0,0). \end{cases} \tag{B.3.7}$$

Subtracting N removes the zero-separation self-pair contribution, while setting the zero mode to zero removes the instantaneous mean density. This is the discrete counterpart of the structure-factor decomposition for a point process.

The self-term-corrected periodic covariance is obtained through the discrete Wiener–Khinchin relation,

$$C_{\rho\rho}(\Delta x, \Delta z; t_m) = \mathrm{fftshift}\left\{\mathrm{Re}\left[\mathrm{ifft2}\big(P_\rho(p,q;t_m)\big)\right]\right\}, \tag{B.3.8}$$

and is normalised separately in each retained frame:

$$R_{\rho\rho}(\Delta x, \Delta z; t_m) = \frac{C_{\rho\rho}(\Delta x, \Delta z; t_m)}{C_{\rho\rho}(0,0;t_m)}. \tag{B.3.9}$$

The FFT therefore evaluates the required circular correlation directly; no windowing, zero-padding, or edge correction is applied [27].

Frames without sufficient excess count variance are excluded before normalisation. For $M = N_x^\rho N_z^\rho$ cells, define

$$\mu_m = \frac{N}{M}, \qquad V_m = \frac{1}{M}\sum_{\alpha,\beta}\left(H_{\alpha\beta}(t_m) - \mu_m\right)^2, \tag{B.3.10}$$

and the operational support score

$$S_m = \frac{V_m - \mu_m}{\left[\left(\mu_m + 2\mu_m^2\right)/M\right]^{1/2}}. \tag{B.3.11}$$

A frame is retained when $S_m \geq 3$. This Poisson-reference score is used only to prevent normalisation by unresolved excess variance; because the total particle number is fixed, it is not interpreted as a formal hypothesis-test statistic or p-value.

One grid is selected for each analysis window and then held fixed across all frames. The shared collisional–collisionless grid is the coarser of the grids supported independently by the two legs. Refinement is accepted only while it retains all frames supported by the coarsest grid and targets at least four positive-lag cells before the spanwise $1/e$ crossing. The window average uses only the common supported frame set

$$\mathcal{S} = \left\{m : S_m^{\mathrm{coll}} \geq 3\right\} \cap \left\{m : S_m^{\mathrm{nocoll}} \geq 3\right\}. \tag{B.3.12}$$

For either leg j, the reported correlation map is

$$\overline{R}_{\rho\rho}^{\,j}(\Delta x, \Delta z) = \frac{1}{|\mathcal{S}|}\sum_{m\in\mathcal{S}} R_{\rho\rho}^{j}(\Delta x, \Delta z; t_m). \tag{B.3.13}$$

Thus, framewise normalised correlations are averaged; raw covariance maps are not averaged before normalisation.

The directional profiles are the positive-lag axial cuts

$$R_x(\Delta x) = \overline{R}_{\rho\rho}(\Delta x, 0), \qquad R_z(\Delta z) = \overline{R}_{\rho\rho}(0, \Delta z), \qquad \Delta x, \Delta z \geq 0. \tag{B.3.14}$$

The streamwise and spanwise coherence lengths, ℓ_x and ℓ_z, are obtained by linear interpolation across the first positive-lag crossings of $R_x = 1/\mathrm{e}$ and $R_z = 1/\mathrm{e}$, respectively. A crossing within the first positive-lag cell is treated as an upper resolution bound, not as a resolved length. If no crossing occurs before the largest non-duplicated periodic lag, the reported value is a lower bound.

The ratio ℓ_x/ℓ_z measures coherence along the coordinate axes and is not, in general, the geometric aspect ratio of a tilted structure. The code therefore also evaluates the second-moment tensor of the positive part of $\overline{R}_{\rho\rho}$ as a separate orientation diagnostic.

The calculation is implemented with `NumPy` using `histogram2d`, `fft2`, `ifft2`, and `fftshift`. The built-in validation suite confirms rejection of uniform random fields, recovery of prescribed correlation lengths, resolution improvement under grid refinement, correct treatment of unresolved and over-domain crossings, common frame support between matched legs, and recovery of imposed structural tilt.

B.3.1.3 Computation of Pair-Axis Geometry Statistics

For each unordered pair $i < j$ contributing to Fig. 4.7, the minimum-image wall-parallel separation, distance, and folded pair-axis angle are

$$\Delta\mathbf{r}_{ij}(t) = \left(\Delta x_{ij}(t), \Delta z_{ij}(t)\right), \qquad r_{ij}(t) = \|\Delta\mathbf{r}_{ij}(t)\|, \qquad \theta_{ij}(t) = \operatorname{atan2}\left(\Delta z_{ij}(t), \Delta x_{ij}(t)\right) \bmod \pi. \tag{B.3.15}$$

Thus $\theta_{ij} \in [0, \pi)$, with 0° denoting the streamwise direction and 90° the spanwise direction. Separations are normalised by the mean planar spacing

$$s = \sqrt{\frac{LW}{N}}.$$

Panel (a) uses all pair observations satisfying $s \le r_{ij} \le 5s$, pooled over the analysis frames. If H_b is the number of observations in angular bin I_b of width $\Delta\theta$, and $M = \sum_b H_b$, the normalised pair-axis distribution is

$$g_\theta(\theta_b) = \frac{\pi H_b}{M\,\Delta\theta} = \frac{H_b}{\overline{H}}, \qquad \frac{1}{\pi}\int_0^\pi g_\theta(\theta)\,\mathrm{d}\theta = 1, \tag{B.3.16}$$

where the second expression applies to the equal-width bins used here and $\overline{H} = M/N_{\mathrm{bins}}$. Isotropy corresponds to $g_\theta = 1$. For the polar display, the distribution is repeated with period π over $[0, 2\pi)$; this repetition has no effect on the statistics.

Panels (b) and (c) quantify pair-axis order as a function of separation. For scale r_k, pair observations are collected in the factor-two shell

$$B_k = \left[\frac{r_k}{2},\, r_k\right], \tag{B.3.17}$$

and plotted at the geometric shell centre $r_{\mathrm{mid},k} = r_k/\sqrt{2}$, normalised by s. Define

$$\mathcal{P}_k = \left\{(i, j, t) : i < j,\ r_{ij}(t) \in B_k\right\}, \qquad M_k = |\mathcal{P}_k|, \tag{B.3.18}$$

where t runs over the analysis frames. The corresponding nematic pair-axis order parameter is

$$Q_{2,k} = \frac{1}{M_k}\sum_{(i,j,t)\in\mathcal{P}_k} \exp\bigl(2\mathrm{i}\,\theta_{ij}(t)\bigr)\,. \tag{B.3.19}$$

Its magnitude and director are

$$S_k = |Q_{2,k}|, \qquad \Theta_k = \frac{1}{2}\arg Q_{2,k}, \tag{B.3.20}$$

with Θ_k defined modulo π and reported on $(-\pi/2, \pi/2]$. Panel (b) reports S_k, while panel (c) reports Θ_k; marker area is scaled with S_k so that shells with weak orientational order are visually de-emphasised.

B.3.1.4 Pair-axis averaging and statistical references.

The pair-axis statistics in Fig. 4.7 use all supported pairs from 25 matched frames spanning $t = 104$–200, with angles folded onto $[0, \pi)$. In panel (a), the angular histogram contains 36 equal bins over the shell $r \in [s, 5s] = [0.0838, 0.419]$. Each frame is normalised by its mean bin count before the curves are averaged with equal frame weights. Consequently, $g_\theta = 1$ is the exact isotropic reference represented by the grey circle. The repeated curve over $[0, 2\pi)$ is for polar visualisation only. In panel (b), $S = |\langle e^{2\mathrm{i}\theta}\rangle|$ is calculated independently in the shells $r/s \in [0.5, 1], [1, 2], \ldots, [16, 32]$, plotted at their geometric centres, and averaged equally over frames. The coloured bands denote one sample standard deviation across frames, not confidence intervals.

Panel (c) instead pools the complex moments using pair-count weights,

$$(Q_{2,k}^{\mathrm{pool}} = \sum_m M_k(t_m) Q_{2,k}(t_m) / \sum_m M_k(t_m))$$

before computing $S_k^{\mathrm{pool}} = |Q_{2,k}^{\mathrm{pool}}|$ and $\Theta_k^{\mathrm{pool}} = \frac{1}{2}\arg Q_{2,k}^{\mathrm{pool}}$. The director is withheld when the pooled magnitude lies below its isotropic noise floor. Panel (a) confidence intervals are obtained from 2000 moving-block bootstrap replicates with block length two; the same blocks are applied to both legs when

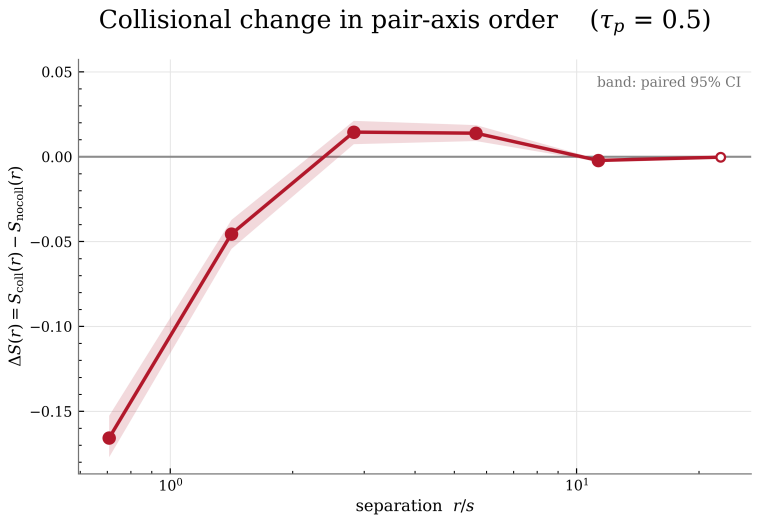

Figure B.5: **Paired collision-induced change in pair-axis order for $\tau_p = 0.5$.** The horizontal line denotes the null value $\Delta S = 0$, and the shaded band is the paired moving-block-bootstrap 95% confidence interval. Shell definitions and observation times match Fig. 4.7. The open outermost marker denotes an unresolved shell below the finite-sample order floor and is excluded from interpretation.

estimating ΔS, preserving the matched temporal structure. Because pairs sharing a particle are not independent, the analytical noise floor is a lower bound on the true threshold, and because only 25 frames are available, the bootstrap intervals are interpreted as conditional on the adopted block length.

Paired uncertainty in scale-dependent pair-axis order The collision-induced change in pair-axis order was quantified as $\Delta S(r) = S_{\mathrm{coll}}(r) - S_{\mathrm{nocoll}}(r)$ using the same 25 matched frames and contiguous logarithmic shells $[r_h/2, r_h]$ as in Fig. 4.7; each value is plotted at $r_h/\sqrt{2}$. The shell-wise order was evaluated in every frame, averaged separately for the two legs, and then differenced. Uncertainty was estimated with a paired moving-block bootstrap using identical temporal blocks in both legs. Figure B.5 confirms that the collision effect changes with scale. At the two finest shell centres, $r/s \simeq 0.71$ and 1.41, collisions reduce the order by approximately 0.166 and 0.046, respectively, with both confidence intervals below zero. The difference becomes small but positive at $r/s \simeq 2.83$ and 5.66, before becoming indistinguishable from zero at larger separations. The outermost shell is not interpreted because its order has reached the finite-sample noise floor. Collisions therefore redistribute pair-axis order across separation scales rather than uniformly strengthening or weakening the carrier-imposed alignment.

B.3.1.5 Cluster-axis statistics.

Connected components are identified using periodic DBSCAN with the production neighbourhood scale and minimum-sample criterion fixed before the collision-on and collision-off fields are compared. Particle coordinates within each component are unwrapped by propagating minimum-image displacements through its connectivity graph. Consistency is checked across independent paths. Components that wind around the periodic domain or cannot be unwrapped unambiguously are identified separately and are not assigned a Euclidean gyration axis.

If $\lambda_{1,c} \geq \lambda_{2,c}$ are the eigenvalues of $\mathbf{G}_c$, the reliability of the principal axis is quantified by

$$a_c = \frac{\lambda_{1,c} - \lambda_{2,c}}{\lambda_{1,c} + \lambda_{2,c}}. \tag{B.3.21}$$

Components below the prescribed minimum size or shape-anisotropy threshold are excluded from axis statistics because their directors are ill-conditioned. Their abundance and particle fraction remain part of the cluster-population summary.

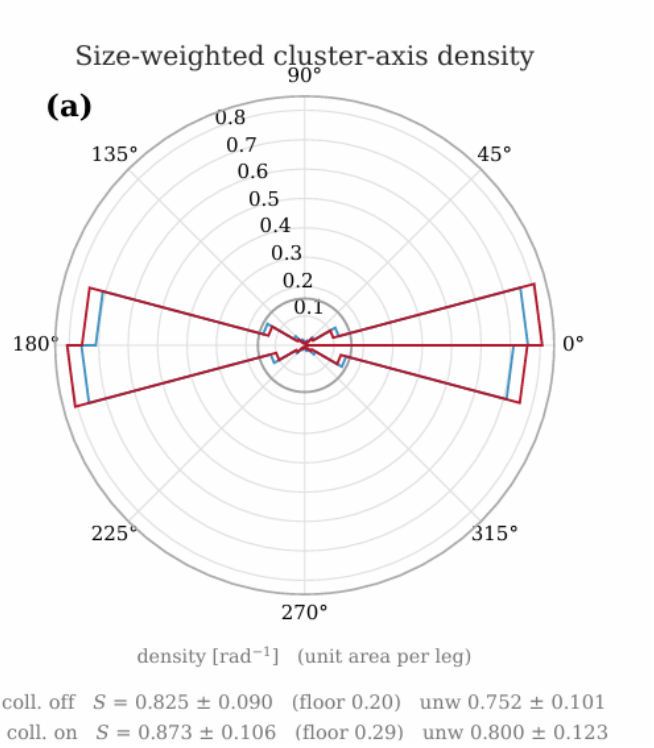

Figure B.6: **Size-weighted particle-cluster orientation for $\tau_p = 0.5$.** Cluster-axis densities are pooled over 25 matched frames spanning $t = 104$–200; blue and red denote the collision-off and collision-on legs, respectively. Cluster axes are head–tail symmetric, and each cluster is weighted by its particle population. The grey circle denotes the isotropic angular density.

The weighted cluster order is

$$Q_{2,\mathrm{cl}}^{(w)} = \frac{\sum_c w_c \exp(2\mathrm{i}\Theta_c)}{\sum_c w_c}, \tag{B.3.22}$$

with $w_c = 1$ for equal-cluster weighting and $w_c = N_c$ for particle-mass weighting. The complete DBSCAN, periodic-unwrapping, component-selection, and axis-fitting pipeline is applied to matched random configurations. This complete-pipeline null captures finite-sample, rectangular-domain, and detection-induced orientational biases that would be missed by imposing $S = 0$ directly.(See Fig. B.6)

B.3.2 Averaging and uncertainty for pair-kinematic statistics

Figure 4.10 uses $M = 25$ matched frames at

$$t_m = 104 + 4m, \qquad m = 0, \ldots, 24.$$

For panels (a)–(c), the collision-on and collision-off legs use identical radial-bin boundaries, and quantitative comparisons are restricted to $r \geq \sigma$. Collision-off values at $r < \sigma$ correspond to ghost-particle overlap and are shown only descriptively.

For radial bin B_k, let $\mathcal{P}_{mk}$ denote the set of N_{mk} unordered pairs in frame m. The framewise quantities are

$$W_{mk} = \frac{1}{N_{mk}} \sum_{(i,j)\in\mathcal{P}_{mk}} |w_{r,ij}|,$$

$$F_{mk}^{-} = \frac{1}{N_{mk}} \sum_{(i,j)\in\mathcal{P}_{mk}} \mathbf{1}(w_{r,ij} < 0),$$

$$C_{v,mk} = \frac{N_{mk}^{-1} \sum_{(i,j)\in\mathcal{P}_{mk}} \mathbf{v}_i \cdot \mathbf{v}_j}{N^{-1} \sum_{i=1}^{N} |\mathbf{v}_i|^2}$$

.

For any such statistic Y_{mk}, the plotted curve is the equal-frame mean

$$\overline{Y}_k = \frac{1}{M_k} \sum_{m\in\mathcal{M}_k} Y_{mk}, \tag{B.3.23}$$

and its shading is the corresponding sample standard deviation across frames,

$$s_{Y,k} = \left[\frac{1}{M_k - 1} \sum_{m\in\mathcal{M}_k} (Y_{mk} - \overline{Y}_k)^2 \right]^{1/2}. \tag{B.3.24}$$

Thus, the bands represent frame-to-frame variability rather than confidence intervals.

For panel (d), the fluctuation velocities, directions, and framewise polar order $P'(t_m)$ are defined in Sec. 4.4.1. The plotted relative-to-isotropic angular density is

$$g_{\phi'}(\phi) = 2\pi p_{\phi'}(\phi), \tag{B.3.25}$$

so that $g_{\phi'} = 1$ corresponds to isotropy. The framewise densities are averaged with equal frame weights, and the shading denotes one sample standard deviation across frames.

The mean polar order in simulation leg ℓ is

$$\langle P' \rangle_\ell = \frac{1}{M} \sum_{m=0}^{M-1} P'_\ell(t_m), \tag{B.3.26}$$

and the paired collision effect is

$$\Delta\langle P' \rangle = \frac{1}{M} \sum_{m=0}^{M-1} \left[P'_{\mathrm{coll}}(t_m) - P'_{\mathrm{nocoll}}(t_m) \right]. \tag{B.3.27}$$

Its 95% confidence interval is obtained from 2,000 paired moving-block-bootstrap replicates with block length two retained frames, using identical temporal block indices in the two simulation legs. With only 25 retained frames, this interval is interpreted as conditional on the adopted block length.

For N' independent isotropically distributed directions, the large-N' polar-order magnitude satisfies

$$\Pr(P' > p) \simeq \exp(-N' p^2),$$

giving the 95th-percentile finite-sample isotropic reference

$$P'_{\mathrm{iso},95} = \sqrt{\frac{-\ln(0.05)}{N'}}. \tag{B.3.28}$$

For $N' \simeq 15{,}000$, $P'_{\mathrm{iso},95} \simeq 0.0141$. This reference characterises the absolute polar-order magnitude expected from a finite isotropic sample; it is not a confidence interval or significance threshold for $\Delta\langle P' \rangle$.

B.3.3 Angular Collision-Selection Estimators

Discrete accumulators. Let

$$t_0 < \cdots < t_{M-1}, \qquad [t_a, t_b] = [104, 200],$$

denote the saved particle states used in the analysis. Pair-state quantities are integrated in time using nearest-frame, weights. The cell edges are

$$e_0 = t_a, \qquad e_M = t_b, \qquad e_m = \frac{t_{m-1} + t_m}{2}, \quad 1 \le m \le M-1, \tag{B.3.29}$$

with

$$\Delta t_m = e_{m+1} - e_m, \qquad \sum_{m=0}^{M-1} \Delta t_m = t_b - t_a = 96. \tag{B.3.30}$$

The primary calculation uses all $M = 4{,}801$ frames at spacing $h = 0.02$; hence the two endpoint weights are $h/2$ and all interior weights are h.

The axial interval $[0, \pi)$ is divided into $K = 18$ equal-width angular bins B_k. For unordered pair $i < j$, define

$$\chi_{ij,k}^{(m)} = \mathbb{I}\big(\sigma \le r_{ij}(t_m) \le 1.5\sigma\big)\, \mathbb{I}\big(\theta_{ij}(t_m) \in B_k\big)\,. \tag{B.3.31}$$

The accumulated near-contact pair-time, inward radial-motion exposure, and signed radial-motion exposure are then

$$T_k = \sum_{m=0}^{M-1} \Delta t_m \sum_{i<j} \chi_{ij,k}^{(m)}, \tag{B.3.32}$$

$$F_k = \sum_{m=0}^{M-1} \Delta t_m \sum_{i<j} \chi_{ij,k}^{(m)} \big[-w_{r,ij}(t_m)\big]_+, \tag{B.3.33}$$

$$W_k = \sum_{m=0}^{M-1} \Delta t_m \sum_{i<j} \chi_{ij,k}^{(m)} w_{r,ij}(t_m), \tag{B.3.34}$$

where

$$[-w_r]_+ = \max(-w_r, 0).$$

Thus every near-contact pair state contributes to T_k and W_k, whereas separating states contribute zero to F_k.

For the set C of retained collision events, let θ_e denote the contact-axis angle associated with event e. The event count in bin k is

$$C_k = \sum_{e \in C} \mathbb{I}(\theta_e \in B_k)\,. \tag{B.3.35}$$

All collision events are retained; repeated collisions of the same unordered pair therefore contribute as separate events.

Derived angular statistics. With $T = \sum_k T_k$, $C = \sum_k C_k$ and $F = \sum_k F_k$, the angular masses of near-contact pair-time, collision events and inward-motion exposure are $p_{\mathrm{near},k} = T_k/T$, $p_{\mathrm{coll},k} = C_k/C$ and $p_{\mathrm{flux},k} = F_k/F$. The bins have equal width, so the relative-to-isotropic distributions of Fig. 4.11(a) are $K p_{\mathrm{near},k}$ and $K p_{\mathrm{coll},k}$, each of unit value for an isotropic axial distribution. The pair-time-weighted inward and signed radial velocities are $A_{-,k} = F_k/T_k$ and $\langle w_r \mid B_k \rangle_T = W_k/T_k$, with global mean $\overline{A}_- = F/T = \sum_k p_{\mathrm{near},k} A_{-,k}$.

Relative collision selection and the yield per unit inward exposure are

$$R_k = \frac{p_{\mathrm{coll},k}}{p_{\mathrm{near},k}} = \frac{C_k/T_k}{C/T}, \qquad \mathcal{E}_{\mathrm{flux},k} = \frac{p_{\mathrm{coll},k}}{p_{\mathrm{flux},k}} = \frac{C_k/F_k}{C/F}, \tag{B.3.36}$$

so that $R_k = 1$ denotes collision occurrence proportional to the available near-contact pair-time. Since

$p_{\mathrm{flux},k} = p_{\mathrm{near},k} A_{-,k}/\overline{A}_-$,

$$\boxed{R_k = \mathcal{E}_{\mathrm{flux},k} \frac{A_{-,k}}{\overline{A}_-}} \tag{B.3.37}$$

exactly, separating selection into the inward motion supplied per unit available pair-time and the residual event yield per unit inward exposure.

Temporal uncertainty and effective support. The temporal rows of C_k, T_k, F_k, W_k, and $T_{-,k}$ are resampled jointly by an overlapping, non-circular moving-block bootstrap. The same row indices are applied to every accumulator, and all probability masses, normalisations, and ratios are recomputed within each replicate. The primary calculation uses 2,000 replicates and blocks of duration 2.0 time units. With selected-frame spacing 0.02, each block contains 100 consecutive temporal rows. The 96-time-unit record therefore contains approximately 48 block lengths; this describes temporal coverage and is not interpreted as 48 independent observations.

Let $\mathcal{S}$ denote the supported angular-bin set. The implementation requires, in each reported bin,

$$C_k \geq 50, \qquad N_k^{\mathrm{obs}} \geq 200, \qquad N_{-,k}^{\mathrm{obs}} \geq 100, \qquad T_k > 0, \qquad F_k > 0, \tag{B.3.38}$$

where

$$N_k^{\mathrm{obs}} = \sum_m \sum_{i<j} \chi_{ij,k}^{(m)}, \qquad N_{-,k}^{\mathrm{obs}} = \sum_m \sum_{i<j} \chi_{ij,k}^{(m)} \mathbb{I}\big(w_{r,ij}(t_m) < 0\big) \tag{B.3.39}$$

are unweighted saved-state observation counts used only to screen support. They do not replace the quadrature-weighted accumulators in the estimators.

Pointwise percentile intervals are used for the distributions and radial kinematics in Fig. 4.11(a,c). For $Q \in \{R, \mathcal{E}_{\mathrm{flux}}\}$, replicate b gives the maximum supported-bin log-deviation

$$D_Q^{(b)} = \max_{k \in \mathcal{S}} \left| \log Q_k^{(b)} - \log \widehat{Q}_k \right|. \tag{B.3.40}$$

If $c_{0.95,Q}$ is the 95th percentile of $\{D_Q^{(b)}\}$, the simultaneous band is

$$\widehat{Q}_k \exp(-c_{0.95,Q}) \leq Q_k \leq \widehat{Q}_k \exp(c_{0.95,Q}), \qquad k \in \mathcal{S}. \tag{B.3.41}$$

This multiplicative construction preserves positivity and controls the maximum curvewise deviation over the supported bins. A bootstrap replicate is discarded as a whole if R_k or $\mathcal{E}_{\mathrm{flux},k}$ is undefined or non-positive in any supported bin; the implementation requires at least 90% valid replicates.

Repeated events from the same unordered pair are summarised by the Kish effective count

$$n_{\mathrm{eff}} = \frac{\left(\sum_g m_g\right)^2}{\sum_g m_g^2} \simeq 27{,}945, \tag{B.3.42}$$

where m_g is the collision multiplicity of unordered pair g. This is a descriptive measure of repeated-event concentration, not the sample size of the temporal bootstrap.

Robustness of the angular-selection result. Four sensitivity checks are used to determine whether the angular-selection pattern depends materially on the near-contact shell, repeated-event weighting, temporal block length, or analysis interval.

First, the outer boundary of the near-contact shell is varied from the reference value 1.5σ to 1.2σ and 2σ. For each choice, the pair-time and inward-motion accumulators are recomputed from the corresponding shell; the collision-event population itself is unchanged.

Second, sensitivity to repeated collisions of the same unordered pair is examined by comparing the primary all-event estimator with two alternative event-weighting schemes. Let

$$\mathcal{G}_C = \{g : g \text{ contributes at least one event to } C\},$$

and let

$$m_g = \#\{e \in C : g(e) = g\}$$

be the number of retained events generated by pair g. The pair-balanced angular event count is

$$C_k^{\mathrm{pb}} = \sum_{e \in C} \frac{1}{m_{g(e)}} \mathbf{1}(\theta_e \in B_k), \tag{B.3.43}$$

so that every colliding unordered pair has total event weight one.

For the one-event-per-pair sensitivity check, let $e_g^{(1)}$ denote the earliest retained collision of pair g within the analysis interval $[104, 200]$. The corresponding event count is

$$C_k^{\mathrm{first}} = \sum_{g \in \mathcal{G}_C} \mathbf{1}\left(\theta_{e_g^{(1)}} \in B_k\right). \tag{B.3.44}$$

This is an earliest-retained-event statistic for the analysis window, not a claim that $e_g^{(1)}$ is the first collision in the complete history of the pair.

Writing

$$C^{(a)} = \sum_k C_k^{(a)}, \qquad a \in \{\mathrm{pb}, \mathrm{first}\},$$

the corresponding sensitivity curves are

$$R_k^{(a)} = \frac{C_k^{(a)}/C^{(a)}}{T_k/T}. \tag{B.3.45}$$

The near-contact pair-time denominator is kept unchanged, so these curves specifically test the sensitivity of the angular-selection result to event-side weighting. They should not be interpreted as new absolute collision rates.

Third, the simultaneous confidence bands are recomputed using temporal block durations of 1, 2, and 4 time units, while retaining the same joint moving-block resampling construction used for the primary estimator.

Fourth, temporal persistence is assessed by recomputing the estimators over the full interval $t \in [104, 200]$ and separately over the non-overlapping early and late intervals

$$[104, 152) \qquad \text{and} \qquad [152, 200].$$

All event counts, pair-time accumulators, and normalising totals are recomputed within the corresponding interval.

For any angular sector $\mathcal{K}$ formed from a specified set of angular bins, the integrated relative selection is

$$R_{\mathcal{K}} = \frac{\sum_{k \in \mathcal{K}} C_k/C}{\sum_{k \in \mathcal{K}} T_k/T}. \tag{B.3.46}$$

Thus, sector aggregation preserves the pair-availability correction; it is not an unweighted average of the

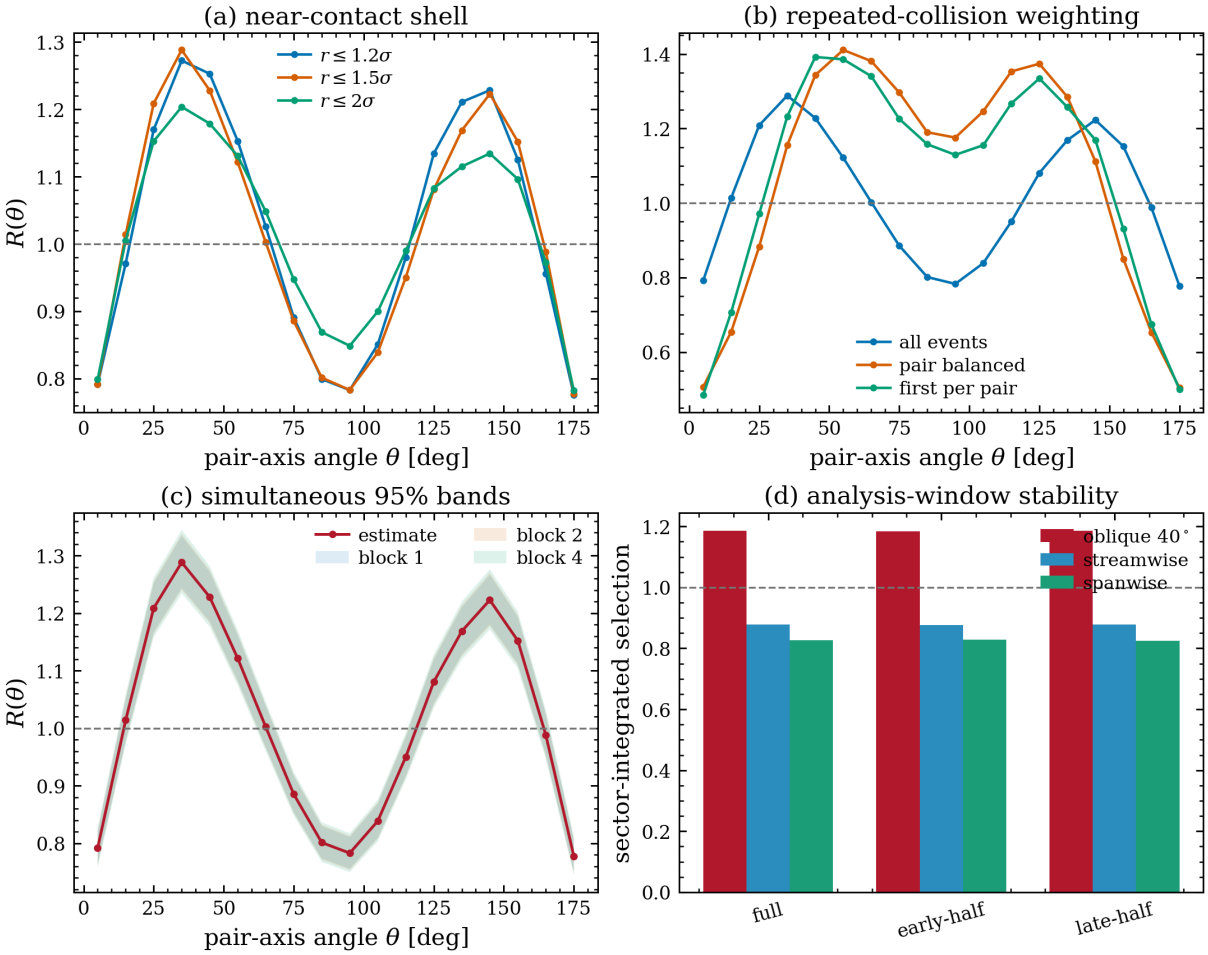

Figure B.7: **Robustness of angular collision selection for $\tau_p = 0.5$. (a)** Relative selection $R(\theta)$ for outer near-contact shell boundaries 1.2σ, 1.5σ, and 2σ. **(b)** Primary all-event selection and the alternative pair-balanced and earliest-retained-event-per-pair event-weighting estimates. **(c)** Simultaneous 95% bands obtained using temporal block durations of 1, 2, and 4 time units. **(d)** Sector-integrated relative selection over the complete analysis interval and its early and late halves. Unity denotes collision occurrence proportional to the corresponding near-contact pair-time exposure.

individual R_k.

The qualitative excess at oblique orientations relative to the streamwise direction persists when the outer shell boundary is varied between 1.2σ and 2σ, and the same ordering is present in both halves of the retained record [Fig. B.7(a,d)]. The conclusion drawn from the simultaneous bands is also insensitive to temporal block durations between 1 and 4 time units [Fig. B.7(c)].

The pair-balanced and earliest-retained-event-per-pair curves retain the suppression near the streamwise axis, although the detailed angular shape changes, particularly towards the spanwise direction [Fig. B.7(b)]. Thus, the preferential selection away from the streamwise axis is not a feature unique to the repeated-event weighting of the primary estimator. The detailed production curve and the flux-normalised yield reported in the main text nevertheless remain explicitly event-weighted quantities.

An independent numerical check of the particle-timestep dependence of the angular-selection statistics is reported in Table B.2.

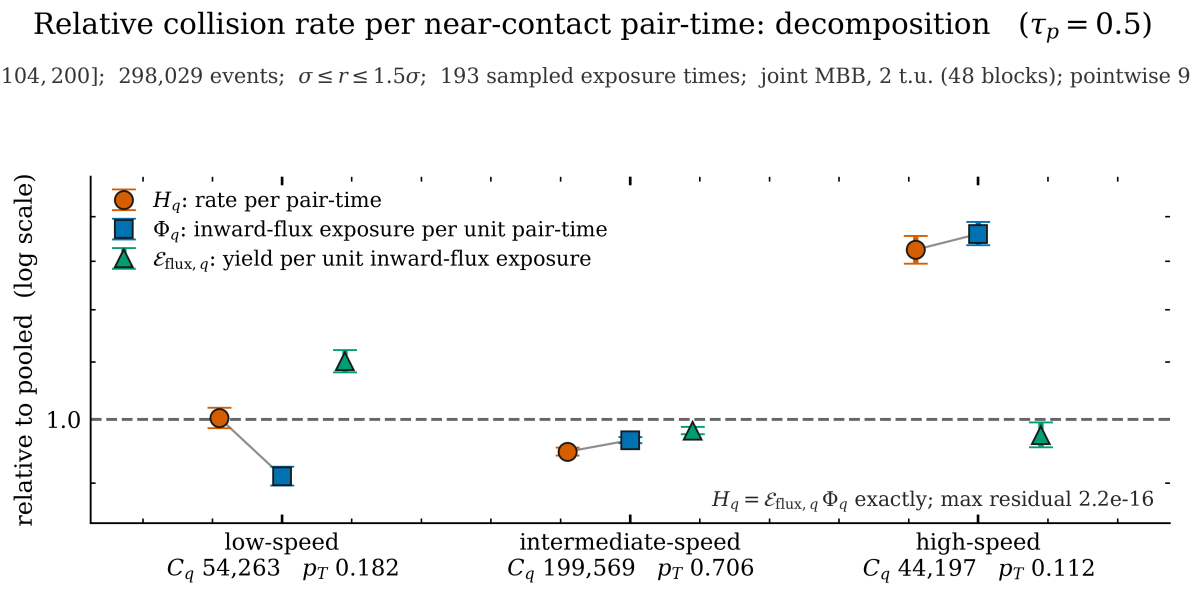

Figure B.8: **Carrier-speed-conditioned collision yield for $\tau_p = 0.5$.** Markers show the relative collision yield per unit inward-flux exposure, $\mathcal{E}_{\mathrm{flux},q}$; unity denotes collision occurrence proportional to the exposure supplied by the class. Statistics comprise 298,029 events and 193 exposure rows over $t = 104$–200. Red error bars and grey rectangles denote pointwise and simultaneous 95% intervals, respectively, obtained by joint circular moving-block resampling with 2.0-time-unit blocks. Class labels report $p_{\mathrm{flux},q}$ and $p_{\mathrm{coll},q}$. Changing the temporal pairing window changes the yields by at most 0.02; classification on the previous or next carrier snapshot changes them by less than 10^{-4}. Bilinear spatial classing is retained as a separate sensitivity check.

B.3.4 Environmental conditioning, sampling, and uncertainty

B.3.4.1 Carrier-Speed Inward-Flux Decomposition

This appendix gives the estimator supporting the carrier-speed result in subsec. 4.5.4. All quantities are relative finite-shell statistics evaluated over $\sigma \le r \le 1.5\sigma$; they are not absolute collision probabilities, hazards, or contact-limit kernels. For environmental class q, define the accumulated near-contact pair-time, inward-flux exposure, and collision count by

$$T_q = \sum_m \Delta t_{\mathrm{exp}} \sum_{i<j} \chi_q(\boldsymbol{x}_{ij}^{\mathrm{mid}}, t_m)\mathbb{I}(\sigma \le r_{ij} \le 1.5\sigma), \tag{B.3.47}$$

$$F_q = \sum_m \Delta t_{\mathrm{exp}} \sum_{i<j} \chi_q(\boldsymbol{x}_{ij}^{\mathrm{mid}}, t_m)\mathbb{I}(\sigma \le r_{ij} \le 1.5\sigma)[-w_{r,ij}]_+, \tag{B.3.48}$$

$$C_q = \sum_e \chi_q(\boldsymbol{x}_e, t_e), \tag{B.3.49}$$

where $[a]_+ = \max(a, 0)$, $\boldsymbol{x}_{ij}^{\mathrm{mid}}$ is the periodic pair midpoint, and $\boldsymbol{x}_e$ is the logged event position.

Particle states are stored every 0.02 time units. Retaining every 25th state gives

$$\Delta t_{\mathrm{exp}} = 25(0.02) = 0.5, \qquad M = 1 + \frac{200 - 104}{0.5} = 193. \tag{B.3.50}$$

Thus, 193 is the number of exposure rows, not the number of events or independent realisations. The estimators are formed from pooled sums over these rows, not from averages of rowwise ratios.

Each event is classified on the carrier snapshot nearest its own event time, giving a maximum carrier-association offset of 0.1 time units for the 0.2-unit carrier cadence. It is additionally assigned to the nearest exposure time, with maximum offset 0.25, solely to place its count in the appropriate bootstrap

row. This row assignment does not determine the event's carrier class.

Define

$$p_{T,q} = \frac{T_q}{T}, \qquad p_{\mathrm{flux},q} = \frac{F_q}{F}, \qquad p_{\mathrm{coll},q} = \frac{C_q}{C},$$

where $T = \sum_q T_q$, $F = \sum_q F_q$, and $C = \sum_q C_q$. The relative collision rate, inward-motion supply, and residual yield are

$$J_q = \frac{p_{\mathrm{coll},q}}{p_{T,q}}, \tag{B.3.51}$$

$$\Phi_q = \frac{p_{\mathrm{flux},q}}{p_{T,q}}, \tag{B.3.52}$$

$$\mathcal{E}_{\mathrm{flux},q} = \frac{p_{\mathrm{coll},q}}{p_{\mathrm{flux},q}}. \tag{B.3.53}$$

They satisfy the exact identity

$$J_q = \mathcal{E}_{\mathrm{flux},q}\Phi_q. \tag{B.3.54}$$

The low-, intermediate-, and high-speed classes have

$$p_{T,q} = 0.182,\ 0.706,\ 0.112, \qquad J_q \simeq 1.00,\ 0.95,\ 1.32, \qquad \Phi_q \simeq 0.91,\ 0.97,\ 1.36.$$

The corresponding residual yields are approximately 1.10, 0.98, and 0.97. Hence the elevated high-speed rate is explained by inward-motion supply, while the low-speed class has the largest yield after that supply is controlled.

B.3.4.2 One-particle environmental sampling

For particle frame k at time t_k, let $t_f^\star$ be the nearest carrier save time; the carrier plane at $y = a$ carries $N_g = N_x N_z = 256 \times 128 = 32{,}768$ points. For an observable X,

$$\begin{aligned}
\langle X\rangle_{E,k} &= \frac{1}{N_g}\sum_{j=0}^{N_z-1}\sum_{i=0}^{N_x-1} X_{ji}(t_f^\star), & s_{E,k} &= \left[\frac{1}{N_g}\sum_{j=0}^{N_z-1}\sum_{i=0}^{N_x-1}\left(X_{ji}(t_f^\star) - \langle X\rangle_{E,k}\right)^2\right]^{1/2},\\
\langle X\rangle_{P,k} &= \frac{1}{N}\sum_{n=1}^{N} \mathcal{I}_{\mathrm{bil}}[X]\left(\mathbf{x}_n(t_k), t_f^\star\right), & B_{X,k} &= \frac{\langle X\rangle_{P,k} - \langle X\rangle_{E,k}}{s_{E,k}},
\end{aligned} \tag{B.3.55}$$

with $N = 15{,}000$ and $\mathcal{I}_{\mathrm{bil}}$ periodic bilinear interpolation. Here $s_{E,k}$ is the population standard deviation of the plane, computed on each distinct carrier plane and never pooled across times, so $B_{X,k}$ is a standardised sampling bias measured in units of the ambient field fluctuation rather than of a standard error.

The reported statistic at nominal time t_c is the arithmetic mean of the 101 framewise biases in $t_c - 1 \le t \le t_c + 1$; at particle and carrier cadences of 0.02 and 0.2, such a window spans 11 distinct carrier planes, so consecutive frames may share a plane while sampling it at different particle positions. Both legs are processed identically and $\Delta B_X = B_X^{\mathrm{coll}} - B_X^{\mathrm{nocoll}}$.

Intervals for ΔB_X use 1,000 paired non-circular moving-block replicates with blocks of duration 0.5 (25 frames), the same block indices being applied to both legs. Detection floors follow from 250 random periodic translations of the evolved particle clouds, evaluated on every fifth frame of the window (21 frames) with the same translation applied to both legs; this preserves particle number, clustering and all pair separations while destroying alignment with the carrier field. Writing $B_X^{(d)}$ for a shift-null window

mean, the floor is the Gaussian-equivalent standard deviation of that null distribution,

$$f_X = \frac{Q_{97.5}\big(B_X^{(d)}\big) - Q_{2.5}\big(B_X^{(d)}\big)}{2\,\Phi^{-1}(0.975)}, \qquad 2\,\Phi^{-1}(0.975) = 3.919928, \tag{B.3.56}$$

and the figures display $\pm 2 f_X$. Separate floors are retained for each leg and for the paired difference.

B.3.5 Validation of Event-Centred Observables and Isolated-Pair References

This appendix documents the estimators and reduced-dynamics references used in the event-conditioned analysis of Section 4.5.5. It distinguishes the event-resolved collision time from quantities available only at the saved-state cadence, defines the treatment of repeated events and pair-axis memory, and specifies the collision-only and collision–drag references. These references are diagnostic reduced models; they are not counterfactual realisations of the same collisions with selected physical mechanisms removed.

Event weighting and pair dependence. The retained interval $104 \le t_e \le 200$ contains $N_{\mathrm{ev}} = 298{,}029$ collision events generated by $G = 67{,}479$ distinct unordered pairs. The all-event estimand assigns equal weight to each collision, so recurrent pairs contribute in proportion to their collision multiplicity. The first-ever and terminal-event subsets are identified from the complete collision history of each unordered pair before restriction to the retained analysis window.

Repeated events from the same pair are not treated as statistically independent when estimating uncertainty. At each lag, event-level contributions are grouped by unordered pair when computing the pair-cluster-robust standard error. Repeated impacts therefore remain separate observations in the event-weighted mean while their within-pair dependence is retained in the uncertainty estimate. The shading in Fig. 4.14 denotes ± 1 such standard error.

Periodic pair geometry. All pair separations are evaluated using the minimum-image convention in the streamwise and spanwise periodic directions. An explicit branch audit, performed over lags extending to $|\tau| = 50$, detected no minimum-image branch crossing for the audited near-contact pairs. The event-centred statistics are therefore unaffected by periodic-image ambiguity over the reported lag support.

Event times and cadence-resolved states. For collision event e, the collision log provides its event-resolved time t_e and unordered pair identity. Finite-lag positions and velocities are obtained from the saved particle states. If the saved times are t_m, the state associated with requested lag τ is

$$m_e(\tau) = \arg\min_m |t_m - (t_e + \tau)|, \tag{B.3.57}$$

subject to

$$t_m < t_e \quad \text{for } \tau < 0, \qquad t_m > t_e \quad \text{for } \tau > 0.$$

Pre- and post-impact quantities are therefore always evaluated on the corresponding side of contact. Lags extending beyond the available record are omitted rather than assigned an endpoint state.

The saved-state interval is

$$\Delta t_s = 0.02,$$

so the temporal mismatch between the requested lag and the selected state is at most 0.01. No measured radial velocity or separation is assigned at $\tau = 0$; the closest displayed states are at $\tau = \pm 0.02$. For the pair-axis memory, the reference axis $\theta^{s}_{e,0}$ is obtained from the saved state nearest t_e, without a side constraint. This is admissible because the particle positions, and hence the line of centres, are continuous through an instantaneous hard-particle collision.

Collision–drag reference. The second reference retains the same collision law and additionally evolves the relative velocity under isolated homogeneous Stokes relaxation,

$$\frac{\mathrm{d}\Delta\mathbf{v}}{\mathrm{d}\tau} = -\frac{\Delta\mathbf{v}}{\tau_p}. \tag{B.3.58}$$

Using $\Delta\mathbf{v}^-$ on the negative-lag branch and $\Delta\mathbf{v}^+$ on the positive-lag branch gives

$$\Delta\mathbf{v}_{\mathrm{drag}}(\tau) = \Delta\mathbf{v}^{q(\tau)} \exp\!\left(-\frac{\tau}{\tau_p}\right), \tag{B.3.59}$$

$$\mathbf{r}_{\mathrm{drag}}(\tau) = \mathbf{r}_0 + \tau_p \Delta\mathbf{v}^{q(\tau)} \left[1 - \exp\!\left(-\frac{\tau}{\tau_p}\right)\right], \tag{B.3.60}$$

where

$$q(\tau) = \begin{cases} -, & \tau < 0, \\ +, & \tau > 0. \end{cases}$$

The negative-lag branch is a backward continuation of the reduced Stokes equation, not a reconstruction of the measured pre-impact trajectory. Because this continuation amplifies the relative velocity as $\exp(|\tau|/\tau_p)$, it is retained only while

$$\exp\!\left(\frac{|\tau|}{\tau_p}\right) \le 3 \qquad \Longleftrightarrow \qquad |\tau| \le \tau_p \ln 3. \tag{B.3.61}$$

For $\tau_p = 0.5$, the negative-lag limit is

$$\tau_p \ln 3 \simeq 0.549,$$

whereas the positive-lag branch is retained to $3\tau_p = 1.5$, by which point the relative speed has decreased to $e^{-3} \simeq 0.05$ of its immediate post-impact value. Pre/post comparisons involving the collision–drag reference use only lag magnitudes supported on both sides:

$$|\tau| = 0.02,\ 0.05,\ 0.10,\ 0.25,\ 0.50.$$

The isolated references use a flux-weighted proxy for impact kinematics. The event log records the collision time and pair identity but not the relative velocity at impact. The isolated references therefore estimate the joint pre-impact distribution of the normal and tangential relative velocities from 32,056 approaching-pair observations ($w_r < 0$) in the saved shell $\sigma \le r \le 1.5\sigma$. These observations are collected from twelve frames distributed across $104 \le t \le 200$.

A saved frame samples pairs occupying the shell, whereas a collision samples pairs crossing towards contact. Shell observation j is consequently resampled with probability

$$\pi_j = \frac{[-w_{r,j}]_+}{\sum_{\ell=1}^{N_s} [-w_{r,\ell}]_+}, \qquad [a]_+ \equiv \max(a, 0), \qquad N_s = 32{,}056. \tag{B.3.62}$$

This inward-flux weighting retains the empirical pairing of w_r and w_t while reweighting the approaching near-contact population according to its inward crossing flux. It therefore provides a population-level proxy for pre-impact kinematics rather than an event-by-event reconstruction of the logged collisions.

Each of the 200,000 Monte Carlo realisations is initialised at synthetic contact:

$$\boldsymbol{r}_0 = \sigma\widehat{\boldsymbol{n}}, \qquad \Delta\boldsymbol{v}^- = w_r^-\widehat{\boldsymbol{n}} + w_t^-\widehat{\boldsymbol{t}}. \tag{B.3.63}$$

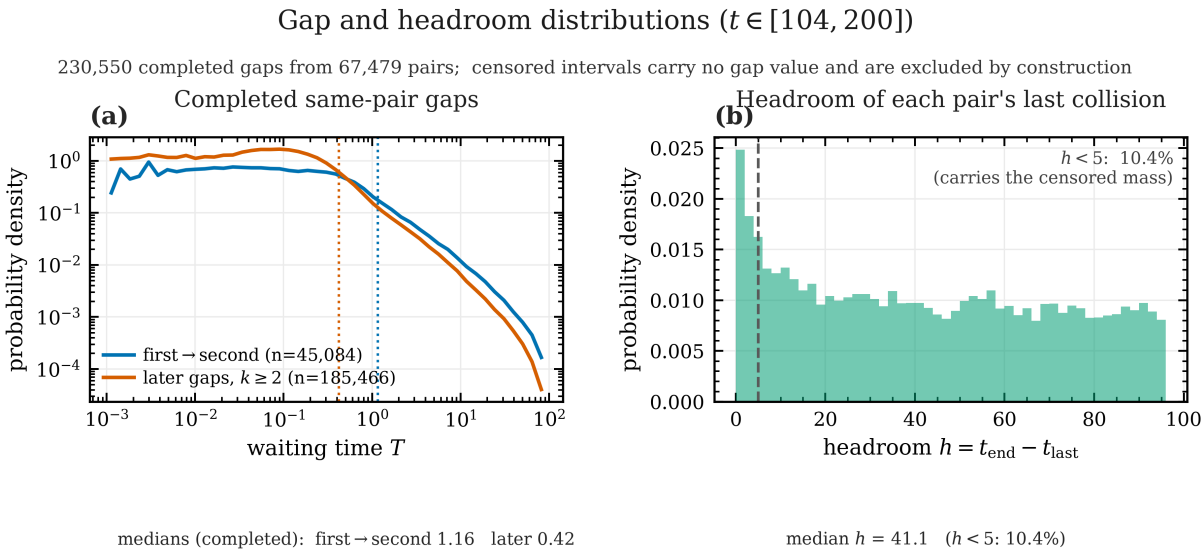

Figure B.9: **Completed same-pair gaps and terminal-event headroom for $\tau_p = 0.5$. (a)** Probability-density estimates of completed first-to-second (median 1.16) and later (median 0.42) same-pair gaps. Censored intervals carry no completed-gap value and are excluded by construction. **(b)** Distribution of the headroom $h_g = t_{\text{end}} - t_g^{\text{term}}$ following each pair's terminal retained collision; median 41.1, with 10.4% of terminal events below $h_g = 5$. Panel (b) audits proximity to the record boundary and is not an estimator of future recurrence probability.

The sampled shell position is discarded. Contact normals are sampled isotropically because the isolated equations are rotationally invariant and the reference concerns motion relative to the initial pair axis, not its absolute orientation in the channel.

For a frictionless collision with normal restitution coefficient ε_n,

$$\Delta \boldsymbol{v}^+ = \Delta \boldsymbol{v}^- - (1 + \varepsilon_n)\,(\Delta \boldsymbol{v}^- \cdot \widehat{\boldsymbol{n}})\,\widehat{\boldsymbol{n}}, \tag{B.3.64}$$

so that

$$w_r^+ = -\varepsilon_n w_r^-, \qquad w_t^+ = w_t^-. \tag{B.3.65}$$

For the present elastic collisions, $\varepsilon_n = 1$. The impulse reverses the normal relative velocity while leaving the tangential component and instantaneous particle positions unchanged.

Terminal-event headroom quantifies proximity to the record end. For the terminal recorded collision of unordered pair g, the available follow-up time is

$$h_g = t_{\text{end}} - t_g^{\text{term}}, \qquad t_{\text{end}} = 200. \tag{B.3.66}$$

The median headroom is 41.1, much larger than the post-impact lag range analysed in Fig. 4.14, and 89.6% of terminal events have $h_g > 5$. The sustained recession of the terminal class is therefore not a truncation effect of the record boundary. Terminal events are only mildly enriched near the end of the record relative to uniform sampling over $t \in [104, 200]$ (10.4% against 5.2% expected for $h_g < 5$), as expected for the last collision of a pair. This audit quantifies the available follow-up only; it does not estimate whether a pair would collide again after $t = 200$ and therefore does not establish permanent escape.

Interpretive scope. The collision-only and collision–drag curves describe the evolution of a flux-weighted proxy for the measured contact population under the specified reduced equations. Both omit the spatially and temporally varying carrier forcing, interactions with neighbouring particles, and subsequent collisions involving either focal particle; the collision-only reference additionally omits Stokes relaxation.

They therefore do not represent how the same realised DNS events would have evolved had the focal collision been removed.

Accordingly, a difference between a measured event-centred observable and an isolated reference identifies the combined influence of dynamics absent from that reference. It cannot be identified with the causal effect of the focal collision alone. A genuine event-level collision counterfactual would require a matched rerun in which the focal impulse is suppressed while the pre-impact state, carrier forcing, and all other model conditions are otherwise retained.

B.3.6 Validation of same-pair multiplicity and spatial recurrence

This appendix documents the estimators of Sec. 4.5.6, separating quantities obtained exactly from the collision-event log from quantities resolved from particle states saved at intervals $\Delta t_{\mathrm{save}} = 0.02$. All results use $\tau_p = 0.5$ and the retained interval $t \in [104, 200]$.

B.3.6.1 Multiplicity and exact waiting times

Collisions are assigned to the unordered pair $g = \{\min(i, j), \max(i, j)\}$. If $t_{g,1} < \cdots < t_{g,M_g}$ are the collision times of g in the retained record, its observed multiplicity is M_g and the plotted tail probability is

$$\widehat{P}(M \geq m) = \frac{1}{G} \sum_{g=1}^{G} \mathbf{1}_{M_g \geq m}, \tag{B.3.67}$$

with $G = 67{,}479$ pairs colliding at least once. The record contains $C = 298{,}029$ collisions, so

$$\sum_g M_g = C, \qquad \sum_g (M_g - 1) = C - G = 230{,}550, \tag{B.3.68}$$

both of which are used as counter checks. Hence 77.4% of events follow an earlier collision of the same pair. The mean, median and maximum multiplicities are 4.42, 2 and 94; the tail values are $\widehat{P}(M \geq 2) = 0.6681$, $\widehat{P}(M \geq 3) = 0.4960$, $\widehat{P}(M \geq 5) = 0.3065$ and $\widehat{P}(M \geq 10) = 0.1152$. No parametric tail model is fitted. Because collisions outside $[104, 200]$ are unavailable, M_g is an observed finite-record property, not an intrinsic property of the pair.

Waiting times $T_{g,k} = t_{g,k+1} - t_{g,k}$ are taken directly from the event times and are unaffected by the saved-state cadence.

B.3.6.2 Resolved maximum excursion

Pair separations $r_g(t_n)$ use the minimum-image convention. For repeat interval (g, k), let $\mathcal{N}_{g,k} = \{n : t_{g,k} < t_n < t_{g,k+1}\}$ be the saved states strictly between the two collisions. Where $\mathcal{N}_{g,k} \neq \varnothing$,

$$\widehat{R}_{\max,gk} = \max_{n \in \mathcal{N}_{g,k}} \frac{r_g(t_n)}{d} \leq R^{\mathrm{true}}_{\max,gk} \equiv \max_{t_{g,k} < t < t_{g,k+1}} \frac{r_g(t)}{d}, \tag{B.3.69}$$

the hat distinguishing the sampled maximum from the continuous-time maximum. The inequality gives $\widehat{F}_R(R) \geq P(R^{\mathrm{true}}_{\max} \leq R)$: the resolved distribution is an upper bound on continuous-time confinement.

Of the 230,550 completed repeat intervals, 228,437 contain at least one interior saved state. The remaining 2,113 (0.9%) complete within one save step; $\widehat{R}_{\max}$ is undefined there and they are excluded rather than assigned the contact value 1. At the measured near-contact relative speed, $w_{\mathrm{nc}} \Delta t_{\mathrm{save}} \sim 10^{-3} \simeq 0.1d$, so these intervals would be expected to fall inside the local population had their interior motion been resolved. Their exclusion therefore biases the reported local fraction downward, opposite in sign to the cadence bound above.

Table B.4: Resolved local-recurrence fraction $\widehat{F}_R(R)$ over the 228,437 supported intervals (left), and its sensitivity at $R = 2$ to the retained saved-state stride (right).

R	$\widehat{F}_R(R)$	Stride	Spacing	$\widehat{F}_R(2)$
1.5	0.8179	1	0.02	0.8788
2.0	0.8788	2	0.04	0.8775
2.5	0.9093	4	0.08	0.8741
5.0	0.9647			

B.3.6.3 Threshold and cadence sensitivity

The main analysis uses $R = 2$ as an operational boundary for the local neighbourhood; the adjacent thresholds in Table B.4 show that the conclusion does not rest on a single sharply chosen value. Coarsening the cadence changes both the supported-interval population and the number of states available to locate a maximum, yet moves $\widehat{F}_R(2)$ by at most 0.0047. None of these thresholds defines a bound state, a mechanically closed cage, or an attractive interaction.

B.3.6.4 Interval and equal-pair weighting

Let $\mathcal{K}_g = \{k : \mathcal{N}_{g,k} \neq \varnothing\}$ and $n_g^{\mathrm{sup}} = |\mathcal{K}_g|$. The within-pair local fraction is

$$c_g(R_c) = \frac{1}{n_g^{\mathrm{sup}}} \sum_{k \in \mathcal{K}_g} \mathbf{1}_{\widehat{R}_{\max,gk} \le R_c}, \tag{B.3.70}$$

from which the interval-weighted and pair-balanced estimators follow:

$$f_{\mathrm{local}}^{\mathrm{int}}(R_c) = \frac{\sum_g n_g^{\mathrm{sup}} c_g(R_c)}{\sum_g n_g^{\mathrm{sup}}}, \qquad f_{\mathrm{local}}^{\mathrm{pair}}(\mathcal{G}; R_c) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} c_g(R_c). \tag{B.3.71}$$

The first weights a pair by its number of supported intervals; the second gives every pair equal weight regardless of how many intervals it supplies.

Table B.5: Local-recurrence fractions at $\widehat{R}_{\max} \le 2$. Brackets are percentile intervals from 400 pair-bootstrap replicates.

Multiplicity class	Pairs	Pair-balanced	Interval-weighted
$M = 2$	11,547	0.728 [0.720, 0.737]	0.728
$3 \le M < 10$	25,694	0.822 [0.819, 0.824]	0.840
$M \ge 10$	7,774	0.919 [0.917, 0.921]	0.928
All supported	45,015	0.815	0.879

The two estimators coincide for $M = 2$, where $n_g^{\mathrm{sup}} = 1$ by construction; this is used as a code check. They separate overall (0.815 against 0.879) because pairs with many collisions supply many intervals and also recur more locally. The monotonic ordering across classes survives equal-pair weighting, so the association between high multiplicity and small inter-collision excursion is not produced by a few heavily sampled pairs. The 45,015 contributing pairs are fewer than the $\widehat{P}(M \ge 2)G \simeq 45{,}084$ recurrent pairs because a small number have no supported interval.